



GumNaaM_Helpers



Mega File for MiD TerM

All Files in ONE FILE

(Handouts_waqar_moaz_junaaid_vu toper_orange_monkey)

Alhamdulillah GumNaaM_Helpers Bs..IT/CS ky Tamam Semesters ki tamam Books ki help krty hn.

Group Join krty hi description Read krna na bholyn

Semester Name	Group Link
Bs..IT/CS...smstr:1	https://chat.whatsapp.com/Ciya2X0DI0l8iQ14m0VNNj
Bs..IT/CS...smstr:2	https://chat.whatsapp.com/Kwh85sRK02z1Qk5B7aMR19
Bs..IT/CS...smstr:3	https://chat.whatsapp.com/ljlpdzi8pIA4lQVNUMmNYE
Bs..IT/CS...smstr:4	https://chat.whatsapp.com/CiGpHsSftir0oyhM47WfKh
Bs..IT/CS...smstr:5	https://chat.whatsapp.com/Hf7NWWhdW06uCW7Mbt1n4Db
Bs..IT/CS...smstr:6	https://chat.whatsapp.com/DyAoT7VFkwy8DuvrYDgm2L
Bs..IT/CS...smstr:7	https://chat.whatsapp.com/ENiyfIDRWatJFPbXCUGYNN
Bs..IT/CS...smstr:8	https://chat.whatsapp.com/Bnkk4n7utpdEuXazHLxRcA

...Paid Service Available for LMS Activities...

Only for those students who are so busy in office or work or job that they can't watch their LMS due to their Busyness

FOR MORE INFO. CONTACT US ONLY WHAT'S APP 0302-1500025

D_C_Chapter_01_Topic 1 to 17.pptx

D_C_Chapter_02_Topic 18 to 28.pptx

D_C_Chapter_03_Topic 29 to 53 (Complete).pptx

D_C_Chapter_04_1Topic 54 to 60.pptx

D_C_Chapter_04_2_Topic 61 to 75.pptx

D_C_Chapter_05_Topic 76 to 88.pptx

D_C_Chapter_06_Topic 89 to 100.pptx

D_C_Chapter_07_Topic 101 to 113.pptx

D_C_Chapter_08_Topic 114 to 123.pptx

D_C_Chapter_09_Topic 124 to 134.pptx

D_C_Chapter_10_Topic 135 to 148.pptx

D_C_Chapter_11_Topic 149 to 165.pptx

D_C_Chapter_12_Topic 166 to 179.pptx

D_C_Chapter_13_Topic 180 to 190.pptx

D_C_Chapter_14_1_Topic 191 to 196.pptx

D_C_Chapter_14_2_Topic 197 to 204.pptx

D_C_Chapter_15_1_Topic 205 to 210.pptx

D_C_Chapter_15_2_Topic 211 to 212.pptx

D_C_Chapter_16_Topic 213 to 220.pptx

Introduction to the Course

**The Need for
Communication**

Trends and Advancements

**What is taught in this
course?**

**What is NOT taught in this
course?**

Tips and Tricks to do well

Text and References

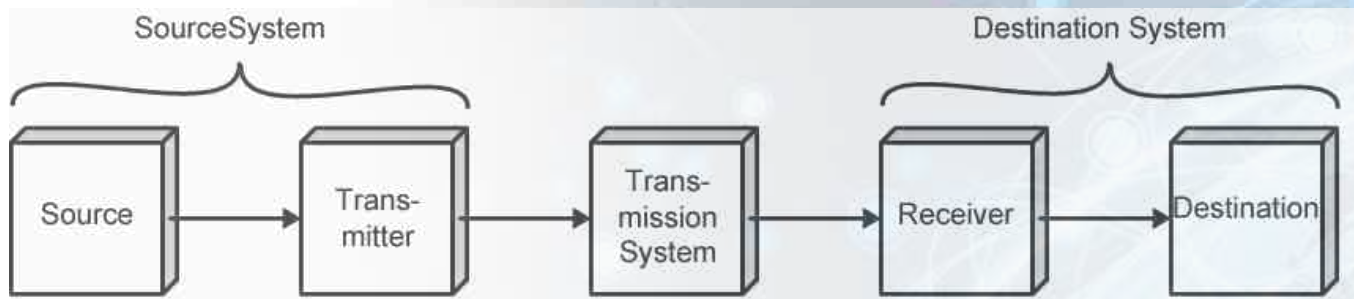
Introduction to Data Communications

Communication: Sharing of Information (Local or remote)

Telecommunications: Communication at a Distance (includes telephony, telegraph, and television etc.)

Data communications: Exchange of data between two devices via some form of transmission media

A Simple Communication Model



(a) General block diagram



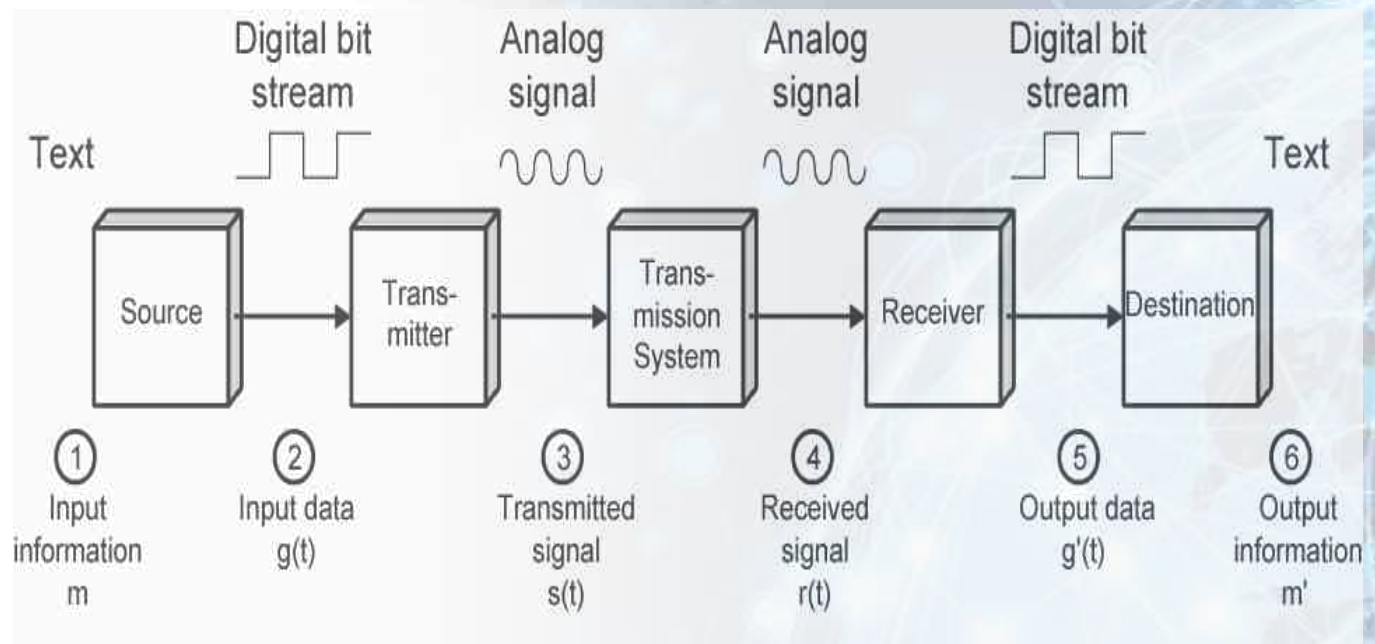
(b) Example

Characteristics of a Data Communication System

Effectiveness of a Data Communication System:

- **Delivery**
- **Accuracy**
- **Timeliness**
- **Jitter**

Characteristics of a Data Communication System

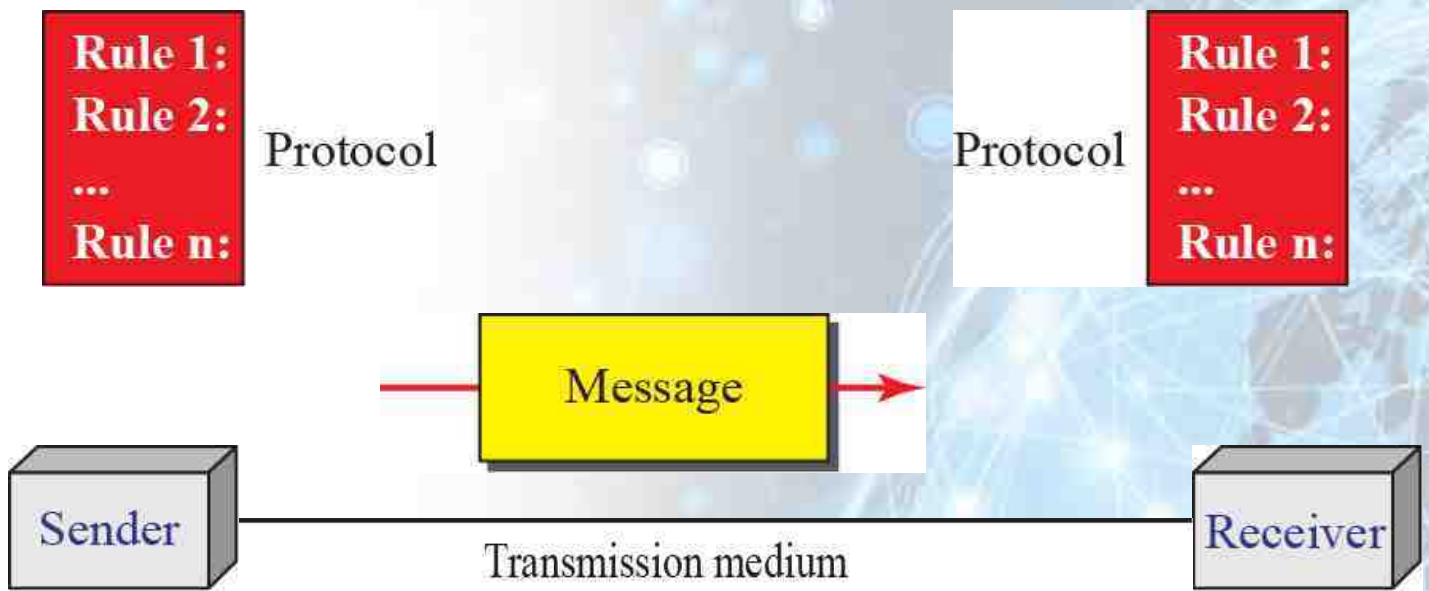


Components of a Data Communication system

A data communications system has five components



Components of Data Communication system



Data Representation and Data Flow

Forms of Information

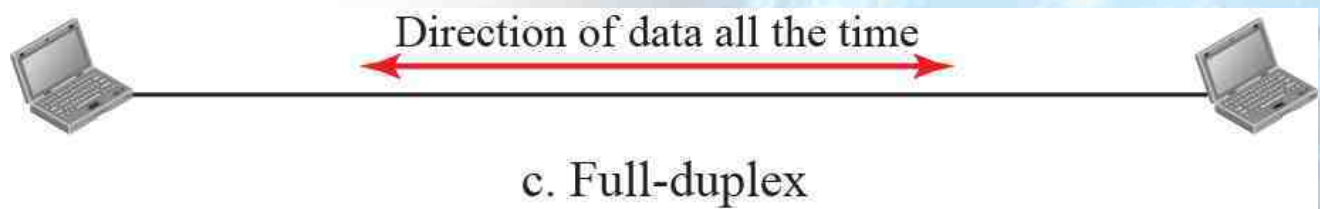
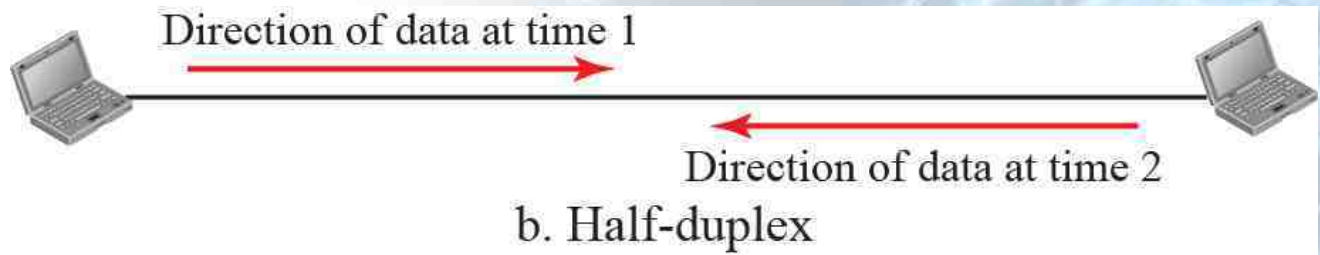
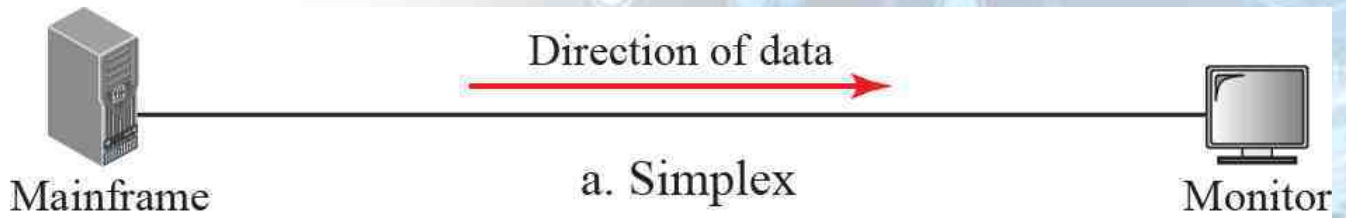
- Text
- Numbers
- Images
- Audio
- Video

Data Representation and Data Flow

Data Flow between two devices:

- **Simplex**
- **Half-Duplex**
- **Full-Duplex**

Data Flow



Networks

- 
- A background graphic showing a globe with a network of glowing blue lines and nodes, representing global connectivity.
- **Network:**
 - Interconnection of a set of devices capable of communication
 - **Host**
 - **Connecting Device**

Network Criteria

A network must be able to meet a certain number of criteria:

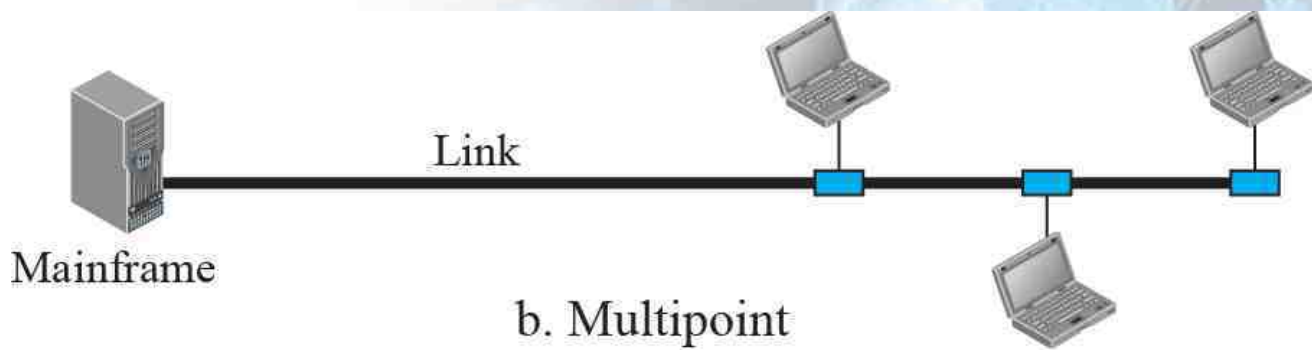
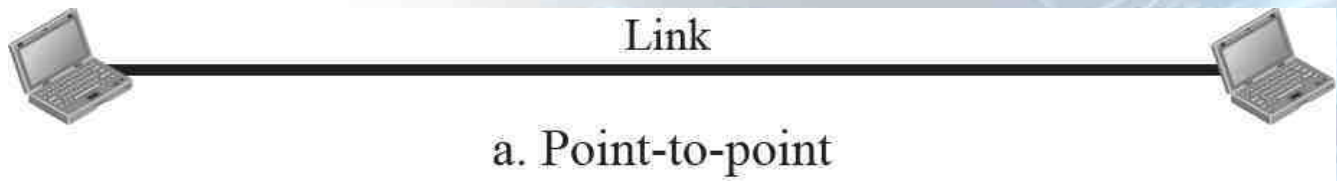
- **Performance**
 - ✓ Throughput
 - ✓ Delay
- **Reliability**
- **Security**

Physical Structures

Physical Network Attributes

- **Link**
- **Type of Connection**
 - ✓ **Point-to-Point**
 - ✓ **Multipoint**

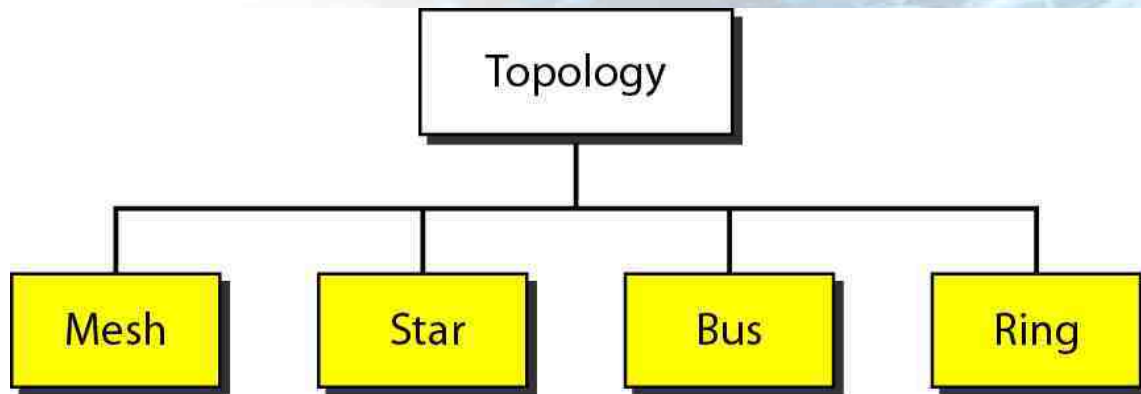
Physical Structures



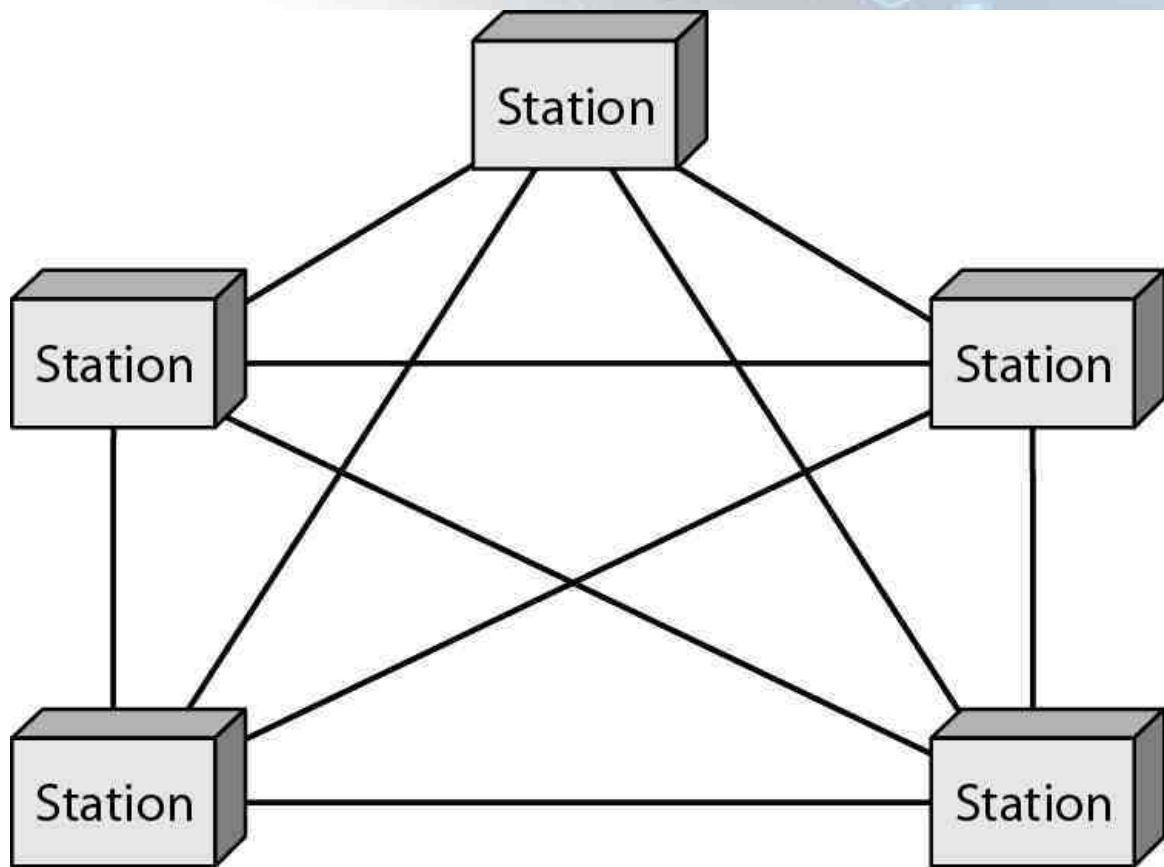
Physical Topologies

- **Physical Layout of Network**
- **Links + Nodes = Topology**
- **Physical Topologies:**
 - ✓ **Mesh**
 - ✓ **Star**
 - ✓ **Bus**
 - ✓ **Ring**

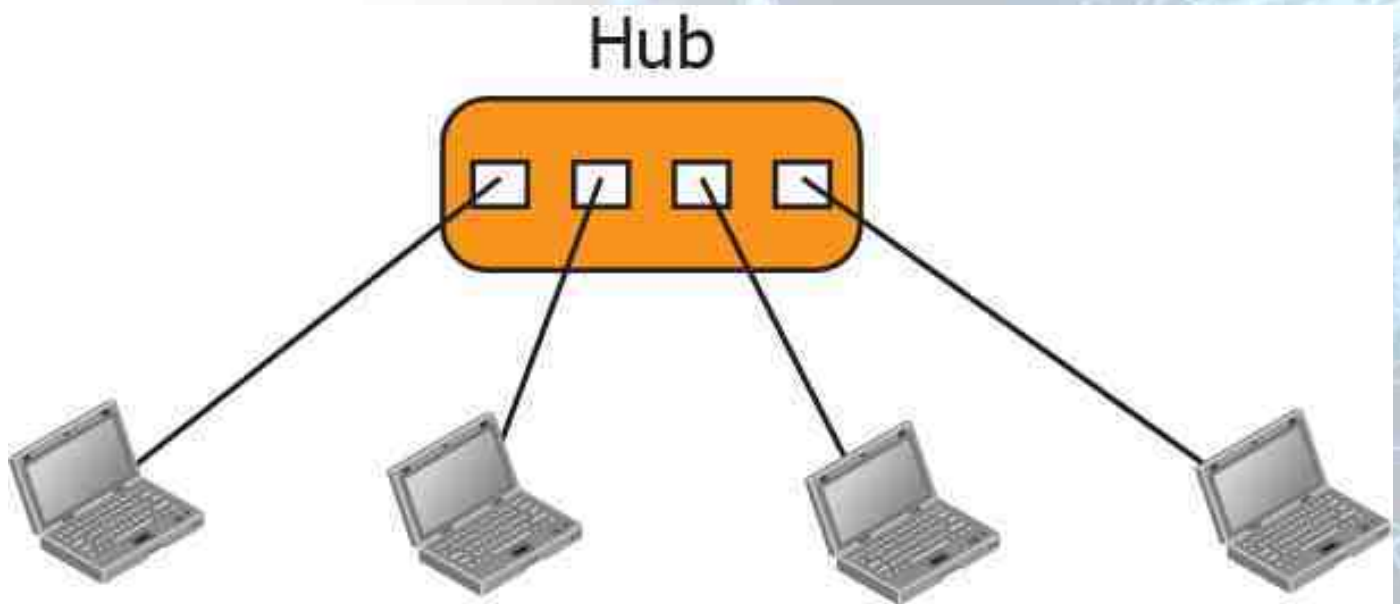
Physical Topologies



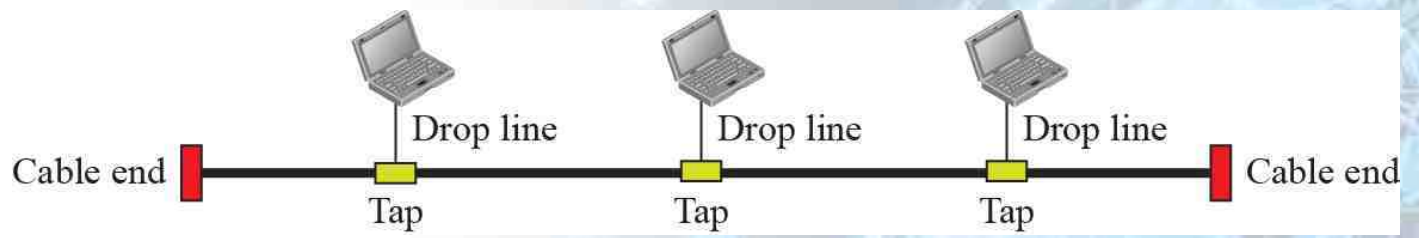
Mesh Topology



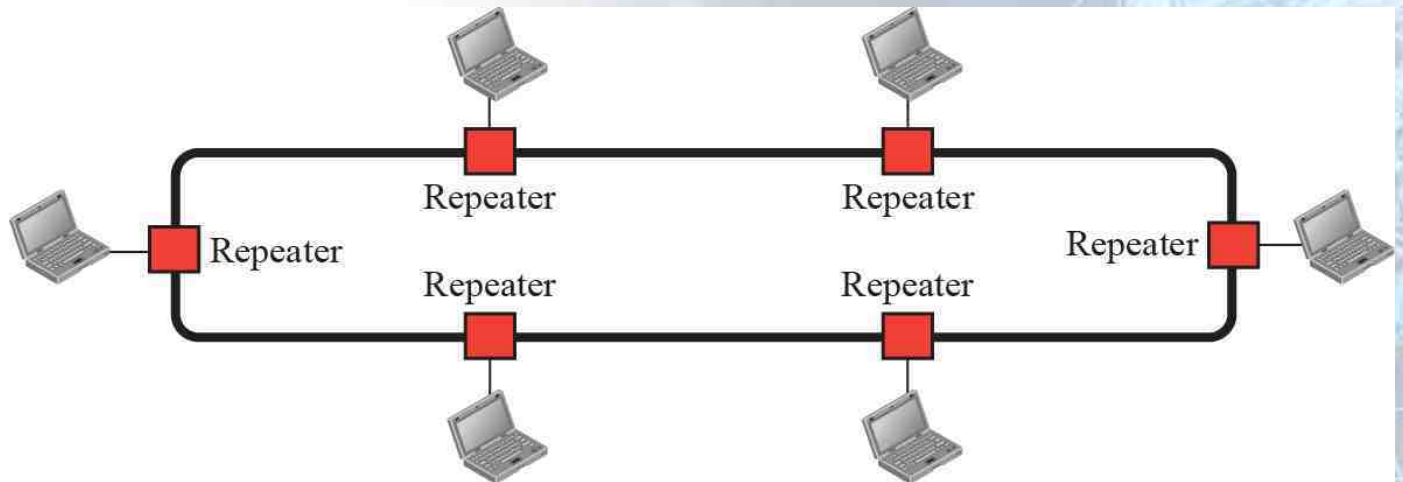
Star Topology



Bus Topology



Ring Topology



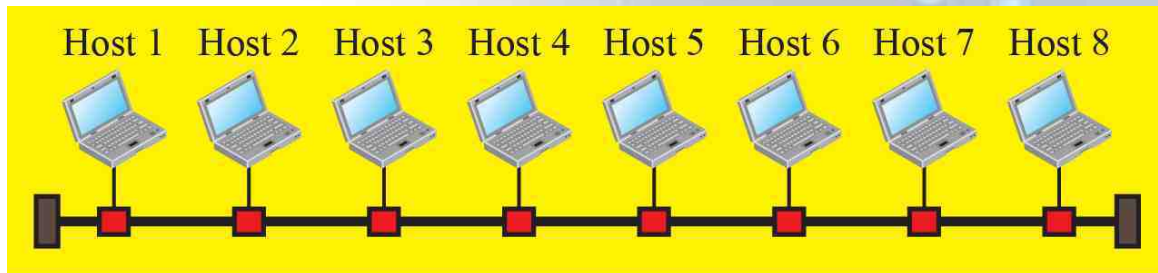
Networks Types

- **Network classification:**
 - ✓ **Size**
 - ✓ **Geographical Coverage**
 - ✓ **Ownership**
- **Local Area Networks (LANs)**
- **Wide Area Networks (WANs)**

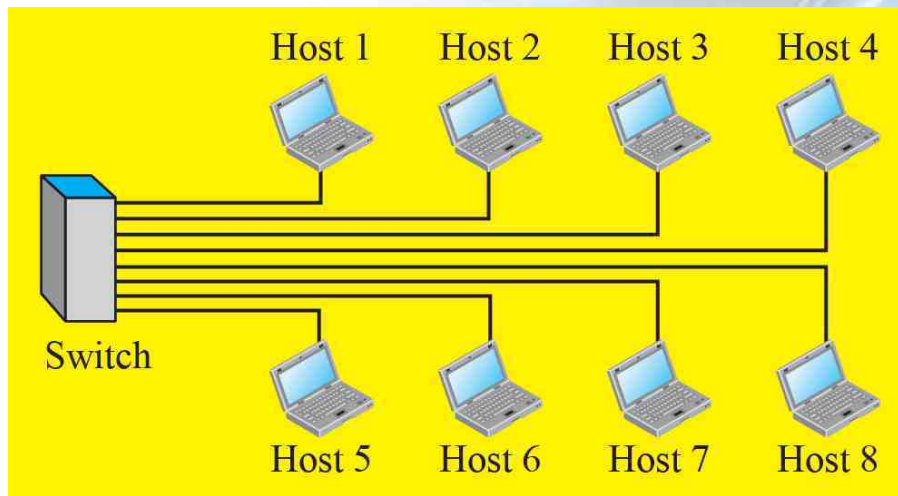
Local Area Networks

- **Usually Privately owned**
- **Connects some hosts in a single office, building, or campus**
- **Can be as simple as two PCs and a printer in someone's home office**
- **Can extend throughout a company**
- **Host Address**

Local Area Networks



a. LAN with a common cable (past)



b. LAN with a switch (today)

Legend

-  A host (of any type)
-  A switch
-  A cable tap
-  A cable end
-  The common cable
-  A connection

Wide Area Network

- Wider geographical span than a LAN
- Spans a town, a state, a country, or even the world
- Interconnects connecting devices such as switches, routers, or modems
- Normally created and run by communication companies

Wide Area Network

- **Point-to-Point WAN**
- **Switched WAN**
- **Internetwork**



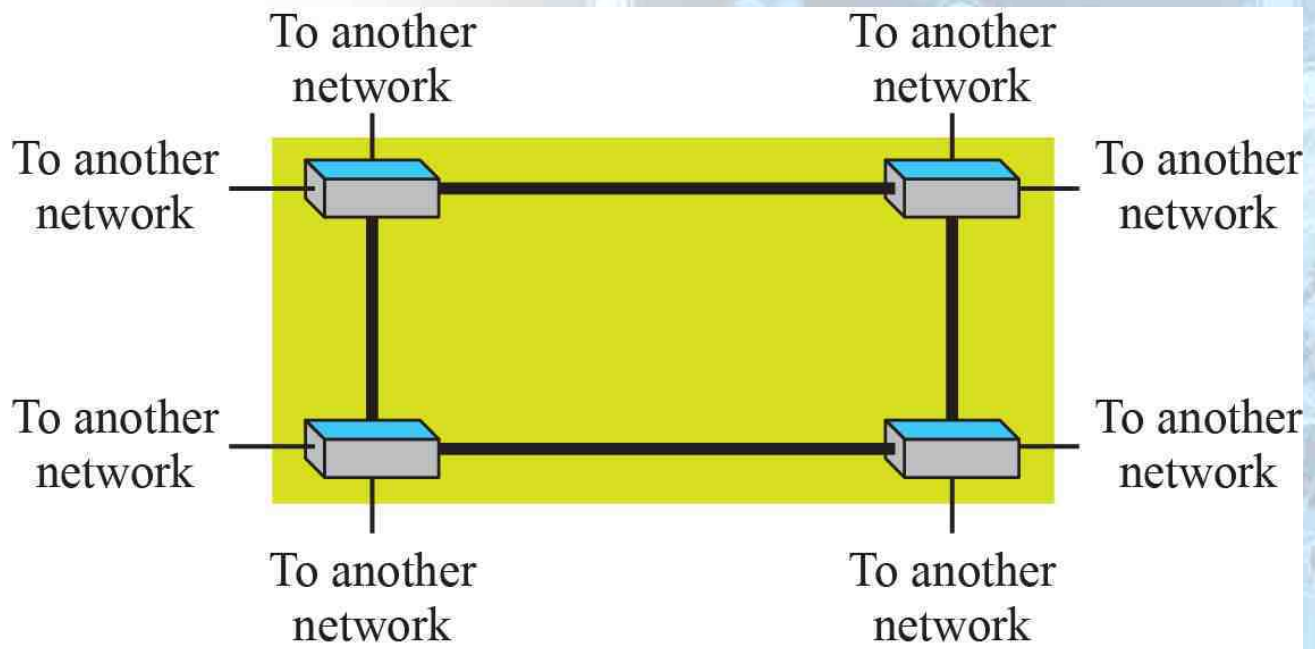
Point-to-Point WANs



Legend

-  A connecting device
-  Connecting medium

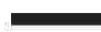
Switched WANs



Legend

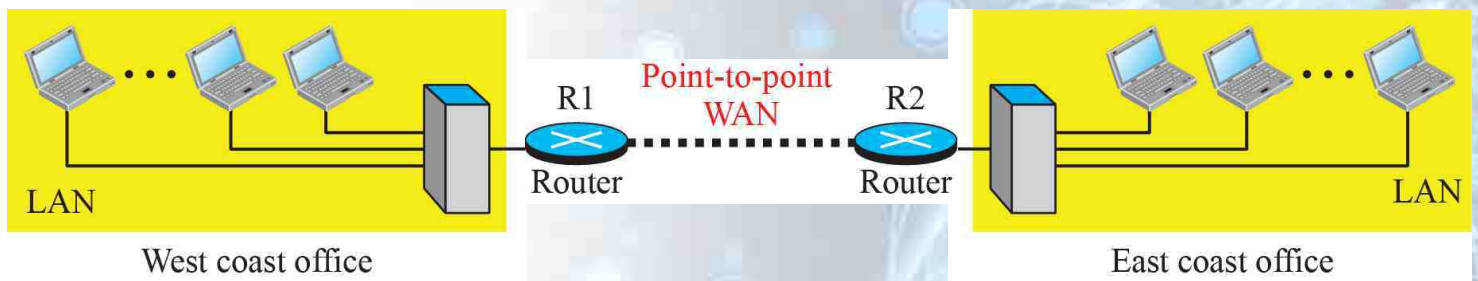


A switch



Connecting medium

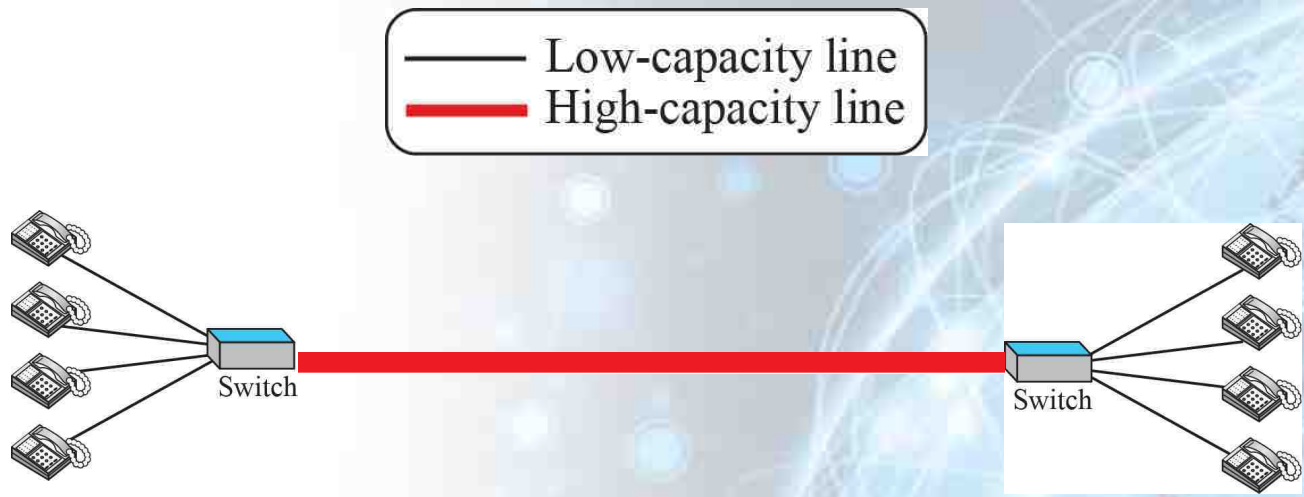
Internetwork



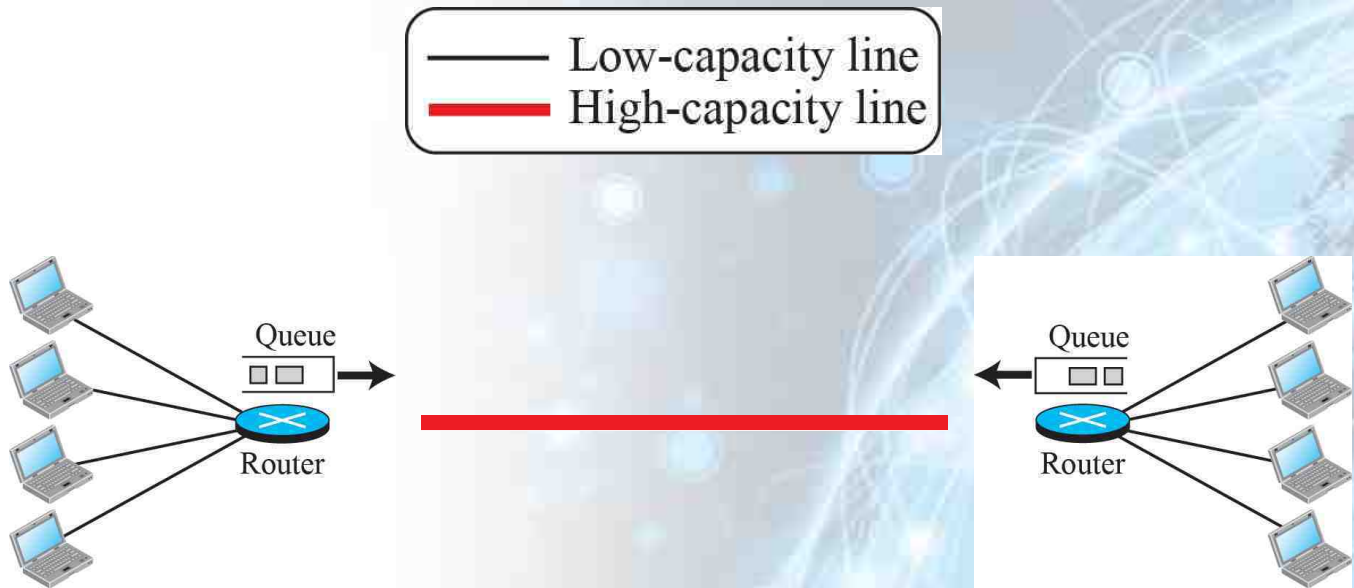
Switching

- **Switching**
 - ✓ **Circuit-Switched Network**
 - ✓ **Packet- Switched Network**

Circuit Switched Network



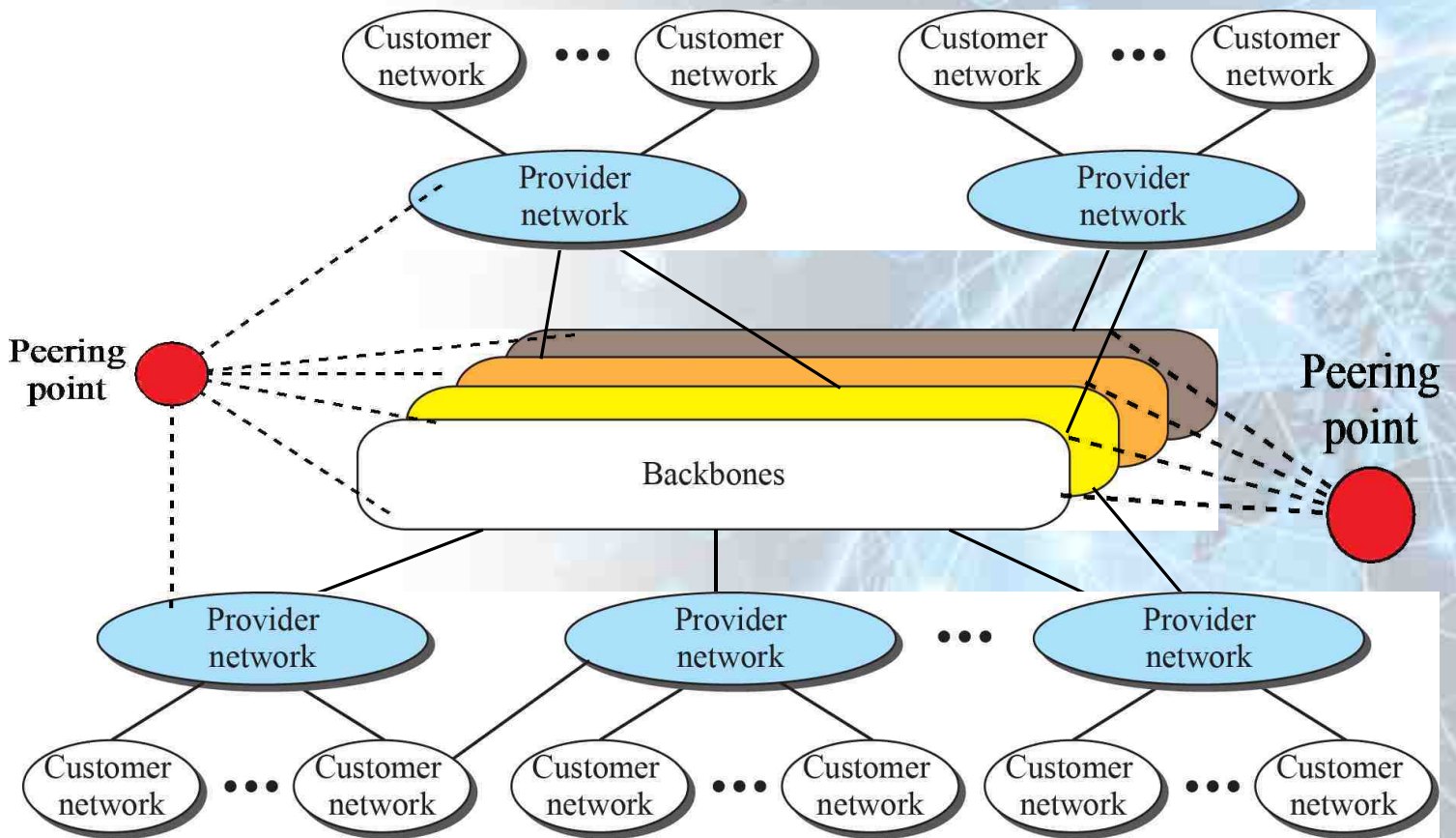
Packet Switched Network



The Internet

- An internet (note the lowercase i) is two or more networks that can communicate with each other
- The Internet (uppercase I), and is composed of thousands of interconnected networks.
- Accessing the Internet

The Internet



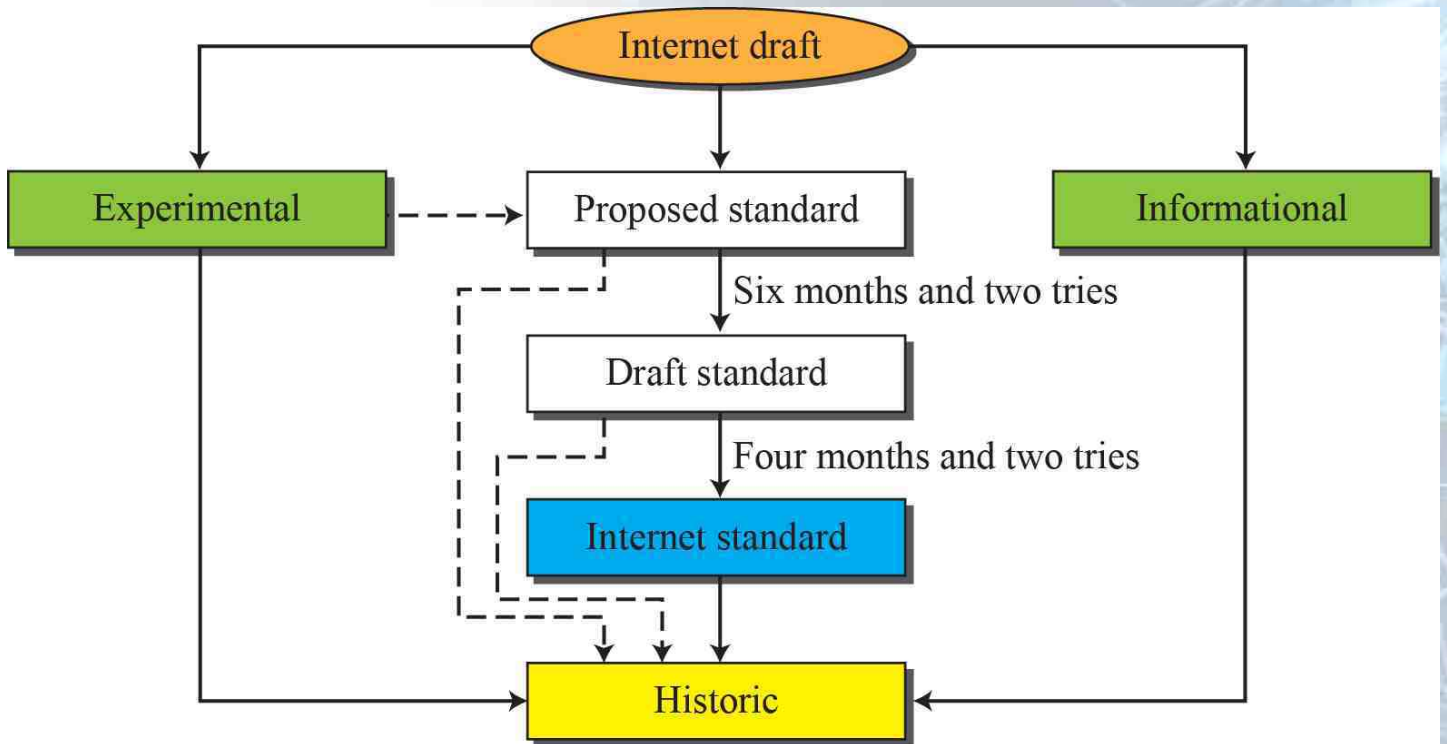
Internet History

- **Telegraph and Telephone networks, before 1960:**
 - ✓ **Constant-rate communication only**
- **ARPANET- Packet Switched**
- **Birth of the Internet & TCP/IP**
- **MILNET**
- **CSNET**
- **NSFNET**
- **Internet Today**

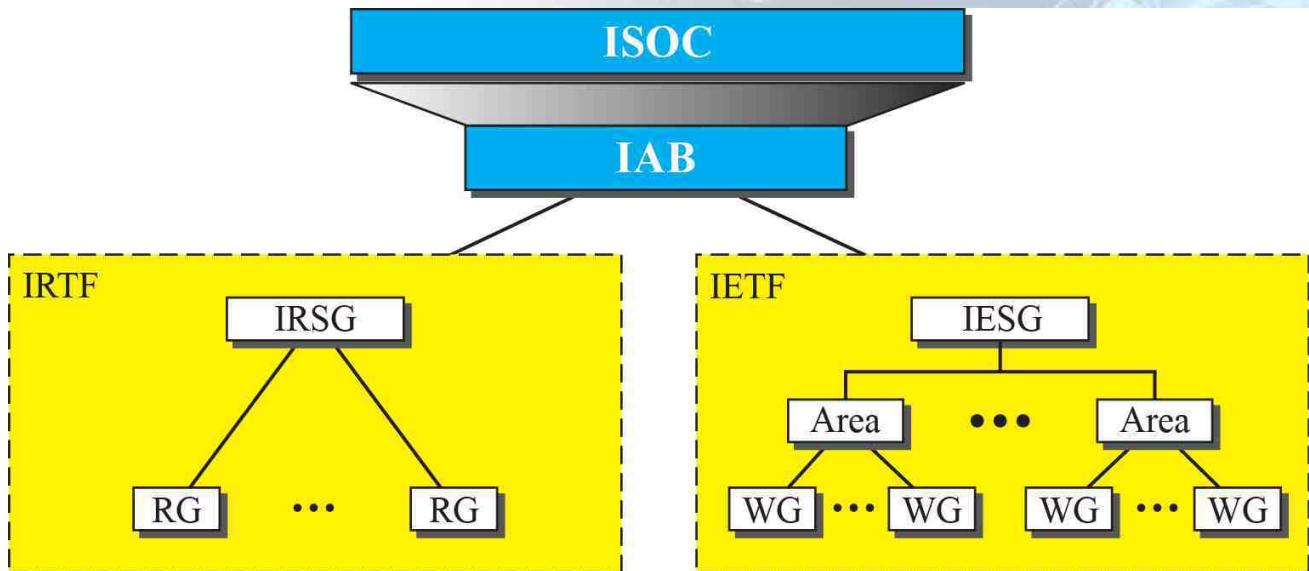
Internet Standards and Administration

- Internet draft
- Request for Comments (RFC)
 - ✓ Proposed Standard
 - ✓ Draft Standard
 - ✓ Internet Standard
 - ✓ Historic
 - ✓ Experimental
 - ✓ Informational

Internet Standards



Internet Administration



Protocol Layering - Introduction

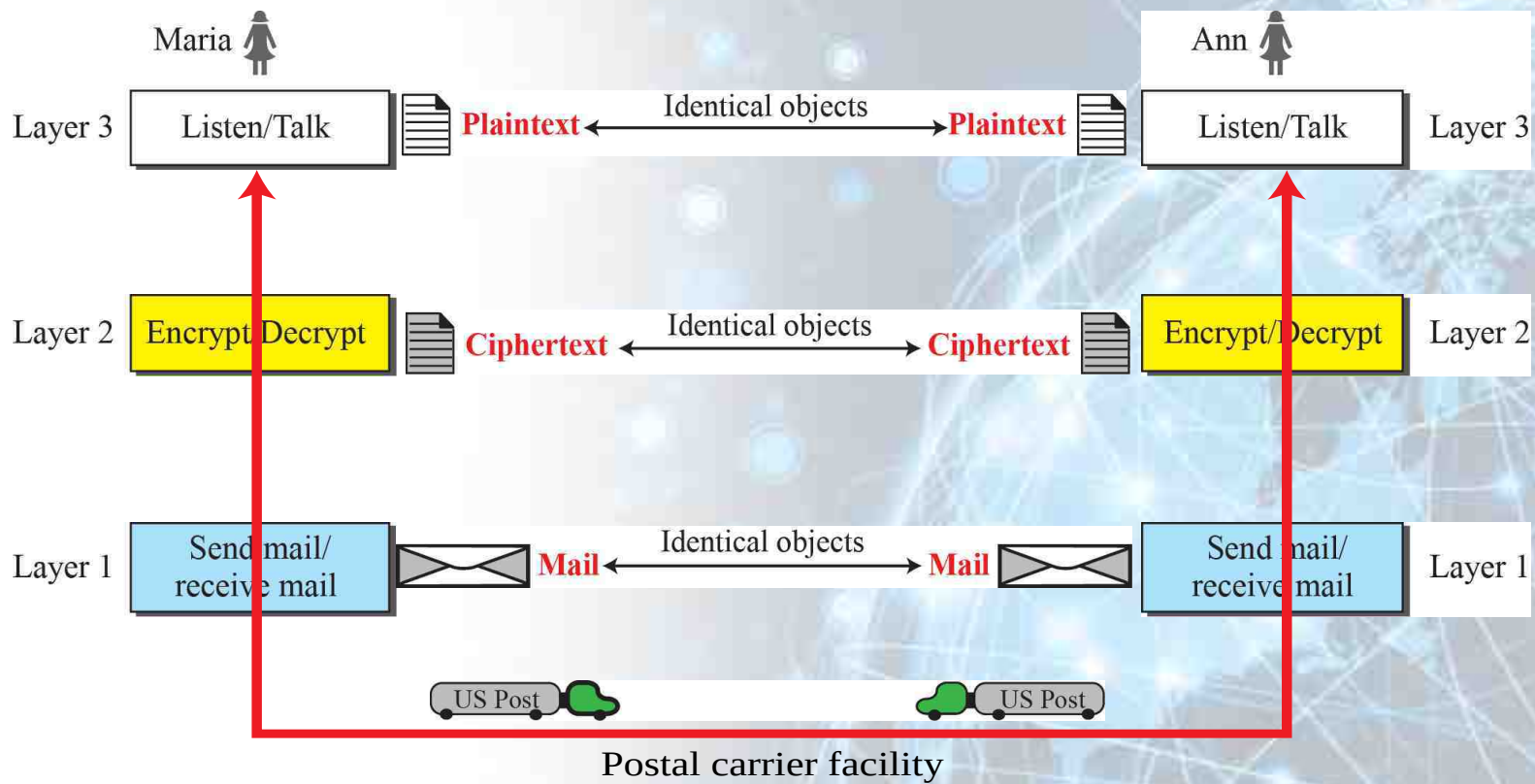
- **Protocol**
Rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively
- **Protocol Layering**
Simple Communication:
only one simple protocol

Complex Communication,
we need a protocol at each layer, or Protocol Layering

Protocol Layering - Example Scenario 1



Protocol Layering - Example Scenario 2



Protocol Layering - Advantages and Disadvantages

- 
- **Advantages**
 - ✓ Modularity
 - ✓ Separation of Service & Implementation
 - ✓ Reduced Complexity & Cost
 - **Disadvantages**
 - ✓ None Really!

Protocol Layering - Principles

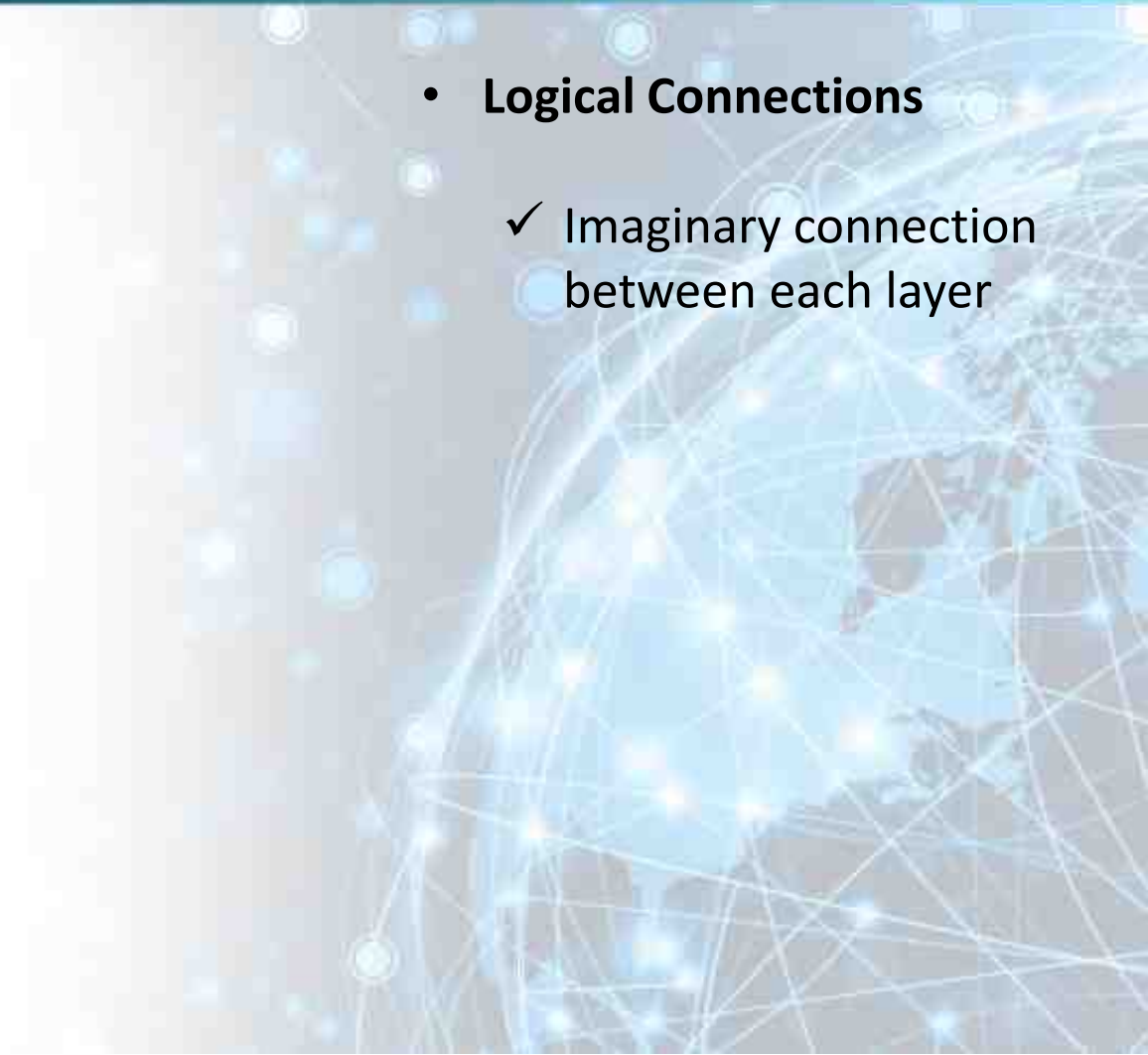
- **Two Principles**

- ✓ Bidirectional Communication → Each Layer performs two opposite tasks in each direction
- ✓ Two objects under each layer at both sites should be identical

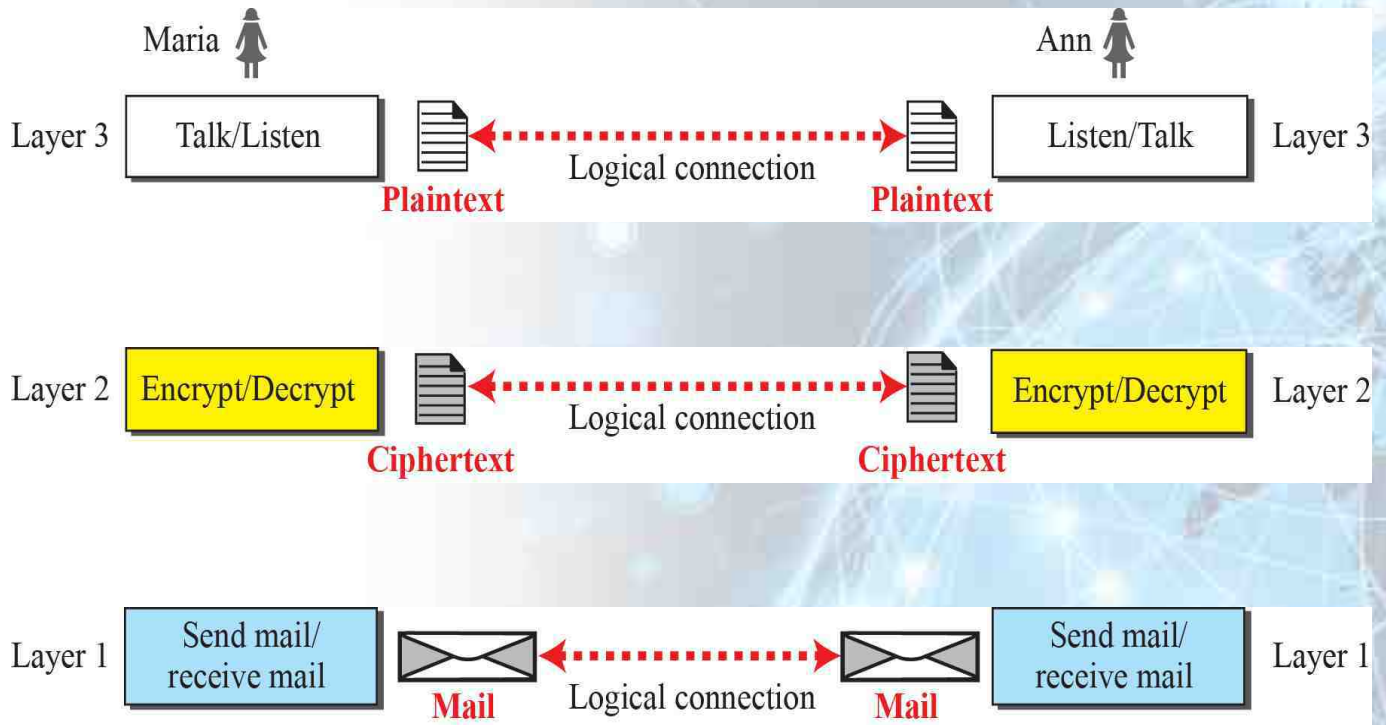
Protocol Layering - Logical Connections

- **Logical Connections**

- ✓ Imaginary connection between each layer



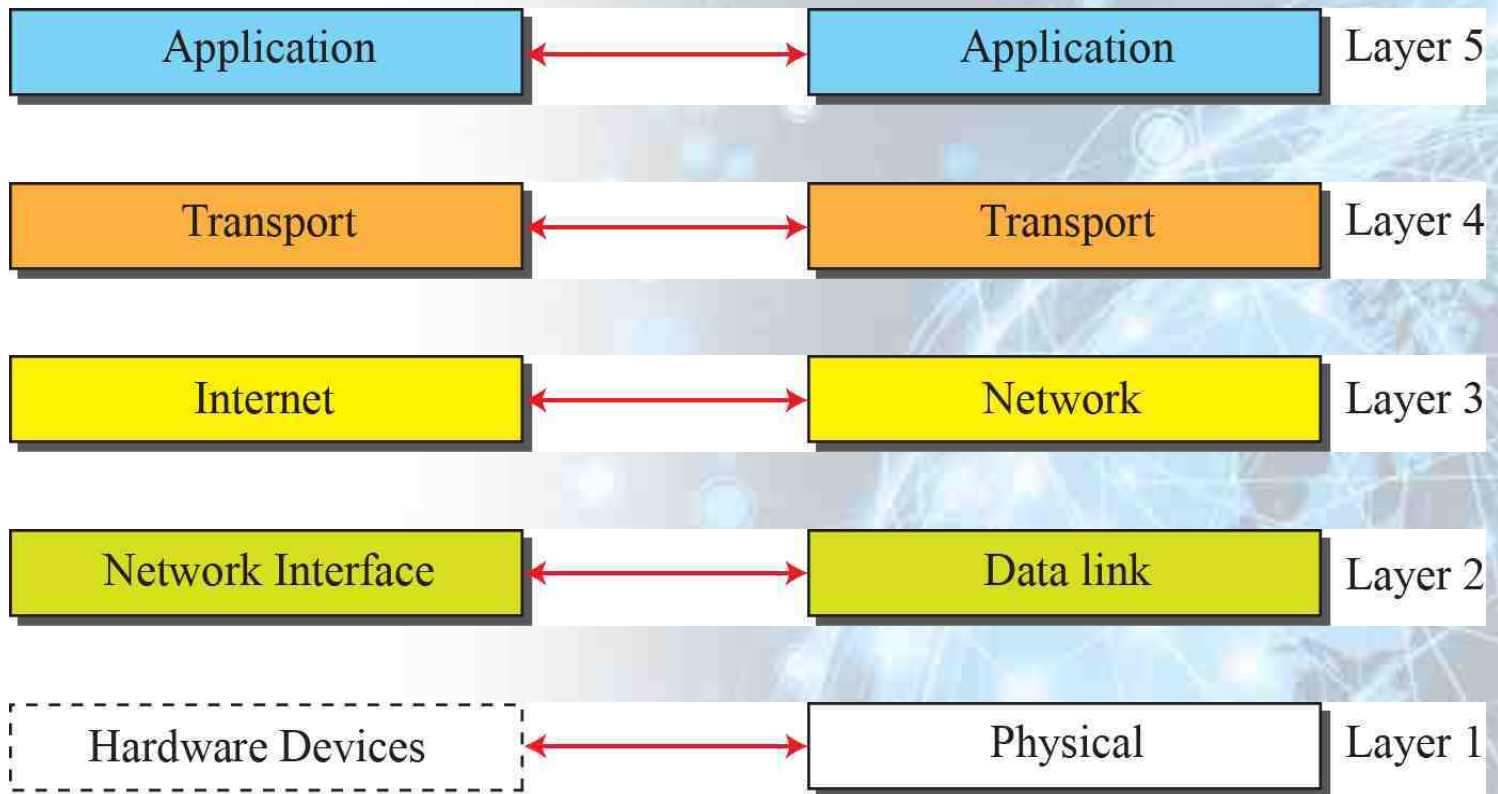
Protocol Layering



TCP/IP Protocol Suite

- **TCP/IP Protocol Suite**
 - ✓ Protocol suite used in Internet today
 - ✓ Each Layer provides specific functionality
 - ✓ Hierarchical Protocol
 - ✓ Presented in 1973 and chosen to be the official protocol of Internet in 1983

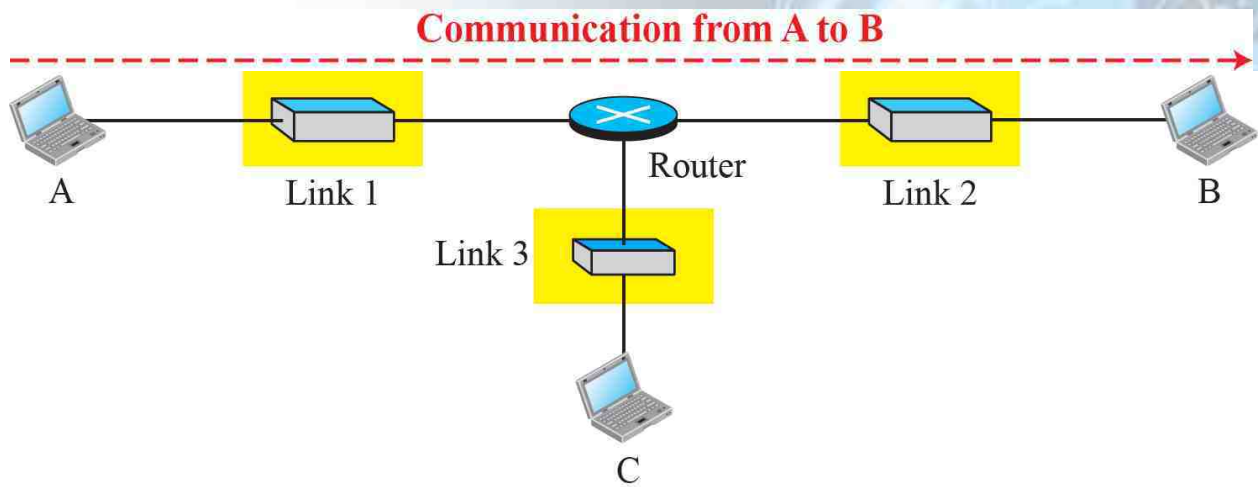
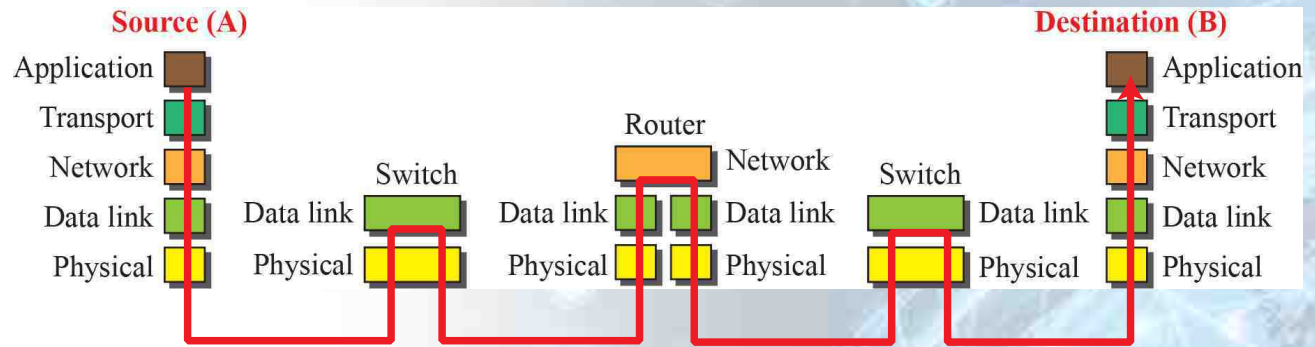
TCP/IP Protocol Suite



a. Original layers

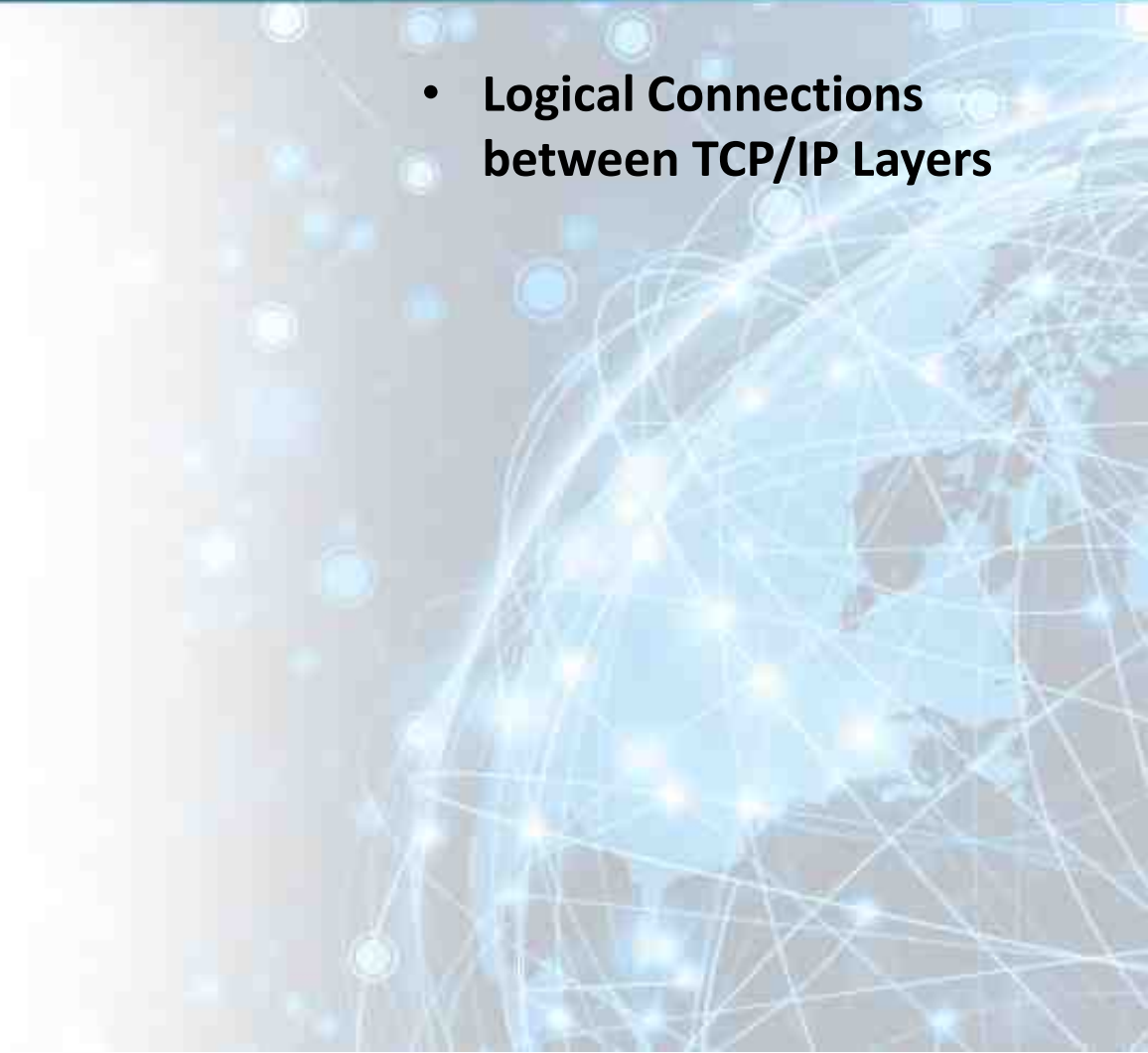
b. Layers used in this book

TCP/IP Protocol Suite - Layered Architecture

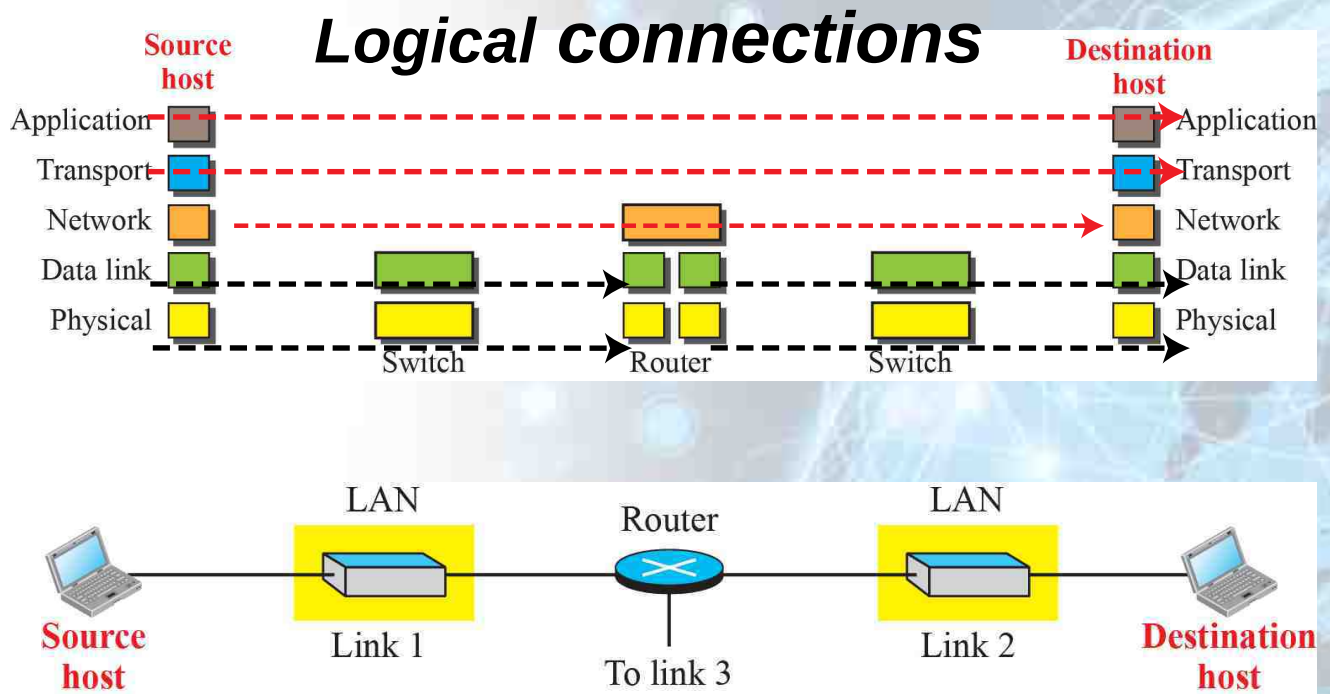


TCP/IP Protocol Suite – Function of Layers

- **Logical Connections between TCP/IP Layers**

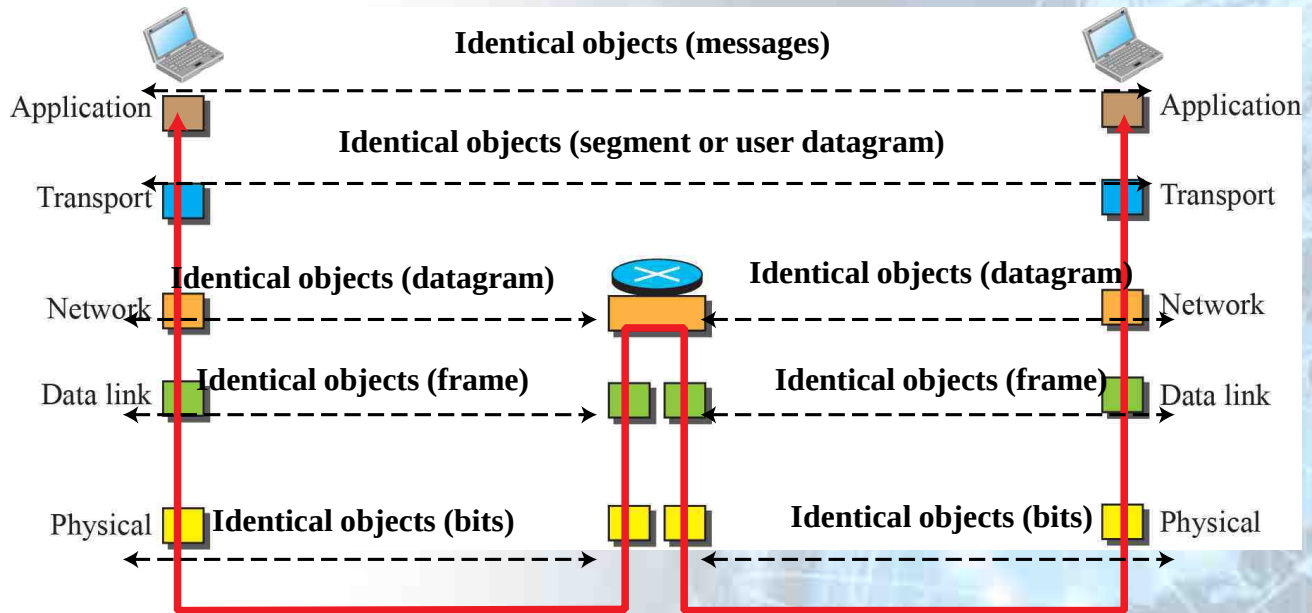


TCP/IP Protocol Suite – Function of Layers

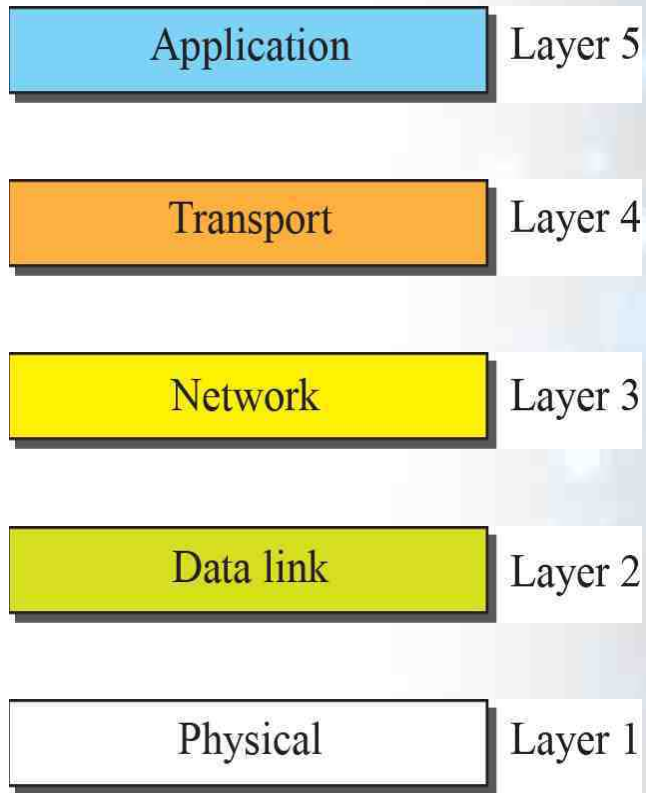


TCP/IP Protocol Suite – Function of Layers

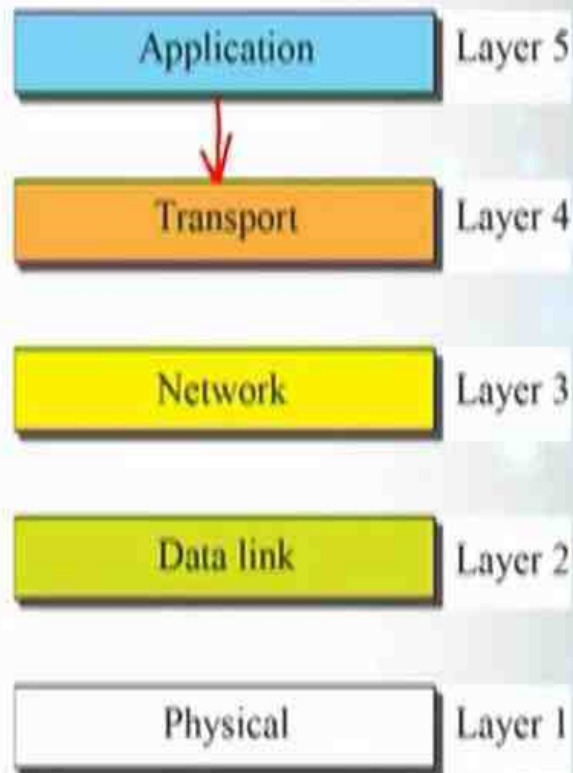
Notes: We have not shown switches because they don't change objects.



TCP/IP Protocol Suite – Layer Description



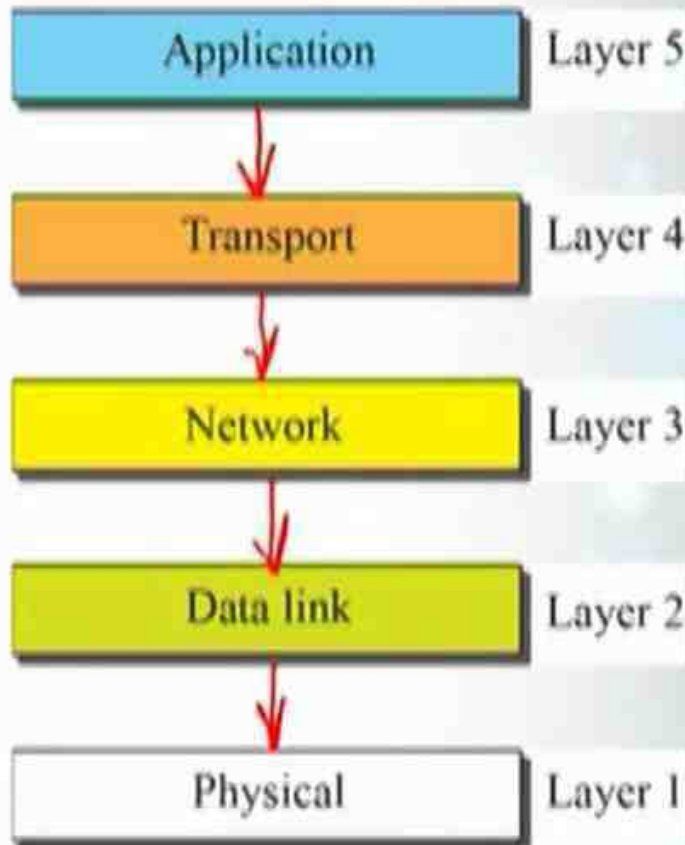
TCP/IP Protocol Suite – Layer Description



Message (object) HTTP, SMTP, FTP,
Segment (User Datagram) TCP, UDP
Datagram → Routing → IP

Double-click to go to fullscreen, ctrl+c

TCP/IP Protocol Suite – Layer Description



Message (object) HTTP, SMTP

Segment (User Datagram) TCP

Datagram → Routing →

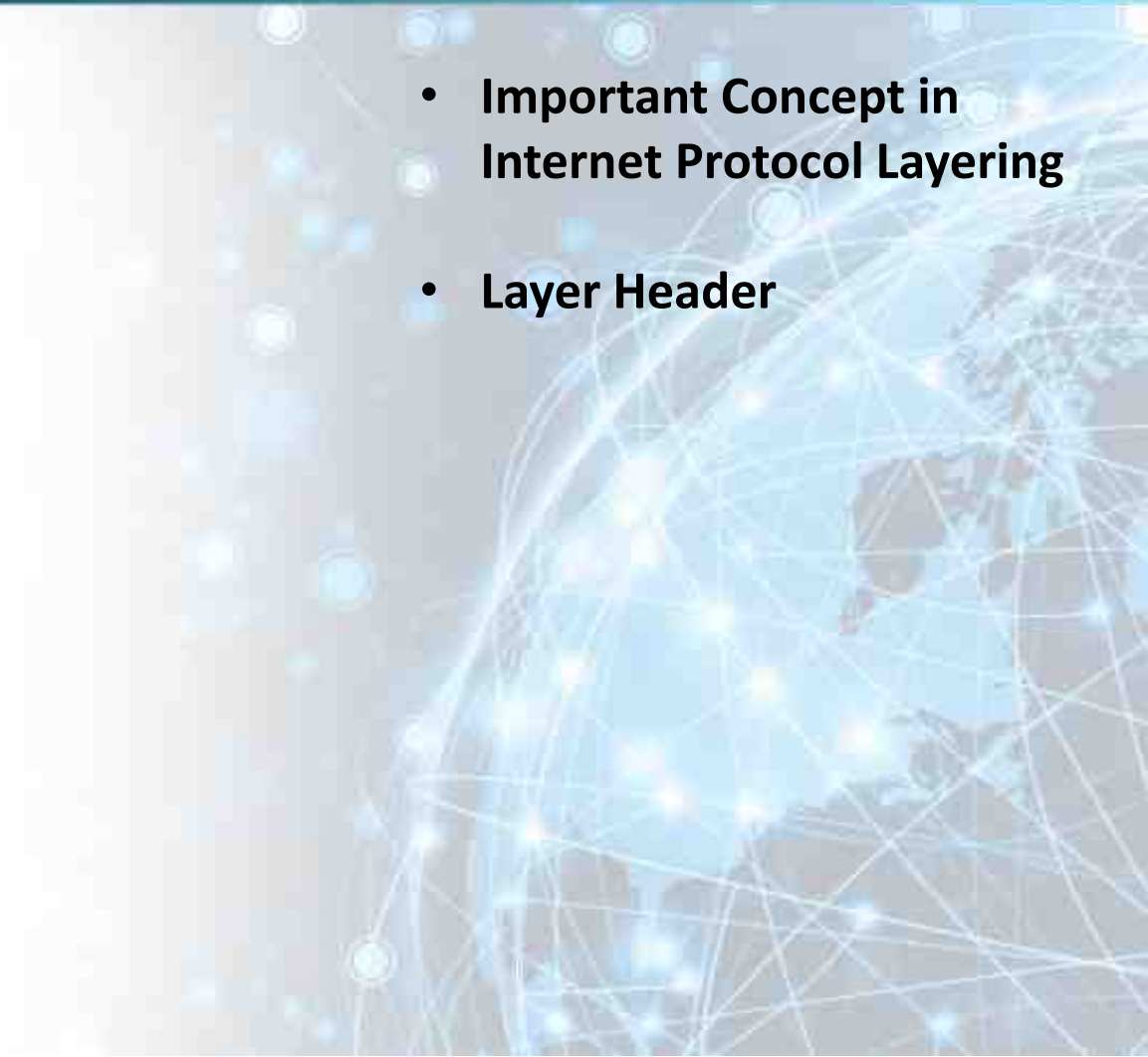
Frame (object) Standard,

Bits (object)

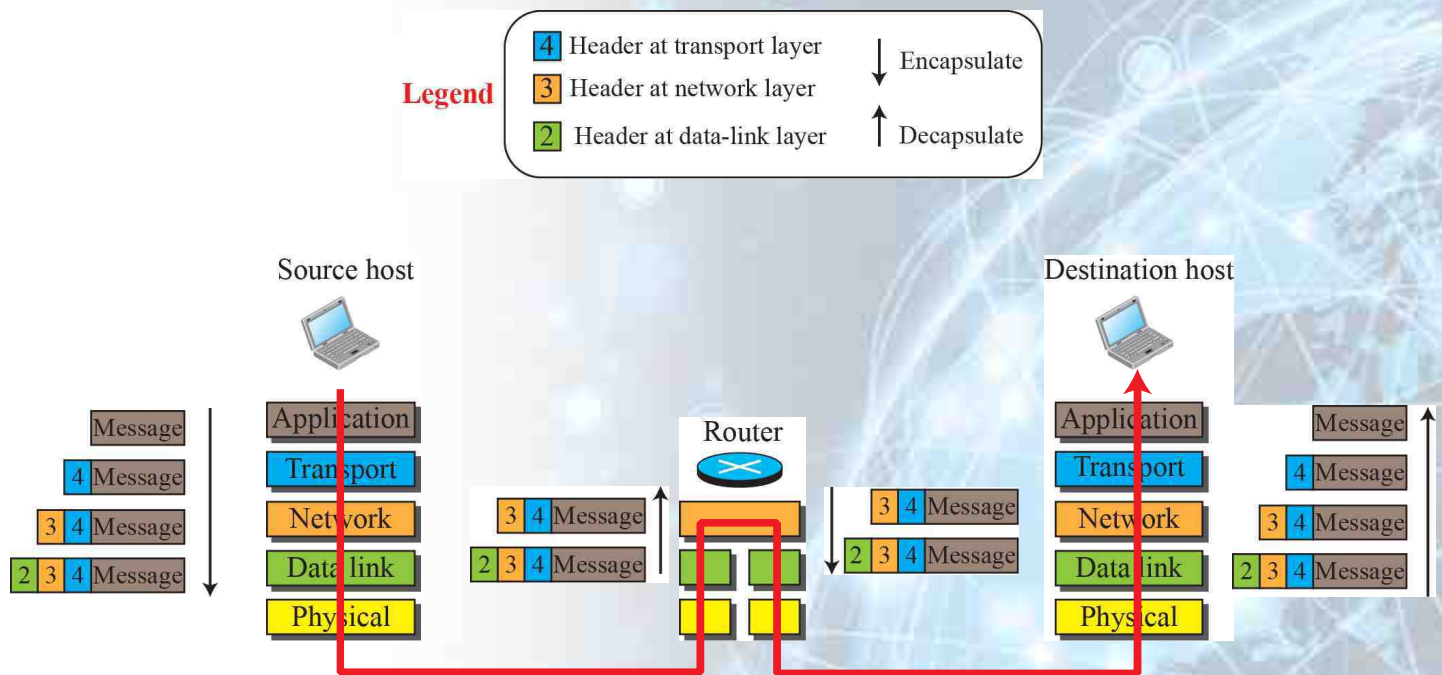


Encapsulation & Decapsulation

- Important Concept in Internet Protocol Layering
- Layer Header



Encapsulation & Decapsulation



Addressing in TCP/IP Protocol Suite

- Every communication needs at least two addresses:
 - ✓ Source Address &
 - ✓ Destination Address
- Addressing by Layer
- Physical Layer is an exception

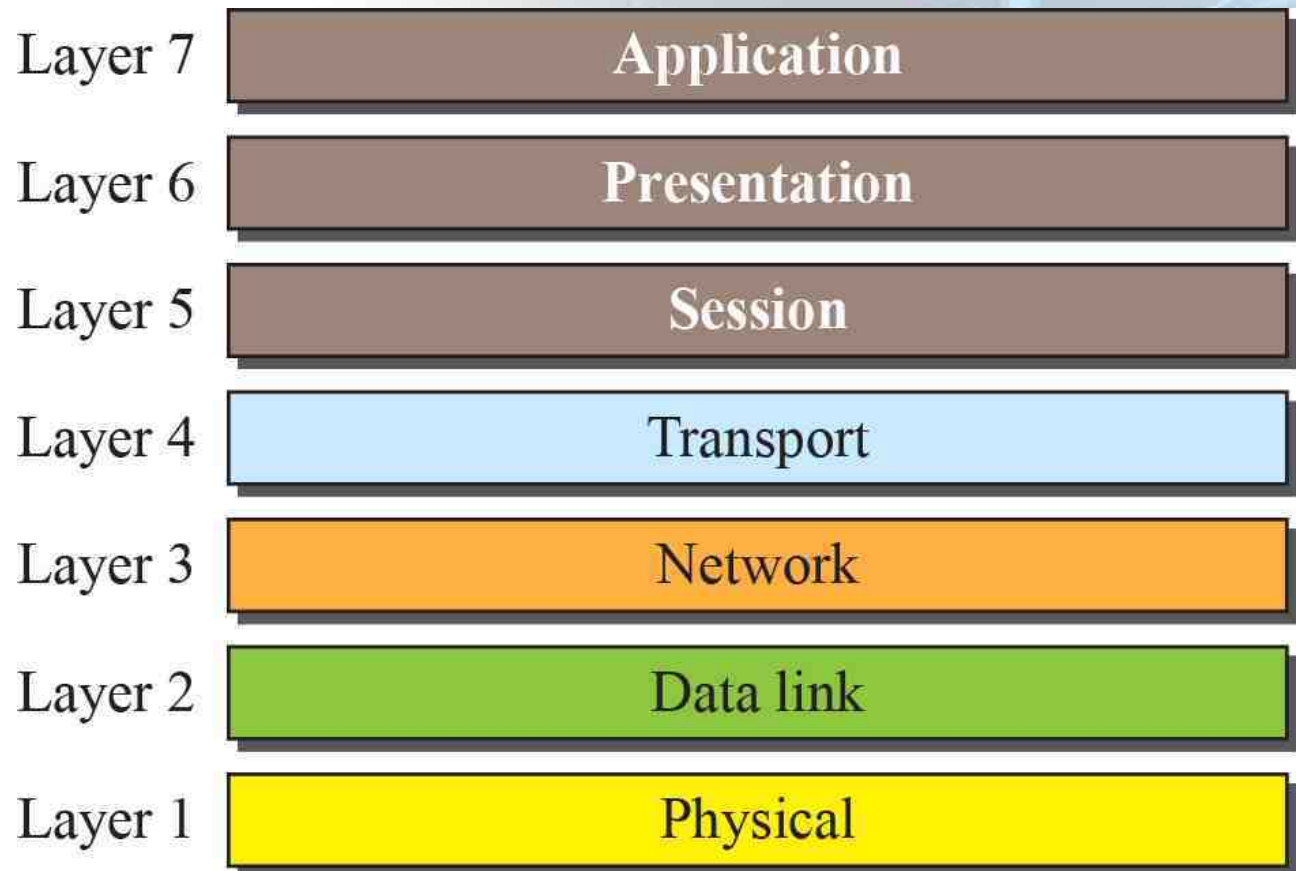
Addressing in TCP/IP Protocol Suite

Packet names	Layers	Addresses
Message	Application layer	Names
Segment / User datagram	Transport layer	Port numbers
Datagram	Network layer	Logical addresses
Frame	Data-link layer	Link-layer addresses
Bits	Physical layer	

The Open System Interconnection (OSI) Model

- International Organization for Standardization (ISO)
- ISO established in 1947
- Close to three-fourths of countries represented
- Introduced OSI Model in late 1970s
- OSI: a 7-Layer Model

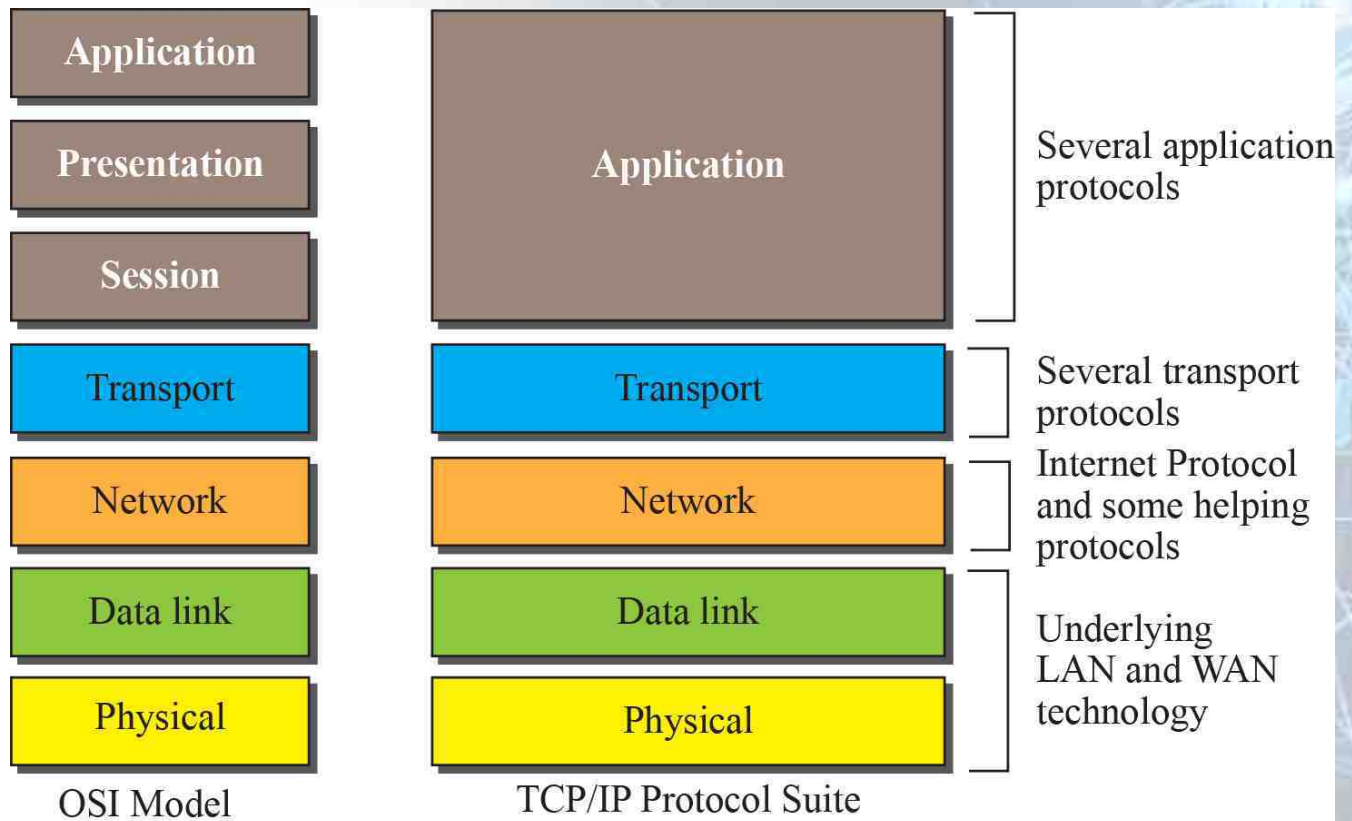
The Open System Interconnection (OSI) Model



OSI Model vs TCP/IP Protocol suite

- Two Layers of OSI missing from TCP/IP
- Application (TCP/IP) = Application + Presentation + Session (OSI)

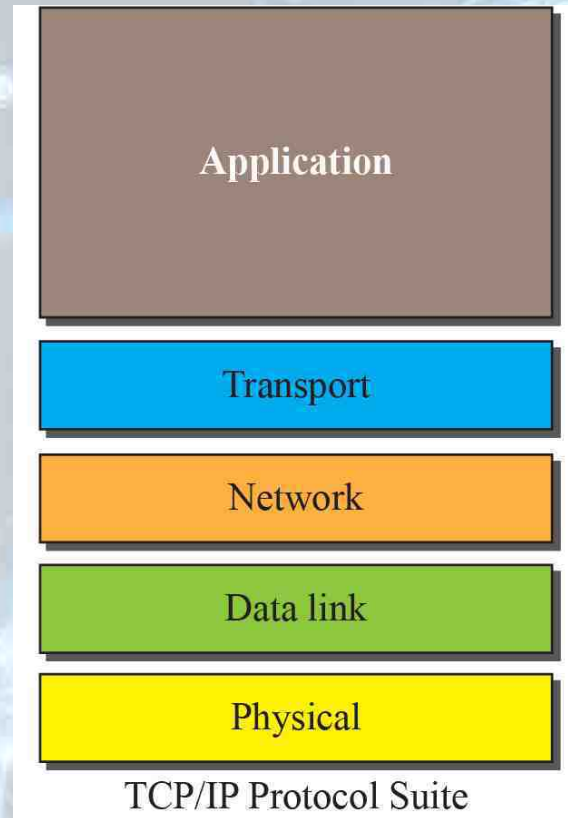
OSI Model vs TCP/IP Protocol suite



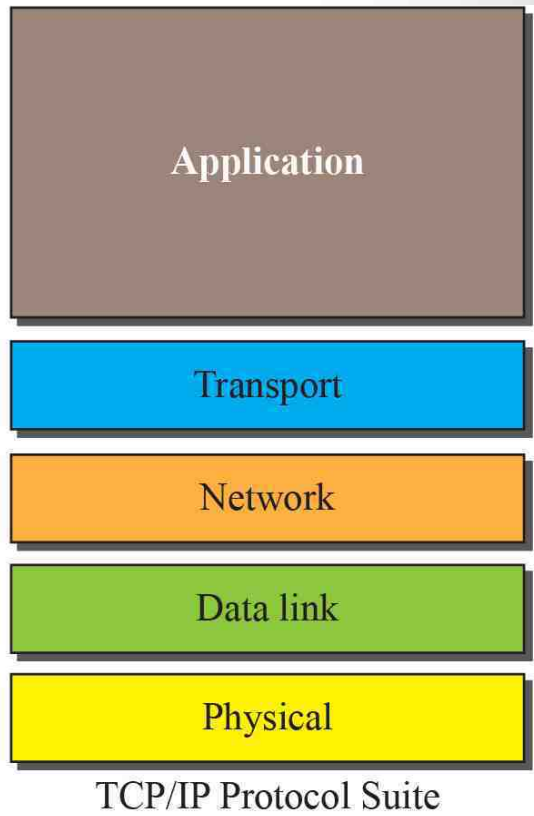
Lack of OSI Model's Success

- Three reasons OSI did not replace TCP/IP:
 - ✓ OSI was completed when TCP/IP was fully in place
 - ✓ Some layers in OSI not fully defined
 - ✓ Performance of TCP/IP better than that of OSI

Data Communication versus Computer Networks

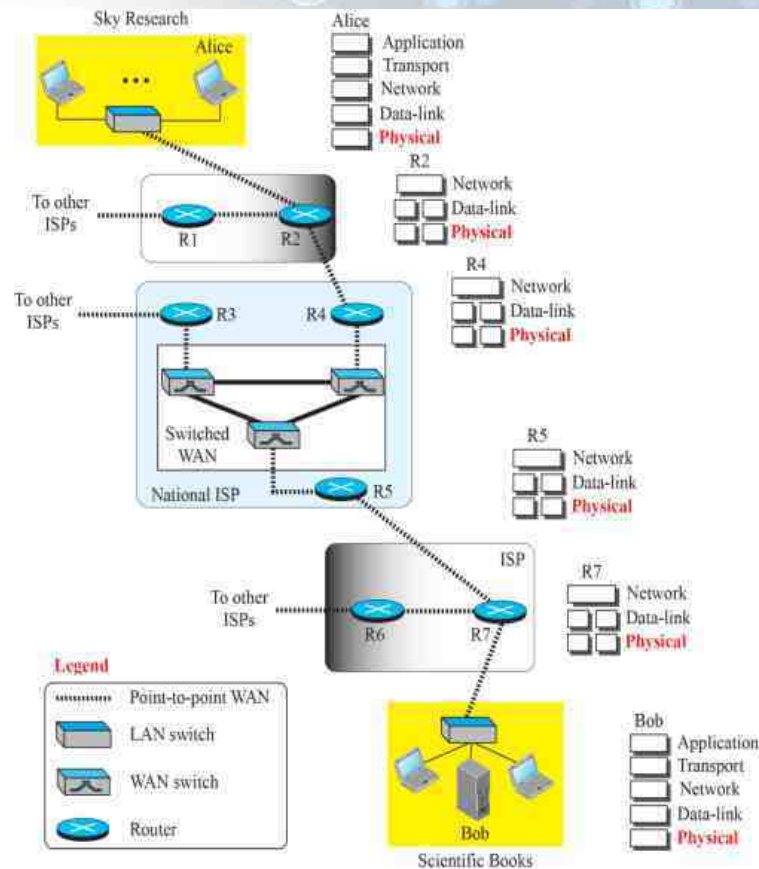


Data Communication versus Computer Networks



- Analog & Digital Transmission
- Transmission Media
- Switching
- Error Detection and Correction
- Media Access and Data Link Control
- Wired and Wireless LANs

Communication at Physical Layer



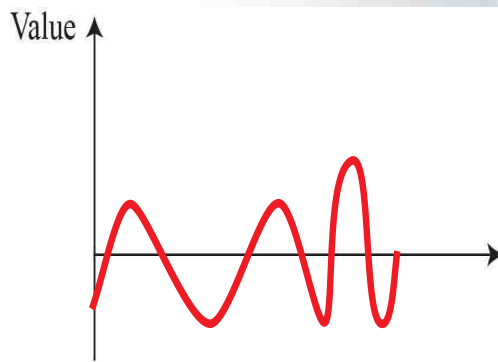
Analog & Digital Data

- **Data → Analog or Digital**
- **Analog Data → Continuous**
- **Digital Data → Discrete**
- **Examples: Analog Clock vs. Digital Clock**
- **Human voice vs. Data in Computer**

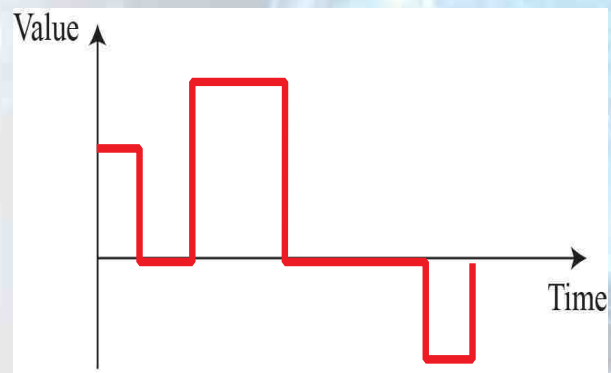
Analog & Digital Signals

- **Signals represent Data**
- **Signals → Analog or Digital**
- **Analog Signal → Infinite Levels of Intensity over time**
- **Digital Signal → Limited number of defined values**

Analog & Digital Signals



a. Analog signal



b. Digital signal

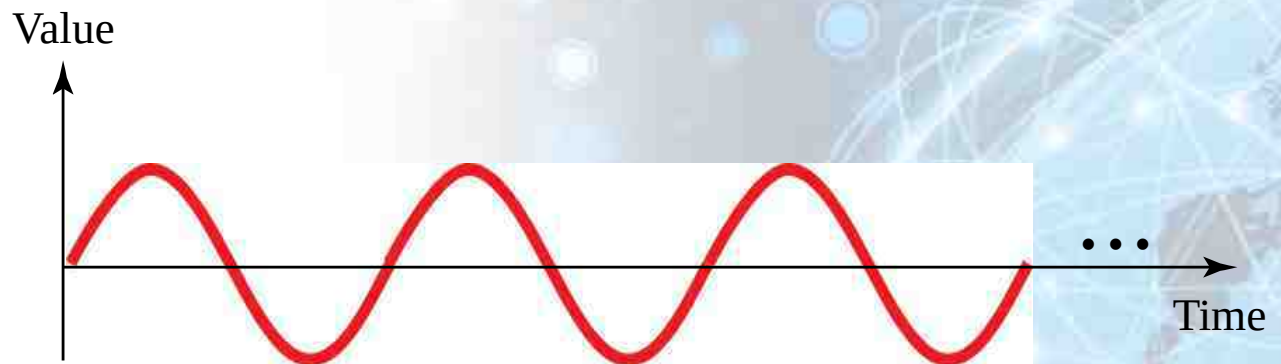
Periodic & Non-periodic Signals

- **Analog/Digital Signal → Periodic or Non-periodic**
- **Periodic Signal → Pattern**
- **Period and Cycle**
- **Non-Periodic → No Pattern**
- **Periodic ANALOG Signals and Non-periodic DIGITAL Signals**

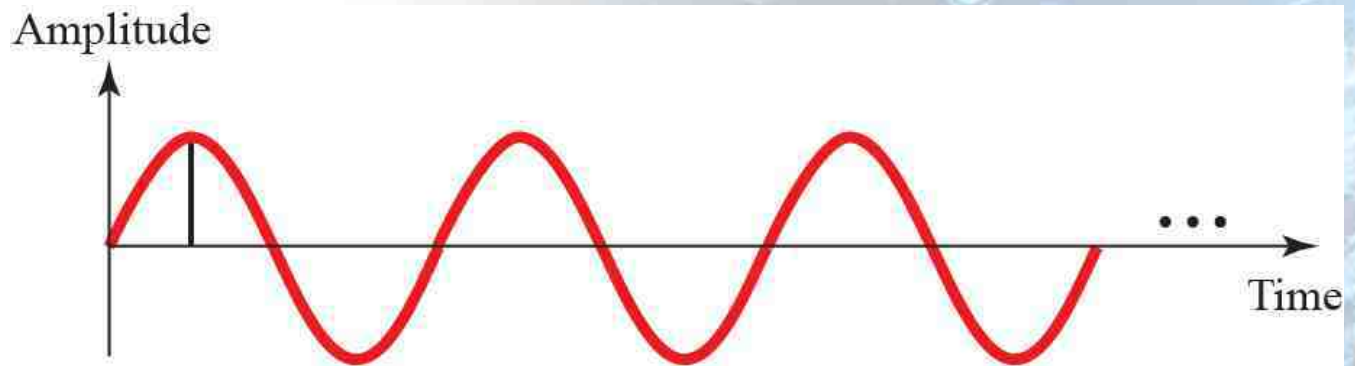
Periodic Analog Signals

- **Periodic Analog Signals**
→ Simple or Composite
- **Simple Periodic Analog signal** → Sine wave
- **Composite Periodic Analog signal** →
Composed of multiple sine waves

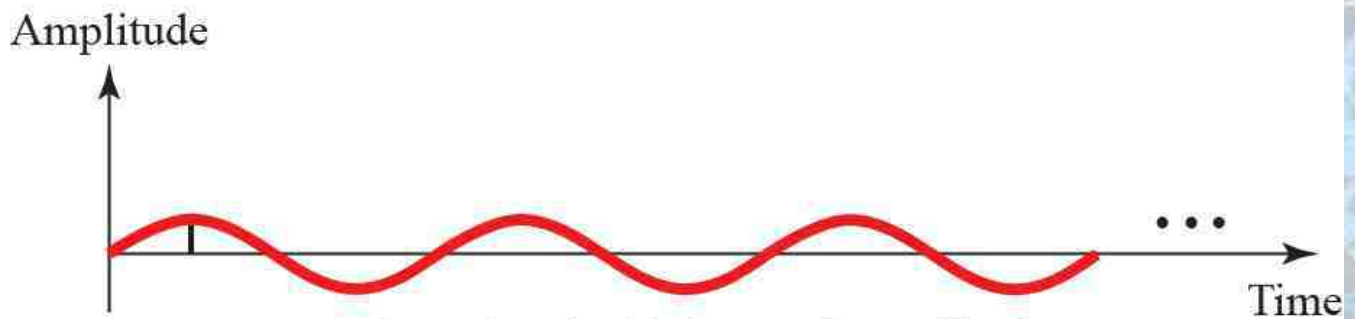
Sine Wave



Sine Wave – Peak Amplitude



a. A signal with high peak amplitude



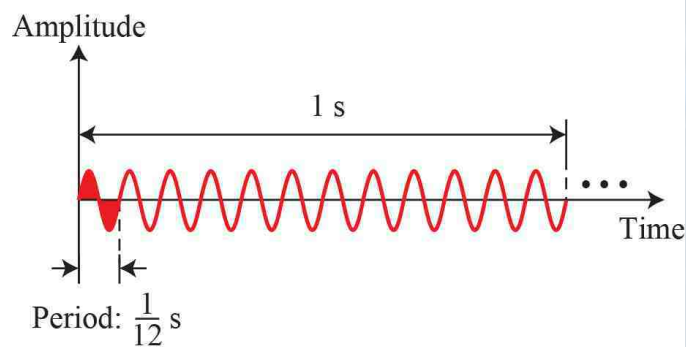
b. A signal with low peak amplitude

Sine Wave –Frequency

- Period (T) → Amount of time required to complete 1 cycle
- Frequency (f) → No. of Periods in 1 sec
- $f = 1/T$ or $T = 1/f$

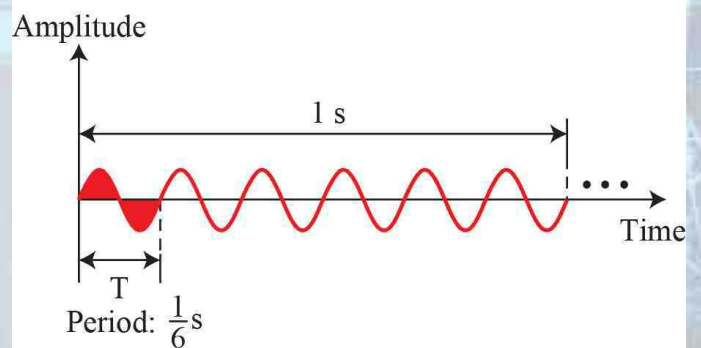
Sine Wave –Frequency

12 periods in 1 s $\rightarrow$ Frequency is 12 Hz



a. A signal with a frequency of 12 Hz

6 periods in 1 s $\rightarrow$ Frequency is 6 Hz



b. A signal with a frequency of 6 Hz

Sine Wave –Frequency

<i>Period</i>		<i>Frequency</i>	
<i>Unit</i>	<i>Equivalent</i>	<i>Unit</i>	<i>Equivalent</i>
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Example

The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:



Example

The period of a signal is 100 ms. What is its frequency in kilohertz?



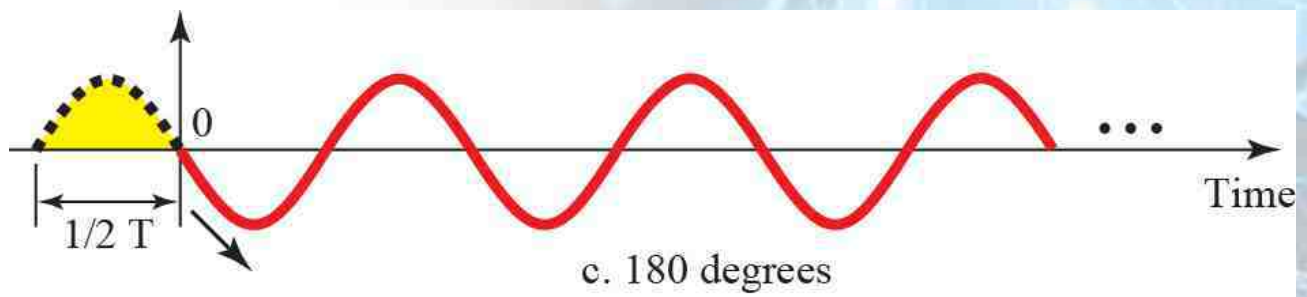
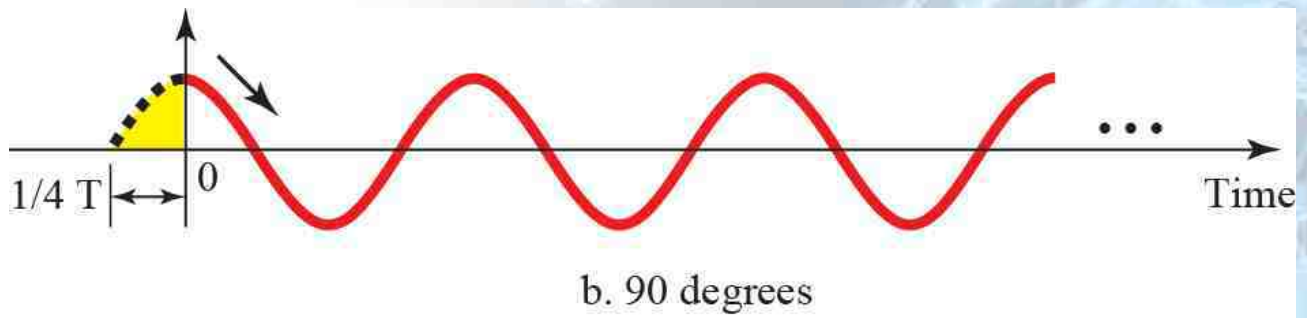
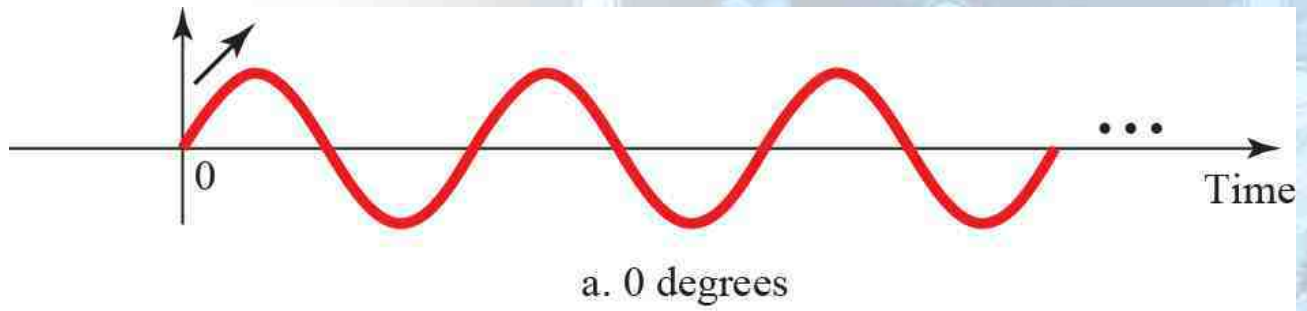
Phase (or Phase Shift)

- **Position of waveform relative to time 0**
- **Phase describes the amount of shift of the wave**
- **Indicates start of the first cycle**

$$360^\circ = 2\pi \text{ radians}$$

$$1^\circ = \frac{2\pi}{360} \text{ rad}$$

Phase



Example

A sine wave is offset $\frac{1}{6}$ cycle with respect to time 0. What is its phase in degrees and radians?

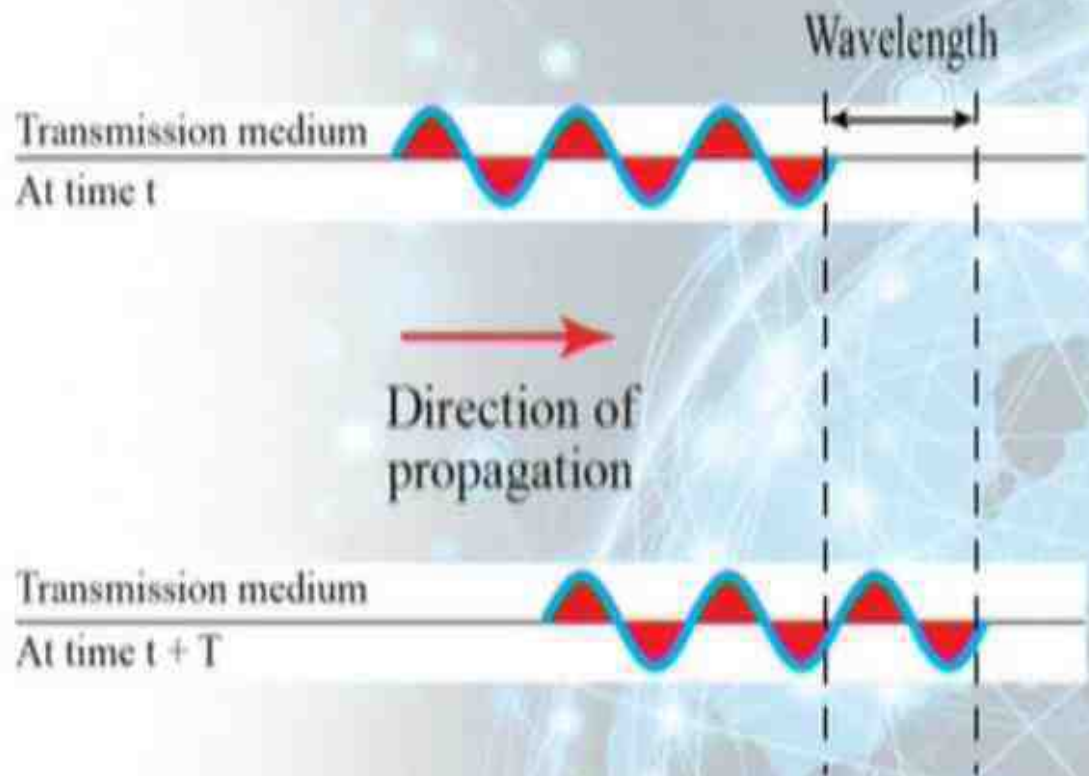


Wavelength

Wavelength is another characteristic of a signal traveling through a transmission medium. Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium (see Figure 3.7).

Wavelength

Figure 3.7



Wavelength

Prop. Speed = c = Light (speed)

frequency = f

Wavelength = λ

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/sec}}{f}$$

$$\lambda = \frac{3 \times 10^8}{4 \times 10^{14}}$$

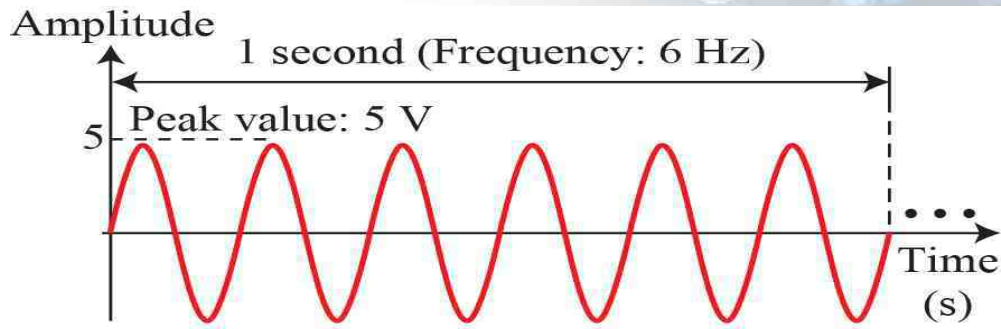
Wavelength

$$\begin{aligned}\lambda &= 0.75 \times 10^{-6} \text{ m} \\ &= \underline{\underline{0.75 \mu\text{m}}}\end{aligned}$$

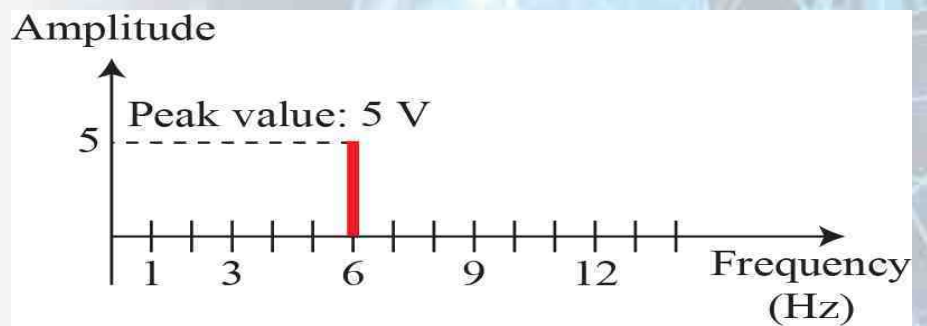
Time & Frequency Domains

A sine wave is comprehensively defined by its amplitude, frequency, and phase. We have been showing a sine wave by using what is called a time domain plot. The time-domain plot shows changes in signal amplitude with respect to time (it is an amplitude-versus-time plot). Phase is not explicitly shown on a time-domain plot.

Time & Frequency Domains



a. A sine wave in the time domain



b. The same sine wave in the frequency domain

Example 3.7

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.9 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

Time & Frequency Domains

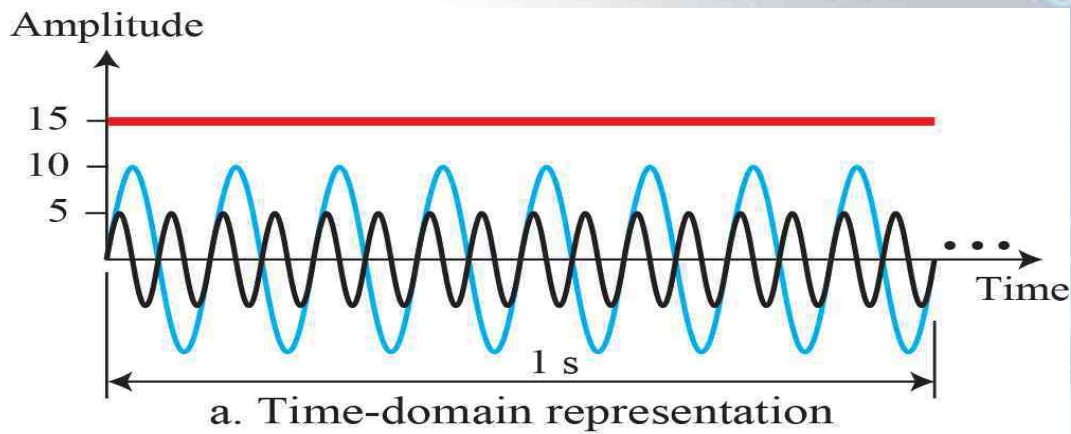
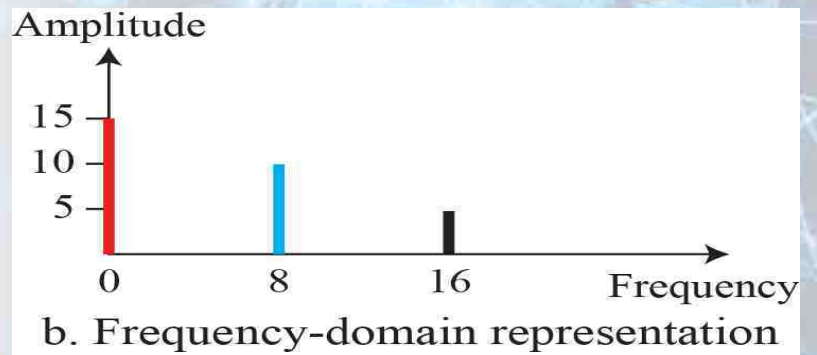


Figure 3.9



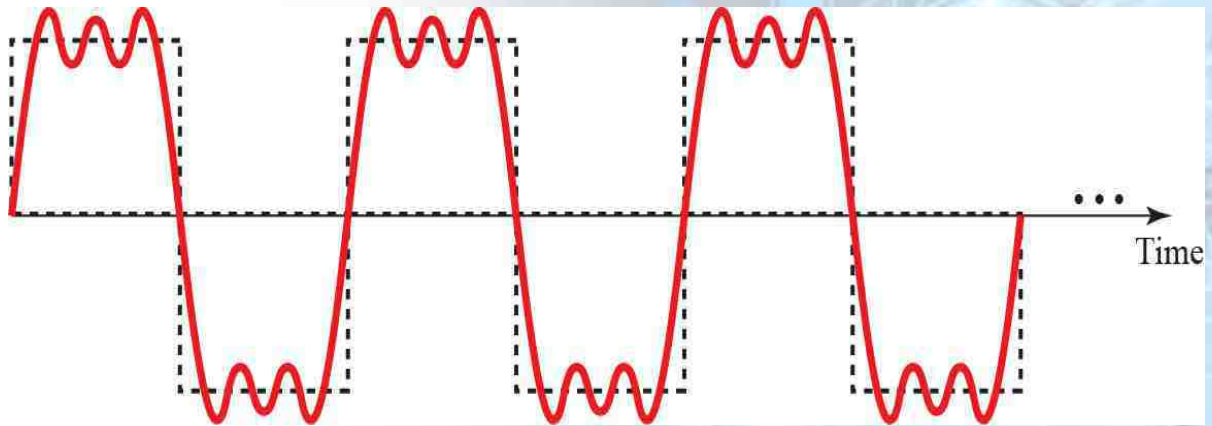
Composite Signals

So far, we have focused on simple sine waves. Simple sine waves have many applications in daily life. We can send a single sine wave to carry electric energy from one place to another. For example, the power company sends a single sine wave with a frequency of 60 Hz to distribute electric energy to houses and businesses.

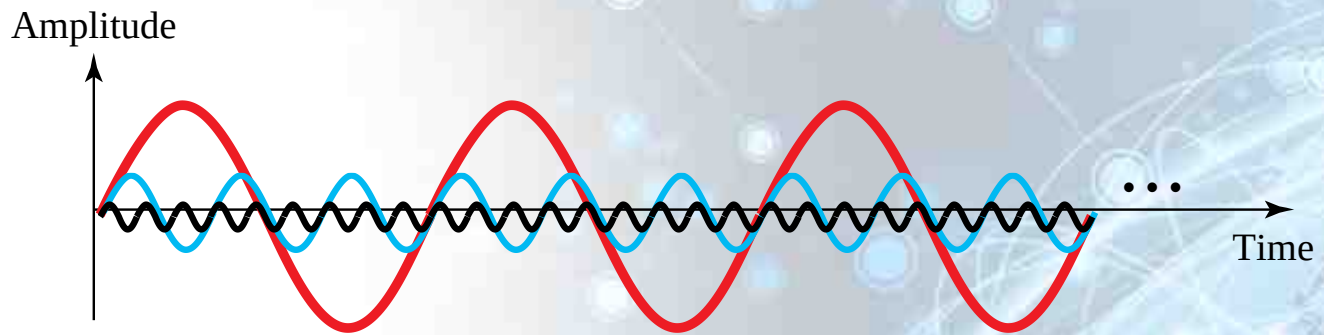
Composite Signals

- **Single Sine Wave can only carry limited information**
- **Composite Signal is made up of multiple simple sine waves**
- **Can be periodic or non-periodic**

A Composite Periodic Signal



Decomposition of Composite Periodic Signal

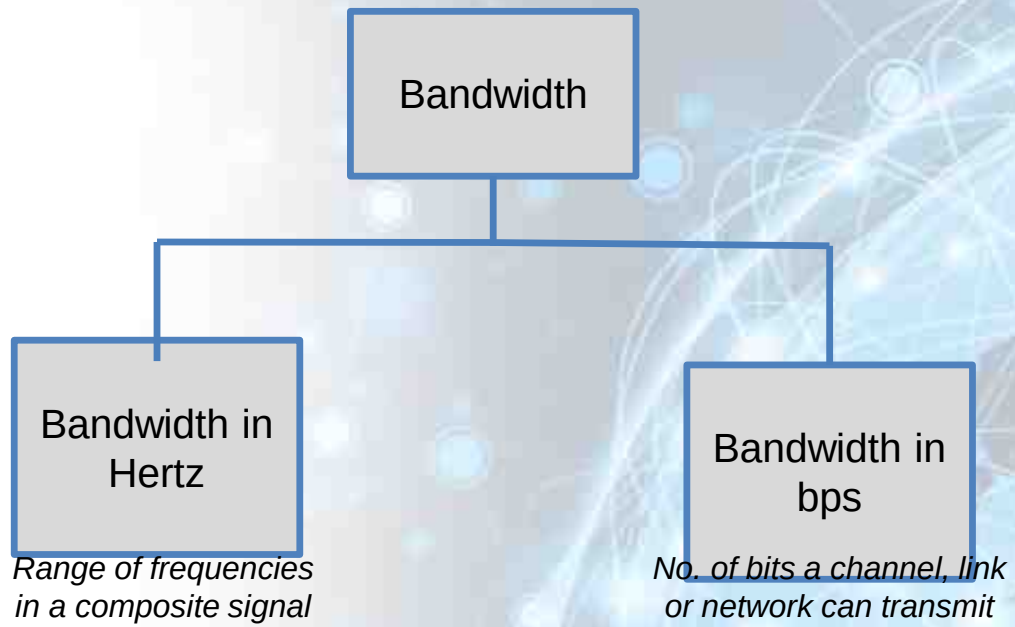


b. Frequency-domain decomposition of the composite signal

Bandwidth

- **An important characteristic that measures Network Performance**
- **Bandwidth can be used in two different contexts with two different measuring values:**
 - **Bandwidth in Hertz**
 - **Bandwidth in bits per second**

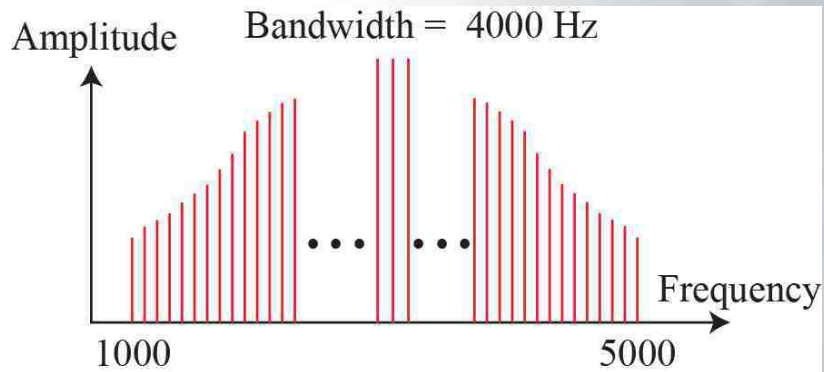
Bandwidth



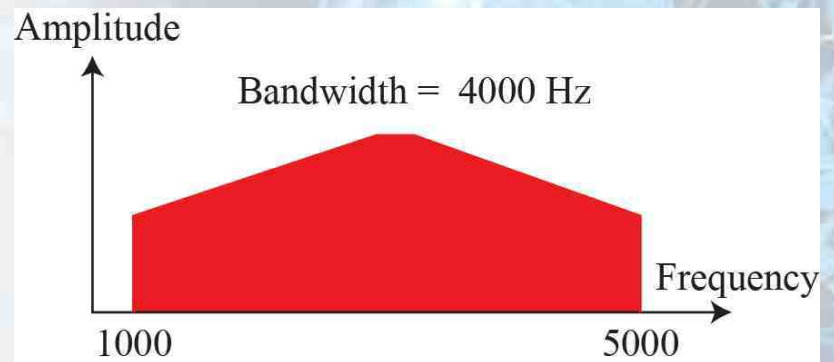
Bandwidth

- Range of frequencies contained in a Composite Signal
- The bandwidth is normally a difference between two frequencies (the highest and the lowest)

Bandwidth of a composite signal



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

Example

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.



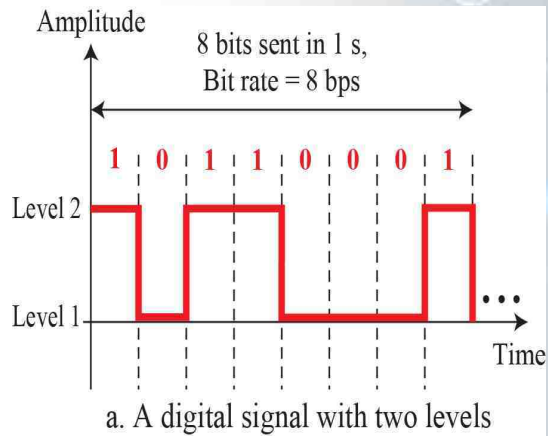
Digital Signals

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels.

Digital Signals

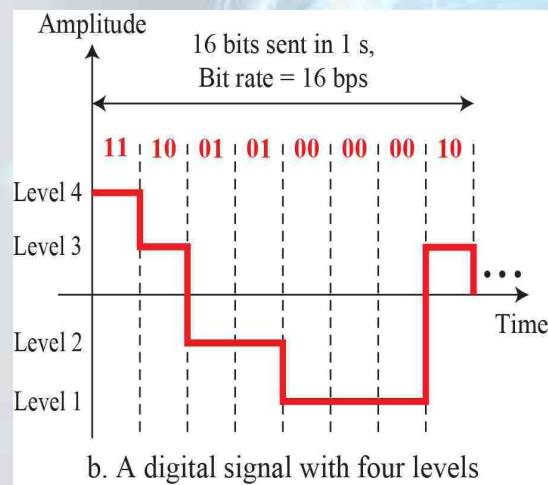
- Information can also be represented by a digital signal
- For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage
- A digital signal can have more than two levels so that we can send more than one bit for each level

Digital Signals



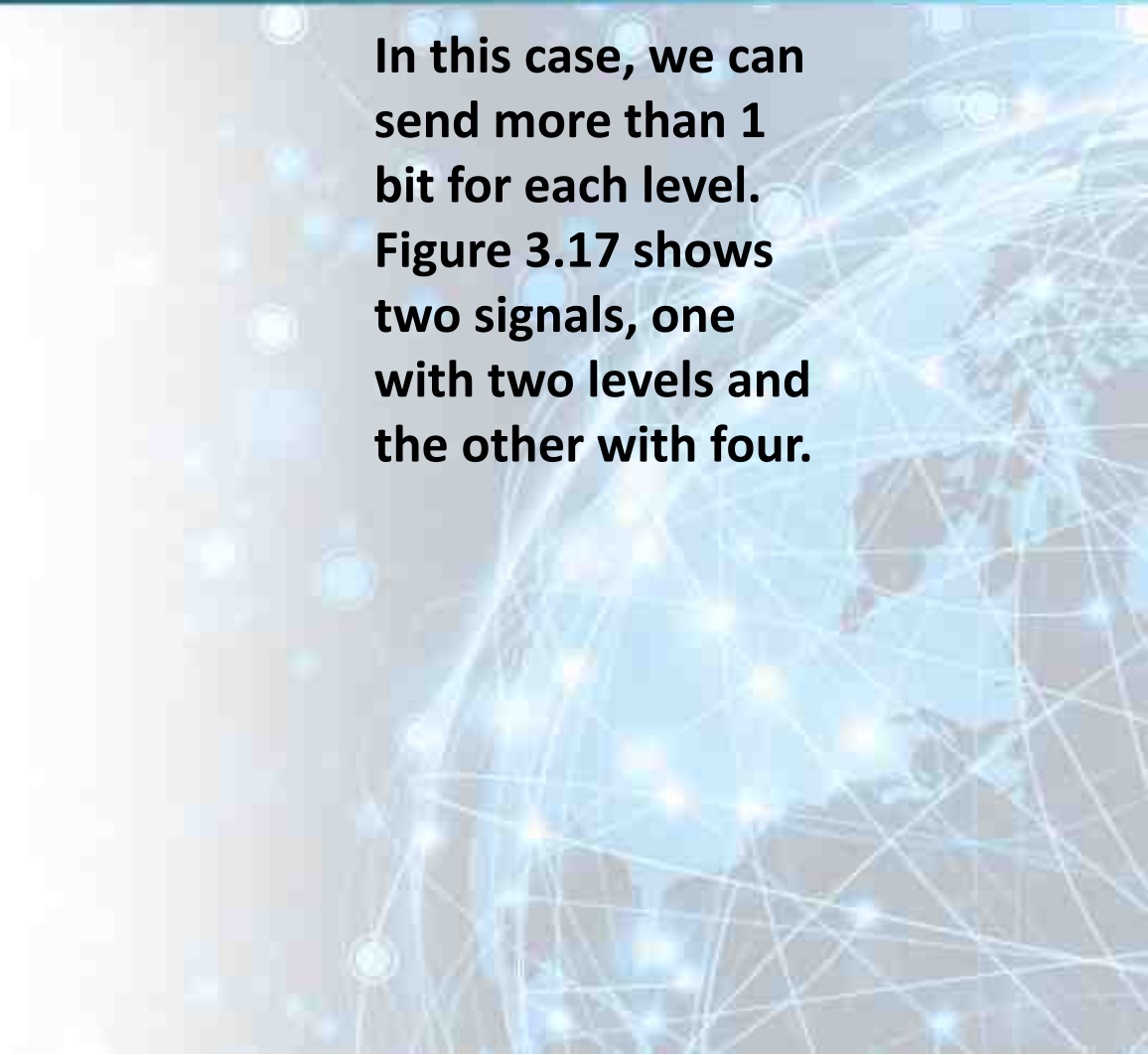
In general, if a signal has L levels, each level needs $\log_2 L$ bits. So, we can send $\log_2 2 = 1$ bit per level

$$\log_2 4 = 2 \text{ bits}$$



Digital Signals

In this case, we can send more than 1 bit for each level. Figure 3.17 shows two signals, one with two levels and the other with four.



Example 3.16

A digital signal has eight levels. How many bits are needed per level? We calculate the number of bits from the following formula. Each

Number of bits per level

$$\text{bits.} = \log_2 8 = 3$$

Example

A digital signal has eight levels. How many bits are needed per symbol?

$$\begin{aligned}\text{No of bits} &= \log_2 L \\ &= \log_2 8 \\ &= 3 \text{ bits}\end{aligned}$$

Example 3.17

A digital signal has nine levels. How many bits are needed per level? We calculate the number of bits by using the formula. Each signal level is represented by 3.17 bits. However, this answer is not realistic. The number of bits sent per level needs to be an integer as well as a power of 2. For this example, 4 bits can represent one level.

Example

A digital signal has nine levels. How many bits are needed per level? We calculate the number of bits by using the formula.

$$\log_2 L = \log_2 9 = 3.17 \text{ bits}$$

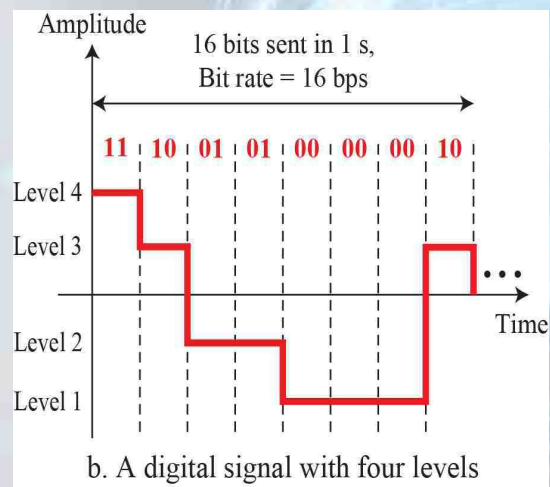
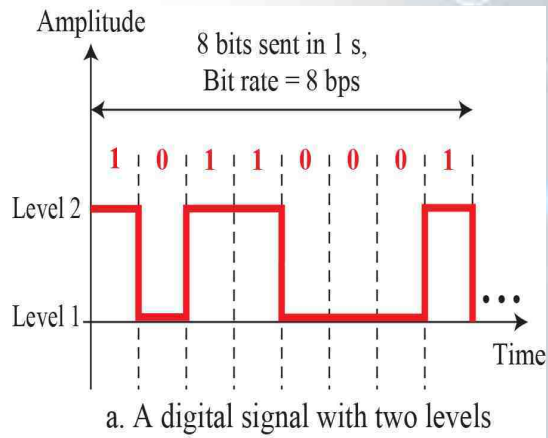
↑ ↑
integer power of 2

4 bits

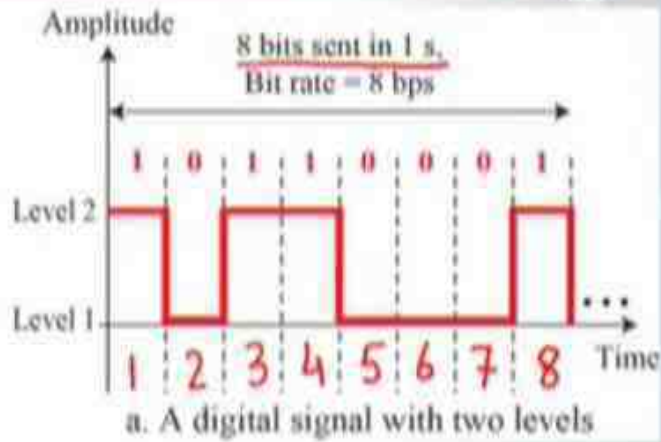
Bit Rate

- **Number of bits sent in 1 second**
- **Bit Rate is expressed in bits per second (bps)**
- **Most digital signals are non-periodic, and thus period and frequency are not appropriate characteristics**

Bit Rate

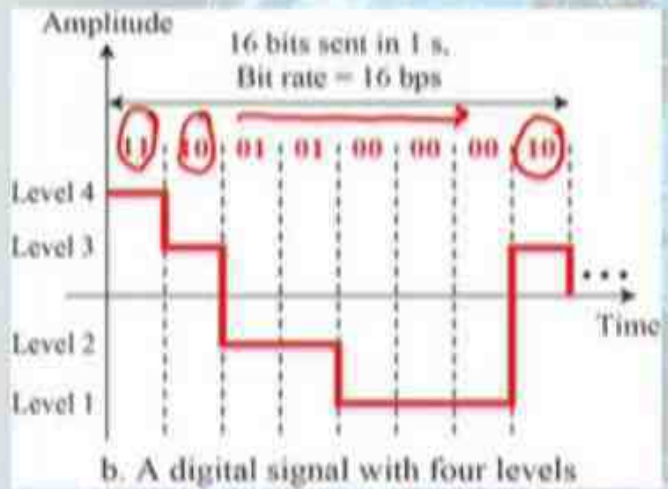


Bit Rate



Bit Rate = 8 bps

BR = 1



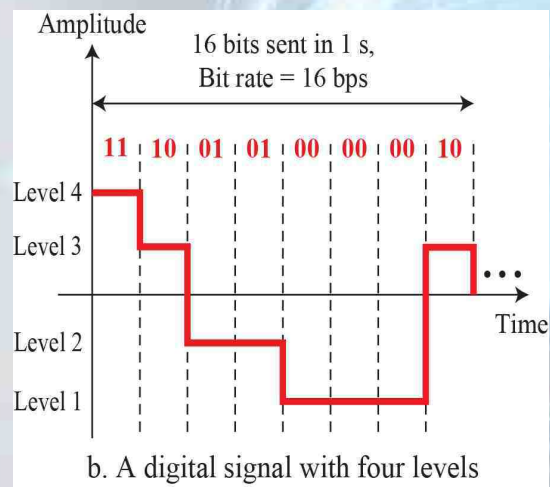
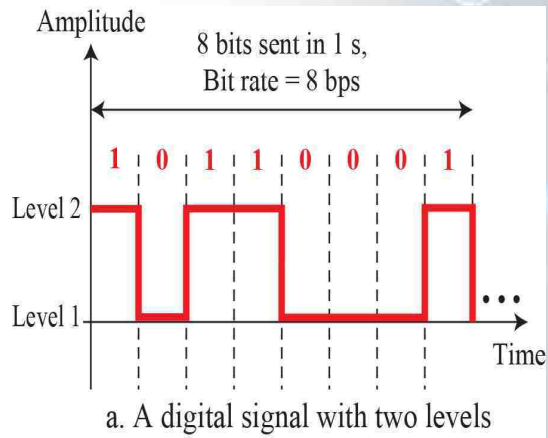
Bit Rate

Most digital signals are nonperiodic, and thus period and frequency are not appropriate characteristics. Another term—bit rate (instead of frequency)—is used to describe digital signals. The bit rate is the number of bits sent in 1s, expressed in bits per second (bps). Figure 3.17 shows the bit rate for two signals.

Bit Rate

- **Number of bits sent in 1 second**
- **Bit Rate is expressed in bits per second (bps)**
- **Most digital signals are non-periodic, and thus period and frequency are not appropriate characteristics**

Bit Rate



Example

Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the cl

$$1 \text{ page} = 24 \text{ lines}$$

$$1 \text{ line} = 80 \text{ characters}$$

$$1 \text{ ch} = 8 \text{ bits}$$

$$\text{Bit Rate} = 100 \times 24 \times 80 \times 8$$

$$= \underline{\underline{1.536 \text{ Mbps}}}$$

Example 3.18

Solution

From Table 3.1 we find the equivalents of 1 ms (1 ms is 10^{-3} s) and 1 s (1 s is 10^6 μ s).

We make the

$$100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bps}$$

subst= 1.536 Mbps ;

Example

A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

$$\begin{aligned} &2 \times 4000 \times 8 \\ &= \underline{\underline{64 \text{ kbps}}} \\ &\quad (64,000 \text{ bps}) \end{aligned}$$

Example 3.19

Solution

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires

$$2 \times 4000 \times 8 = 64,000 \text{ bps}$$

$$= 64 \text{ kbps}$$

Example 3.20

What is the bit rate for high-definition TV (HDTV)?

Solution

HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9 (in contrast to 4 : 3 for regular TV), which means the screen is wider. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second.

Example 3.20

Solution

Twenty-four bits represents one color pixel. We can calculate the bit rate

$$1920 \times 1080 \times 30 \times 24$$

$$= 1,492,992,000 \approx 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.

Bit Length

We discussed the concept of the wavelength for an analog signal: the distance one cycle occupies on the transmission medium. We can define something similar for a digital signal: the bit length. The bit length is the distance one bit occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

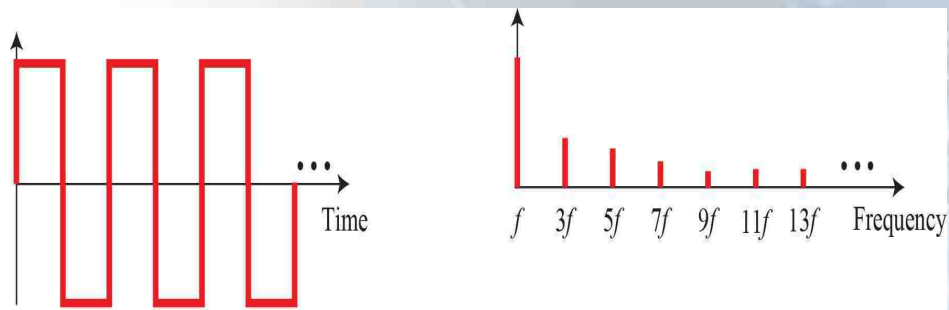
Digital Signal as Composite Analog Signal

- Based on Fourier analysis, a digital signal is a composite analog signal
- A digital signal, in the time domain, comprises connected vertical and horizontal line segments
- Infinite Bandwidth

Digital As Composite Analog

A vertical line in the time domain means a frequency of infinity: a horizontal line in the time domain means a frequency of zero. Going from a frequency of zero to a frequency of infinity implies all frequencies in between are part of the domain.

Digital Signal as Composite Analog Signal



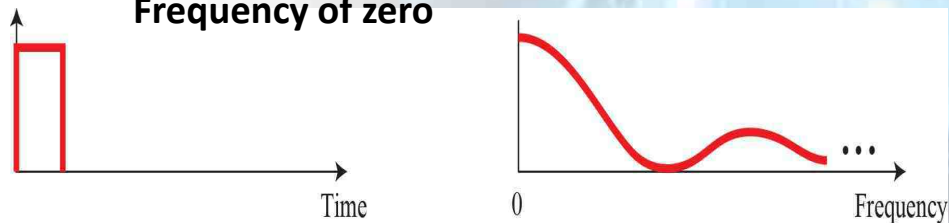
a. Time and frequency domains of periodic digital signal

Vertical line in the time domain:

Frequency of infinity

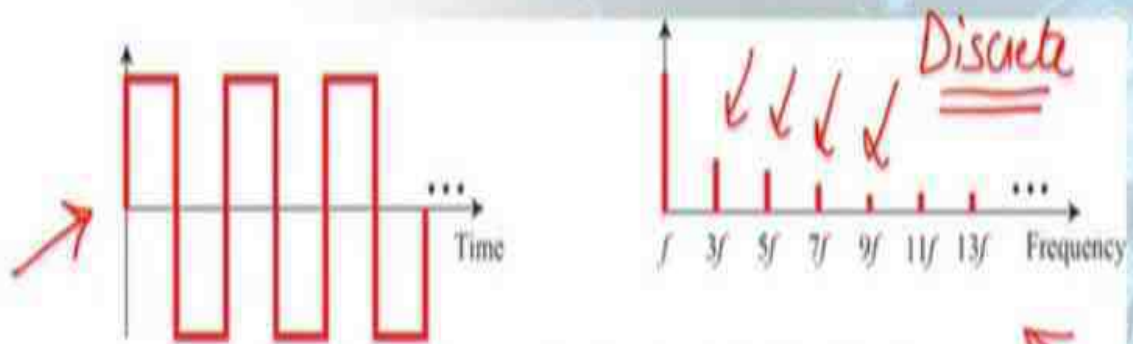
Horizontal line in the time domain:

Frequency of zero



b. Time and frequency domains of nonperiodic digital signal

Digital Signal as Composite Analog Signal



a. Time and frequency domains of periodic digital signal

→ Vertical line in the time domain: Frequency of infinity

→ Horizontal line in the time domain: Frequency of zero

$f = \frac{1}{T}$



b. Time and frequency domains of nonperiodic digital signal

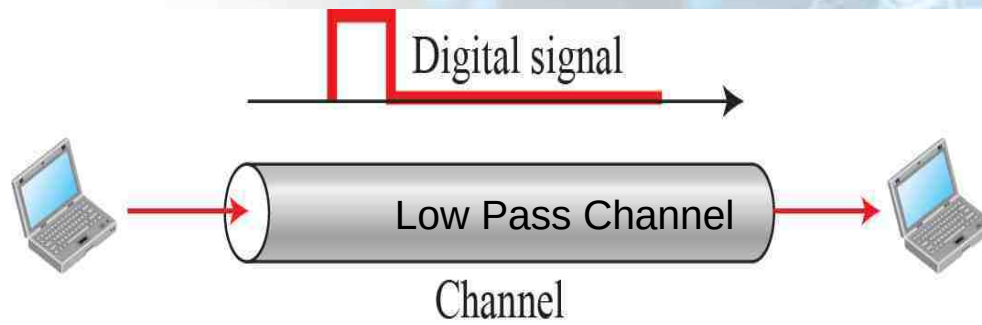
Transmission of Digital Signals

- **Digital signal, periodic or non-periodic, is a composite analog signal with frequencies between zero and infinity (Infinite Bandwidth)**
- **Two approaches for transmission:**
 - ✓ **Baseband Transmission**
 - ✓ **Broadband Transmission**

Transmission of Digital Signals

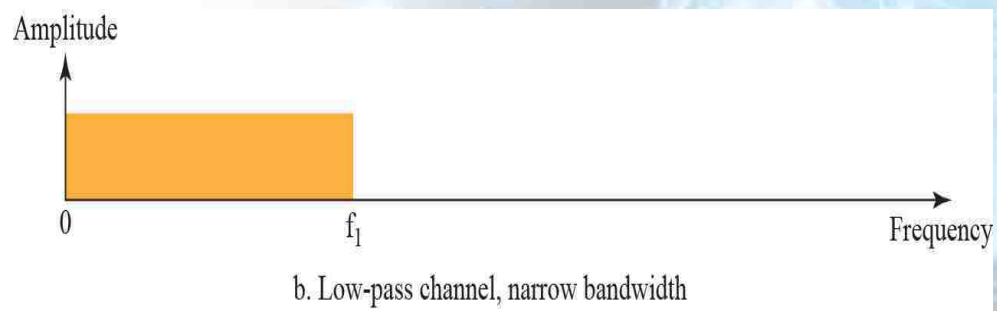
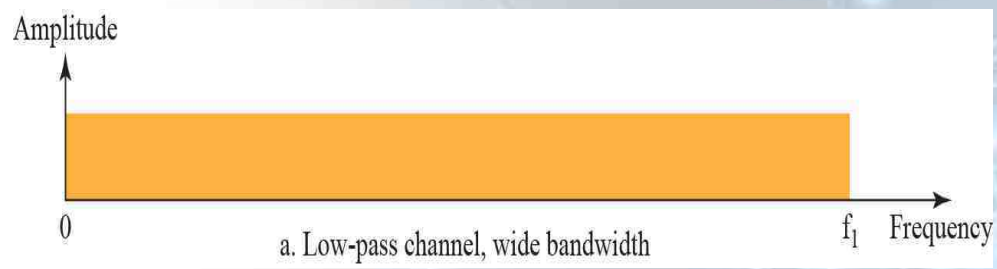
A vertical line in the time domain means a frequency of infinity: a horizontal line in the time domain means a frequency of zero. Going from a frequency of zero to a frequency of infinity implies all frequencies in between are part of the domain.

Baseband Transmission

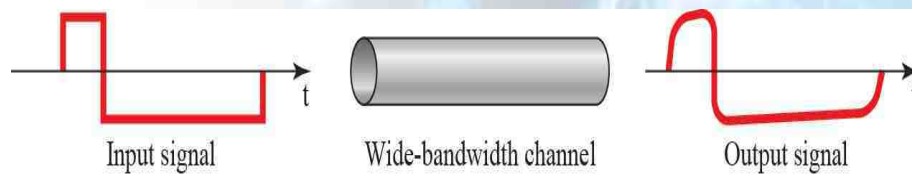
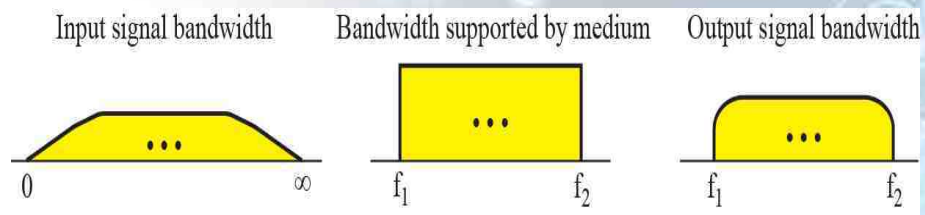


Sending a **Digital Signal** without changing it to an **Analog Signal**

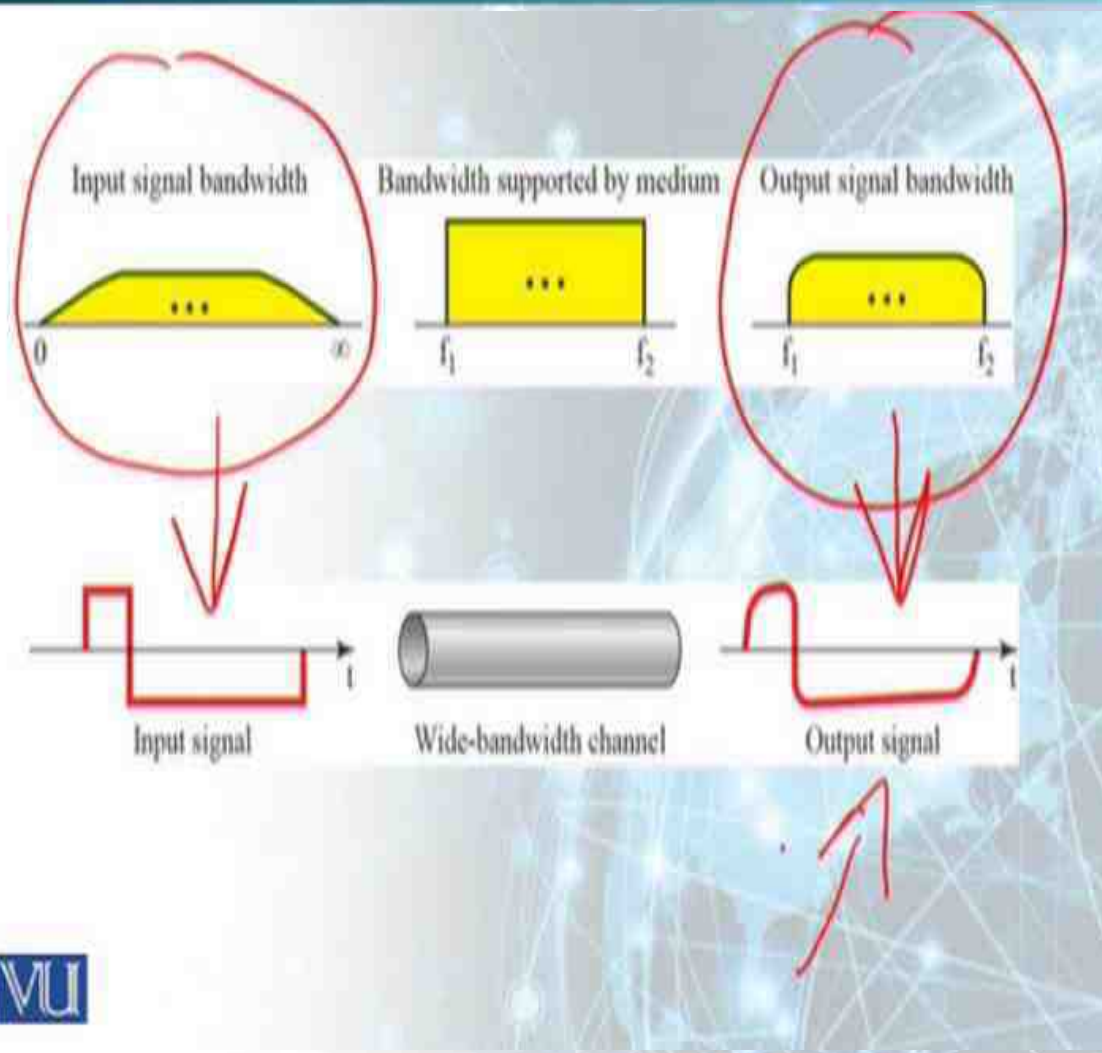
Baseband Transmission



Baseband Transmission



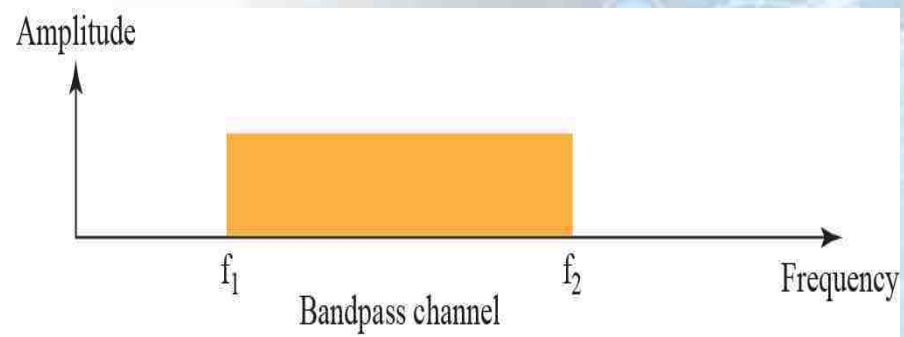
Baseband Transmission



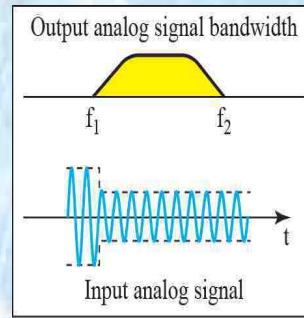
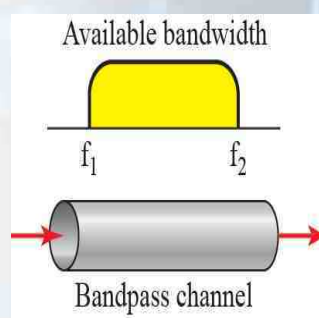
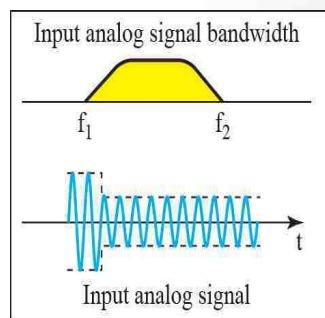
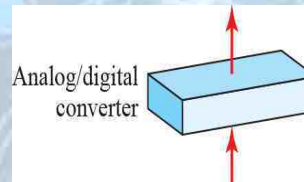
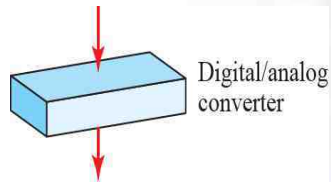
Broadband Transmission (Modulation)

- Changing the Digital signal to an Analog signal for transmission
- Modulation allows us to use a bandpass channel—a channel with a bandwidth that does not start from zero
- More available than a low pass

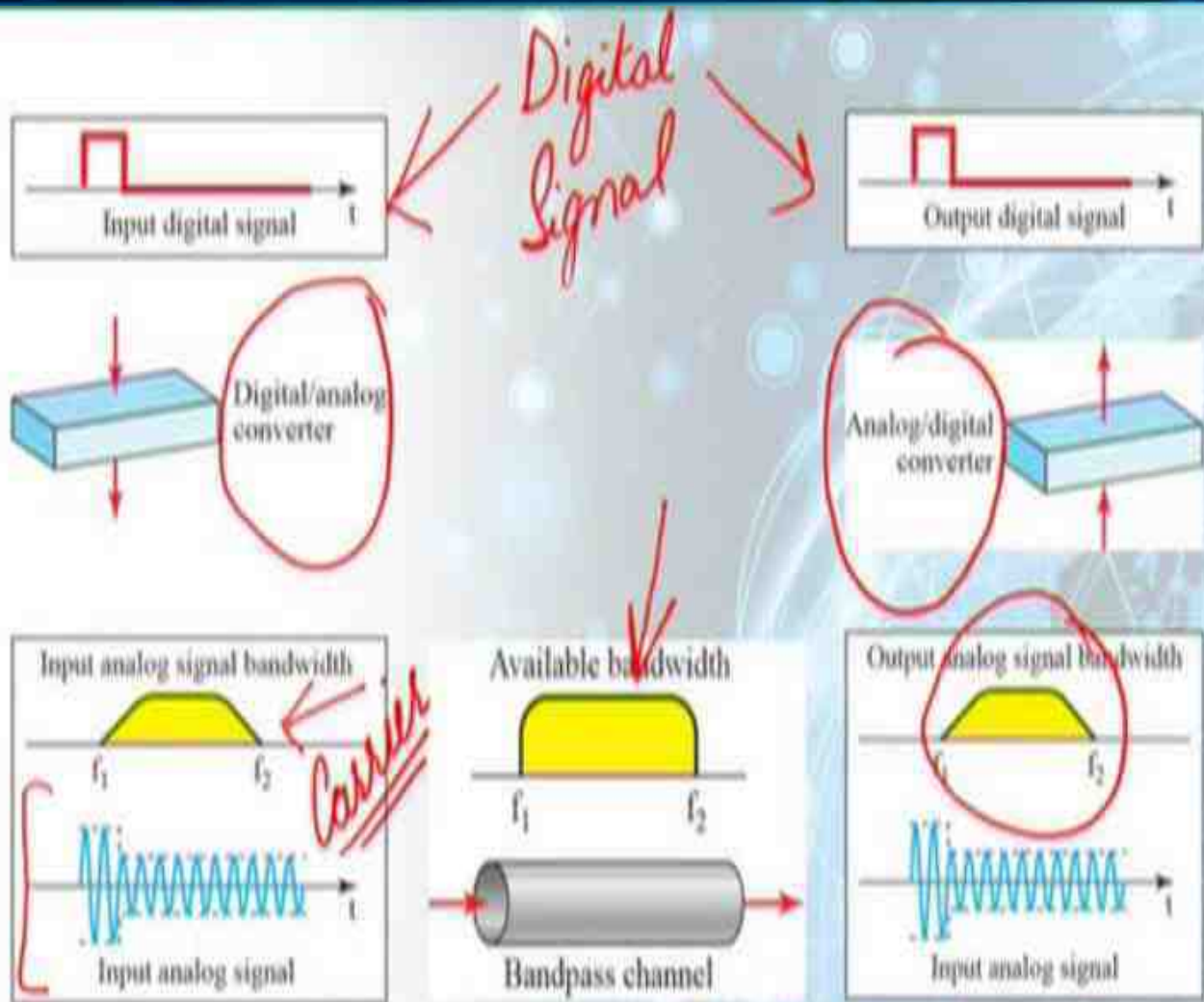
Broadband Transmission (Modulation)



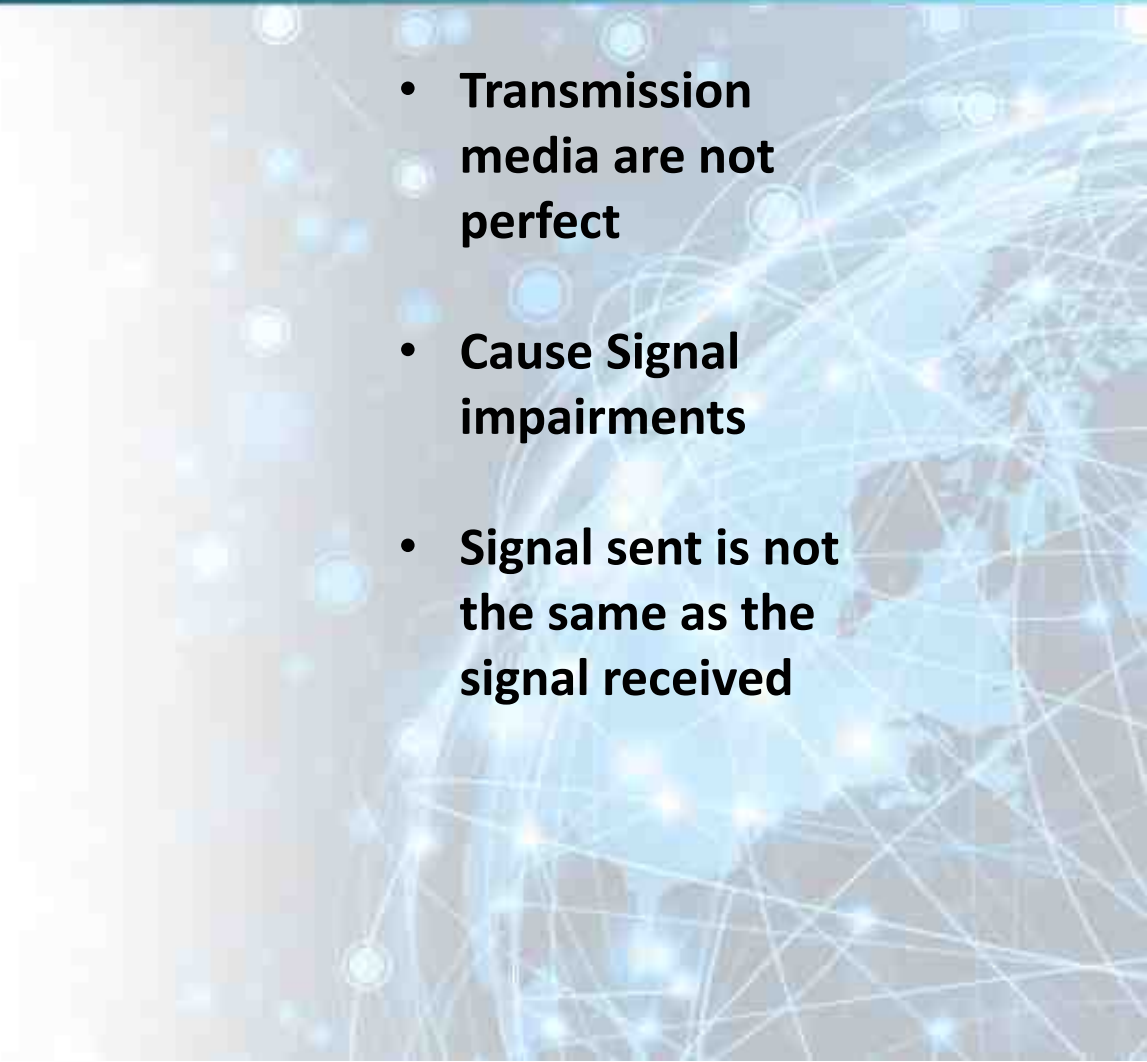
Broadband Transmission (Modulation)



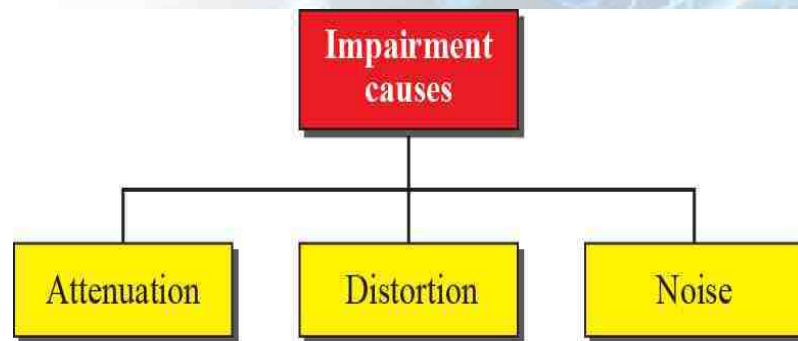
Broadband Transmission (Modulation)



Transmission Impairments

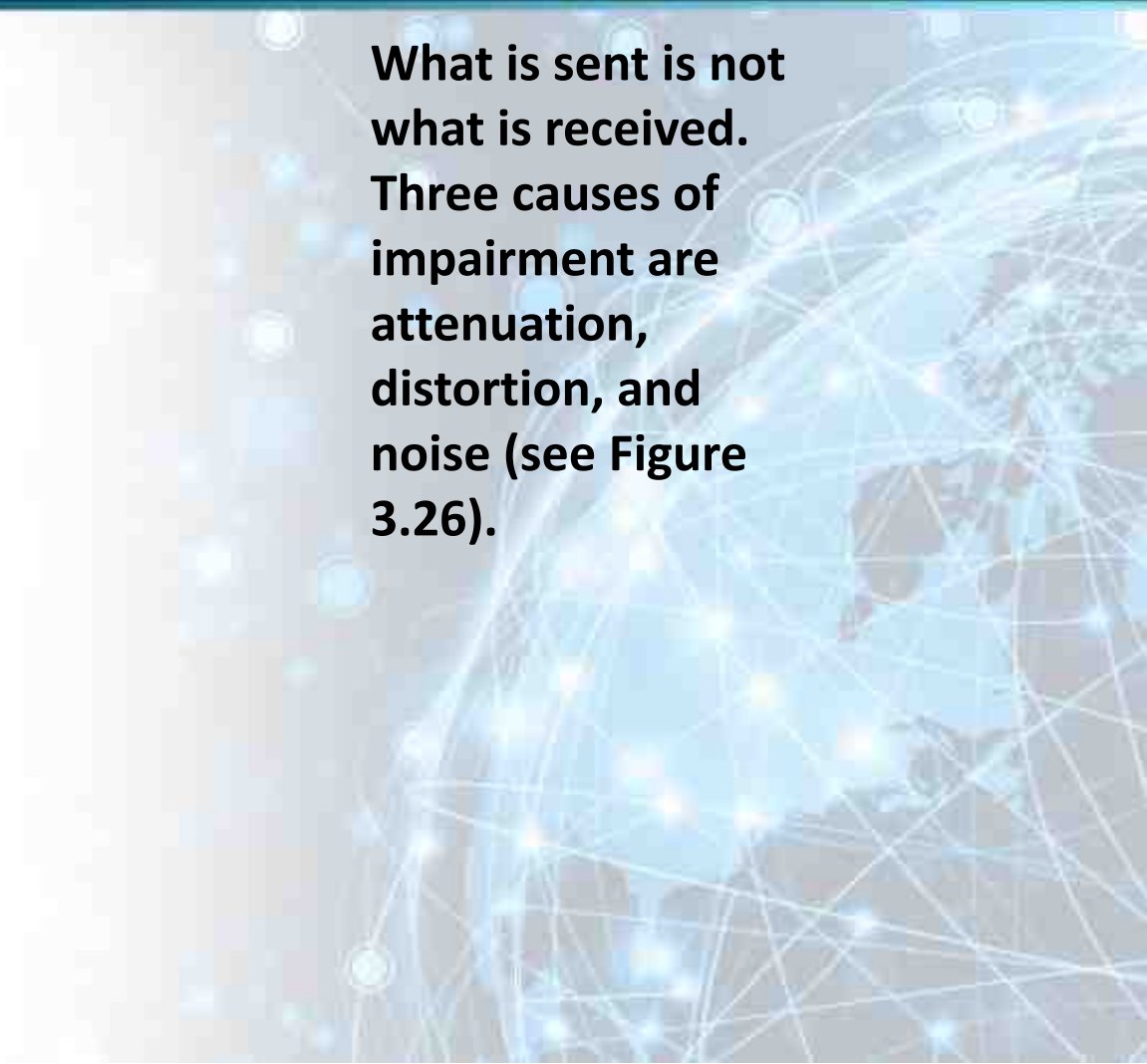
- **Transmission media are not perfect**
 - **Cause Signal impairments**
 - **Signal sent is not the same as the signal received**
- 
- A background graphic featuring a stylized globe with a network of glowing blue lines and nodes, suggesting a global communication or data network. The lines connect various points across the globe, and some nodes are highlighted with larger, brighter circles.

Causes of Transmission Impairment

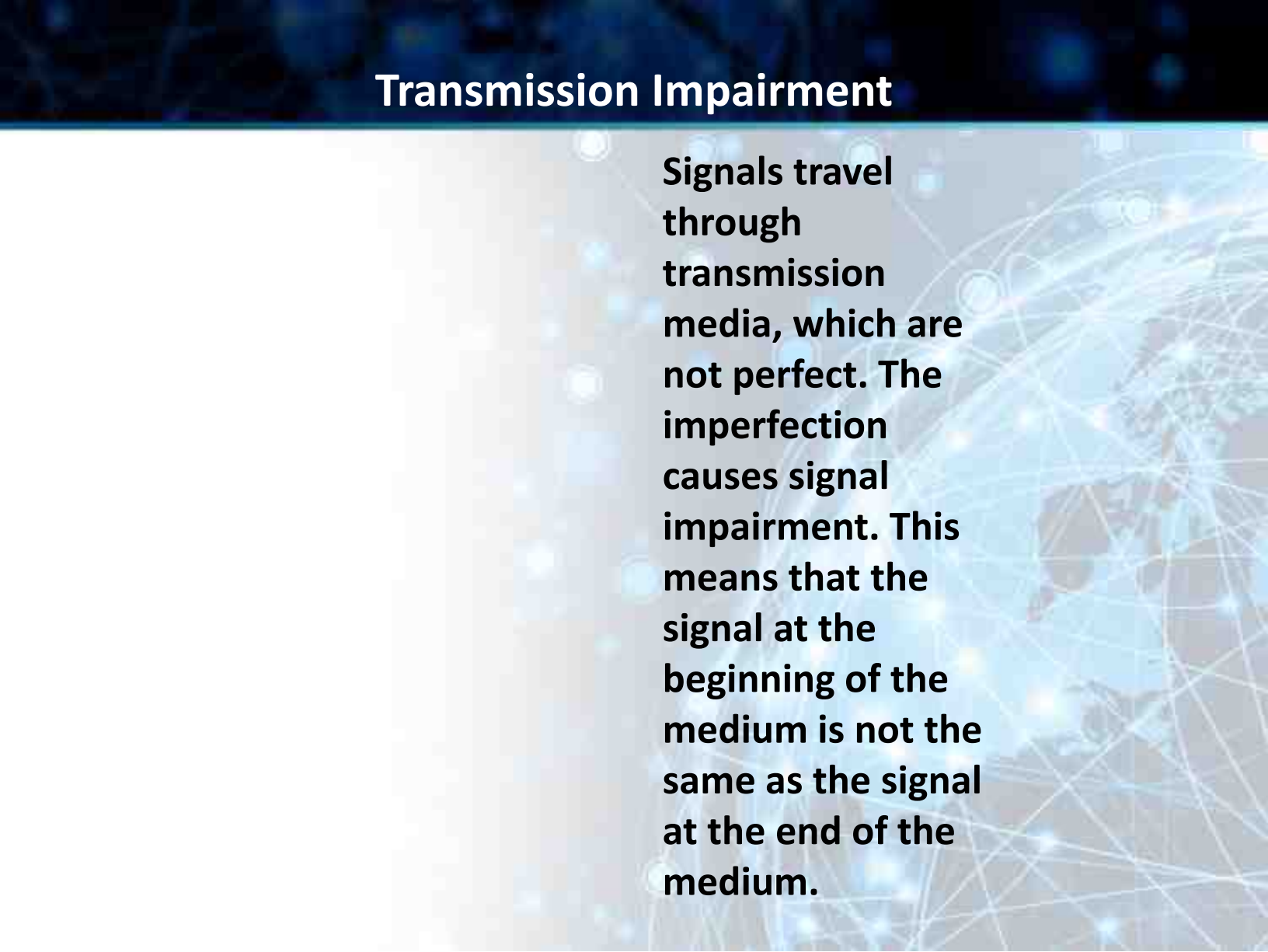


Transmission Impairment

What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise (see Figure 3.26).



Transmission Impairment



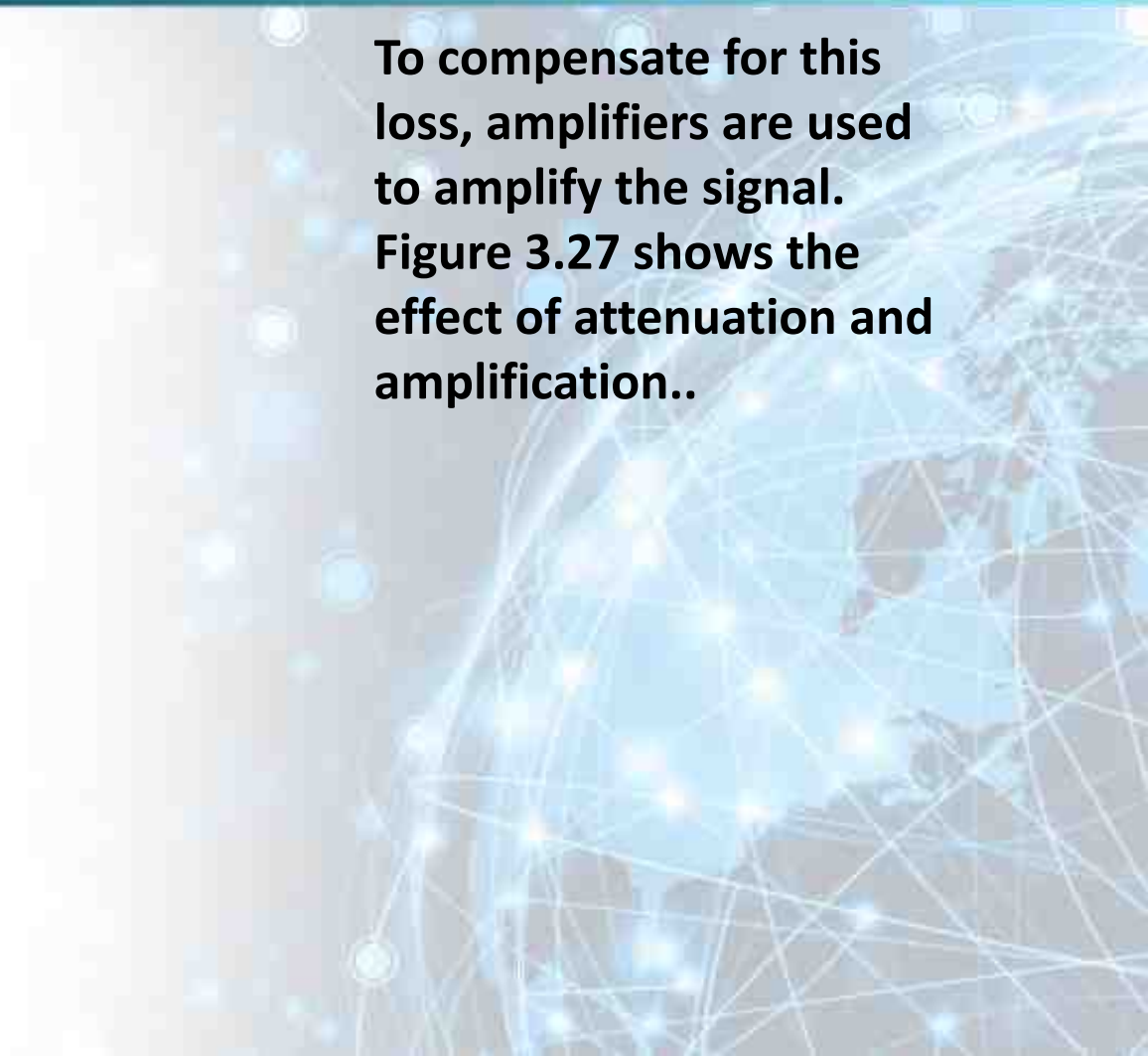
Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.

Attenuation

Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium. That is why a wire carrying electric signals gets warm, if not hot, after a while. Some of the electrical energy in the signal is converted to heat.

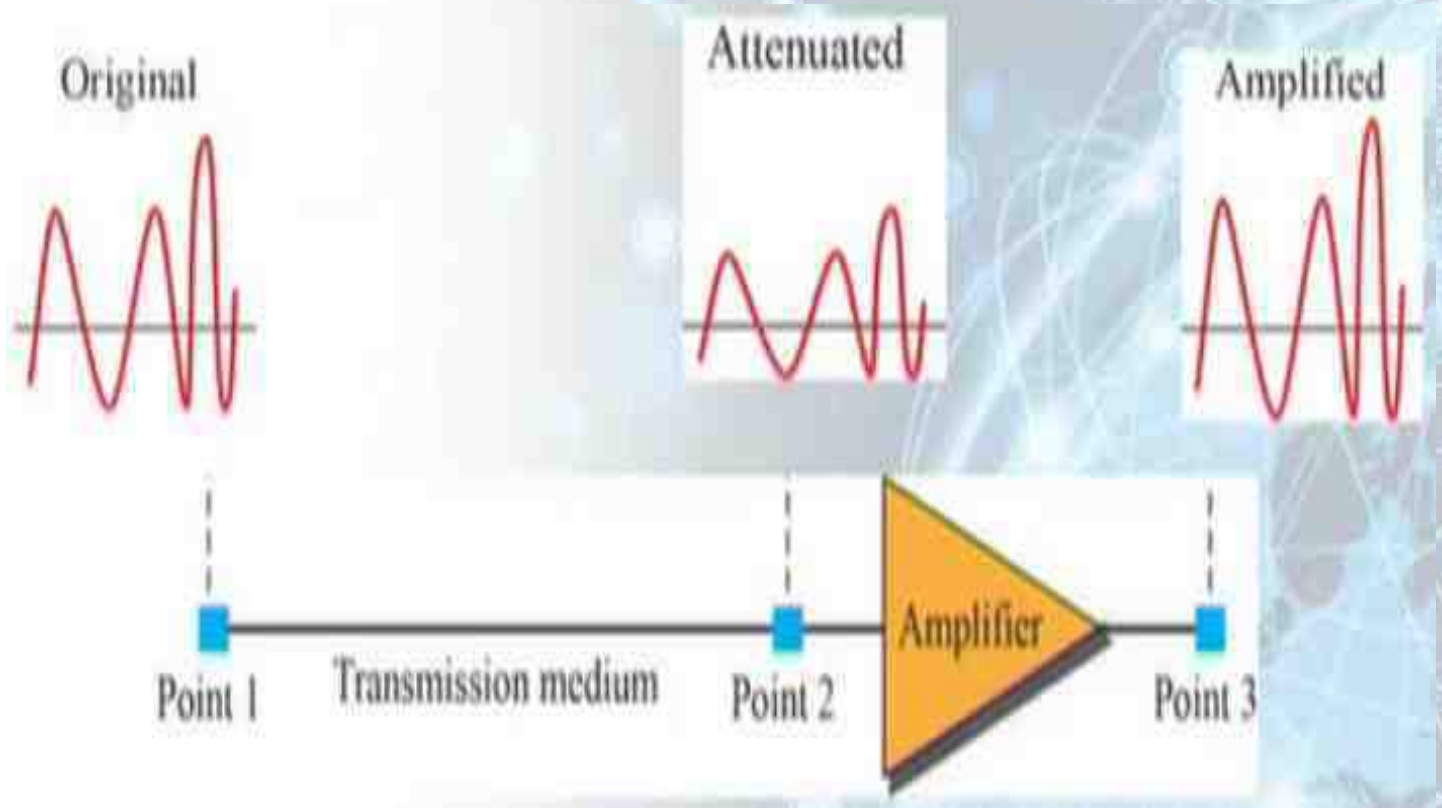
Attenuation

To compensate for this loss, amplifiers are used to amplify the signal. Figure 3.27 shows the effect of attenuation and amplification..



Attenuation and amplification

Figure 3.27



Example 3.26

Suppose a signal travels through a transmission medium and its power is reduced to one half. This means that $P_2 = 0.5 P_1$. In this case, the attenuation (loss of power) can be calculated as

$$\begin{aligned} 10 \log_{10} P_2/P_1 &= 10 \log_{10} (0.5 P_1) / P_1 \\ &= 10 \log_{10} 0.5 = 10 \times (-0.3) = -3 \text{ dB.} \end{aligned}$$

A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.

Example 3.27

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} =$$

$$10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Example 3.28

One reason that engineers use the decibel to measure the changes in the strength of a signal is that decibel numbers can be added (or subtracted) when we are measuring several points (cascading) instead of just two. In Figure 3.28 a signal travels from point 1 to point 4. The signal is attenuated by the time it reaches point 2.

Example 3.28

Between points 2 and 3, the signal is amplified. Again, between points 3 and 4, the signal is attenuated. We can find the resultant decibel value for the signal just by adding the decibel measurements between each set of points. In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

Example 3.29

Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as dB_m and is calculated as $\text{dB}_m = 10 \log_{10} P_m$, where P_m is the power in milliwatts.

Calculate the power of a signal if its $\text{dB}_m = -30$.

Example 3.29

Solution

We can calculate the power in the signal as

$$dB_m = 10 \log_{10} \longrightarrow dB_m = -30 \longrightarrow$$

$$\log_{10} P_m = -3 \longrightarrow P_m = 10^{-3} \text{ mW}$$

Example 3.30

Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as dB_m and is calculated as $\text{dB}_m = 10 \log_{10} P_m$, where P_m is the power in milliwatts.

Calculate the power of a signal if its $\text{dB}_m = -30$.

Example 3.30

Solution

The loss in the cable in decibels is $5 \times (-0.3) = -1.5$ dB. We can

$$\text{dB} = 10 \log_{10} (P_2 / P_1) = -1.5 \quad \text{er}$$

$$\rightarrow (P_2 / P_1) = 10^{-0.15} = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 \text{ mW} = 1.4 \text{ mW}$$

Attenuation and Amplification - Decibel

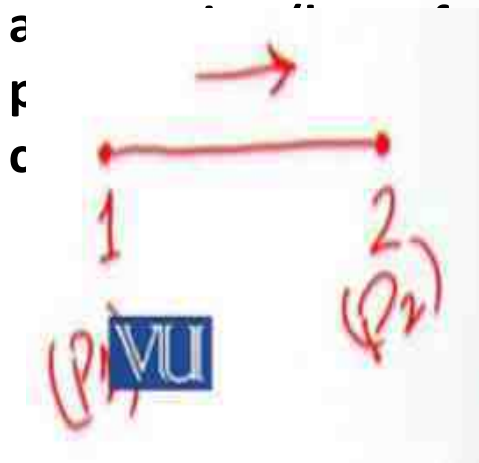
- Unit of Signal strength is Decibel or dB
- Decibel (dB) measures the relative strengths of two signals or one signal at two different points

$$10 \log_{10} P_2/P_1$$

- Decibel is negative if a signal is attenuated and positive if signal is amplified

Example

Suppose a signal travels through a transmission medium and its power is reduced to one half. This means that $P_2 = 0.5 P_1$. In this case, the



$P_2 \rightarrow$ Power at point 2

$P_1 \rightarrow$ Power at point 1

$$P_2 = 0.5 P_1$$

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1}$$

Signal has lost one half the power

$$= 10 \log_{10} 0.5$$
$$= \underline{\underline{-3 \text{ dB}}}$$

Example

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the **amplification** (gain of power) can be calculated as

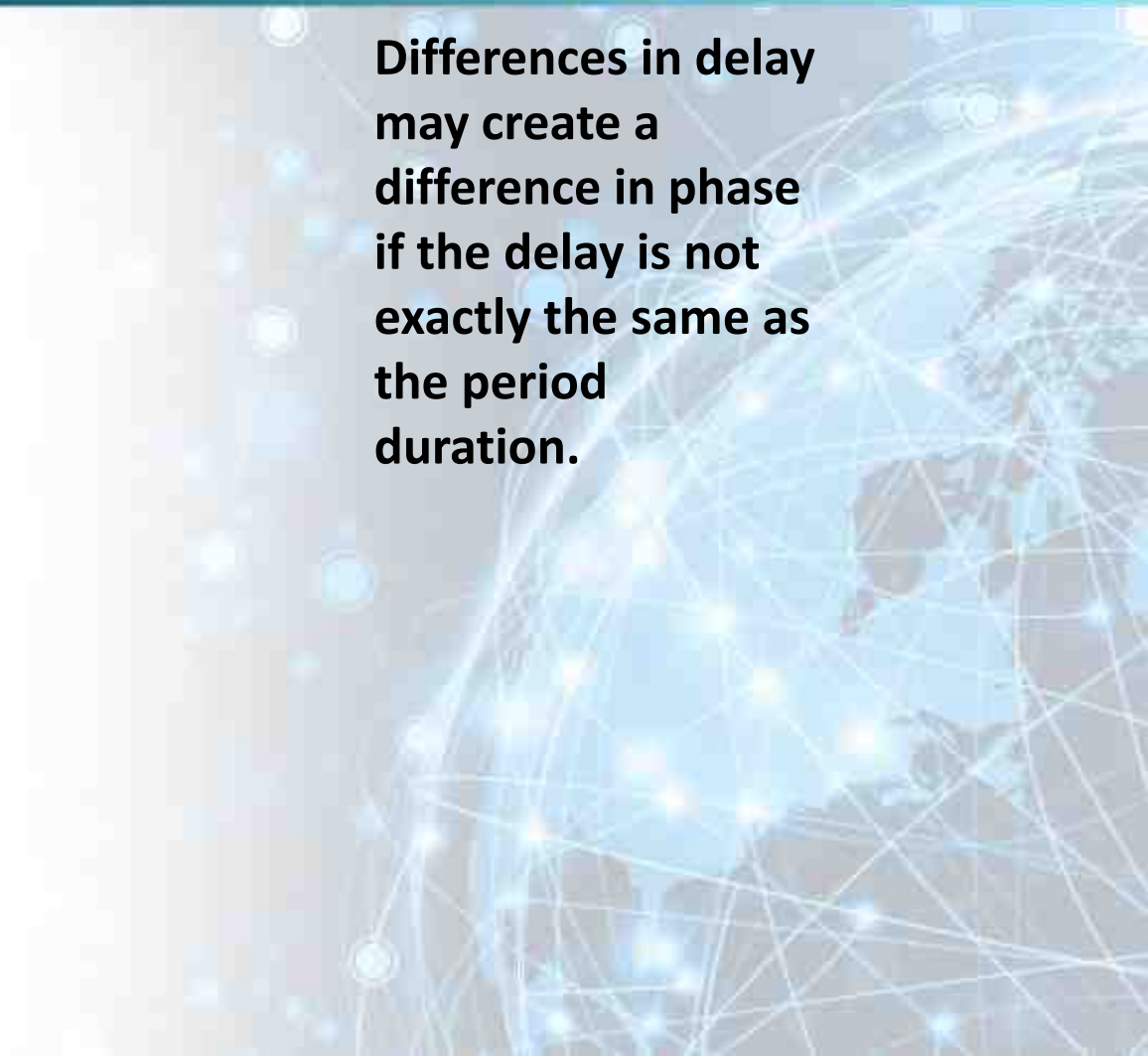
$$\begin{aligned} P_2 &= 10 P_1 \\ 10 \log_{10} \frac{P_2}{P_1} \\ &= 10 \log_{10} \frac{10 P_1}{P_1} \\ &= 10 \text{ dB} \\ &\underline{\underline{\quad\quad\quad}} \end{aligned}$$

Distortion

- Distortion means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies.
- Each signal component has its own propagation speed (see the next section) through a medium and, therefore, its own delay in arriving at the final destination.

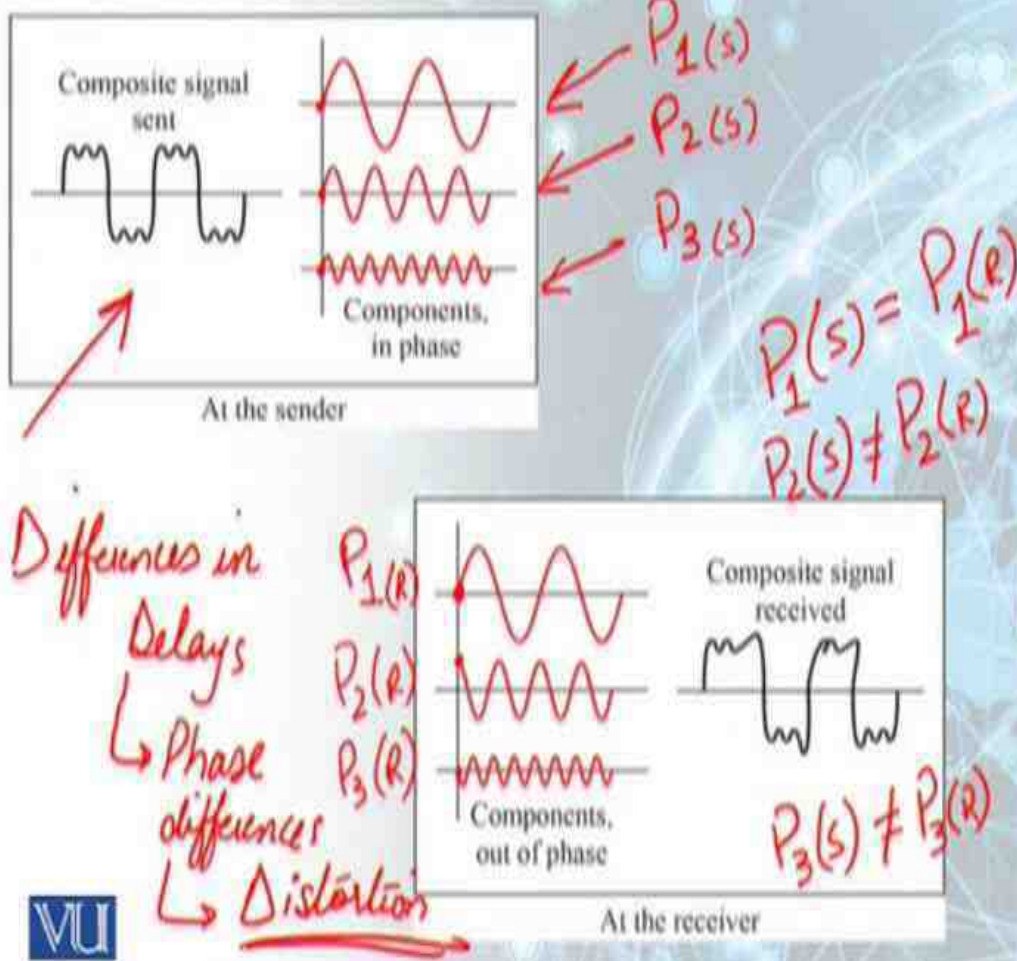
Distortion

Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.



Distortion

Figure 3.2



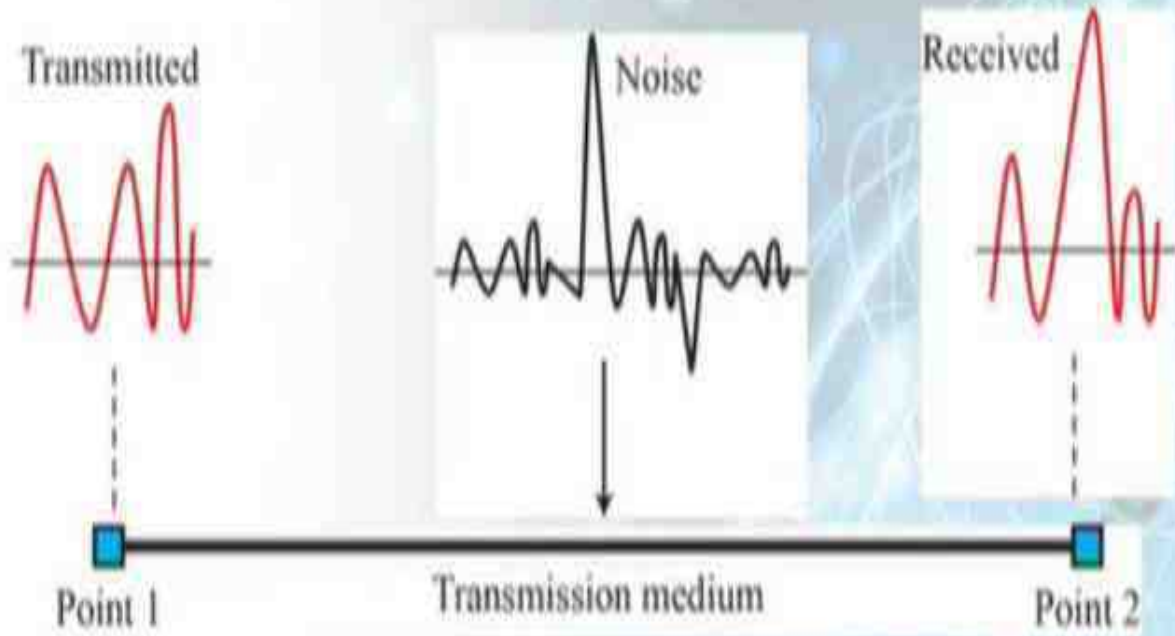
Noise

- Noise is another cause of impairment.
- Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- Thermal noise is the random motion of electrons in a wire, which creates an extra signal not

Noise

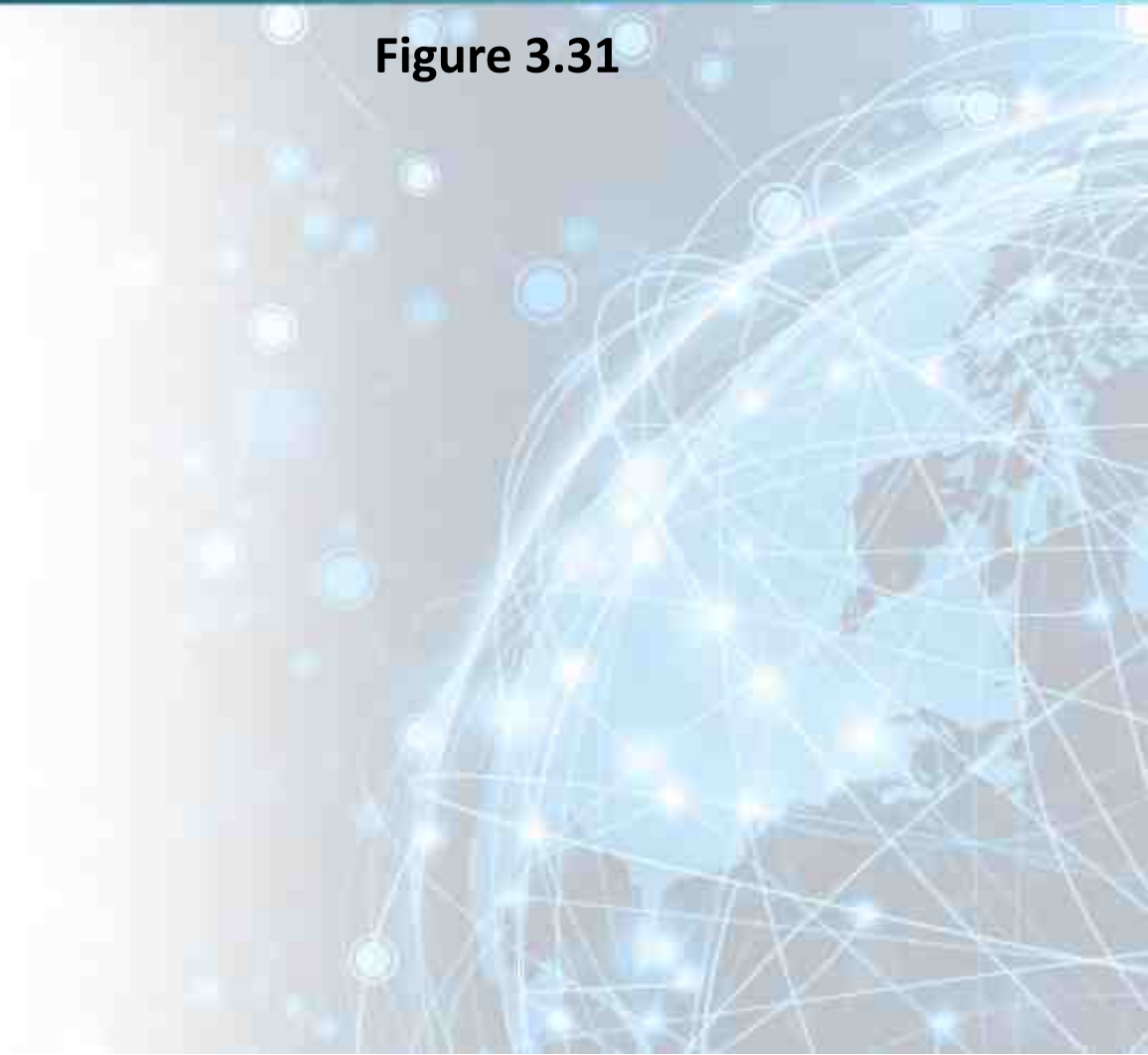
- Induced noise comes from sources such as motors.
- Crosstalk is the effect of one wire on the other.

Noise



Noise

Figure 3.31

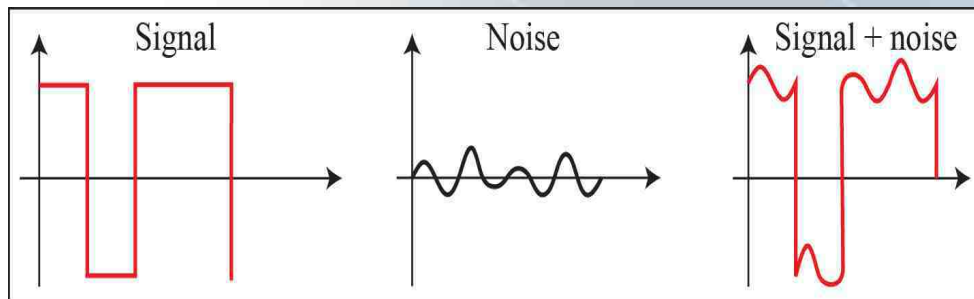


Noise – Signal to Noise Ratio (SNR)

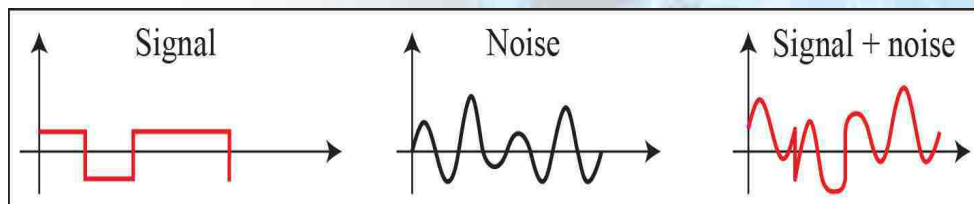
- Signal to Noise Ratio (SNR) is used to find the theoretical bit rate limit of a signal

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$

Noise – Signal to Noise Ratio (SNR)



a. High SNR

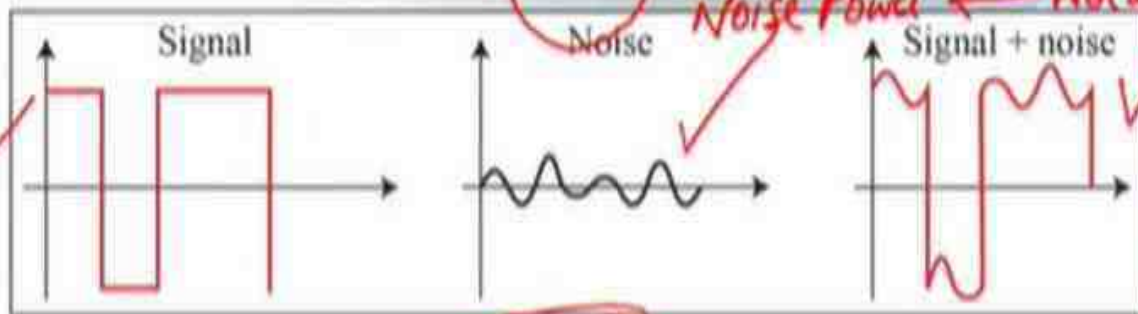


b. Low SNR

Noise – Signal to Noise Ratio (SNR)

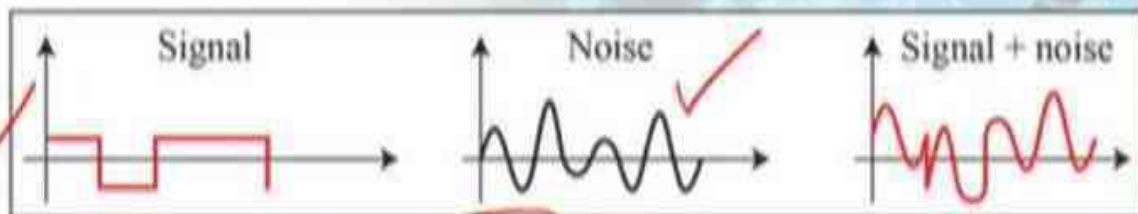
$$\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}}$$

Signal Power ← wanted
Noise Power ← Not wanted



a. High SNR

less corrupted



b. Low SNR

more corrupted

→ $\text{SNR}_{dB} = 10 \log_{10} \text{SNR}$

Example

The power of a signal is 10 mW and the power of the noise is 1 μ W; what are the values of SNR and SN

$$SNR = \frac{10 \text{ mW}}{1 \mu\text{W}} = 10,000$$

$$SNR_{dB} = 10 \log_{10} 10,000 = \underline{\underline{40 \text{ dB}}}$$

Example

The values of SNR and SNR_{dB} for a noiseless channel are calculated as

$$\underline{\text{Noise} = 0}$$

↳ NOT a real life scenario

$$\text{SNR} = \frac{(\text{Sig. Power})}{0} = \infty$$

$$= 10 \log_{10} \infty = \infty$$

Not Real

Example 3.31

The power of a signal is 10 mW and the power of the noise is 1 μ W; what are the values of SNR and SNR_{dB}?

Solution

The values of SNR and SNR_{dB} can be calculated as

$$\text{SNR} = (10,000 \mu\text{W}) / (1 \mu\text{W}) = 10,000 \quad \text{SNR}_{\text{dB}}$$

$$= 10 \log_{10} 10,000 = 10 \log_{10} 10^4 = 40$$

Example 3.32

The values of SNR and SNR_{dB} for a noiseless channel are

Solution

The values of SNR and SNR_{dB} for a noiseless channel are

$$\text{SNR} = (\text{signal power}) / 0 = \infty$$

$$\longrightarrow \text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life; it is an ideal.

Data Rate Limits

- How fast we can send data, in bits per second, over a channel?
- Data Rate depends on 3 factors:
 - ✓ The Bandwidth available
 - ✓ The level of the signals we use
 - ✓ The level of noise

Data Rate Limits

- Two theoretical formulas developed to calculate the data rate:
 - ✓ one by Nyquist for a noiseless channel
 - ✓ another by Shannon for a noisy channel

Noiseless Channel : Nyquist Rate

$$\text{Bit Rate} = 2 \times \text{Bandwidth} \times \log_2 L$$

BW = BW of channel

L = No. of Signal levels

BR = bps

Bit Rate $\propto$ L

$L \uparrow \Rightarrow \text{Bit Rate} \uparrow$

$(L \Rightarrow 0, 1) \quad \underline{L = 2 \text{ levels}}$

- For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate
- Finding balance between Bit rate

m

/

Example

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$BR = 2 \times 3000 \times \log_2 2$$
$$= \underline{6000 \text{ bps}}$$

Example

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The

n
c
a

$$\underline{\underline{BR \propto L}}$$

$$BR = 2 \times 3000 \times \log_2 4$$
$$= 12,000 \text{ bps}$$

Reliability

Noisy Channel : Shannon Capacity

- In reality, we cannot have a noiseless channel; the channel is always noisy
- In 1944, Claude Shannon introduced a formula, to determine the theoretical highest data rate for a noisy

$$\text{Capacity} = \text{Bandwidth}_{(\text{channel})} \times \log_2(1 + \text{SNR})$$

Capacity $\rightarrow$ Capacity of channel
 $\rightarrow$ Bit Rate(max)

Levels $\rightarrow L \times$

Max. BR < Capacity of channel

Example

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$\text{SNR} \approx \text{zero}$$

$$C = B \log_2(1 + \text{SNR})$$

$$= B \log_2 1$$

$$= B \times \text{zero}$$

$$\boxed{C = \text{zero}}$$

→ Cannot receive any data through channel.

Example

Theoretical highest bit rate of a Telephone line with a Bandwidth of 3000 Hz assigned for data communication. SNR is usually 3162. The capacity is calculated as:

$$C = 3000 \times \log_2(1 + 3162)$$
$$= 34,860 \text{ bps}$$

Telephone line

Using Both Limits

- In practice, we need to use both methods to find the limits and signal levels
- Shannon's formula gives us the upper limit while the Nyquist formula gives us the signal levels

Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0)$$

$$= B \log_2 1 = B \times 0 = 0$$

capacity C is calculated as

Example 3.37

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

A background graphic featuring a stylized globe with a network of glowing blue lines and nodes, suggesting a global communication or data network. The globe is partially obscured by the text on the right.

Example 3.38

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot

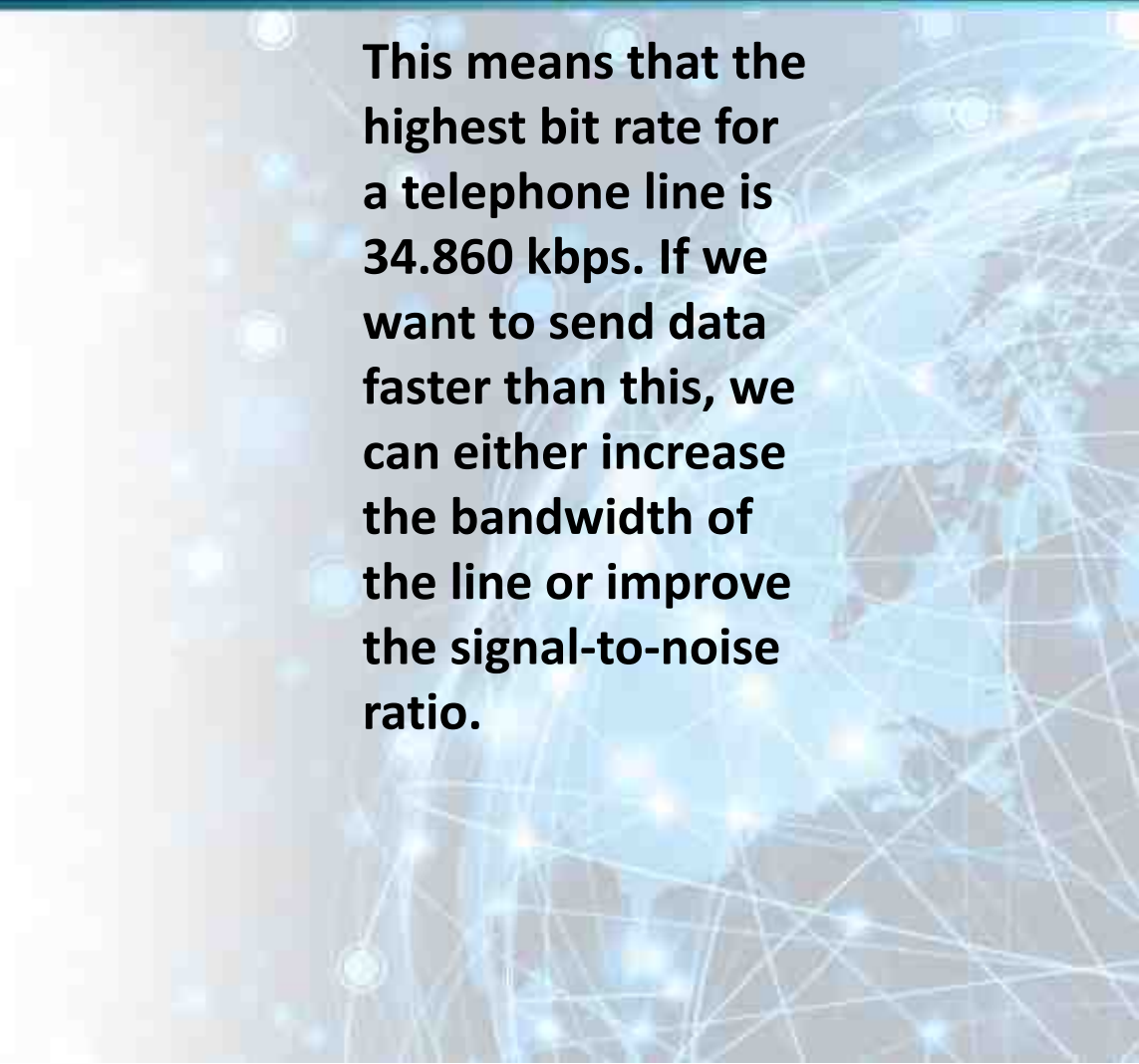
$$C = B \log_2 (1 + \text{SNR}) =$$

$$3000 \log_2 (1 + 3162) =$$

$$3000 \times 11.62 = 34,860 \text{ bps}$$

Example 3.38

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.



Example 3.39

The signal-to-noise ratio is often given in decibels. Assume that $\text{SNR}_{\text{dB}} = 36$ and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

Example 3.39

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} \longrightarrow$$

$$\text{SNR} = 10^{\text{SNR}_{\text{dB}}/10} \longrightarrow$$

$$\text{SNR} = 10^{3.6} = 3981$$

$$C = B \log_2(1 + \text{SNR}) =$$

$$2 \times 10^6 \times \log_2 3982$$

$$= 24 \text{ Mbps}$$

Example 3.40

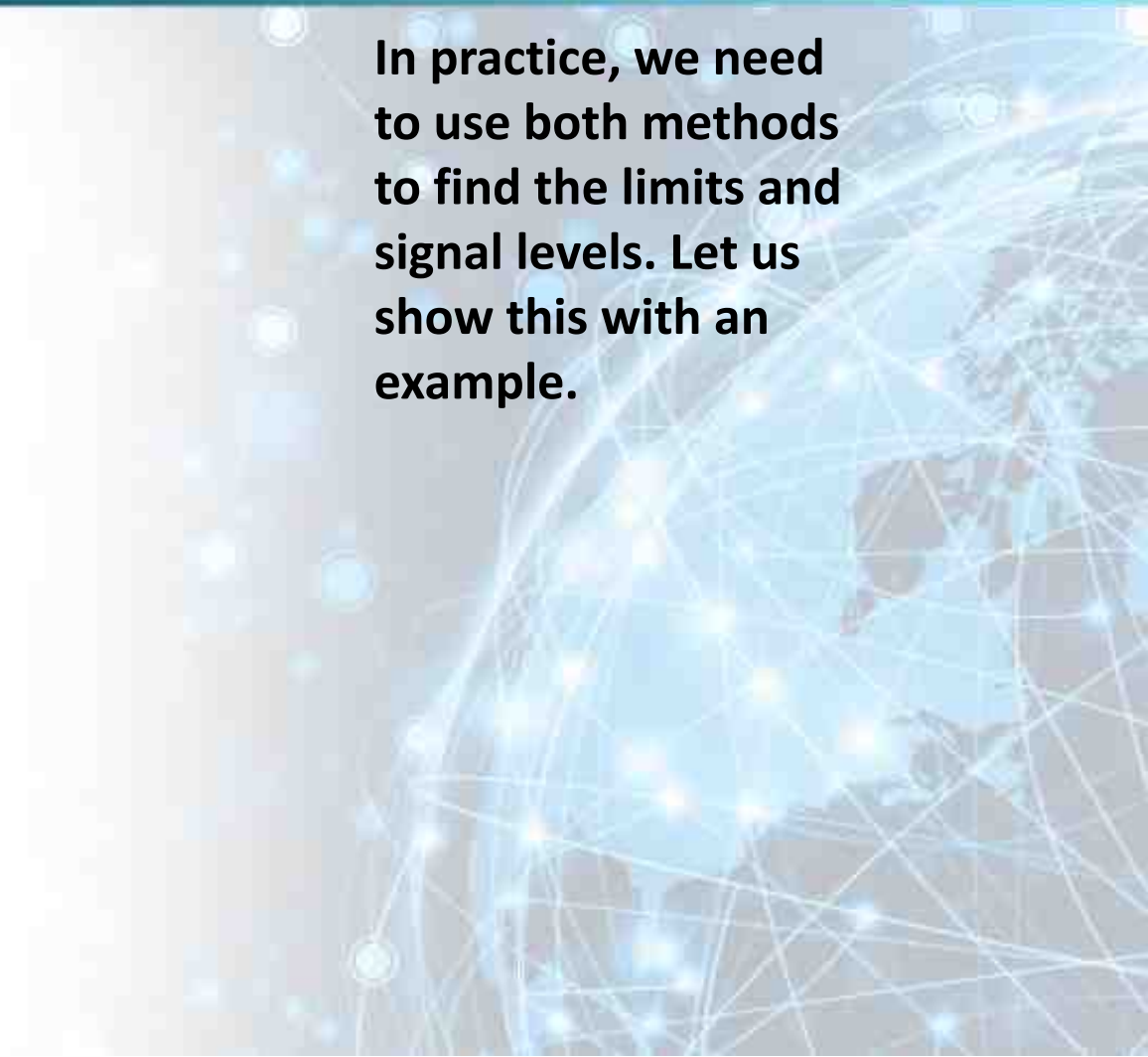
When the SNR is very high, we can assume that $\text{SNR} + 1$ is almost the same as SNR. In these cases, the theoretical channel capacity can be simplified to $C \approx B \log_2 \text{SNR}_{\text{dB}}$. For example, we can calculate the

$$C = 2 \text{ MHz} \times (36 / 3) = 24 \text{ Mbps}$$

of the previous example as

Using Both Limits

In practice, we need to use both methods to find the limits and signal levels. Let us show this with an example.



Example 3.41

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the

$$C = B \log_2(1 + \text{SNR}) = \text{a to bit}$$
$$10^6 \log_2(1 + 63) = 10^6 \log_2 64 =$$
$$6 \text{ Mbps}$$

Example 3.41

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps. Then we use the Nyquist formula

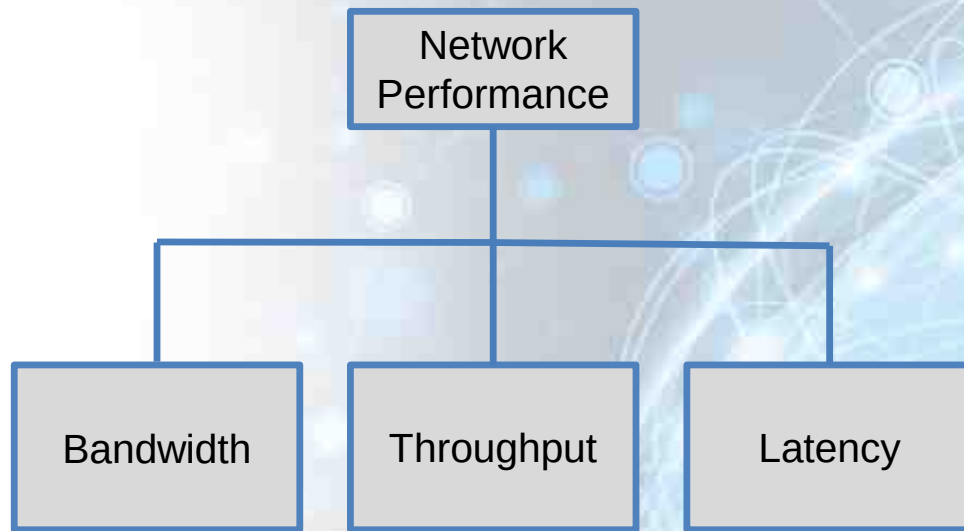
$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L$$

number of levels $\rightarrow L = 4$

Network Performance

- **Data transmission (in form of Signal) over a network and how network behaves is important**
- **More important is the performance of the network**
- **How good is our network?**

Network Performance



- **There are 3 characteristics of network performance**

Bandwidth

- An important characteristic that measures Network Performance
- Bandwidth can be used in two different contexts with two different measuring values:
 - Bandwidth in Hertz
 - Bandwidth in

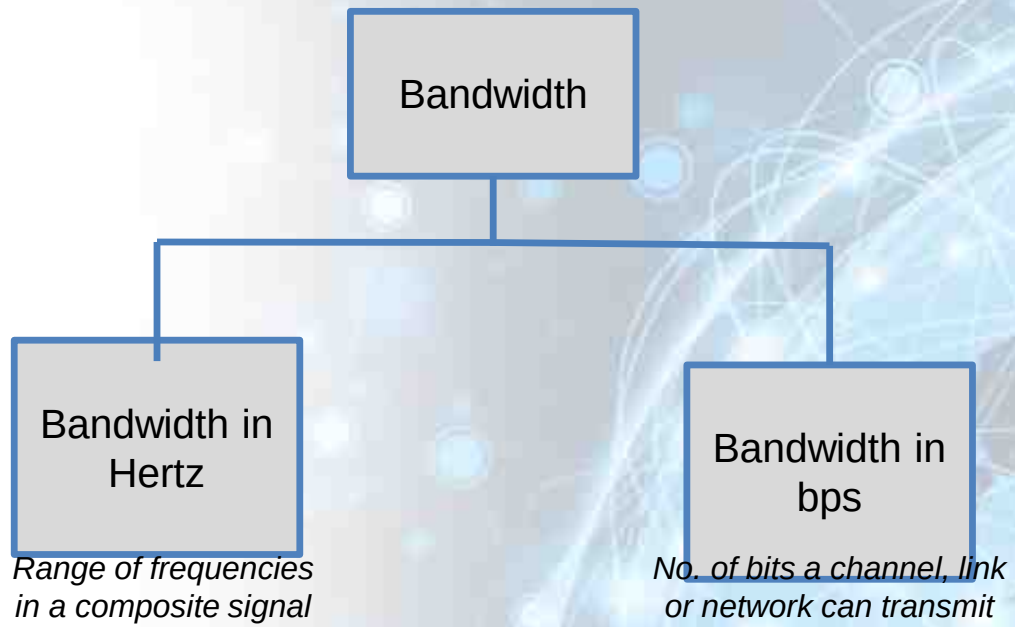
Example 3.42

The bandwidth of a subscriber line is 4 kHz for voice or data. The bandwidth of this line for data transmission can be up to 56,000 bps using a sophisticated modem to change the digital signal to analog.

Example 3.43

If the telephone company improves the quality of the line and increases the bandwidth to 8 kHz, we can send 112,000 bps by using the same technology as mentioned in Example 3.42.

Bandwidth



Throughput

- Measure of how fast we can actually send data through a network.
- Bandwidth is not the same as Throughput
- A link may have a bandwidth of B bps, but we can only send T bps through this link.

Example

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of the network?

$$T = \frac{(12,000 \times 10,000)}{60} = 2 \text{ Mbps}$$

$$\begin{aligned} T &= 2 \text{ Mbps} \\ B &= 10 \text{ Mbps} \end{aligned}$$

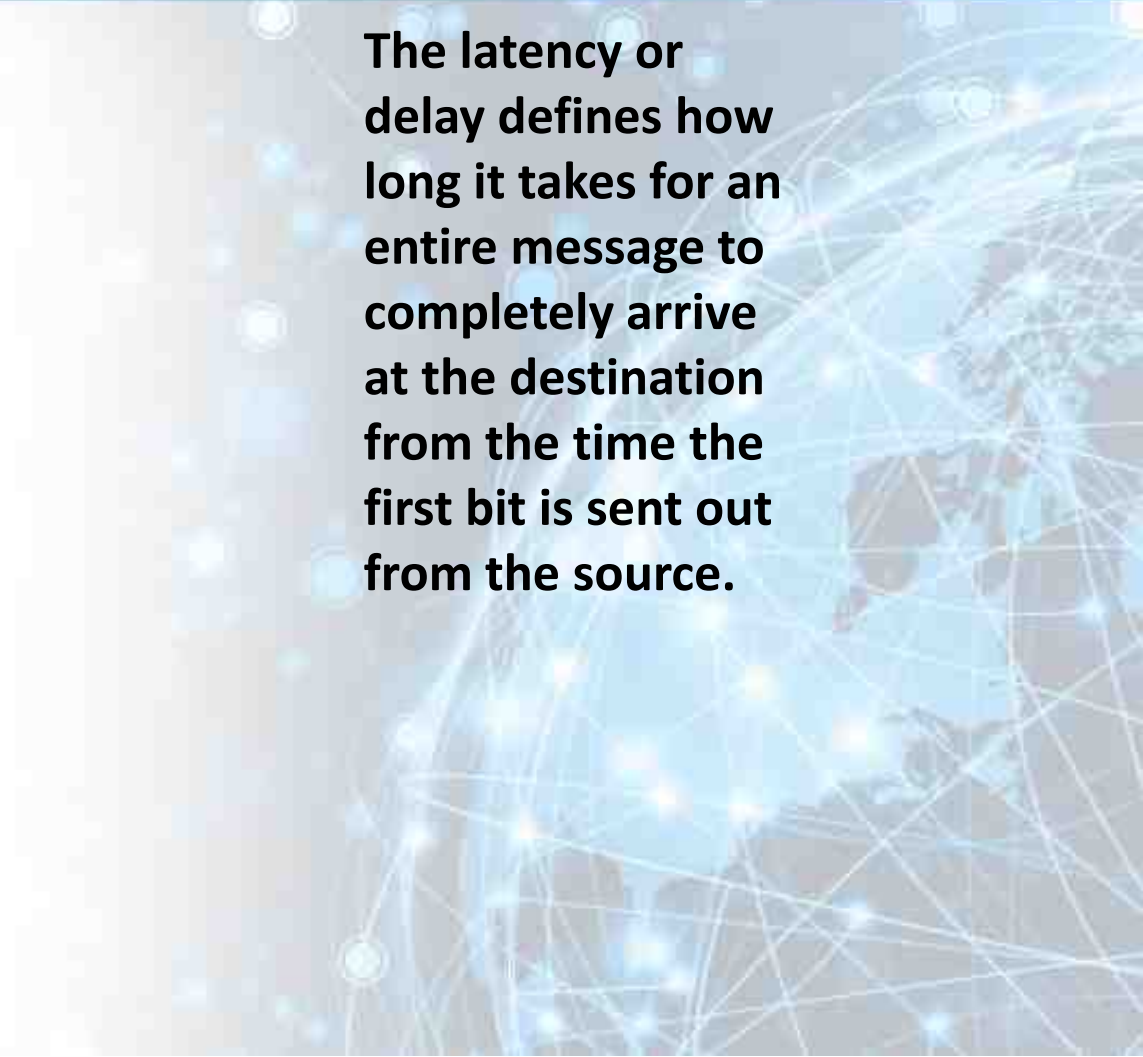
Throughput

The throughput is a measure of how fast we can actually send data through a network.

Although, at first glance, bandwidth in bits per second and throughput seem the same, they are different. A link may have a bandwidth of B bps, but we can only send T bps through this link with T always less than B .

Throughput

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

The background of the slide features a stylized globe with a network of glowing blue lines and nodes, representing global connectivity and data transmission. The lines crisscross the globe, connecting various points, and the nodes are small, bright blue circles. The overall color scheme is light blue and white, with a dark blue header.

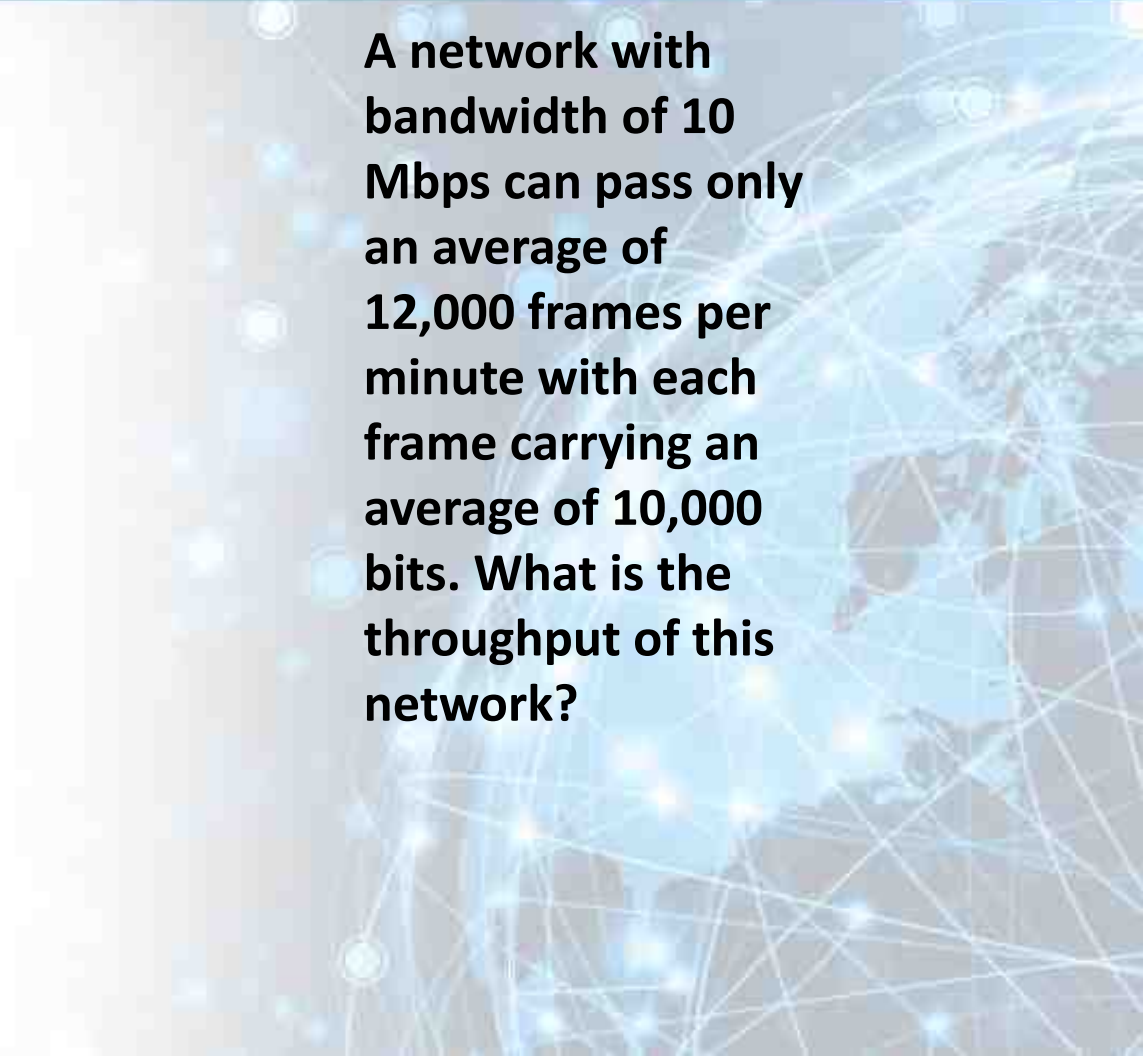
Throughput

We can say that latency is made of four components: propagation time, transmission time, queuing time and processing delay.

Latency =
propagation time +
transmission time +
queuing time +
processing delay

Example 3.44

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

The background of the slide features a stylized globe with a network of glowing blue lines and nodes, representing a global communication network. The lines connect various points across the globe, symbolizing data flow and network connectivity.

Example 3.44

Solution

We can calculate the throughput as

Throughput =

$$(12,000 \times 10,000) / 60 = 2\text{Mbps}$$

The throughput is almost one-fifth of the bandwidth in this case.

Example 3.45

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

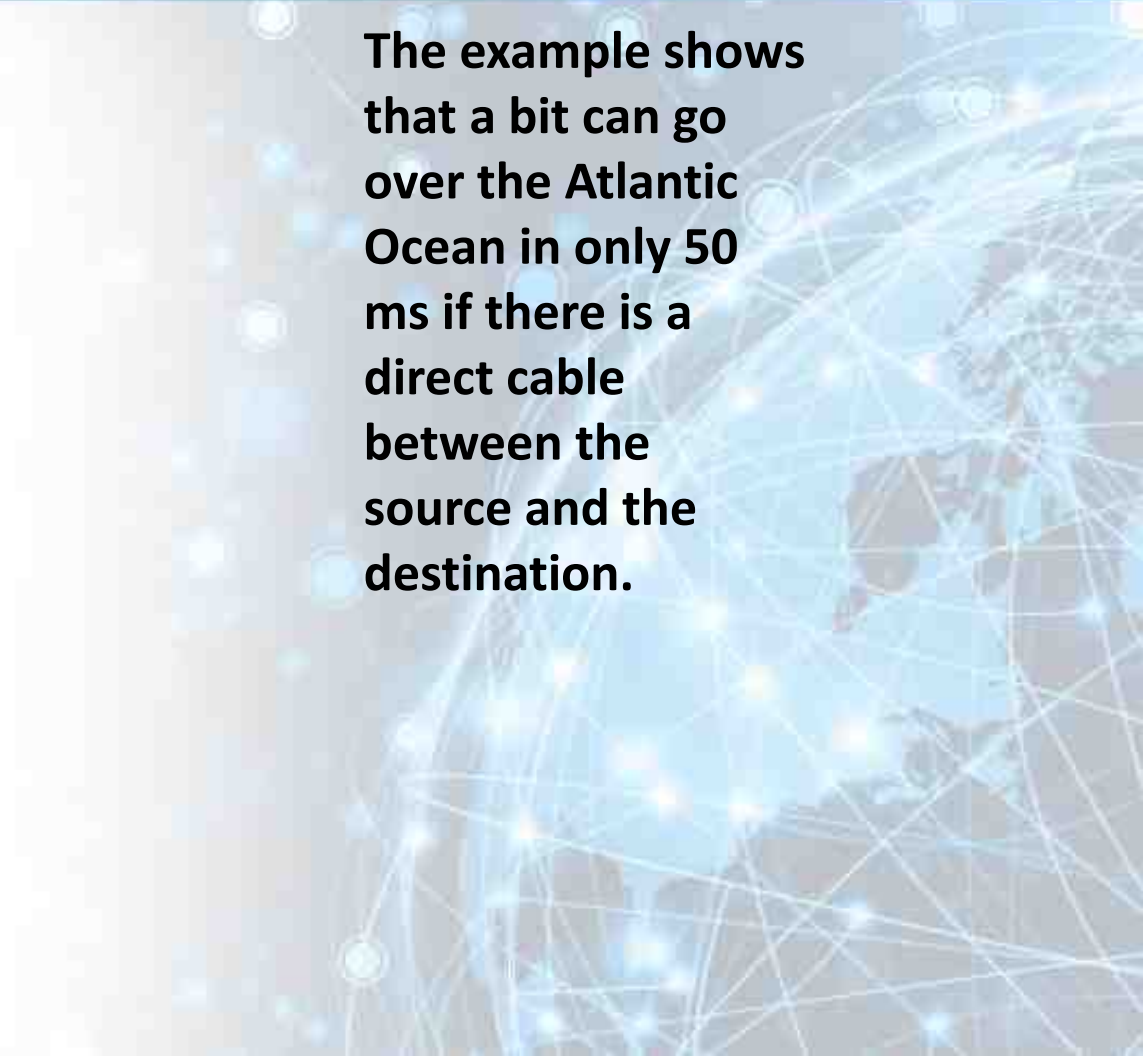
Solution

We can calculate the propagation time as

$$\text{Propagation time} = \frac{(12,000 \times 10,000)}{(2.4 \times 10^8)} =$$

Example 3.45

The example shows that a bit can go over the Atlantic Ocean in only 50 ms if there is a direct cable between the source and the destination.

The background of the slide features a stylized globe with a network of glowing blue lines and nodes, representing global communication or data transmission. The lines connect various points across the globe, with a higher density of connections in the Atlantic Ocean region, visually supporting the text about a direct cable across the Atlantic.

Example 3.46

What are the propagation time and the transmission time for a 2.5-KB (kilobyte) message if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.

Example 3.46

Solution

We can calculate the propagation and transmission time as

$$\begin{aligned} \text{Propagation time} &= \\ &= (12,000 \times 1000) / \\ &= (2.4 \times 10^8) = 50 \text{ ms} \end{aligned}$$

$$\begin{aligned} \text{Transmission time} &= \\ &= (2500 \times 8) / 10^9 = \\ &= 0.020 \text{ ms} \end{aligned}$$

Example 3.46

Solution

Note that in this case, because the message is short and the bandwidth is high, the dominant factor is the propagation time, not the transmission time.

Example 3.47

What are the propagation time and the transmission time for a 5-MB (megabyte) message (an image) if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8

Example 3.47

Solution

We can calculate the propagation and transmission times as

$$\begin{aligned} \text{Propagation time} &= \\ &= (12,000 \times 1000) / \\ &= (2.4 \times 10^8) = 50 \text{ ms} \end{aligned}$$

$$\begin{aligned} \text{Transmission time} &= \\ &= (5,000,000 \times 8) / \\ &= 106 = 40 \text{ s} \end{aligned}$$

We can calculate the propagation and transmission

Latency or Delay

- Latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source

$$\text{Latency} = \text{Propagation Time} + \text{Transmission Time} + \text{Queuing Time} + \text{Processing Delay}$$

$$PT = \frac{\text{Distance}}{\text{Prop Speed}}$$

$$TT = \frac{\text{Message Size}}{BW}$$

QT → Time the message is held

Example

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

$$\begin{aligned} PT &= \frac{(12000 \times 1000)}{2.4 \times 10^8} \\ &= \underline{\underline{50 \text{ msec.}}} \end{aligned}$$

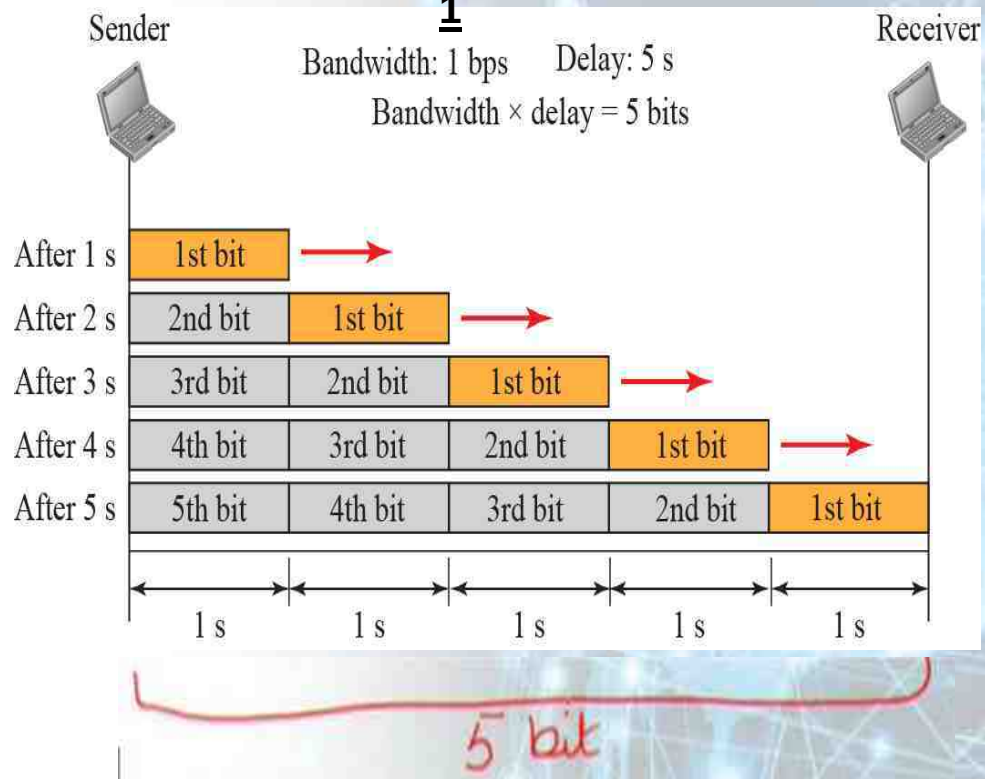
Delay – Bandwidth Delay Product

- Bandwidth and delay are two performance metrics of a link
- Product of the two, The Bandwidth-Delay Product defines the number of bits that can fill a link

Bandwidth-Delay Product

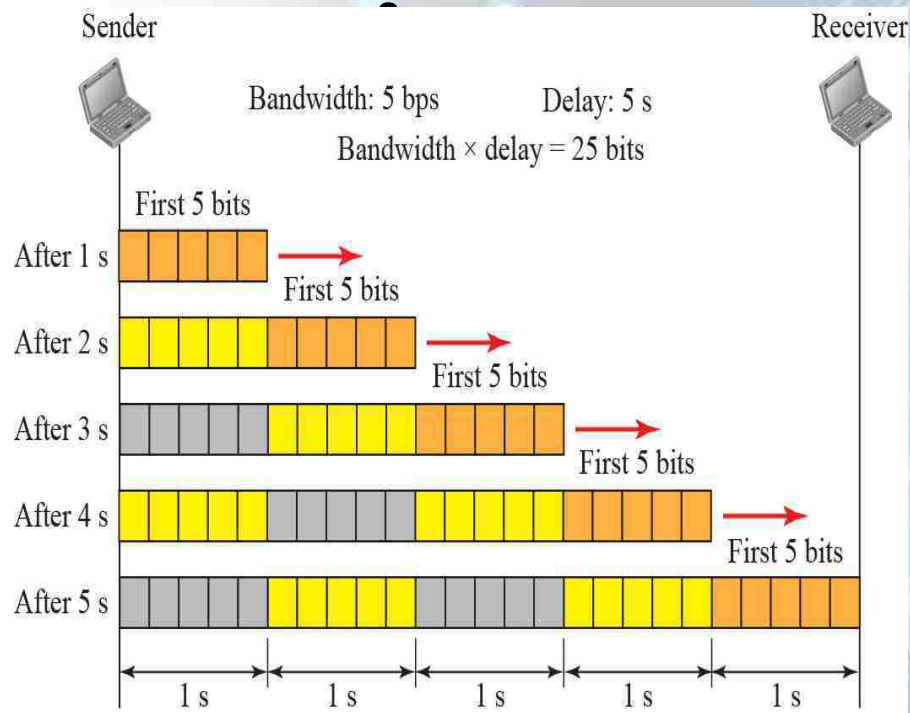
Case

1

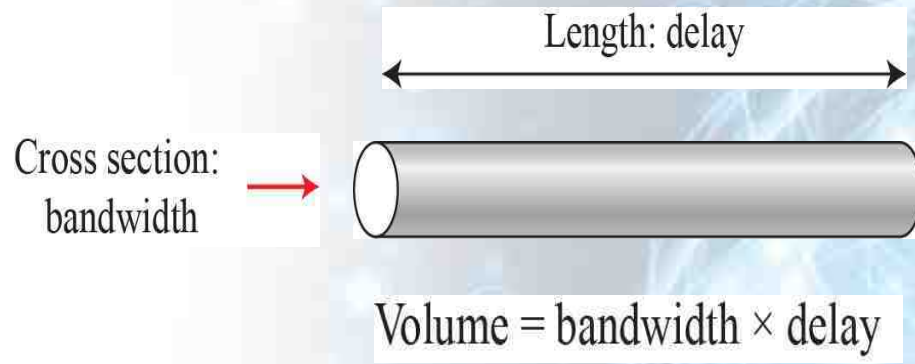


Bandwidth-Delay Product

Case



Bandwidth-Delay Product



Example 3.48

We can think about the link between two points as a pipe. The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay. We can say the volume of the pipe defines the bandwidth-delay product, as shown in Figure 3.34.

Delay - Jitter

- Jitter is a problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example)

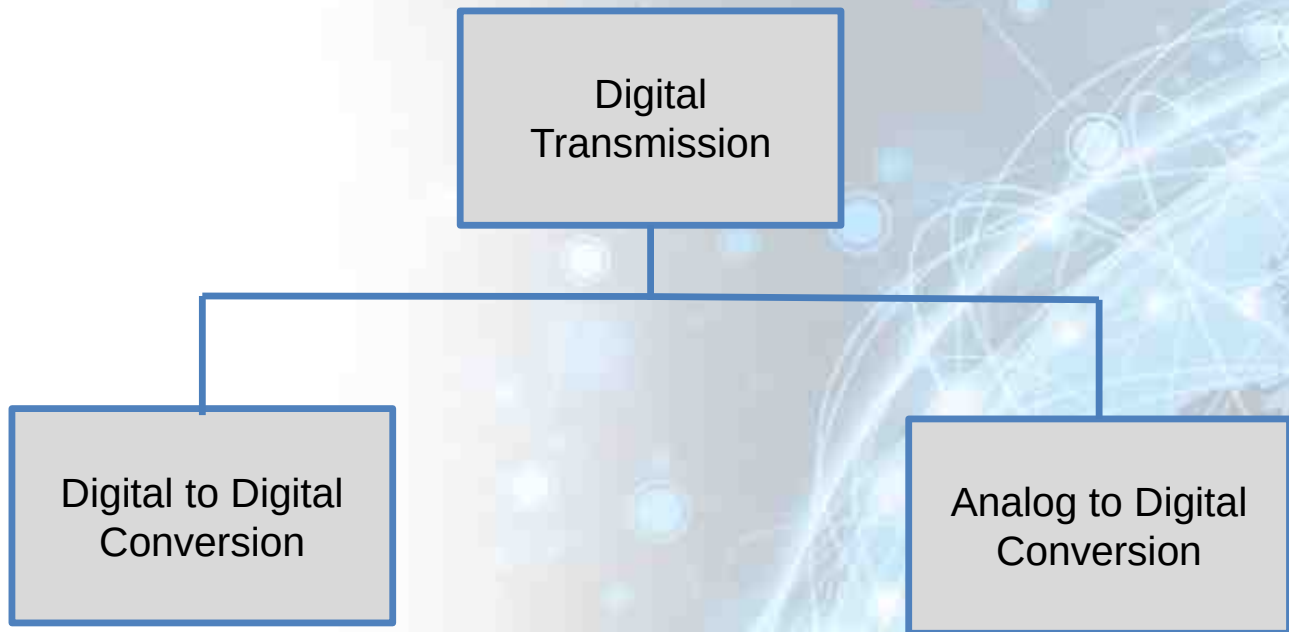
Delay - Jitter

- Delay for first packet is 20 ms for the second is 45 ms, and for the third is 40 ms, then the real-time application that uses the packets endures jitter

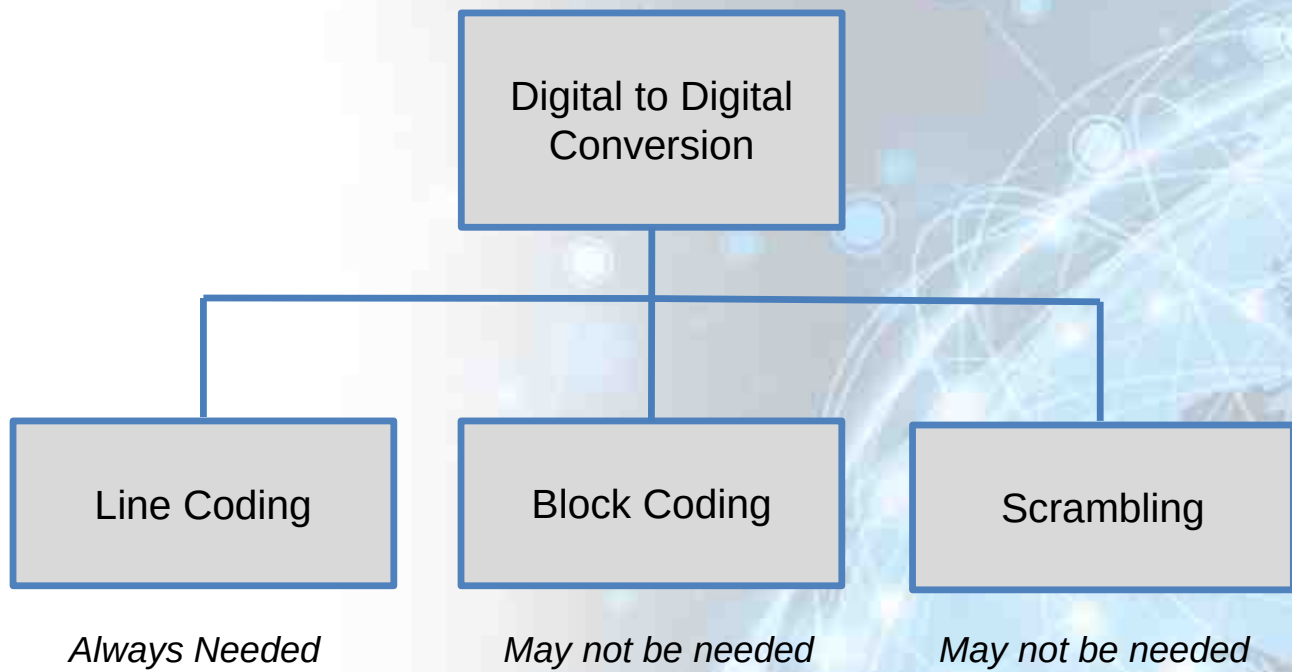
Digital-to-digital Conversion

- Data → Analog or Digital
 - Signals → Analog or Digital
 - Digital Transmission
 - Analog Transmission
- 
- A background graphic featuring a stylized globe with a network of glowing blue lines and nodes, suggesting a global communication or data network. The lines connect various points across the globe, creating a web-like structure. The nodes are represented by small, bright blue circles. The overall color scheme is light blue and white, with a subtle gradient.

Digital Transmission



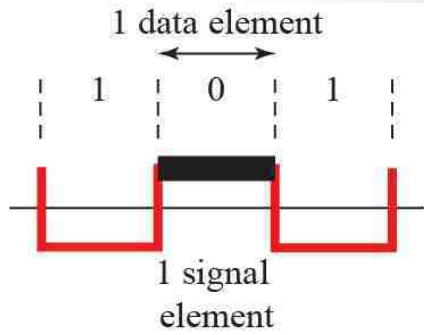
Digital to Digital Conversion



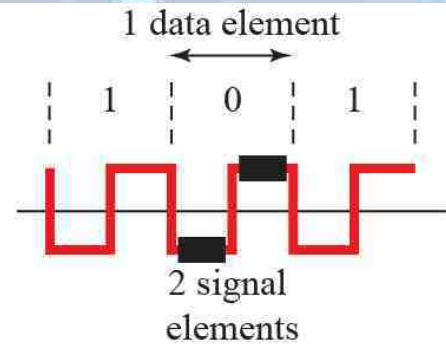
Signal Element versus Data Element

- A Data element is the smallest entity that can represent a piece of information → Bit
- A Signal element is the shortest unit of a digital signal
- Data Elements: Carried
- Signal Elements: Carriers

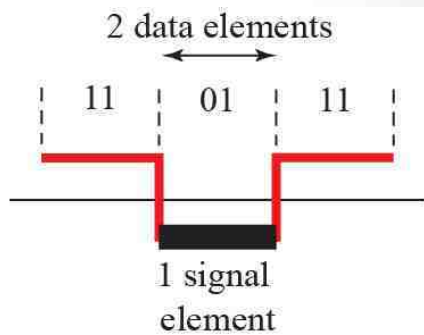
Signal Element versus Data Element



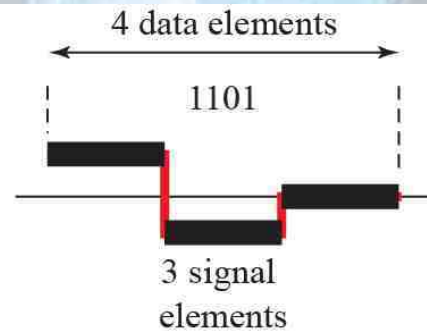
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Data Rate versus Signal Rate

- Data Rate is number of data elements sent in 1 sec (bps)
- Signal Rate is number of signal elements sent in 1 sec (baud)
- Data Rate $\rightarrow$ Bit Rate 
- Signal Rate $\rightarrow$ Pulse Rate, Modulation Rate or Baud Rate 

Example

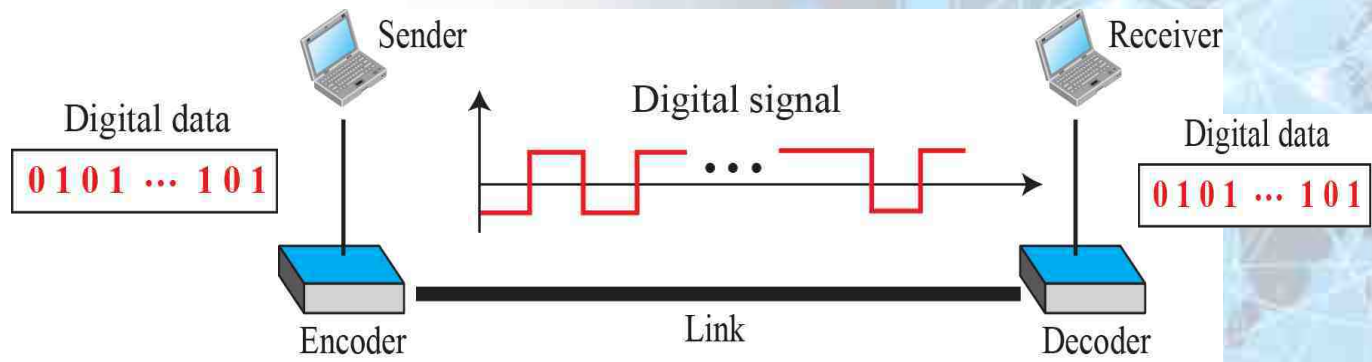
A signal has a signal rate of 100 bauds. What is the Data rate if one data element is carried per signal element?



Line Coding

- **Digital data to Digital signals**
- **Data (Text, Numbers, Pictures, Audio, or Video) is stored in computer memory as sequences of bits**
- **Line coding converts a sequence of Bits to a Digital Signal**

Line Coding and Decoding

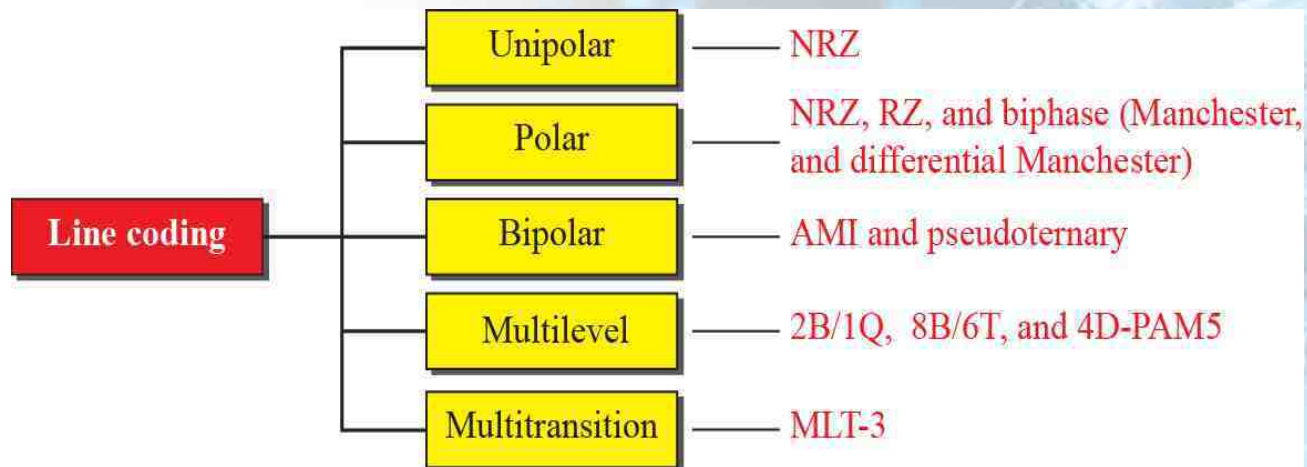


Line Coding Schemes

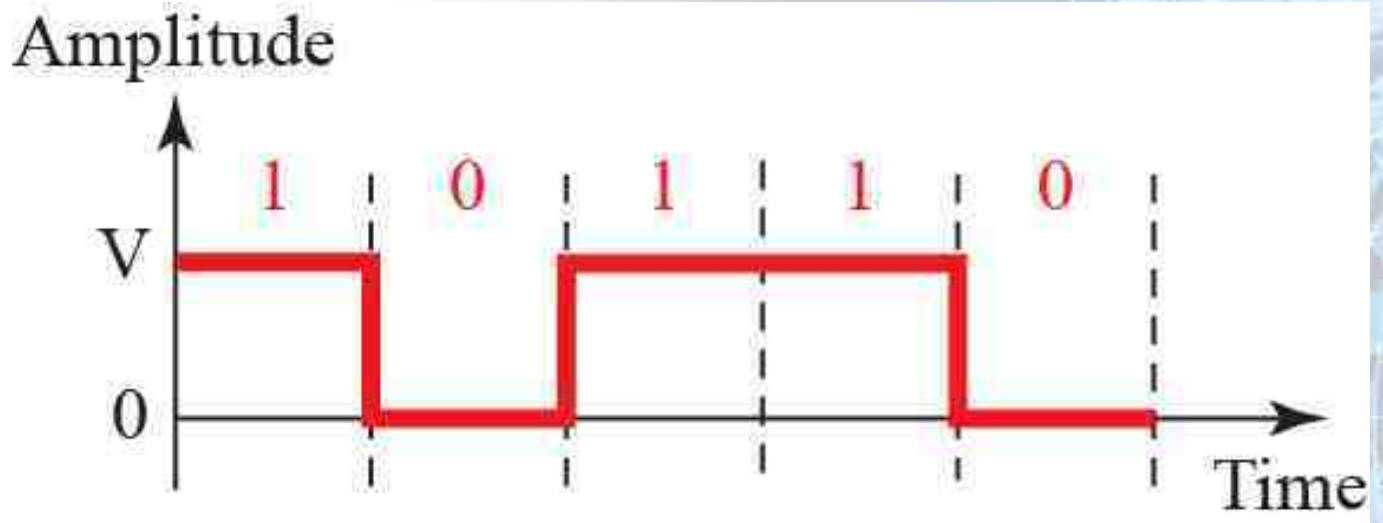
- We can roughly divide line coding schemes into five broad categories



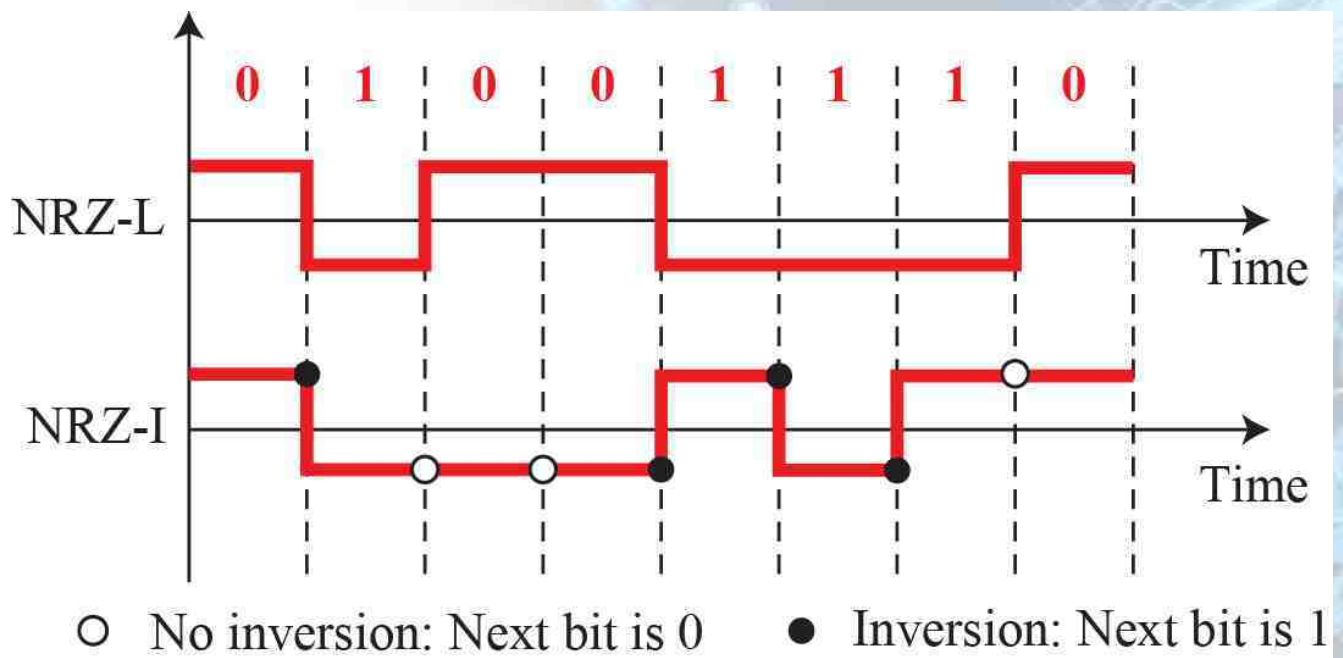
Line Coding Schemes



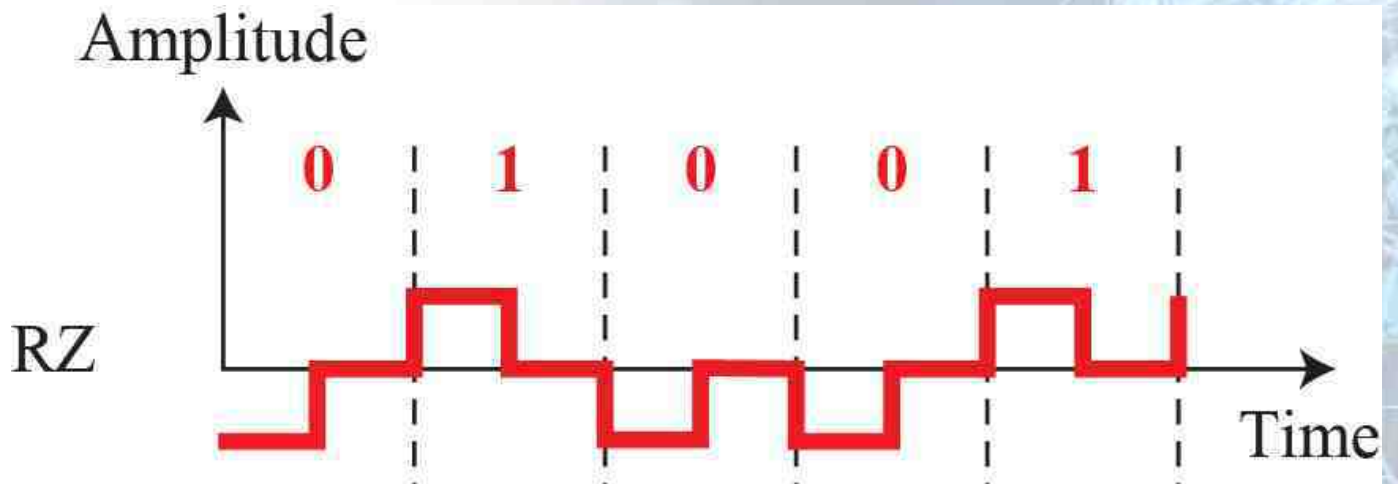
Unipolar NRZ scheme



Polar schemes (NRZ)



Polar schemes (RZ)

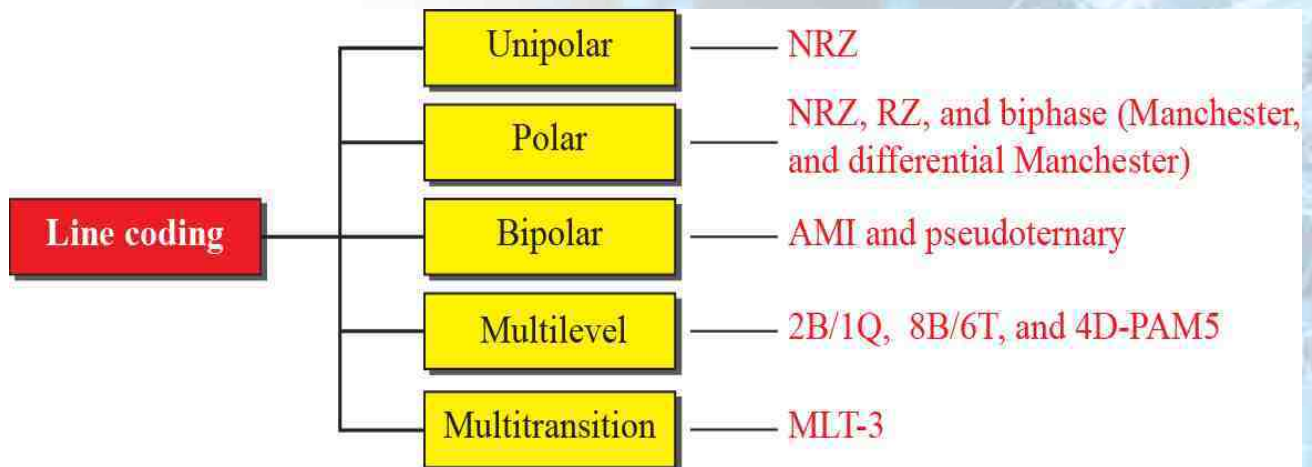


Line Coding Schemes

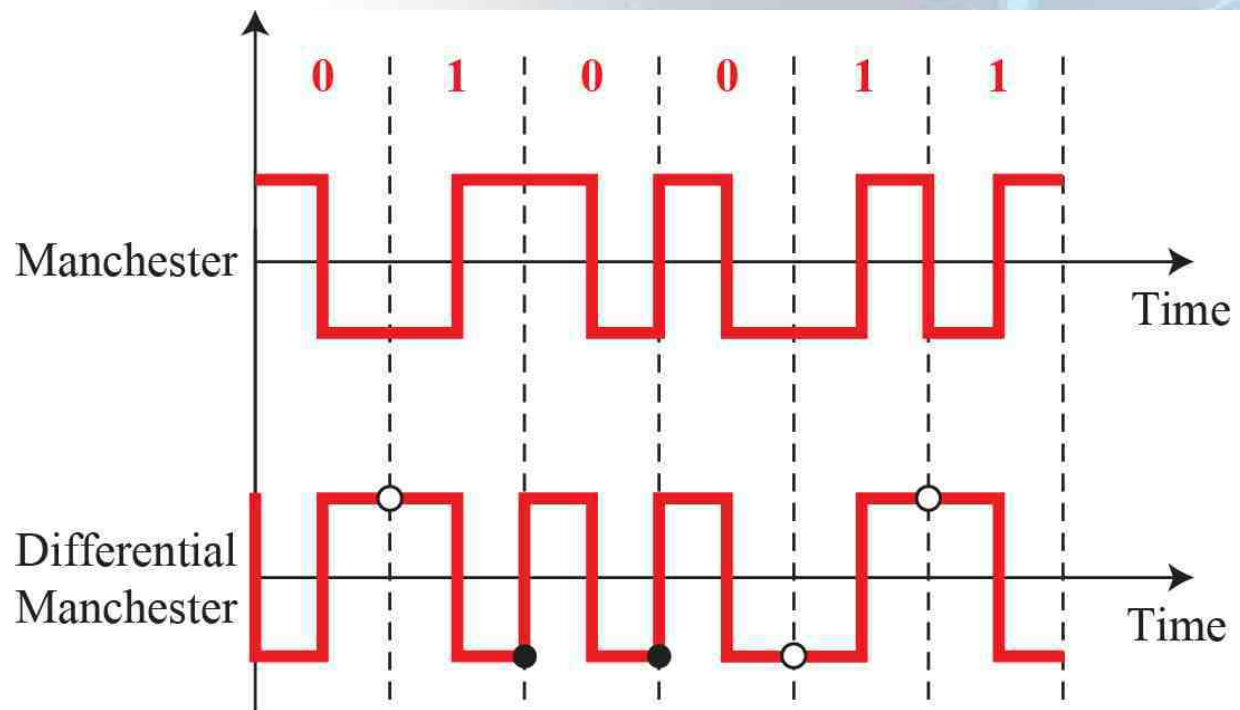
- We can roughly divide line coding schemes into five broad categories



Line Coding Schemes



Polar Biphase



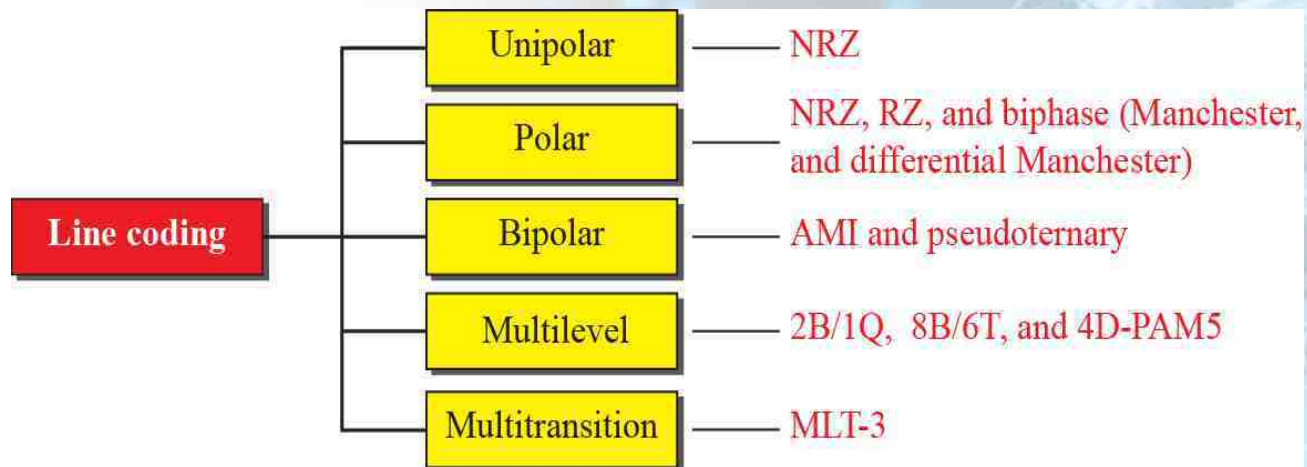
○ No inversion: Next bit is 1 ● Inversion: Next bit is 0

Line Coding Schemes

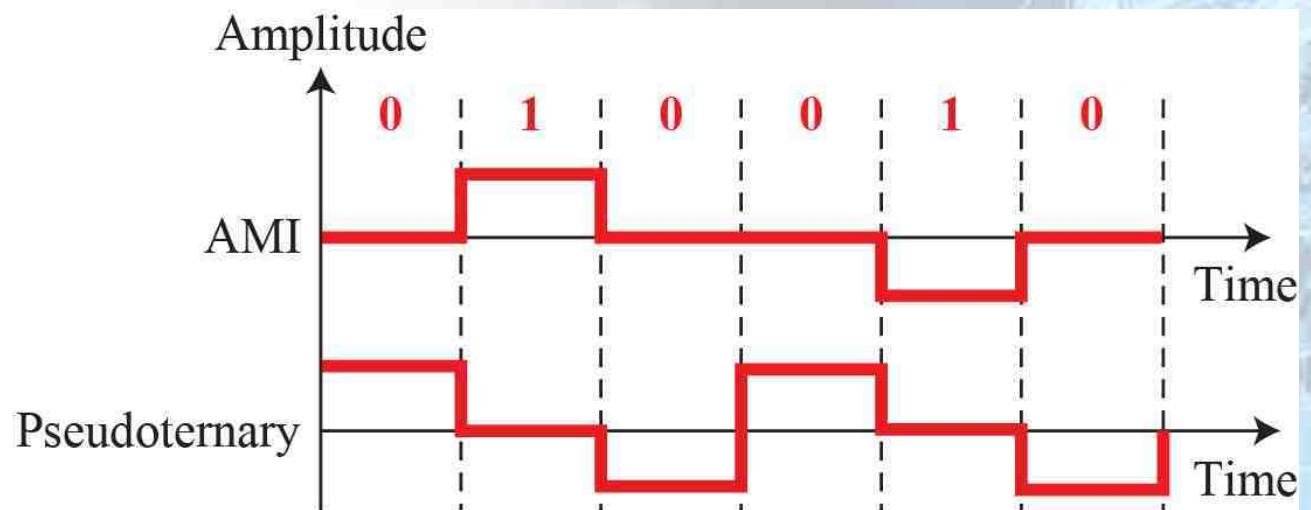
- We can roughly divide line coding schemes into five broad categories



Line Coding Schemes



Bipolar schemes: AMI & Pseudoternary



Multilevel: 2B1Q

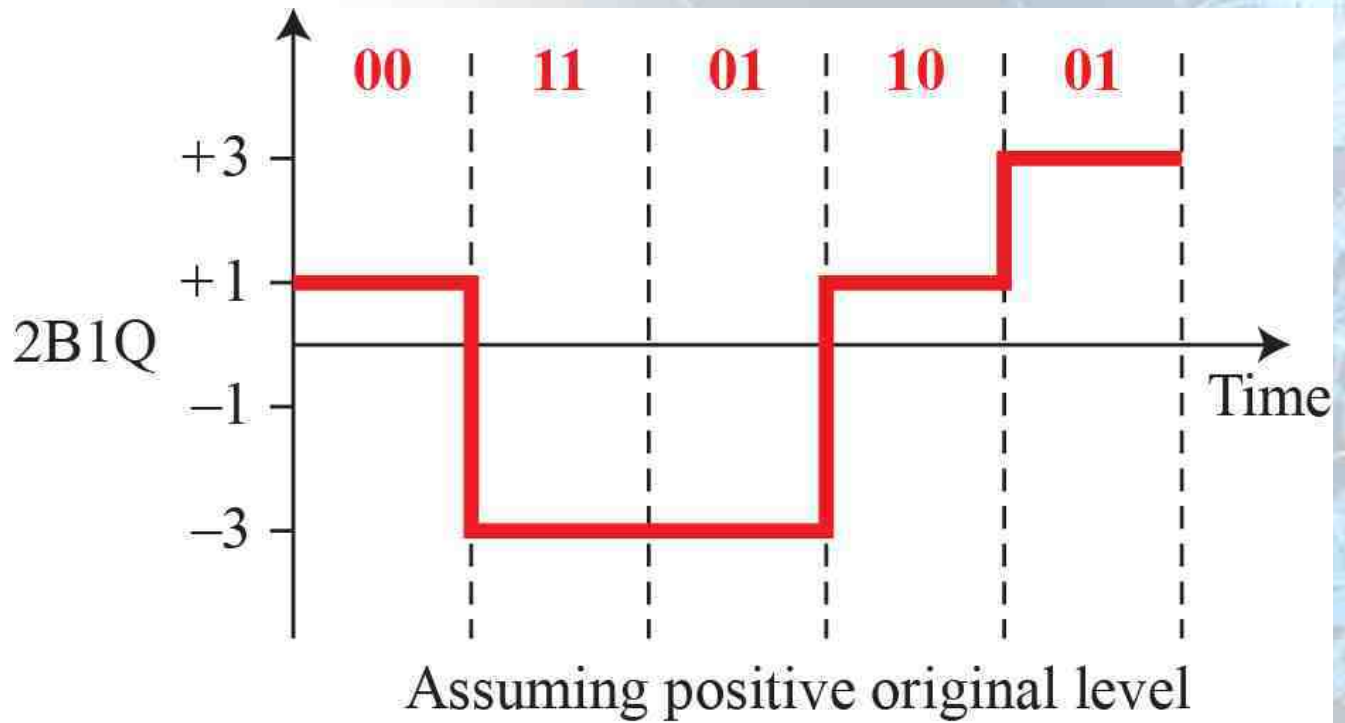


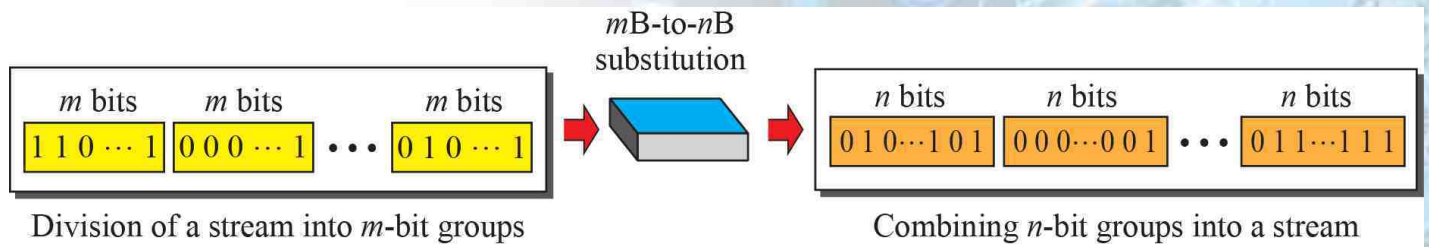
Table 4.1 : Summary of line coding schemes

Category	Scheme	Bandwidth (average)	Characteristics
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Polar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multitransition	MLT-3	$B = N/3$	No self-synchronization for long 0s

Block Coding

- Block coding changes a block of 'm' bits into a block of 'n' bits ($n > m$)
- mB/nB encoding technique
- We need Redundancy to ensure Synchronization
- Block coding gives us redundancy and improves line coding performance

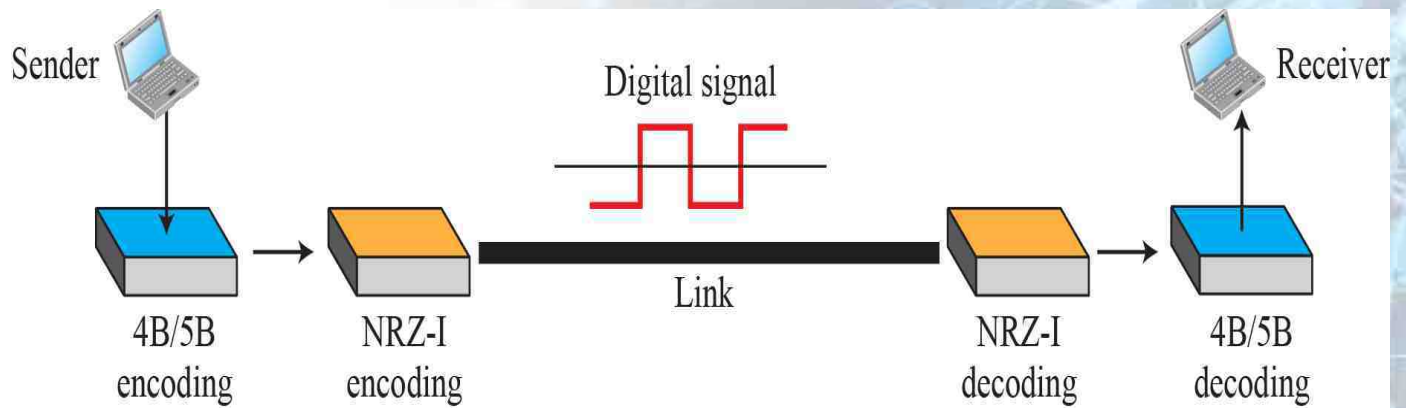
Block coding concept



Block Coding

- Block coding changes a block of 'm' bits into a block of 'n' bits ($n > m$)
- mB/nB encoding technique
- We need Redundancy to ensure Synchronization
- Block coding gives us redundancy and improves line coding performance

Using block coding 4B/5B with NRZ-I line coding



Block Coding

- Block coding changes a block of 'm' bits into a block of 'n' bits ($n > m$)
- mB/nB encoding technique
- We need Redundancy to ensure Synchronization
- Block coding gives us redundancy and improves line coding performance

4B/5B mapping codes

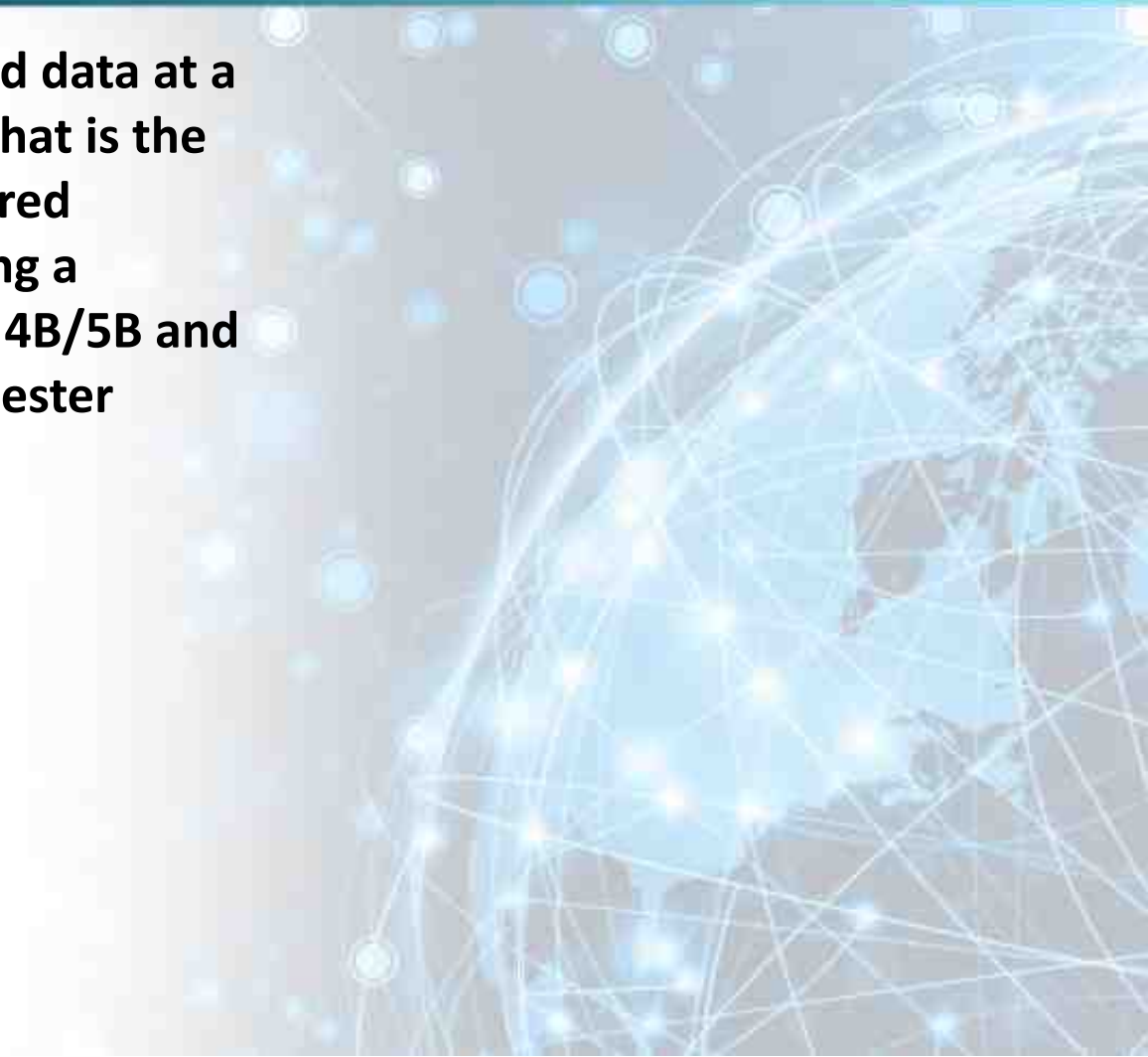
<i>Data Sequence</i>	<i>Encoded Sequence</i>	<i>Control Sequence</i>	<i>Encoded Sequence</i>
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Block Coding

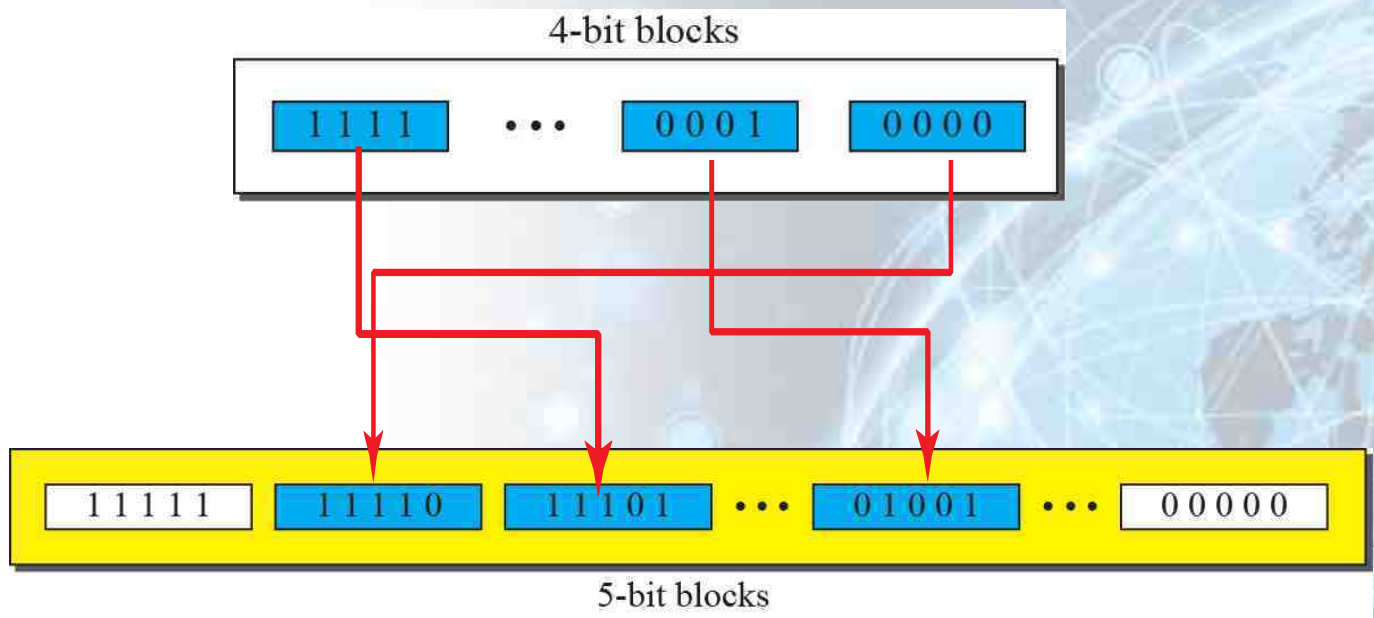
- Block coding changes a block of 'm' bits into a block of 'n' bits ($n > m$)
- mB/nB encoding technique
- We need Redundancy to ensure Synchronization
- Block coding gives us redundancy and improves line coding performance

Example

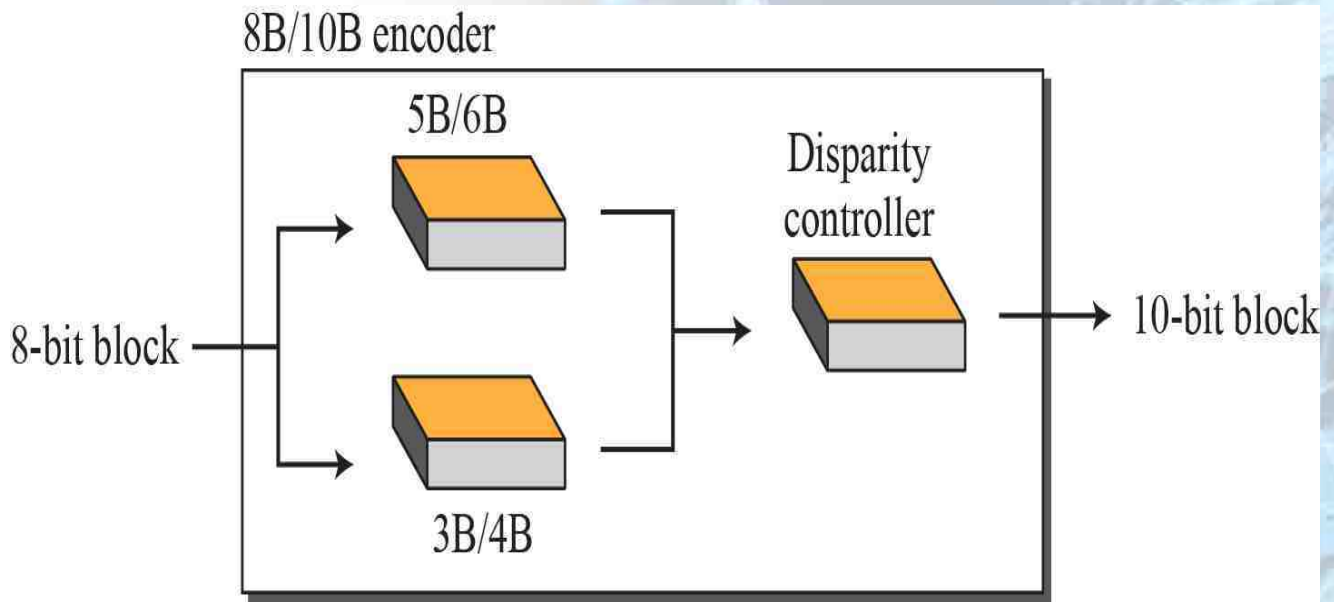
We need to send data at a 1-Mbps rate. What is the minimum required bandwidth, using a combination of 4B/5B and NRZ-I or Manchester coding?



Example



8B/10B block encoding



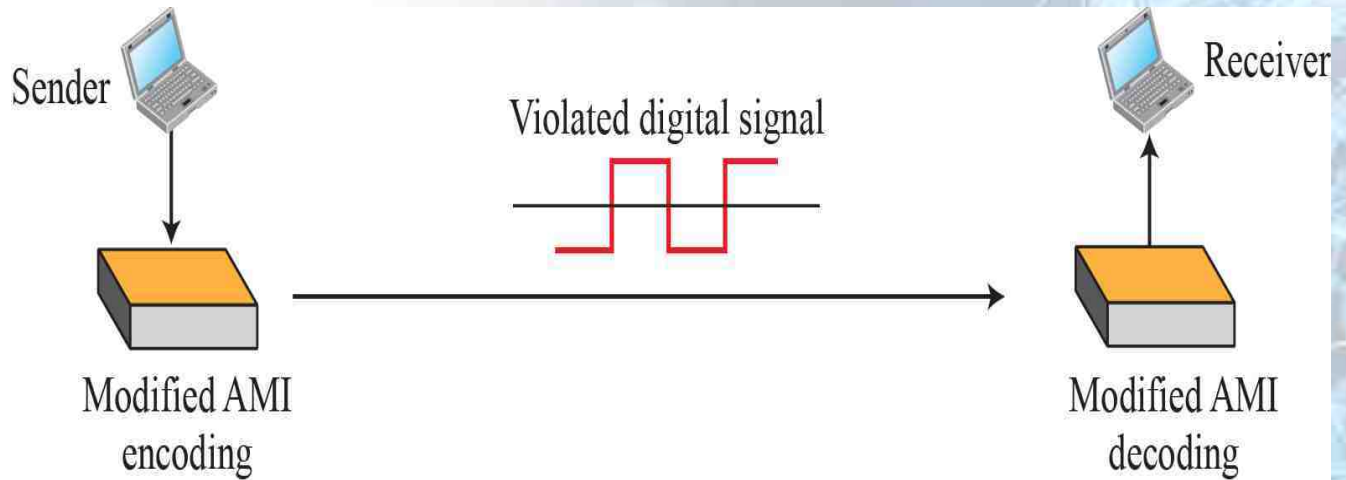
Scrambling

- Biphase schemes suitable for LAN but not for Long Distance
- Block Coding + NRZ-I solves synch issue but has DC component
- Bipolar AMI has a narrow bandwidth (no DC Component) but synch issue (long series of 0s)

Scrambling

- The system needs to insert the required pulses based on the defined scrambling rules

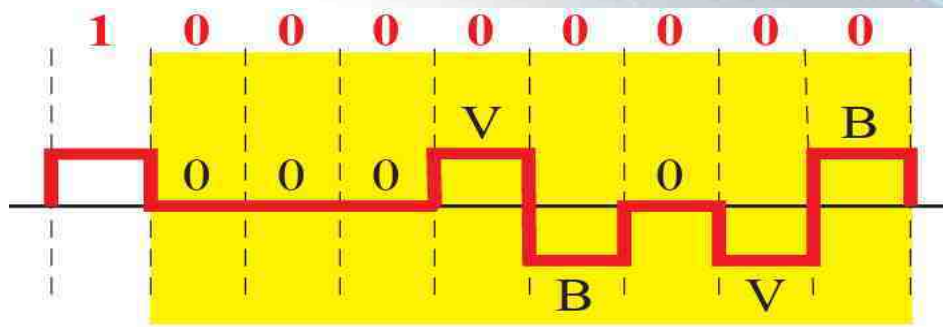
AMI used with scrambling



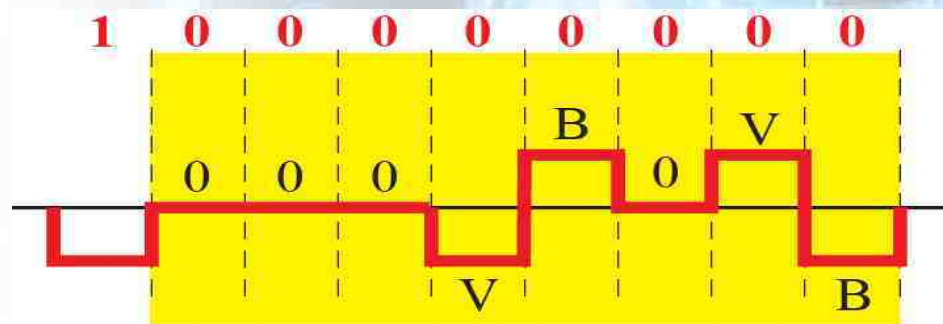
Types of Scrambling Techniques

- Two common scrambling techniques are B8ZS and HDB3
- Bipolar with 8-Zero Substitution (B8ZS)
- High-density bipolar 3-zero (HDB3)

Two cases of B8ZS scrambling technique



a. Previous level is positive.

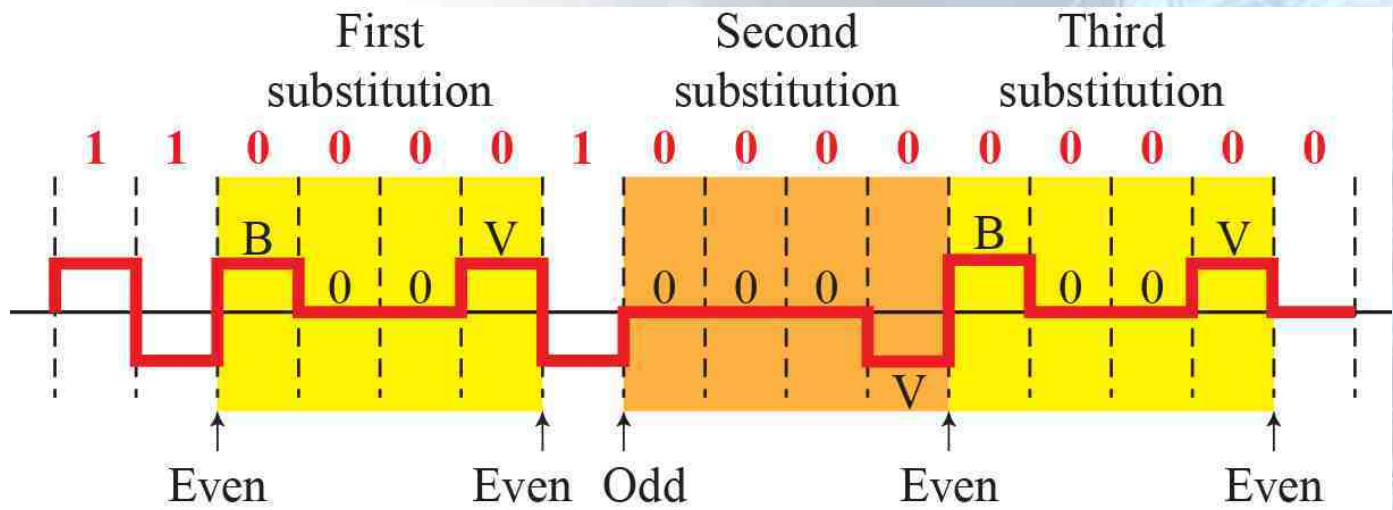


b. Previous level is negative.

Types of Scrambling Techniques

- Two common scrambling techniques are B8ZS and HDB3
- Bipolar with 8-Zero Substitution (B8ZS)
- High-density bipolar 3-zero (HDB3)

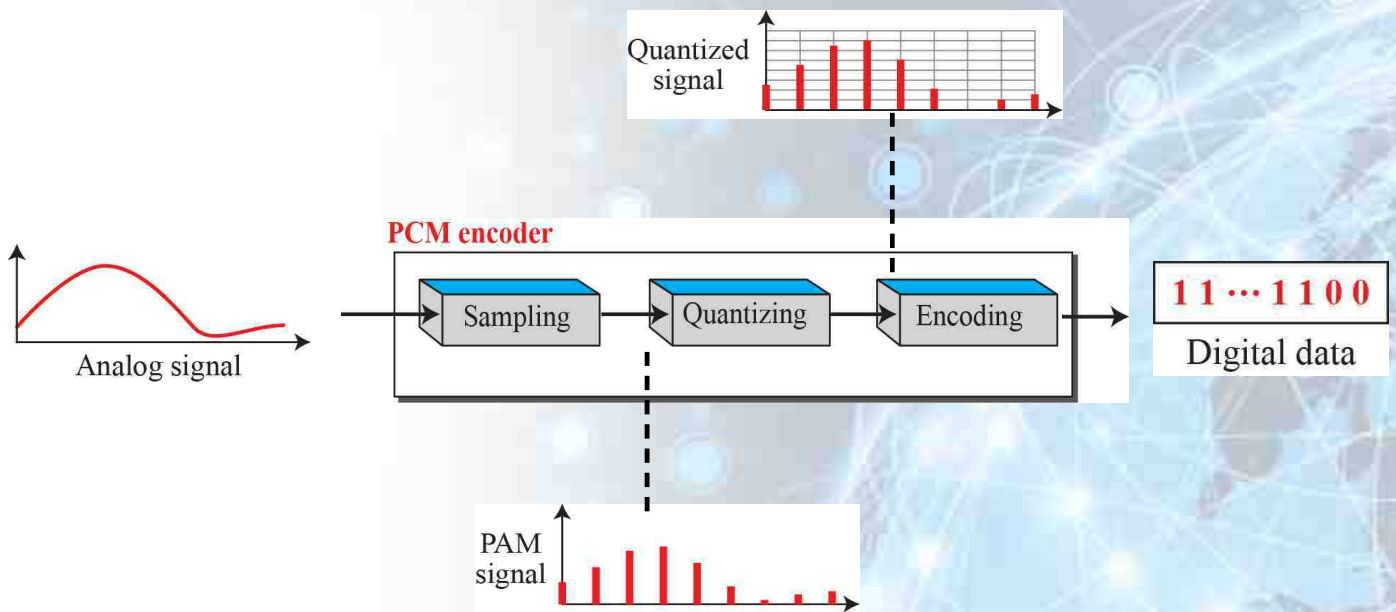
Different situations in HDB3 scrambling technique



Analog-to-digital Conversion

- **Analog Data to Digital Data**
- **Process of Digitization**
- **Two techniques:**
 - ✓ **Pulse Code Modulation (PCM)**
 - ✓ **Delta Modulation (DM)**

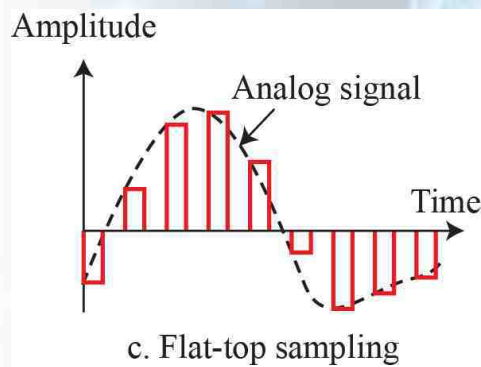
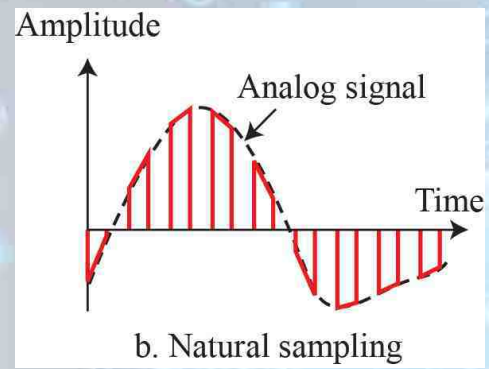
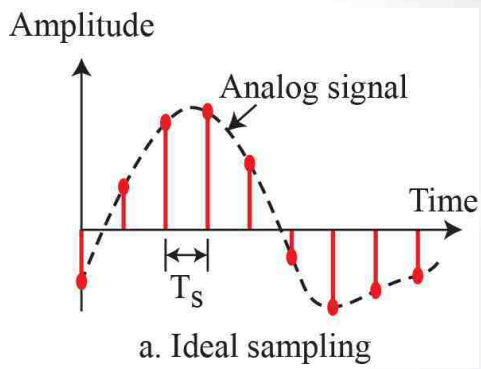
Pulse Code Modulation (PCM)



Pulse Code Modulation (PCM)

- Sampling
 - Quantization
 - Encoding
- 
- A background graphic featuring a stylized globe with a network of glowing blue lines and nodes, suggesting a global communication or data network. The globe is partially obscured by the text and list on the left.

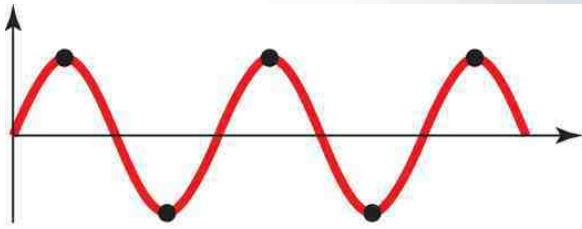
Three different sampling methods for PCM



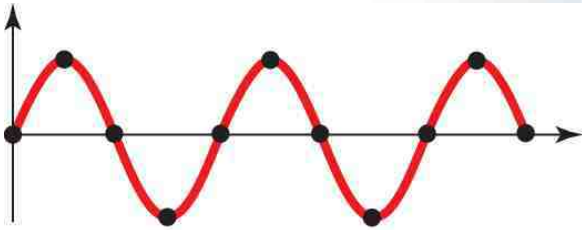
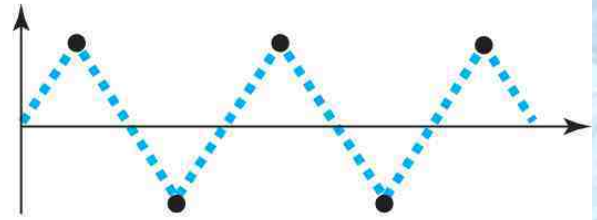
Nyquist Sampling Rate

- Nyquist $\rightarrow f_s = 2f_h$
- Sampling sine wave at three sampling rates:
 - ✓ $f_s = 4f$ (2 times the Nyquist rate)
 - ✓ $f_s = 2f$ (Nyquist rate)
 - ✓ $f_s = f$ (one-half the Nyquist rate)

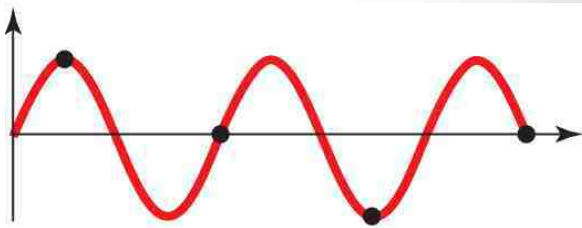
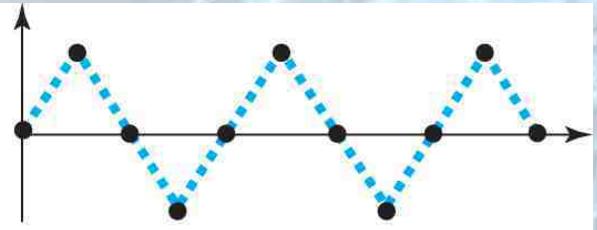
Nyquist Sampling Rate



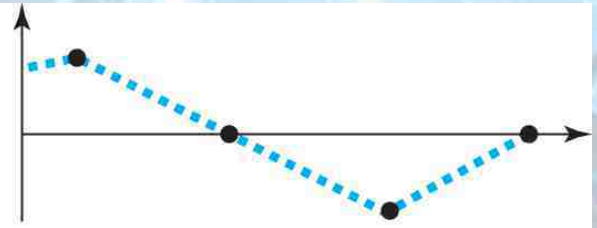
a. Nyquist rate sampling: $f_s = 2 f$



b. Oversampling: $f_s = 4 f$



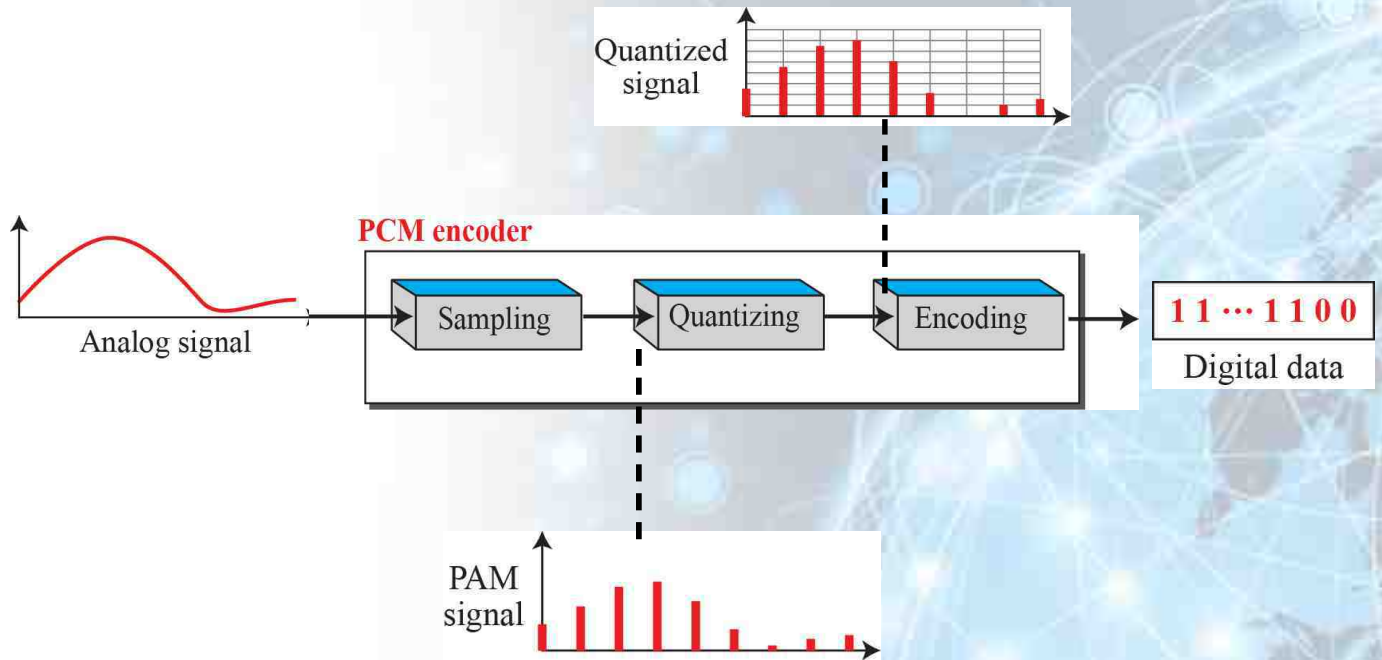
c. Undersampling: $f_s = f$



Pulse Code Modulation (PCM)

- Most common technique
- Employs a PCM Encoder
- A PCM encoder has three processes:
 - ✓ Sampling
 - ✓ Quantization
 - ✓ Encoding

Components of PCM encoder



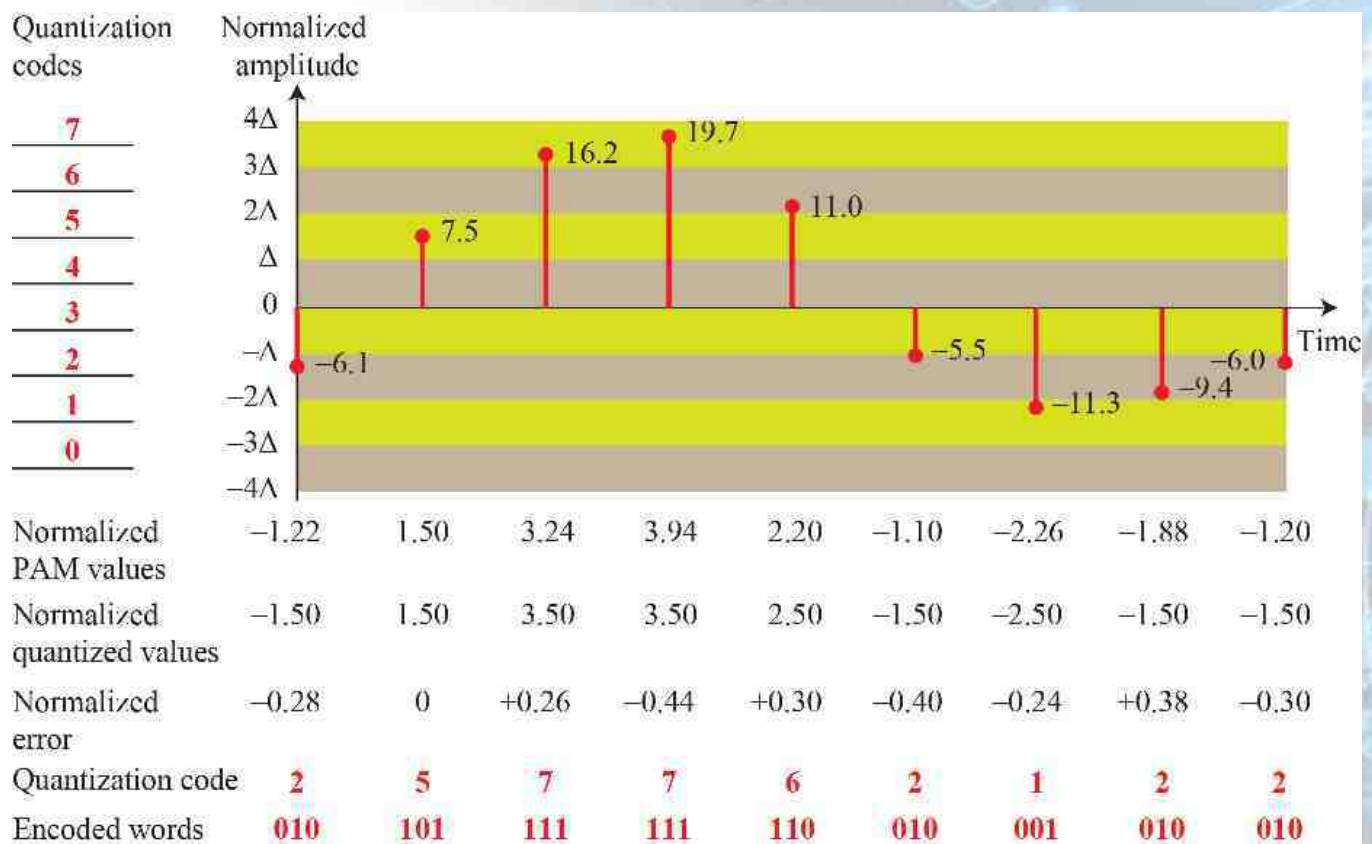
Pulse Code Modulation (PCM)

- Sampling
 - Quantization
 - Encoding
- 
- A background graphic featuring a stylized globe with a network of glowing blue lines and nodes, suggesting a global communication or data network. The globe is partially obscured by a light blue, semi-transparent rectangular area on the left side.

Quantization & encoding of a sampled signal

- Sampling → Series of pulses with amplitude values between min and max signal amplitude
- Infinite set with non-integral values not suitable for encoding
- We quantize the sampling output into certain levels based on range of amplitudes and how much accuracy is needed

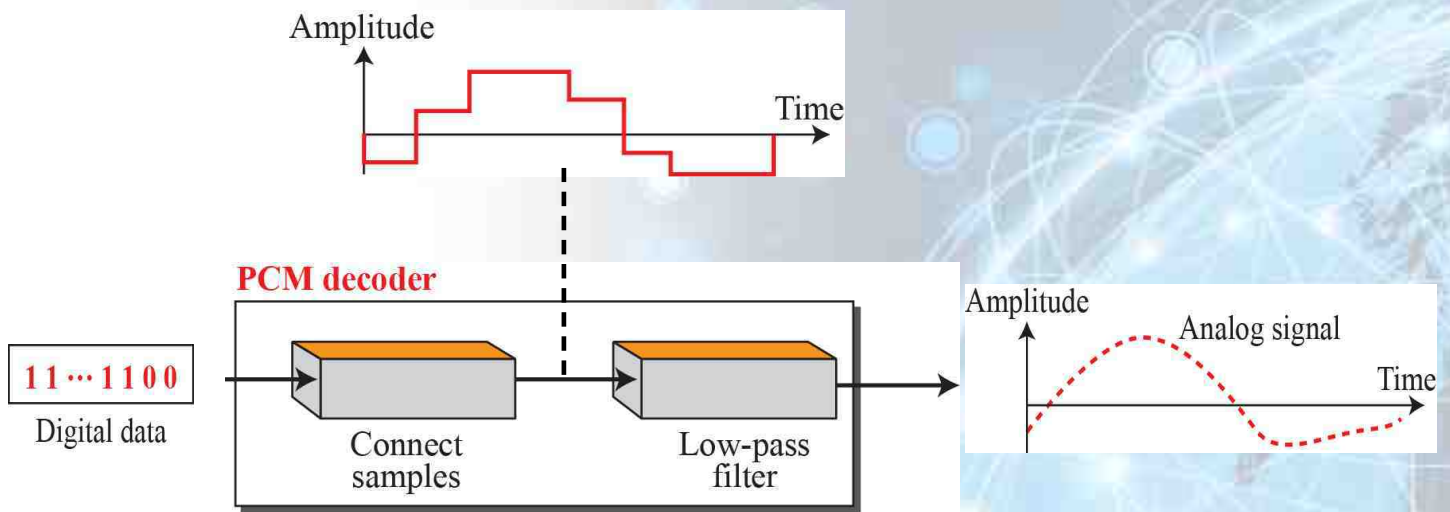
Quantization & encoding of a sampled signal



Pulse Code Modulation (PCM)

- Encoding
 - ✓ Sampling
 - ✓ Quantization
 - ✓ Encoding
- Decoding

Original Signal Recovery- PCM Decoder



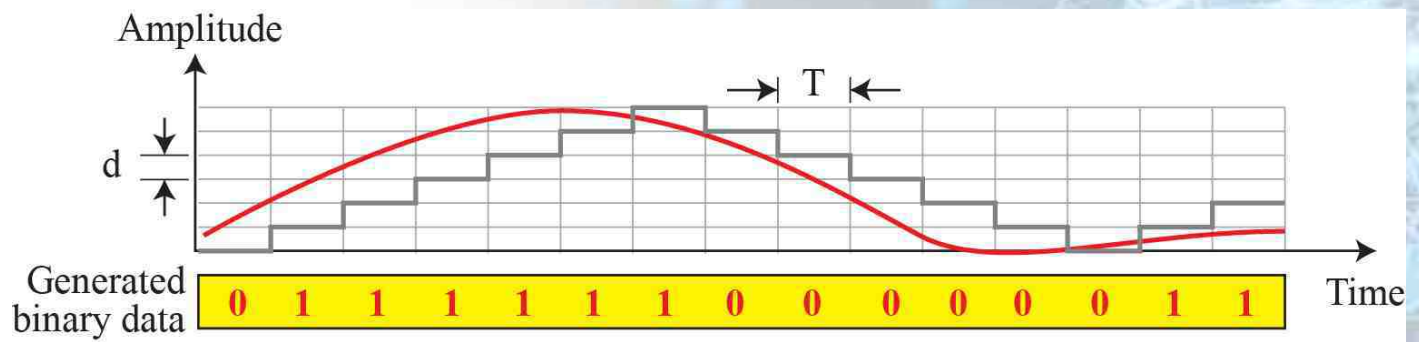
Analog-to-digital Conversion

- **Analog Data to Digital Data**
- **Process of Digitization**
- **Two techniques:**
 - ✓ **Pulse Code Modulation (PCM)**
 - ✓ **Delta Modulation (DM)**

Delta Modulation (DM)

- PCM is a very complex technique
- Delta modulation is a simpler technique
- PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample
- No code words

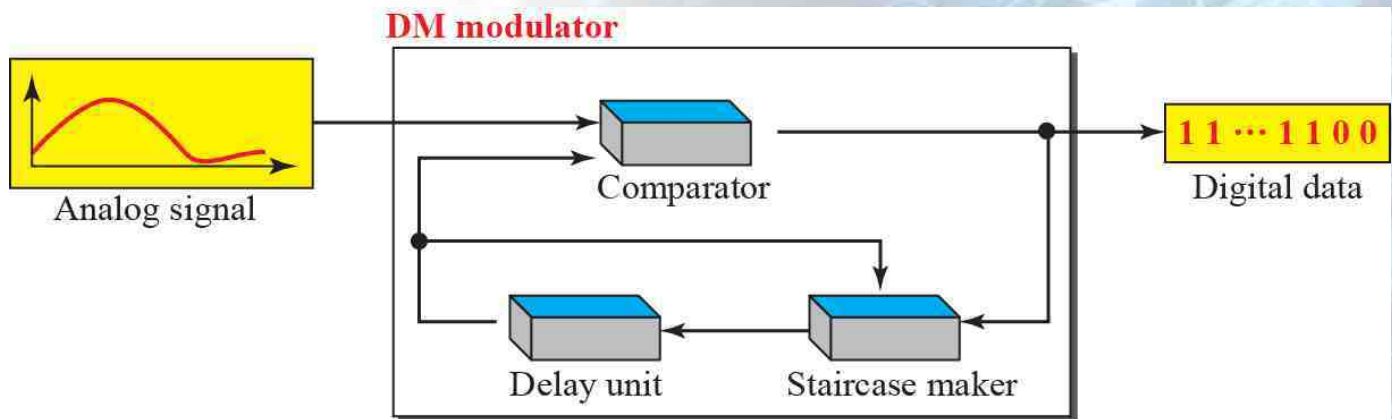
The process of delta modulation



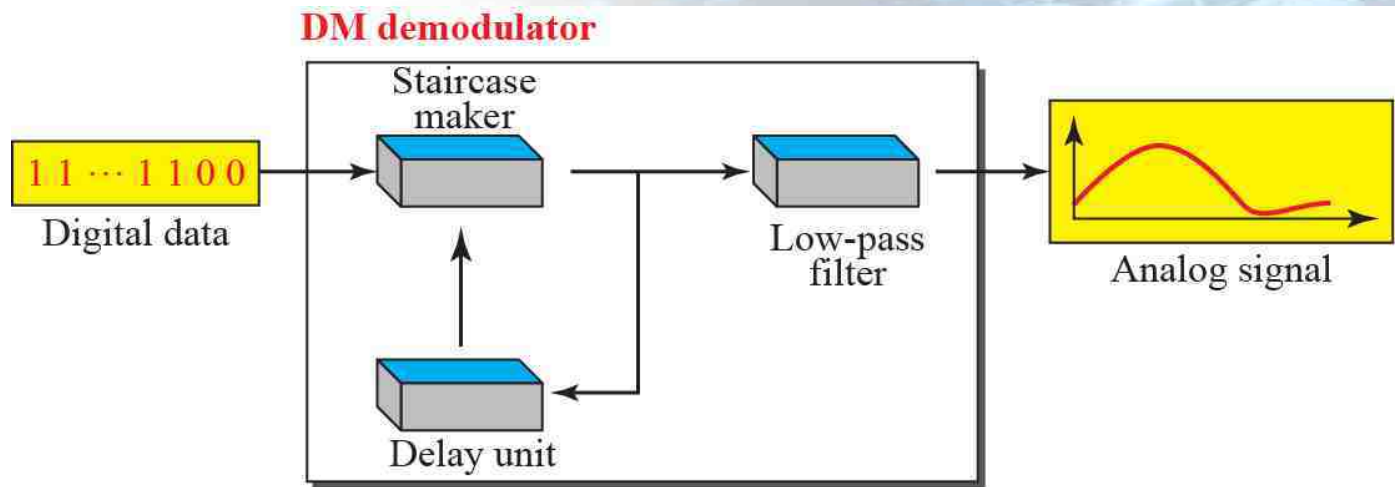
Delta Modulation (DM)

- Delta modulation is a simpler technique
- DM finds the change from the previous sample
- No code words

Delta Modulation Components



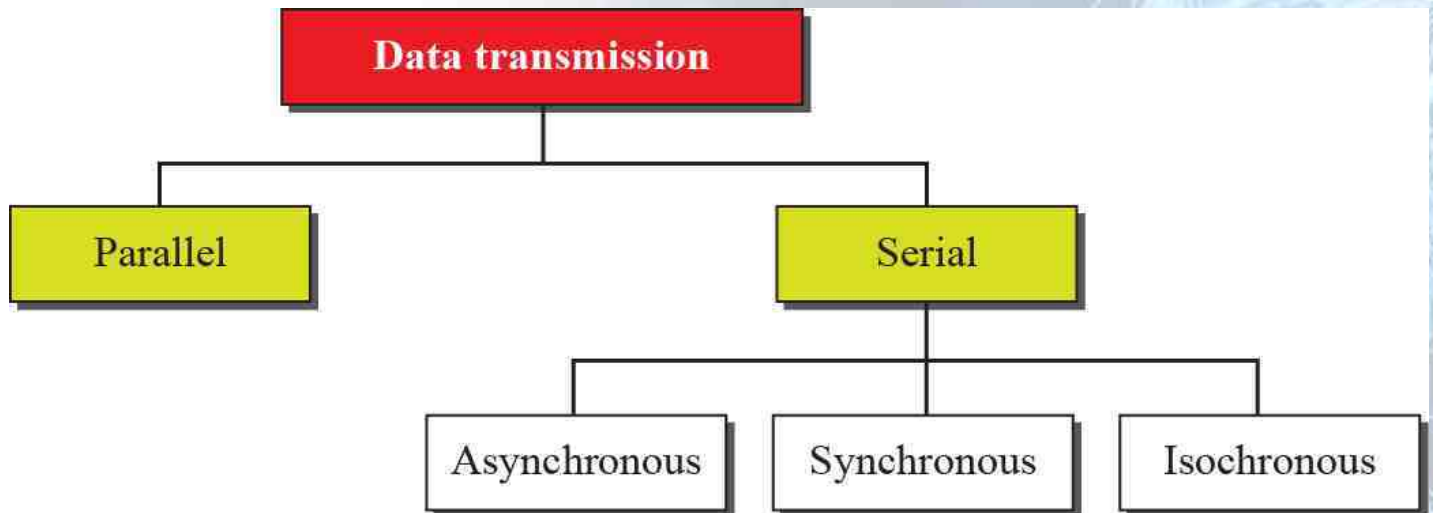
Delta Demodulation Components



Transmission Modes

- **Transmission of Data:**
 - ✓ **Wiring**
 - **Data Stream**
- **Do we send 1 bit at a time; or do we group bits into larger groups and, if so, how?**
- **Parallel or Serial Transmission**

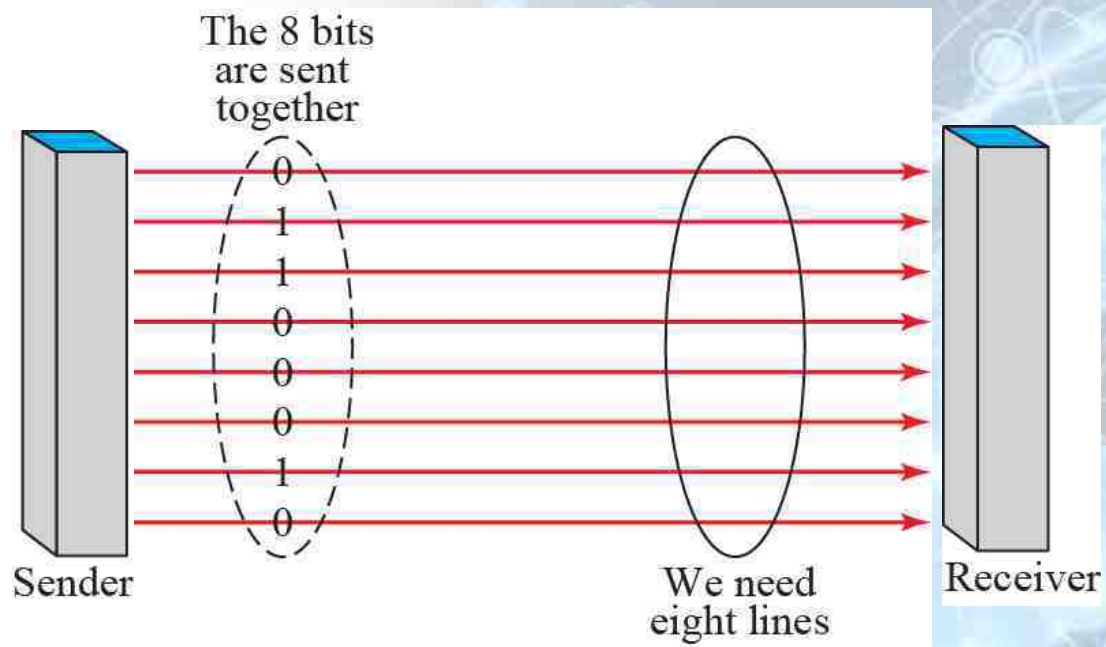
Data transmission modes



Parallel Transmission

- Binary data (1s and 0s) organized in groups of 'n' bits
- We send 'n' bits at a time instead of just one
- 'n' wires required to send 'n' bits at one time

Parallel Transmission

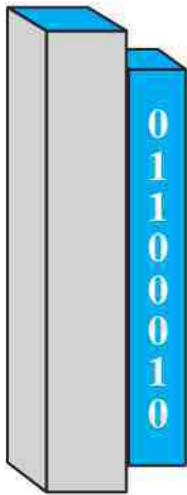


Serial Transmission

- In serial transmission one bit follows another
- Only one communication channel rather than 'n' to transmit data

Serial Transmission

Parallel/serial
converter



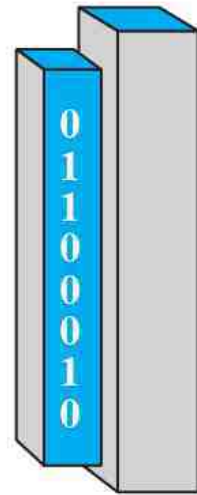
Sender

The 8 bits are sent
one after another.

0 1 1 0 0 0 1 0

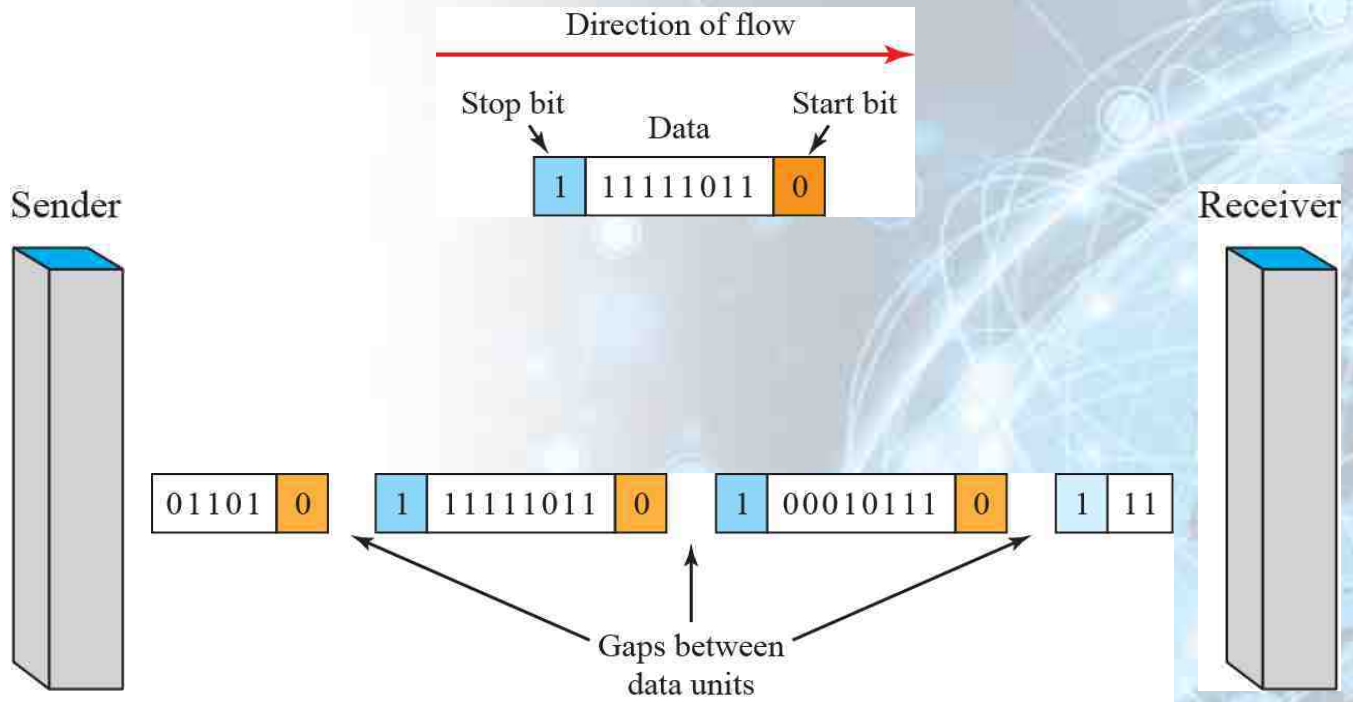
We need only
one line (wire).

Serial/parallel
converter



Receiver

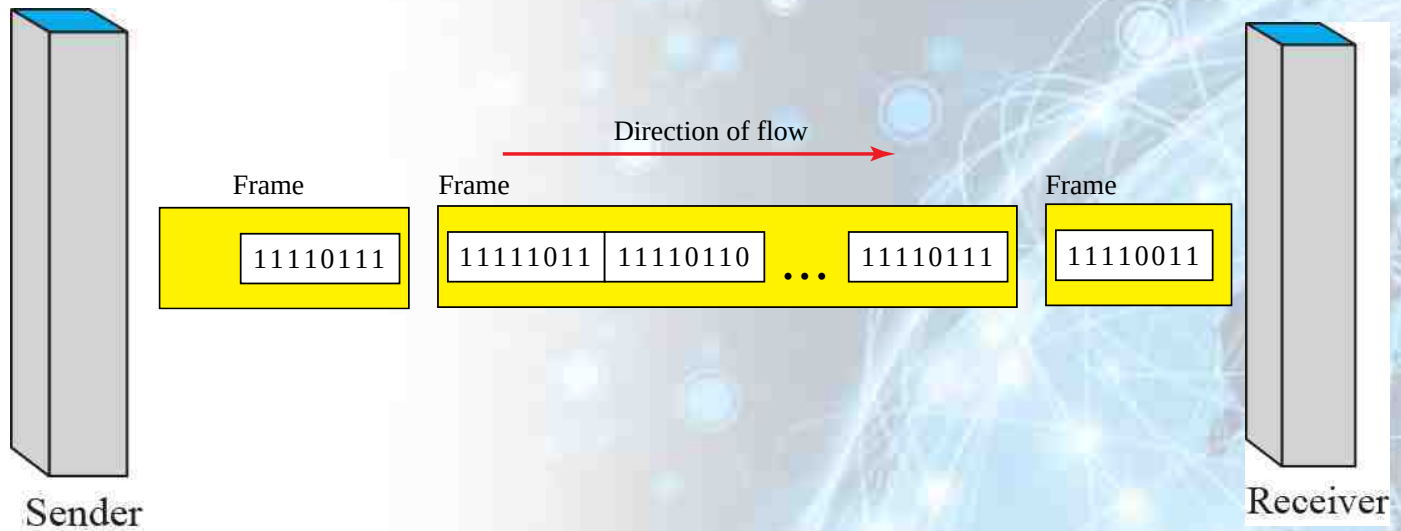
Asynchronous Transmission



Serial Transmission

- In serial transmission one bit follows another
- Only one communication channel rather than 'n' to transmit data

Synchronous Transmission



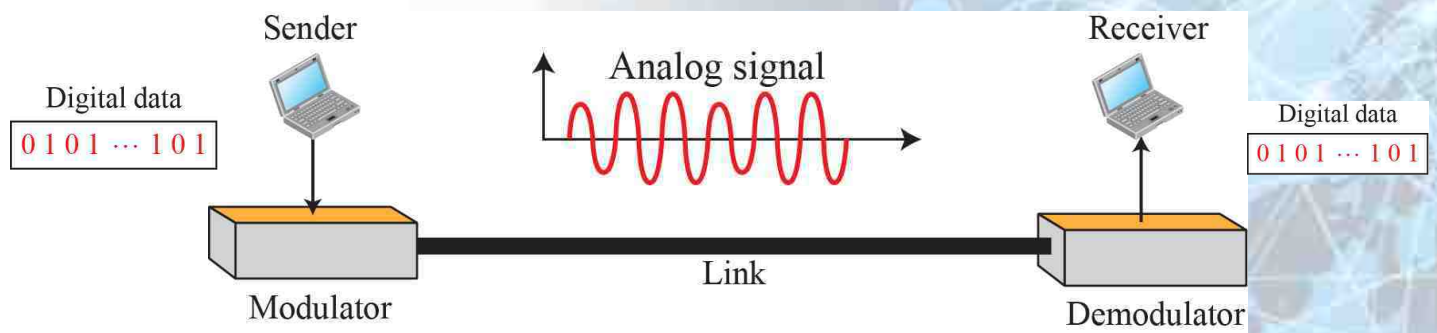
Isochronous Transmission

- **Real time Audio and Video**
- **Synchronization between characters is not enough**
- **Entire stream should be synchronized**
- **Isochronous guarantees fixed rate data**

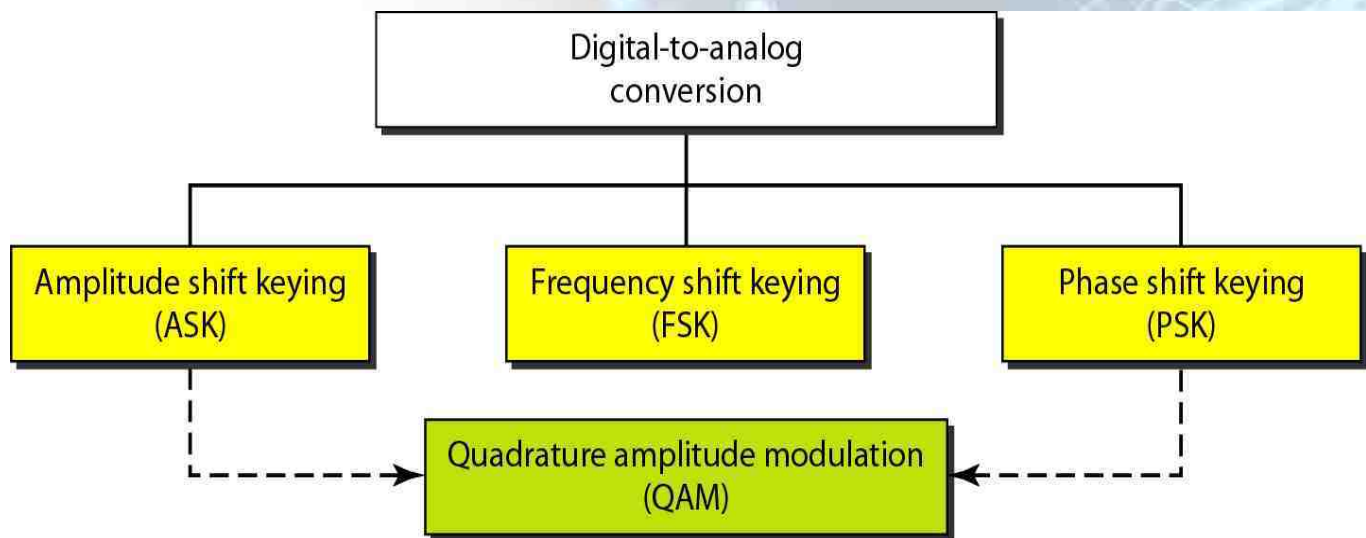
Digital-to-Analog Conversion

- Process of changing one of the characteristics of analog signal based on the information in digital data
- A sine wave is defined by 3 characteristics:
 - ✓ Amplitude
 - ✓ Frequency
 - ✓ Phase
- By changing one of these characteristics, we can use it to represent a digital signal

Digital-to-Analog Conversion



Types of Digital to Analog Conversion



Aspects of Digital to Analog Conversion

- Before we discuss specific methods of digital-to-analog modulation, two basic issues must be reviewed:
 - ✓ Bit and Baud rates and
 - ✓ The Carrier Signal

Aspects of Digital to Analog Conversion

- In Analog Transmission of Digital Data, Baud Rate is less than or equal to the Bit Rate
 - ✓ Data Element vs. Signal Element
 - ✓ Data Rate vs. Signal Rate
 - Bandwidth Required $\propto$ Signal Rate (except FSK)
 - Carrier Signal
- 

Aspects of Digital to Analog Conversion

- Before we discuss specific methods of digital-to-analog modulation, two basic issues must be reviewed:
 - ✓ Bit and Baud rates and
 - ✓ The Carrier Signal

Example

An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate?



Example

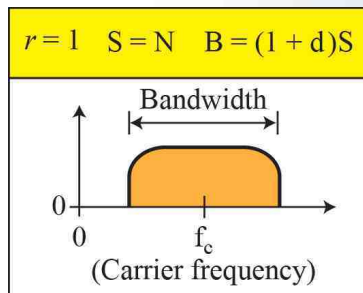
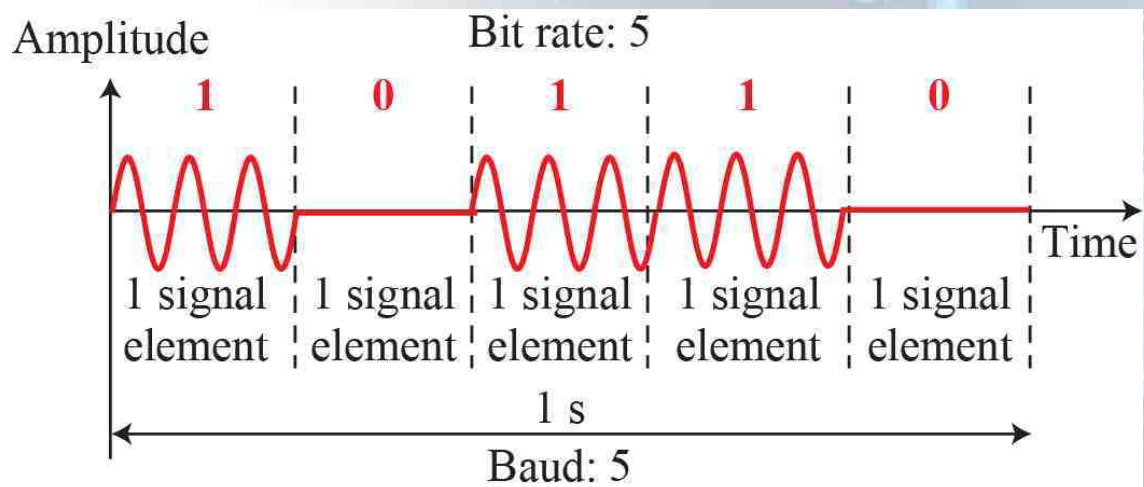
An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?



Amplitude Shift Keying (ASK)

- The amplitude of the carrier signal is varied to create signal elements
- Both frequency and phase remain constant while the amplitude changes
- Binary ASK or On-Off Keying (OOK)

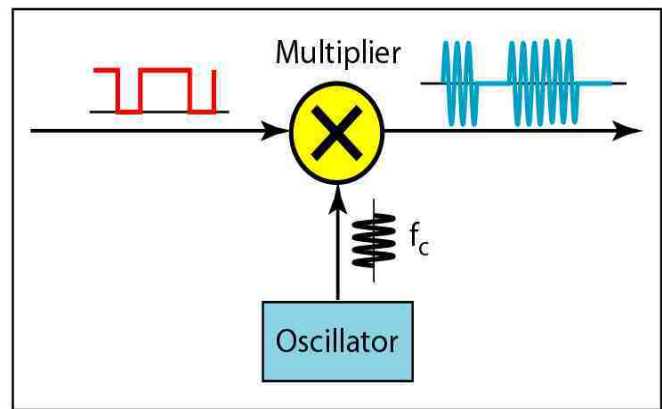
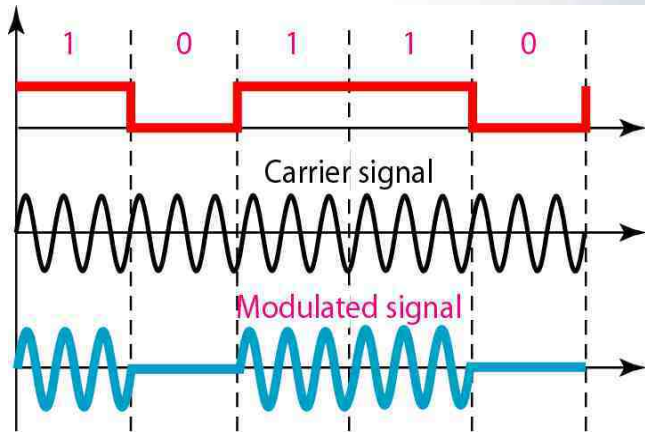
Binary Amplitude Shift Keying (Binary ASK)



Amplitude Shift Keying (ASK)

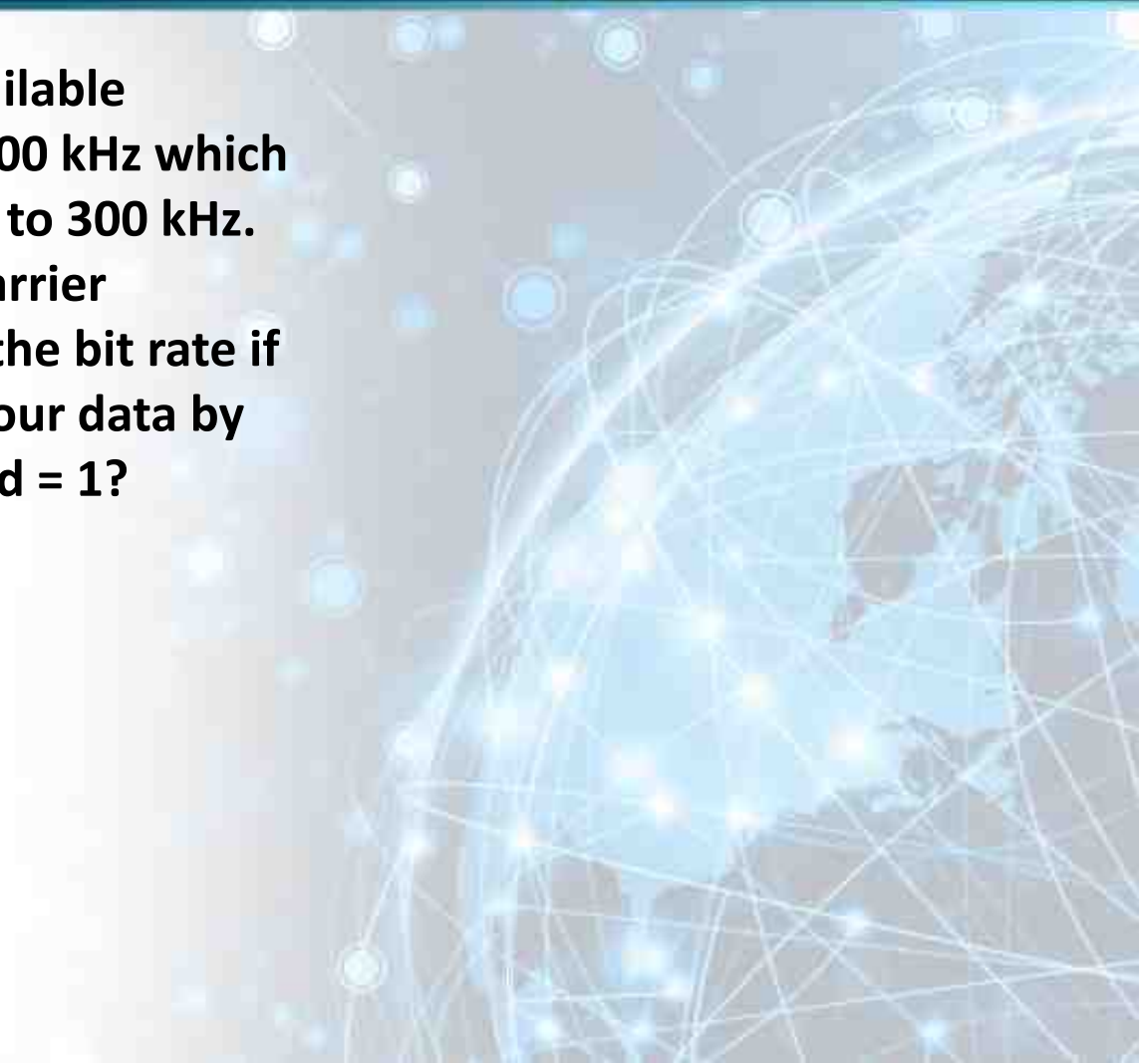
- The amplitude of the carrier signal is varied to create signal elements
- Both frequency and phase remain constant while the amplitude changes
- Binary ASK or On-Off Keying (OOK)

Implementation of Binary ASK



Example

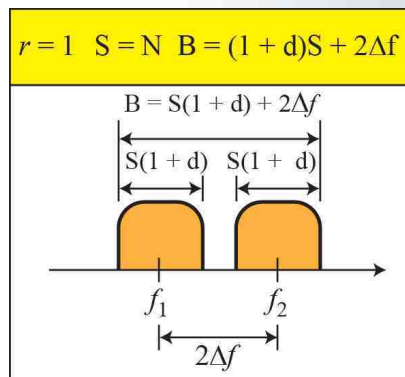
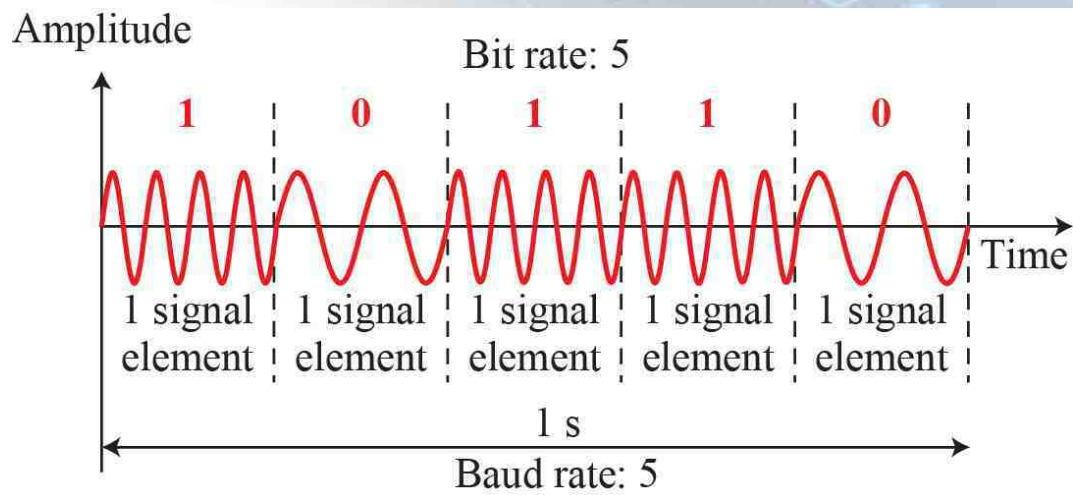
We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with $d = 1$?



Frequency Shift Keying (FSK)

- The frequency of the carrier signal is varied to represent data
- The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes
- Both peak amplitude and phase remain constant

Binary Frequency Shift Keying



Frequency Shift Keying (FSK)

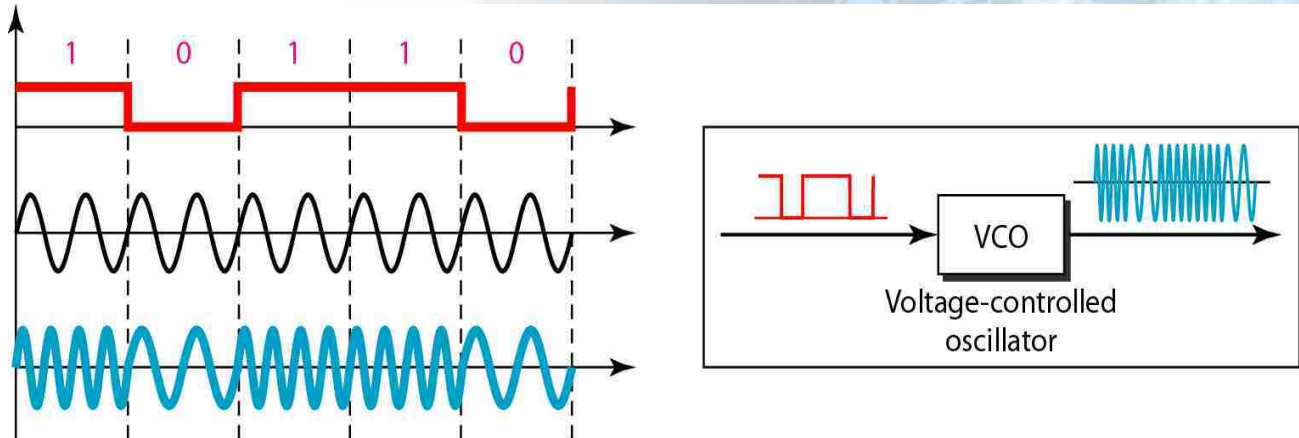
- The frequency of the carrier signal is varied to represent data
- Both peak amplitude and phase remain constant

Example

We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with $d = 1$?



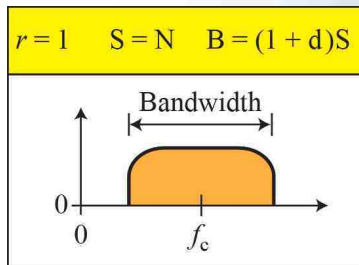
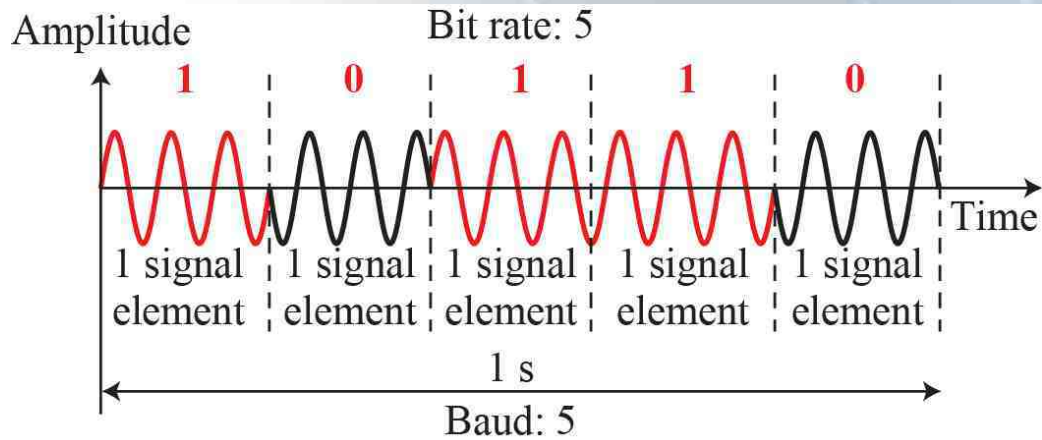
Implementation of BFSK



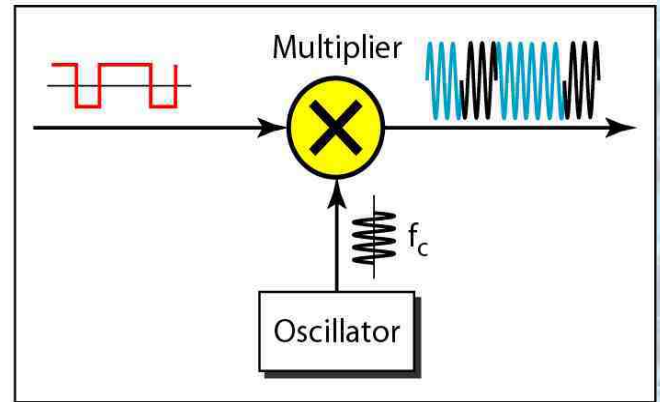
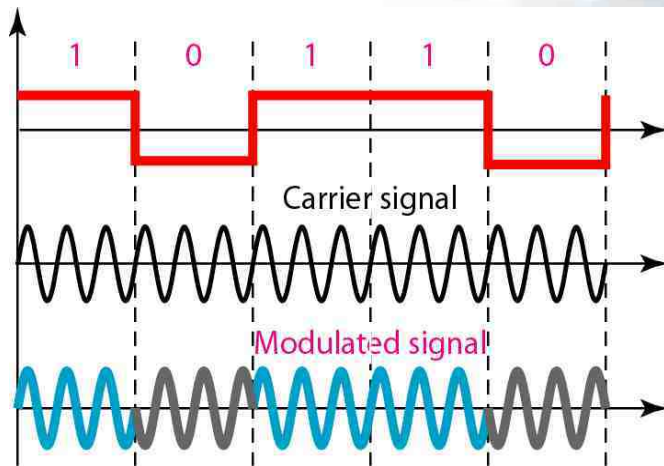
Phase Shift Keying (PSK)

- The phase of the carrier is varied to represent two or more different signal elements
- Both peak amplitude and frequency remain constant
- PSK is relatively common than ASK or FSK

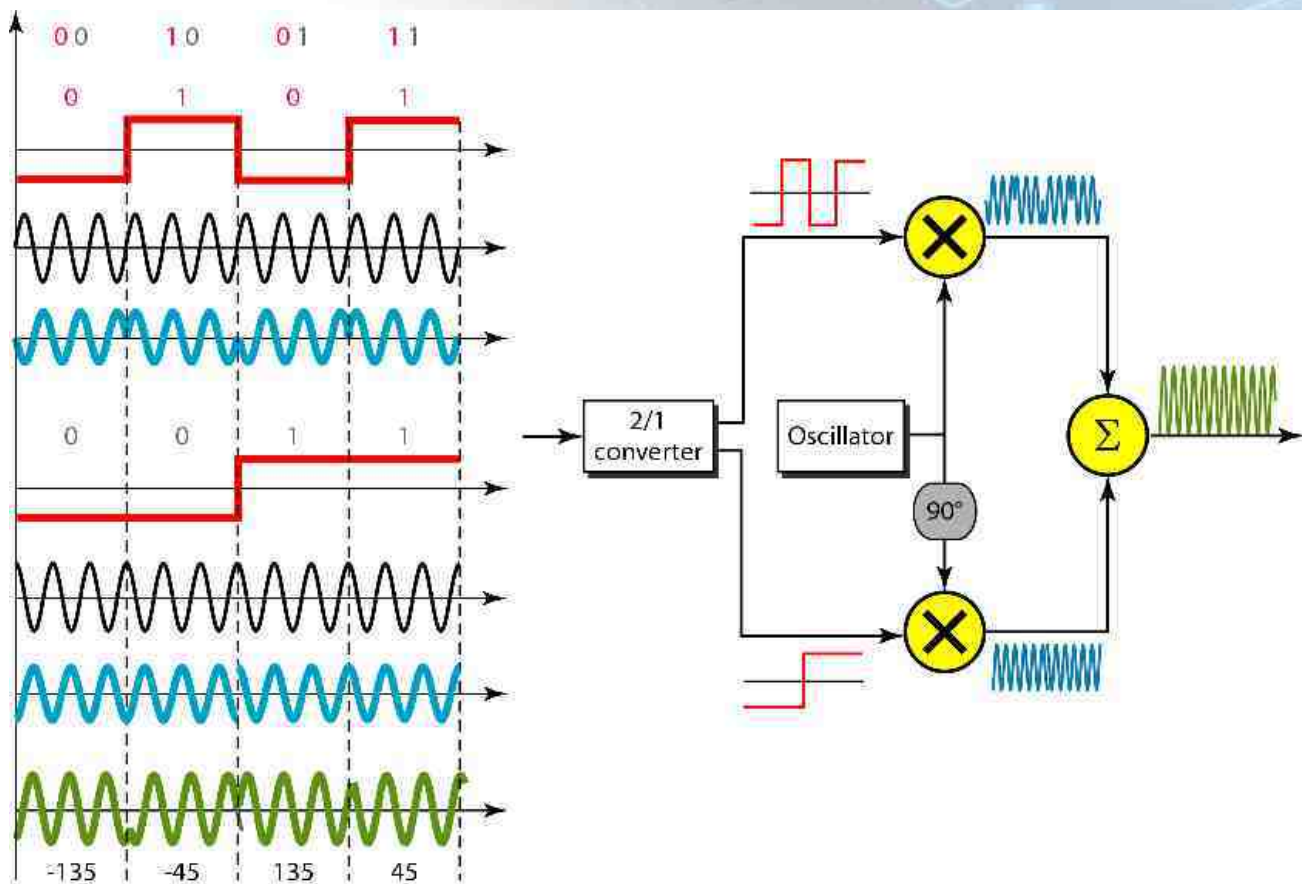
Binary Phase Shift Keying



Implementation of BPSK



QPSK and its Implementation



Example

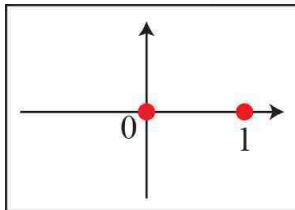
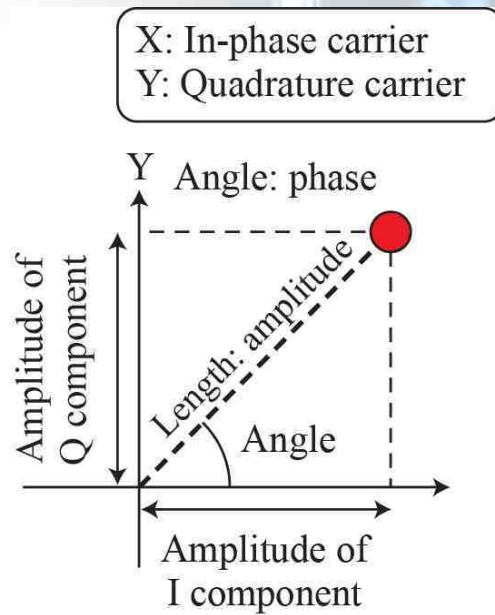
Find the bandwidth for a signal transmitting at 12 Mbps for QPSK. The value of $d = 0$.



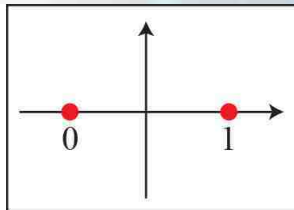
Constellation Diagram

- Helps us define the phase and amplitude of a signal element when we are using two carriers (one in phase and other in quadrature)
- Signal element is represented as a dot

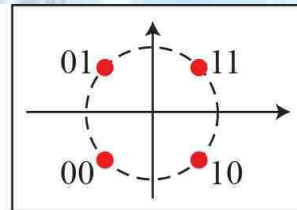
Constellation Diagram



a. BASK (OOK)



b. BPSK

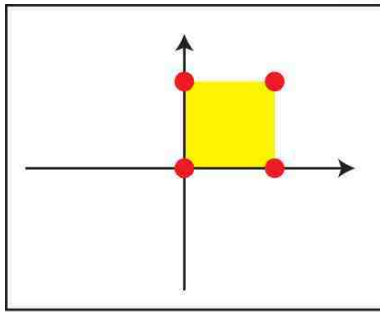


c. QPSK

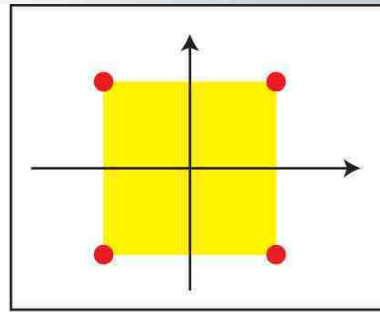
Quadrature Amplitude Modulation (QAM)

- PSK is limited by the ability of the equipment to distinguish small differences in phase which limits its potential bit rate
- We have been altering only one of the three characteristics of a sine wave at a time; but what if we alter two?
- Why not combine ASK and PSK?

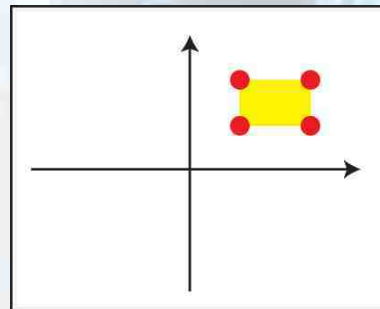
Constellation diagrams for some QAMs



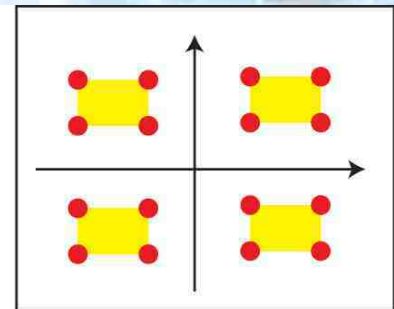
a. 4-QAM



b. 4-QAM



c. 4-QAM

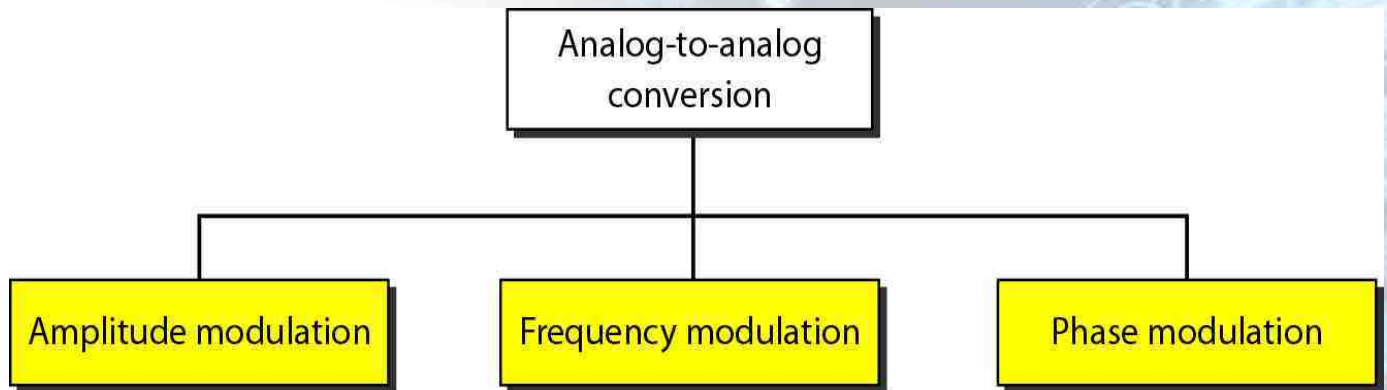


d. 16-QAM

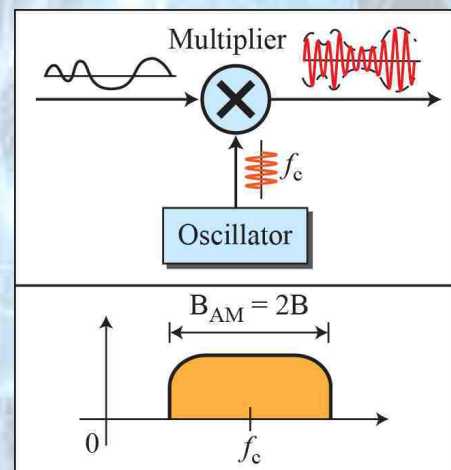
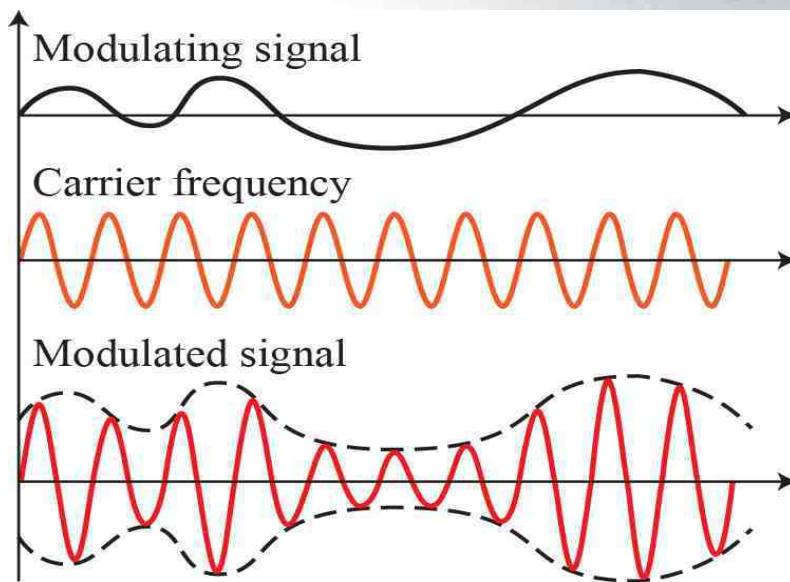
Analog-to-Analog Conversion

- Representation of Analog information by an Analog signal
- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

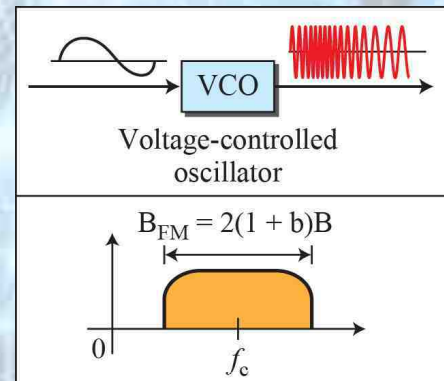
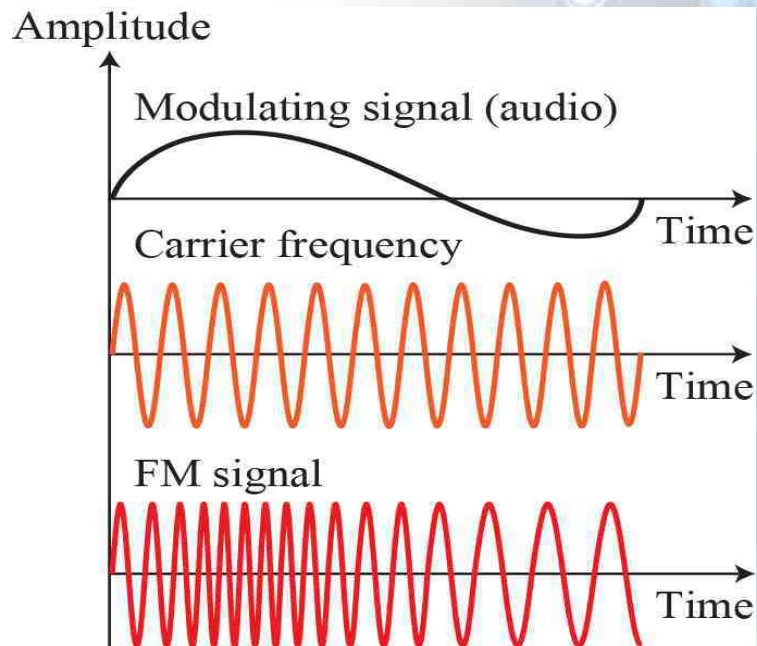
Types of Analog-to-Analog Modulation



Amplitude modulation



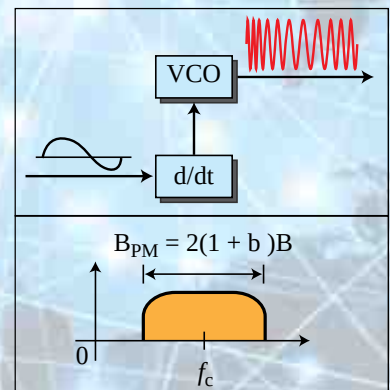
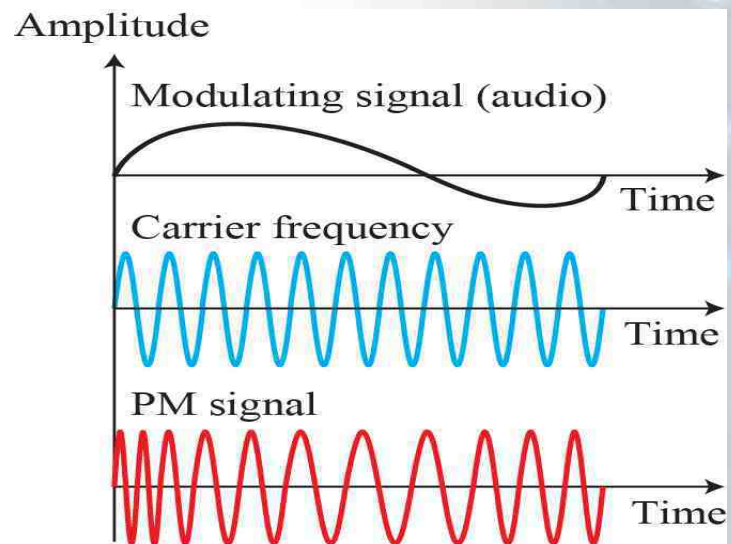
Frequency Modulation



Analog-to-Analog Conversion

- Representation of Analog information by an Analog signal
- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

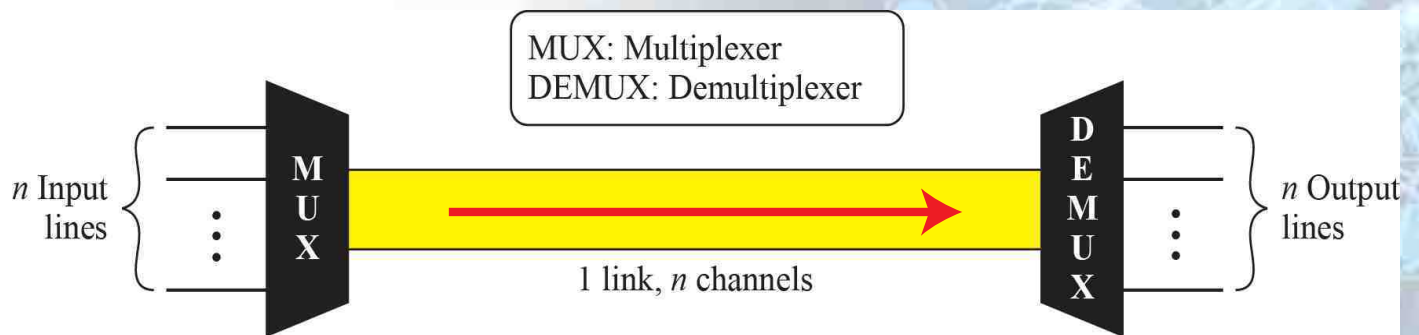
Phase Modulation



Multiplexing

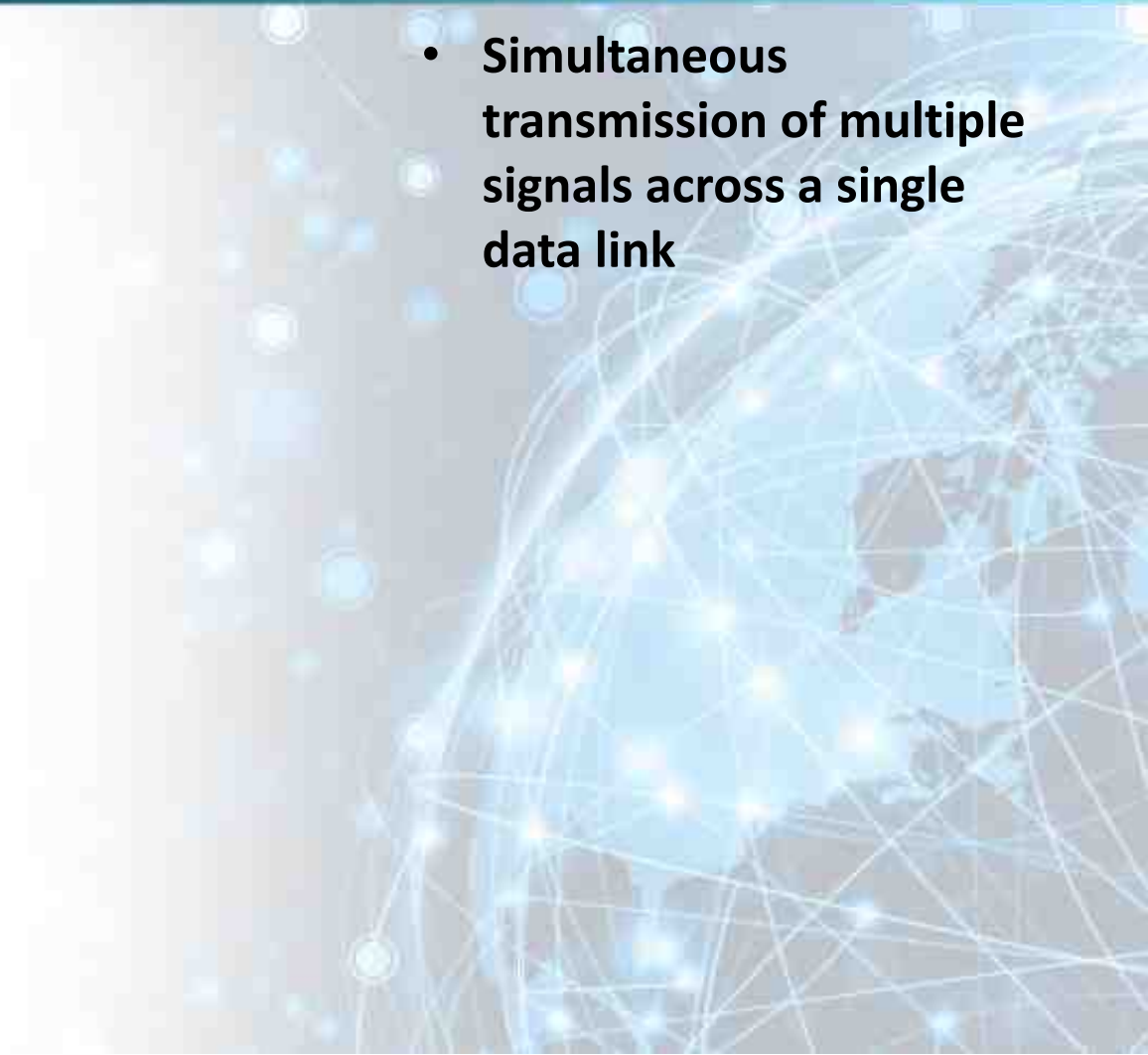
- **Simultaneous transmission of multiple signals across a single data link**
- **As data & telecomm use increases, so does traffic**
 - ✓ **Add individual links each time a new channel is needed**
 - ✓ **Install higher-bandwidth links and use each to carry multiple signals**

Dividing a Link into Channels

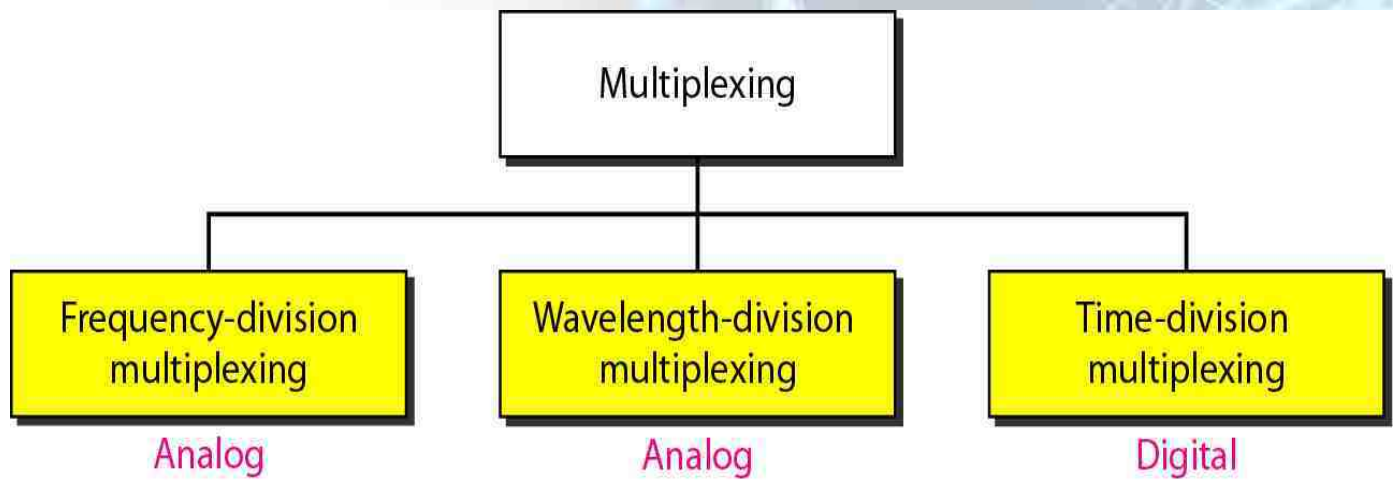


Multiplexing

- **Simultaneous transmission of multiple signals across a single data link**



Categories of Multiplexing

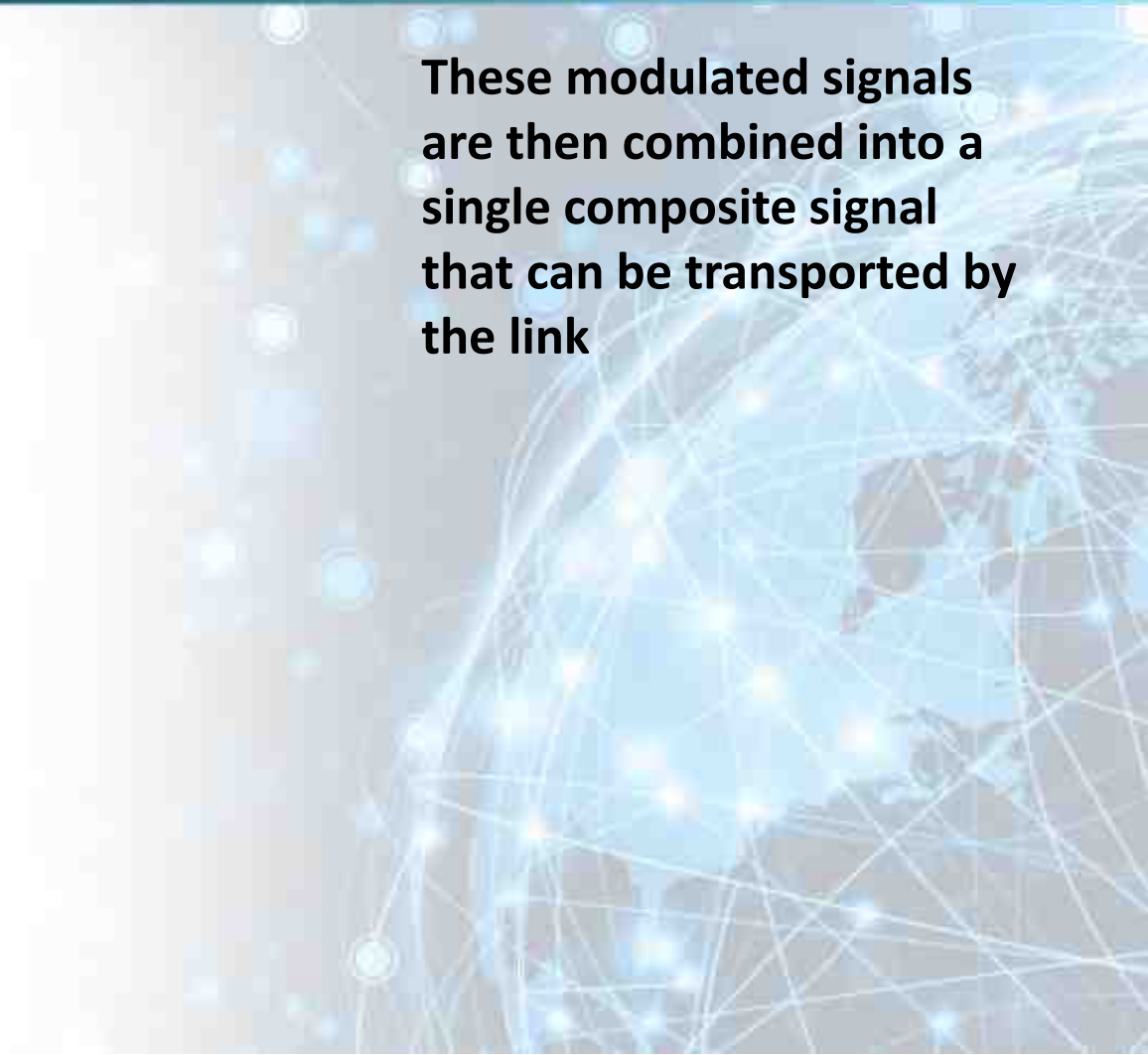


Frequency-Division Multiplexing

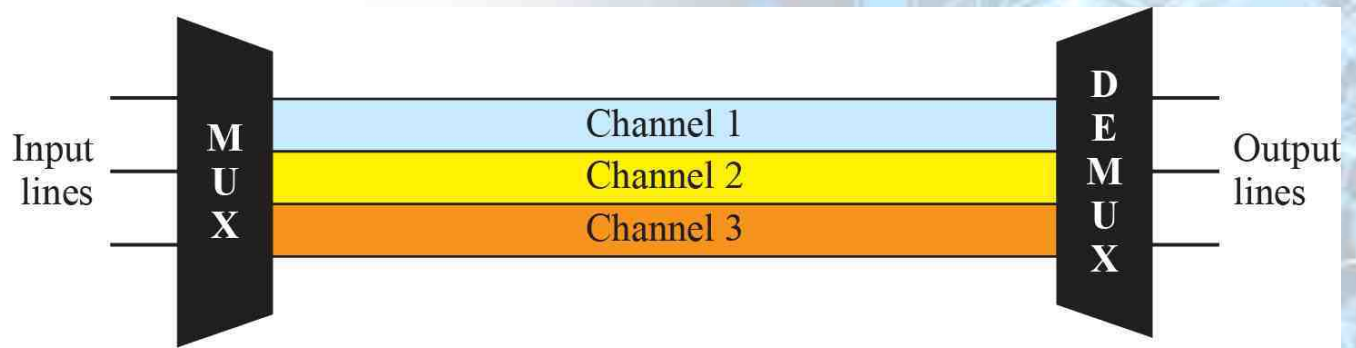
- An analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted
- Signals generated by each sending device modulate different carrier frequencies

Frequency-Division Multiplexing

These modulated signals are then combined into a single composite signal that can be transported by the link



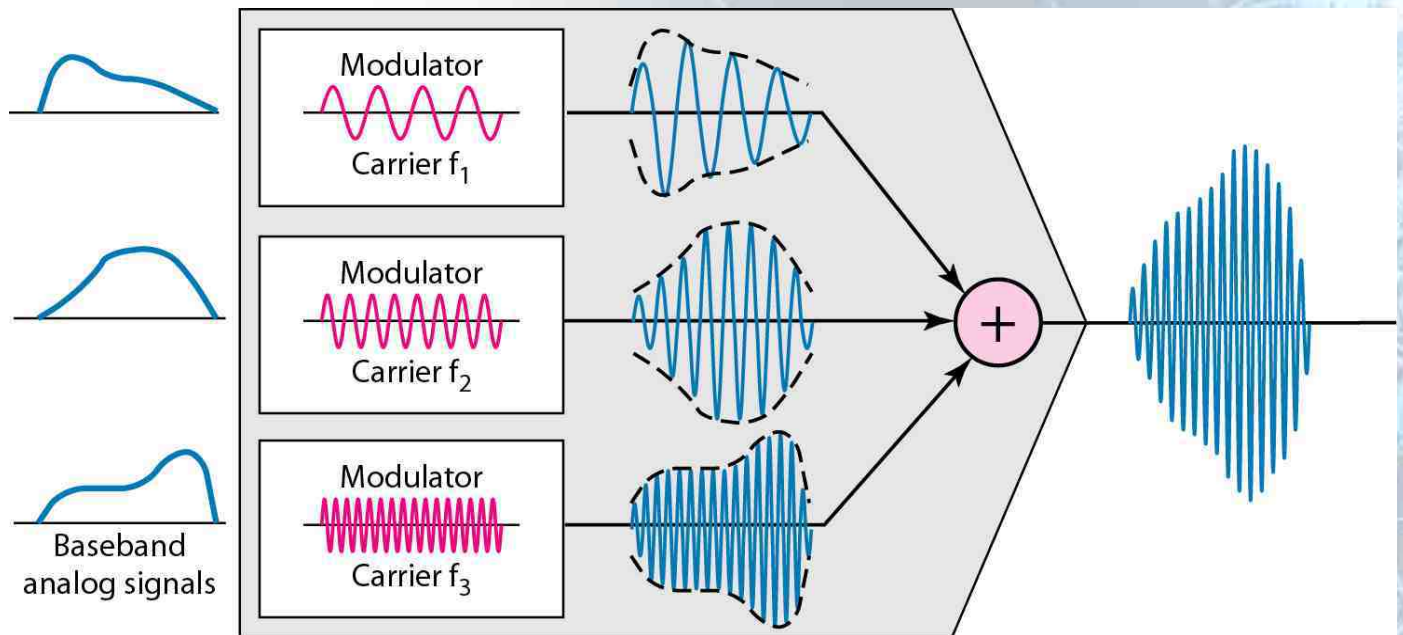
Frequency-Division multiplexing



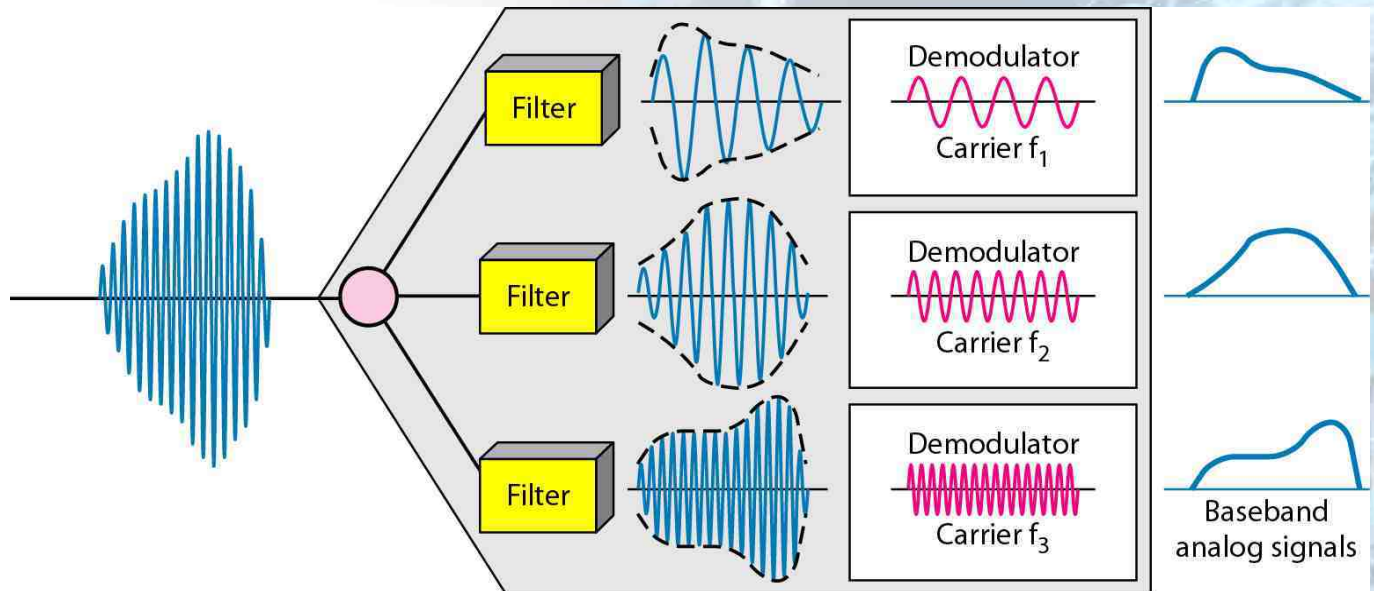
Frequency-Division Multiplexing

- An analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted

FDM Multiplexing

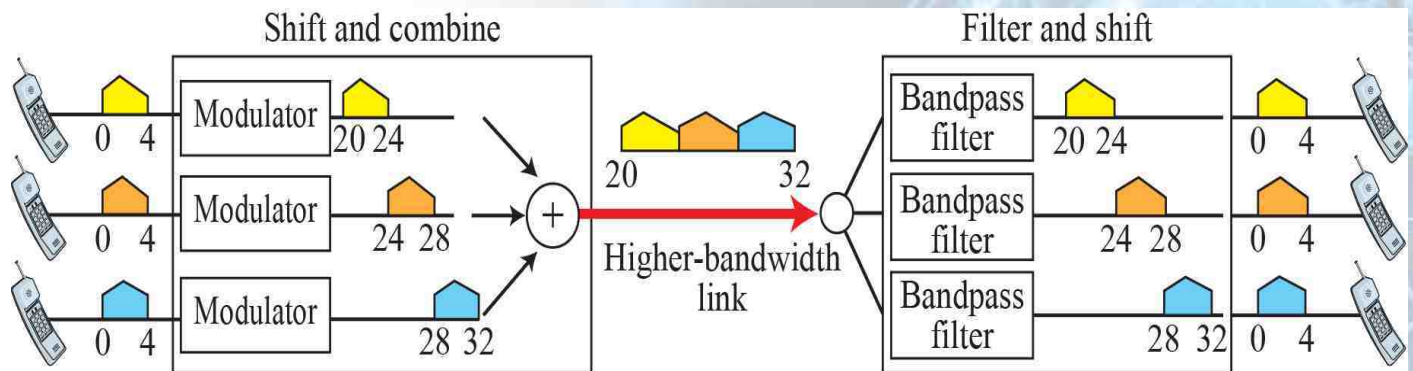


FDM De-Multiplexing



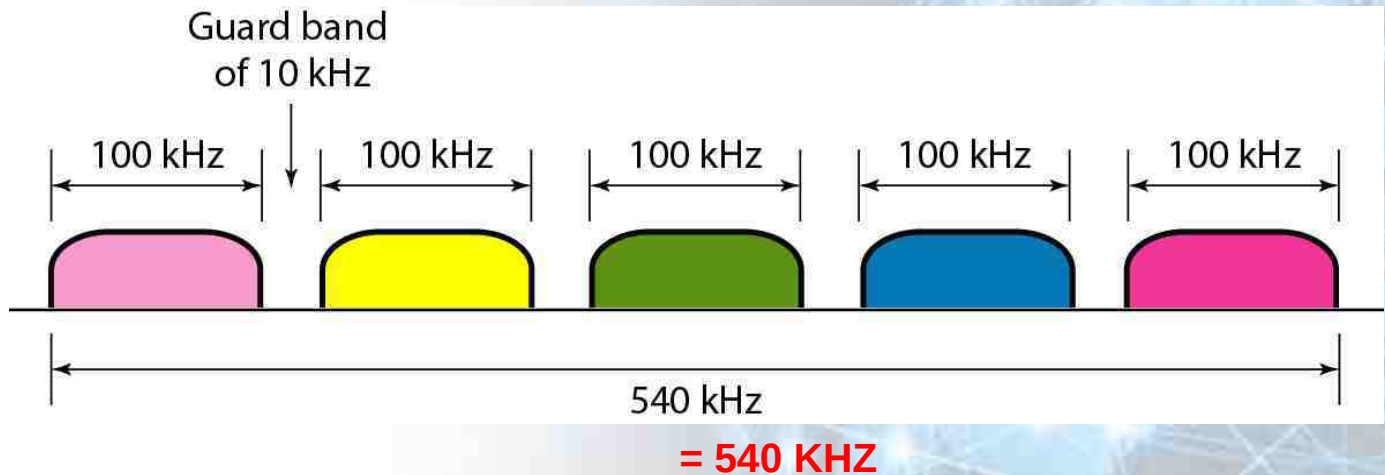
Example

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.



Example

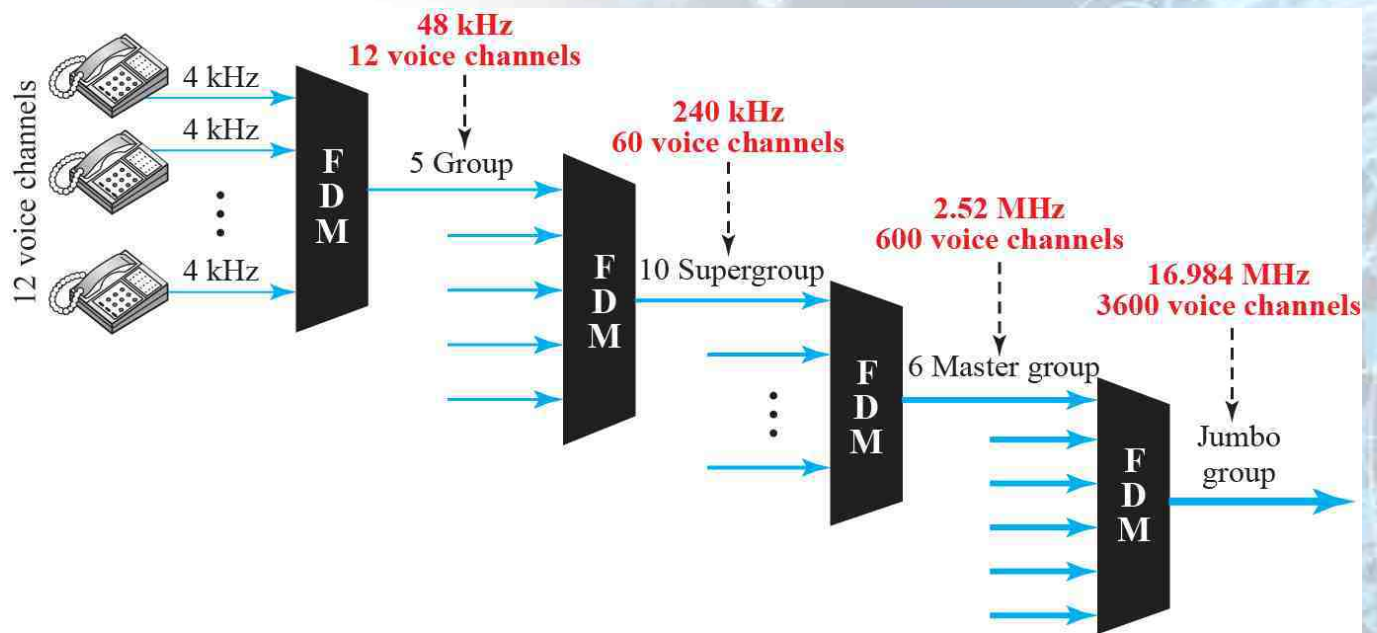
Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?



The Analog Carrier System

- Telephone companies multiplex signals from lower-bandwidth lines on to higher-bandwidth lines
- For Analog, FDM is used

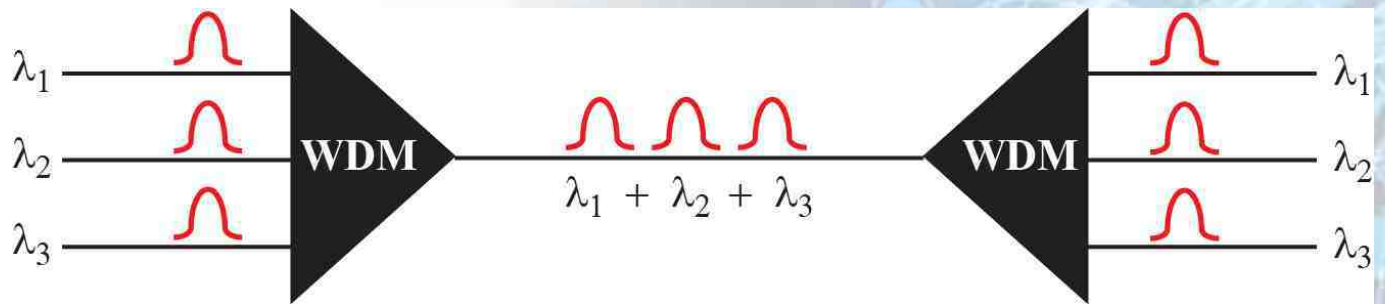
Analog Hierarchy



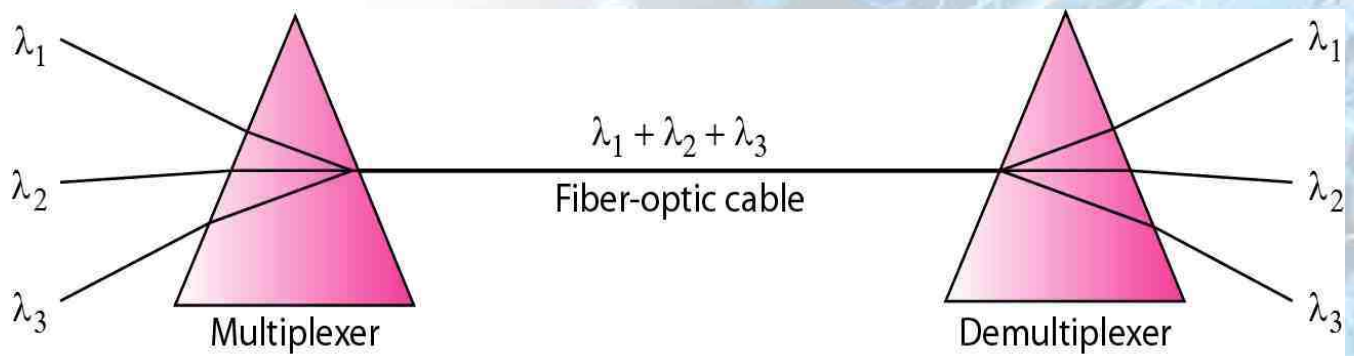
Wavelength-Division Multiplexing

- Designed to use the high-data-rate capability of fiber-optic cable
- Fiber data rate is higher than the data rate of metallic transmission cable
- Using a fiber-optic cable for a single line wastes the available bandwidth
- Multiplexing allows us to combine several lines into one

Wavelength-Division Multiplexing (WDM)



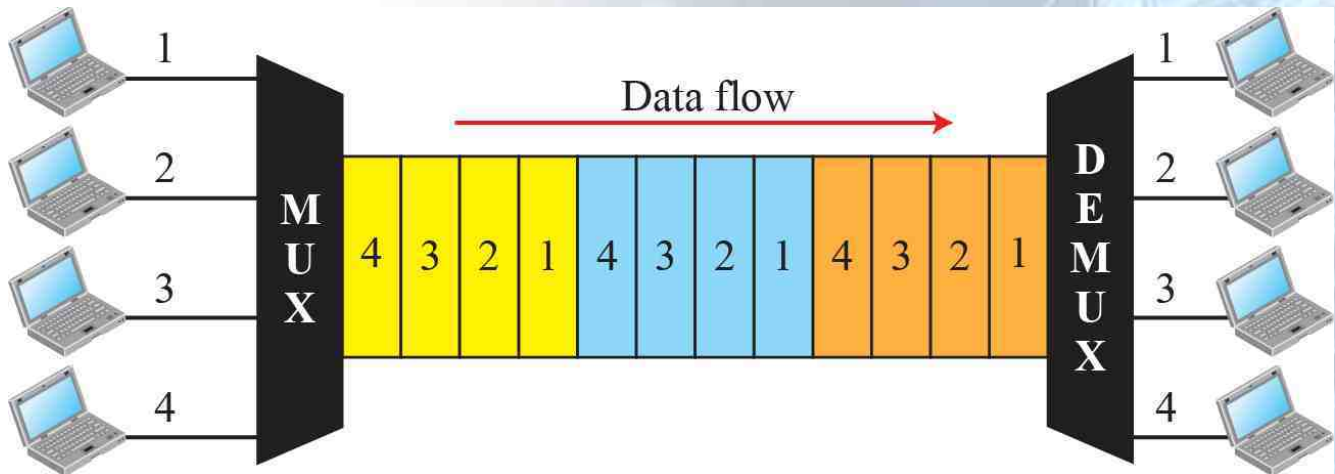
Prisms in Wave-Length Division Multiplexing



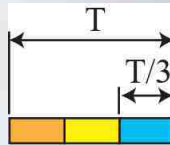
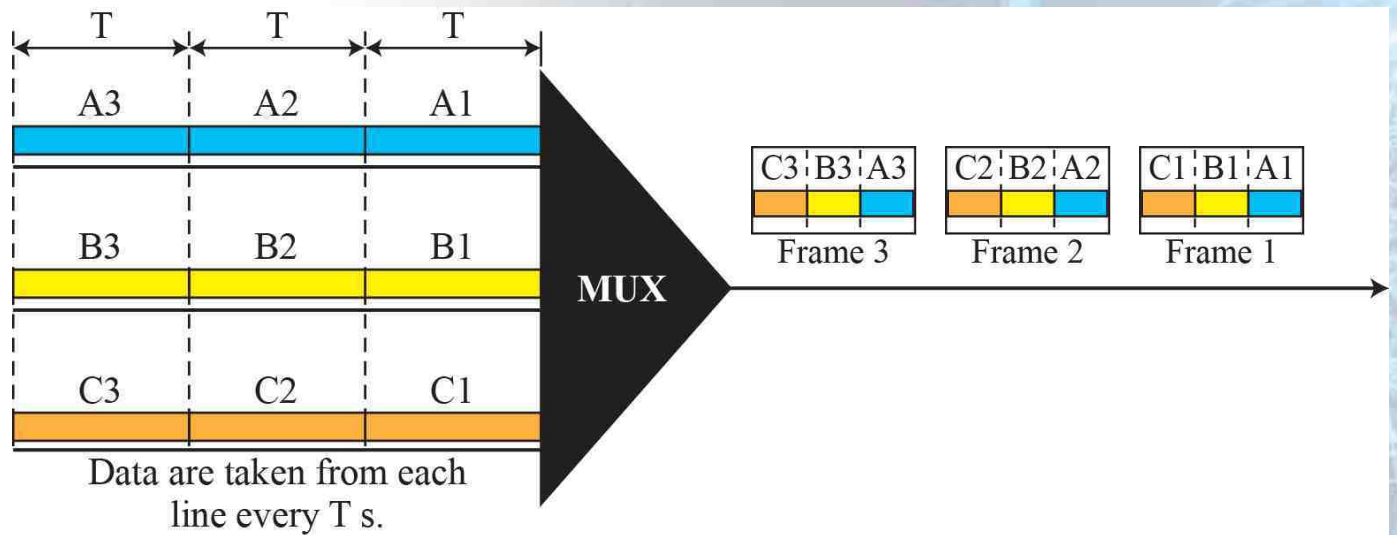
Time-Division Multiplexing

- **Digital process that allows several connections to share the high bandwidth of a link**
- **Time is shared i.e. each connection occupies a portion of time in the link**

TDM



Synchronous Time-Division Multiplexing



Each frame is 3 time slots.
Each time slot duration is $T/3$ s.

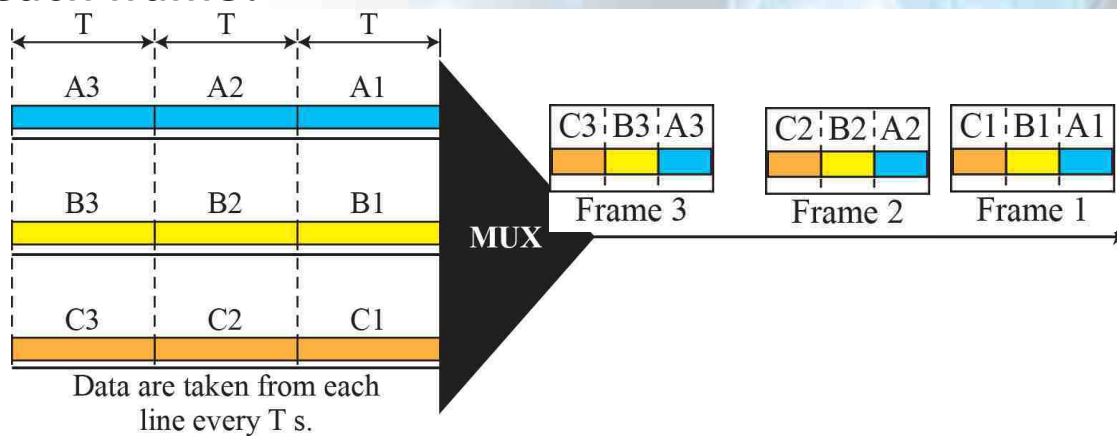
Time-Division Multiplexing

- **Digital process that allows several connections to share the high bandwidth of a link**
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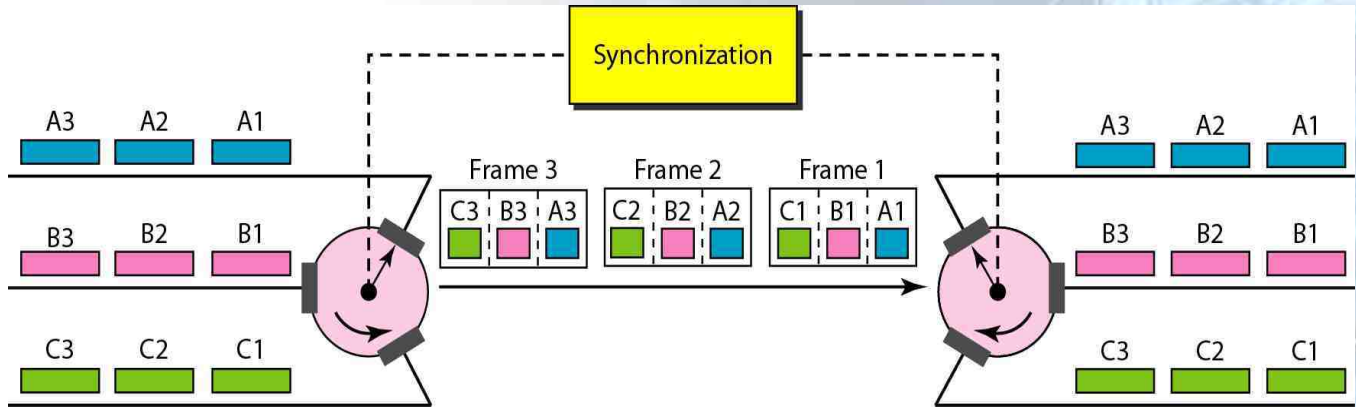
Example

In Figure the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of

- each input slot,
- each output slot, and
- each frame?



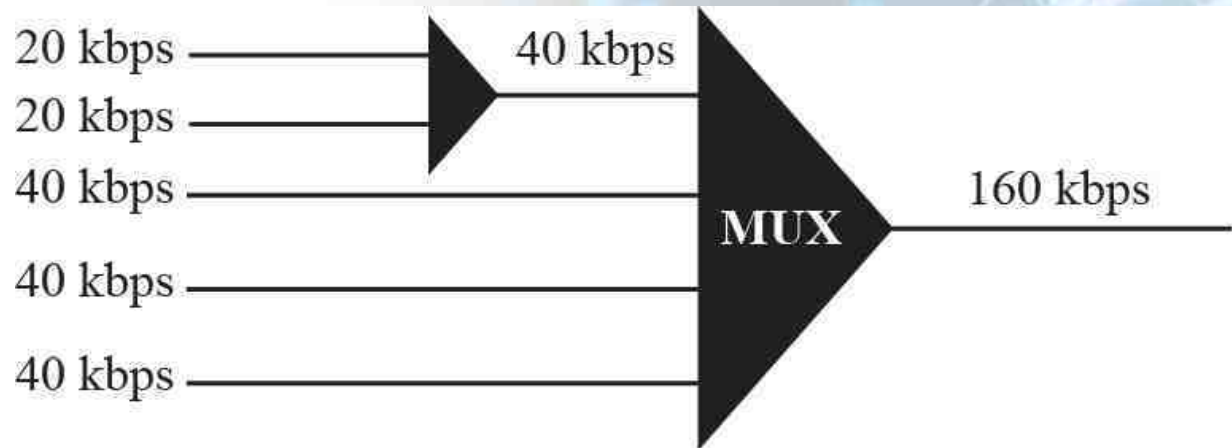
Interleaving



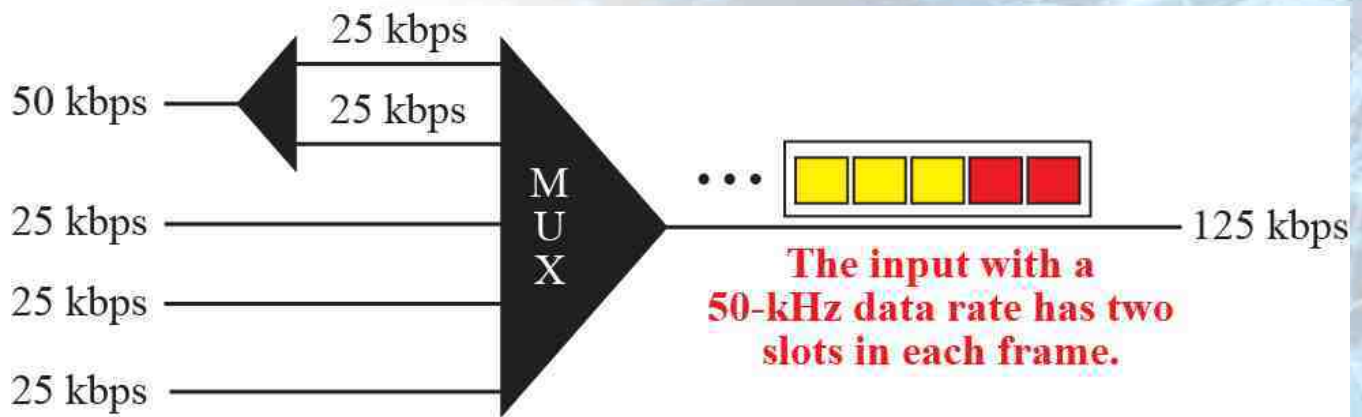
Time-Division Multiplexing

- **Digital process that allows several connections to share the high bandwidth of a link**
- **Time is shared i.e. each connection occupies a portion of time in the link**

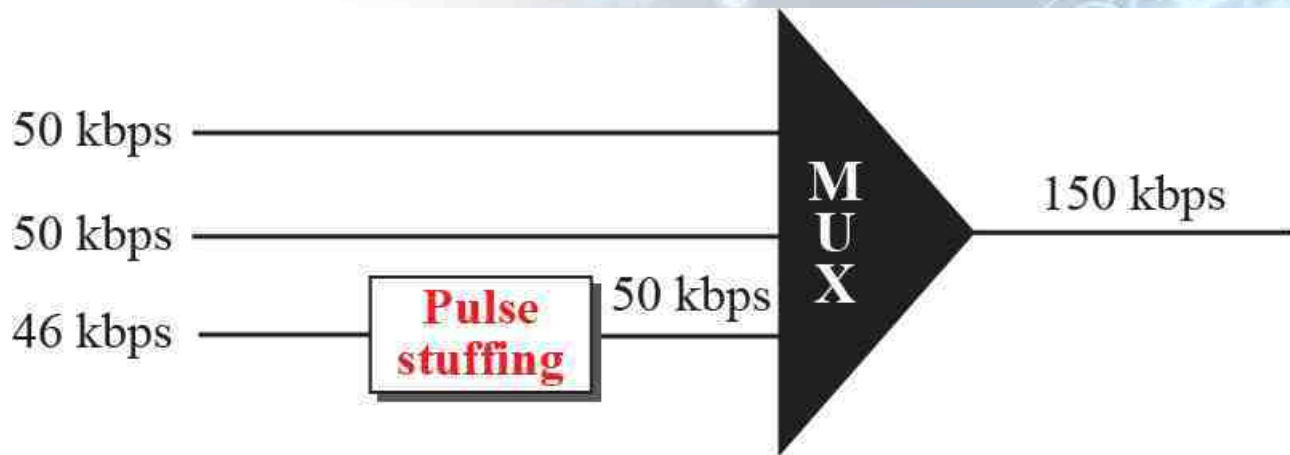
Multilevel Multiplexing



Multiple-Slot Multiplexing



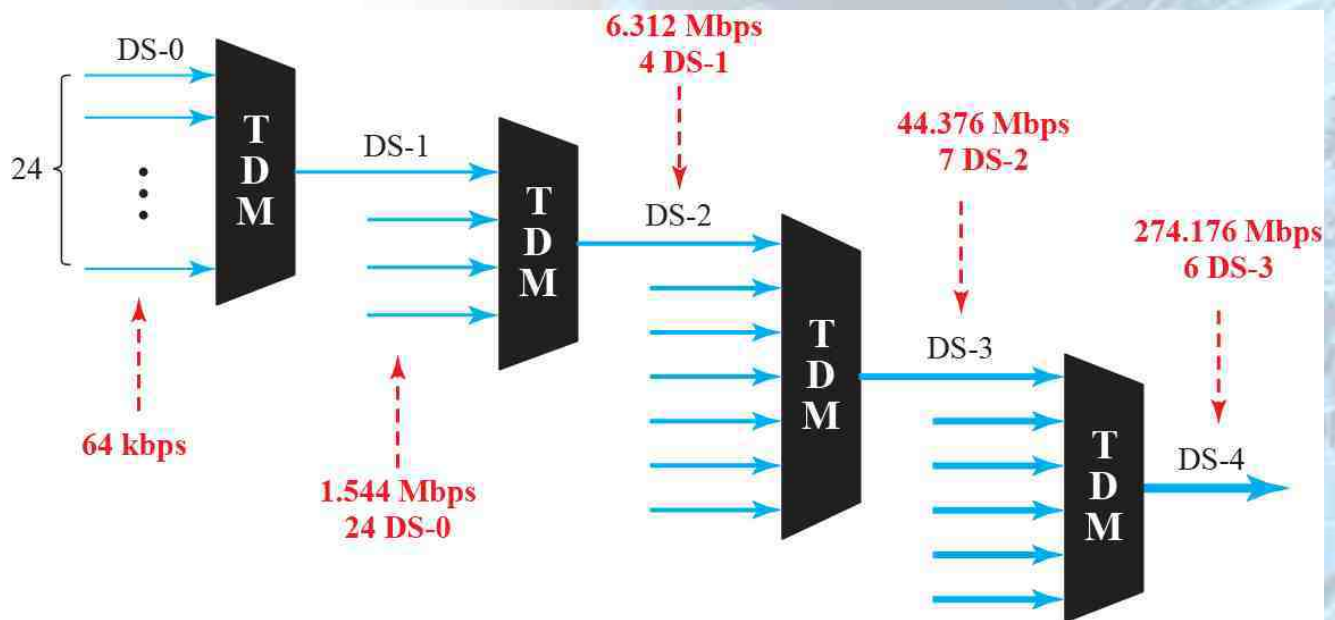
Pulse Stuffing



Time-Division Multiplexing

- **Digital process that allows several connections to share the high bandwidth of a link**
- **Time is shared i.e. each connection occupies a portion of time in the link**

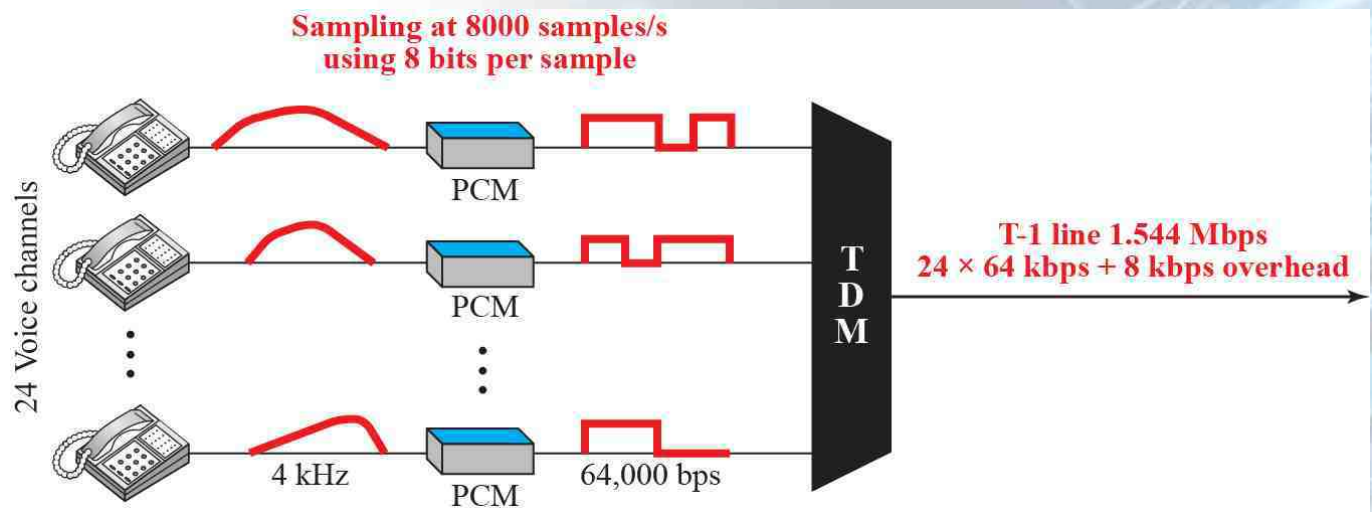
Digital Hierarchy



DS and T Line Rates

<i>Service</i>	<i>Line</i>	<i>Rate (Mbps)</i>	<i>Voice Channels</i>
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

T-1 Line



E Line Rates

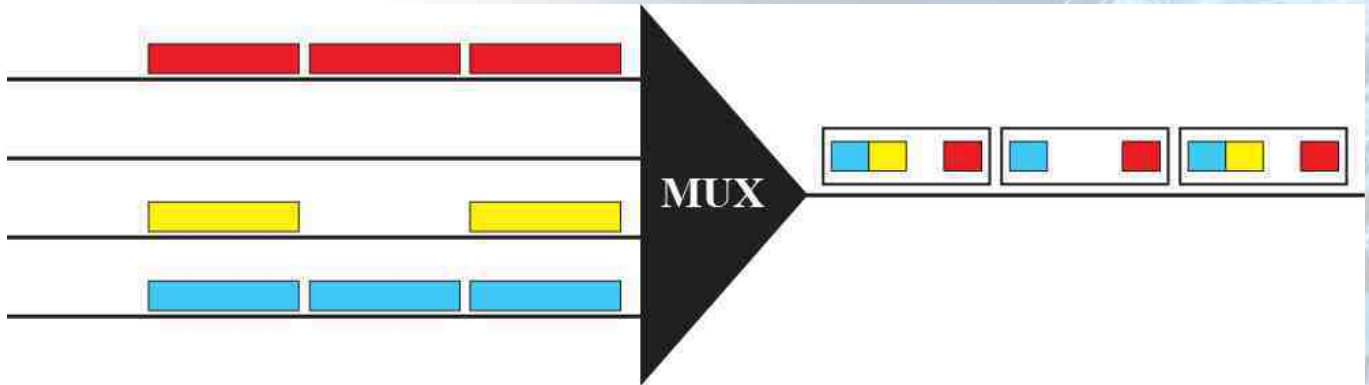
<i>Line</i>	<i>Rate (Mbps)</i>	<i>Voice Channels</i>
E-1	2.048	30
E-2	8.448	120
E-3	34.368	480
E-4	139.264	1920

Time-Division Multiplexing

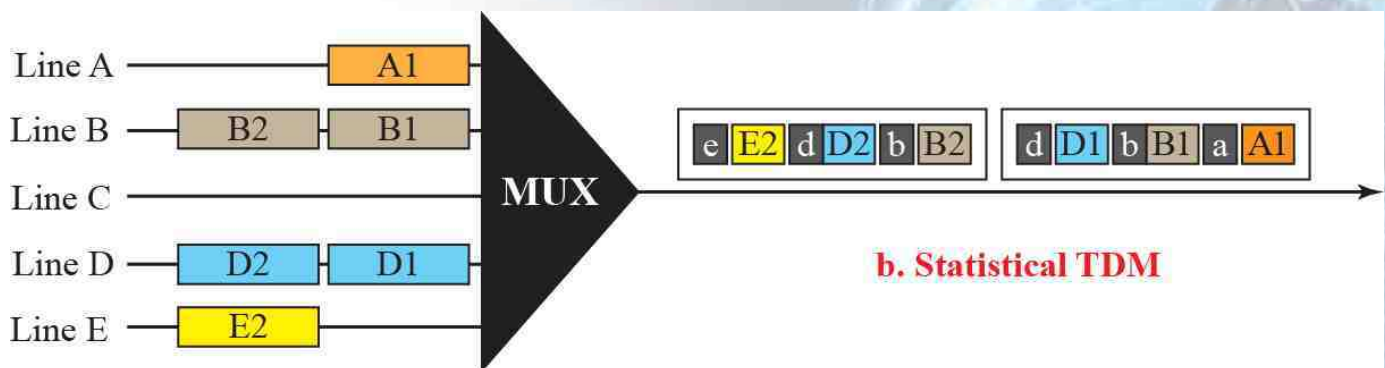
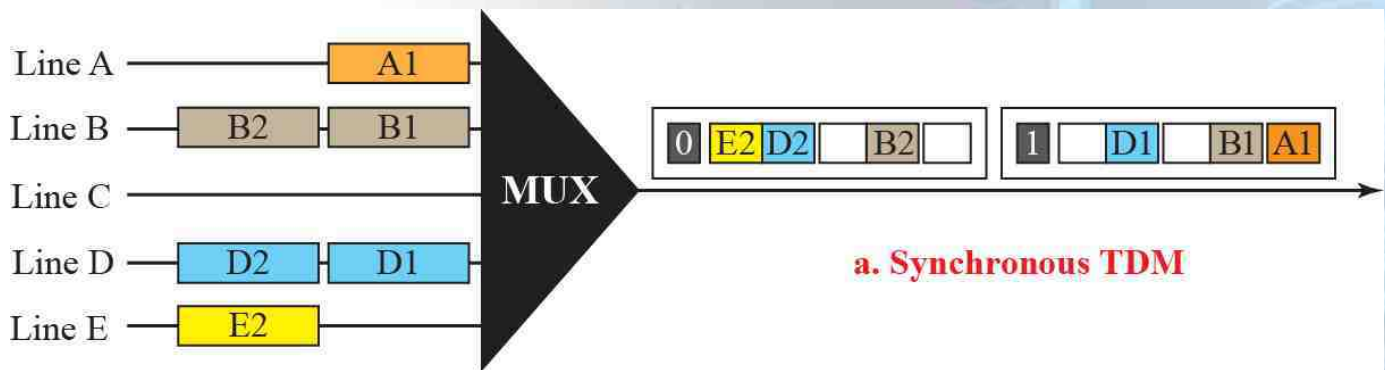
- **Synchronous TDM**
- **Statistical TDM**



Empty slots



Statistical TDM



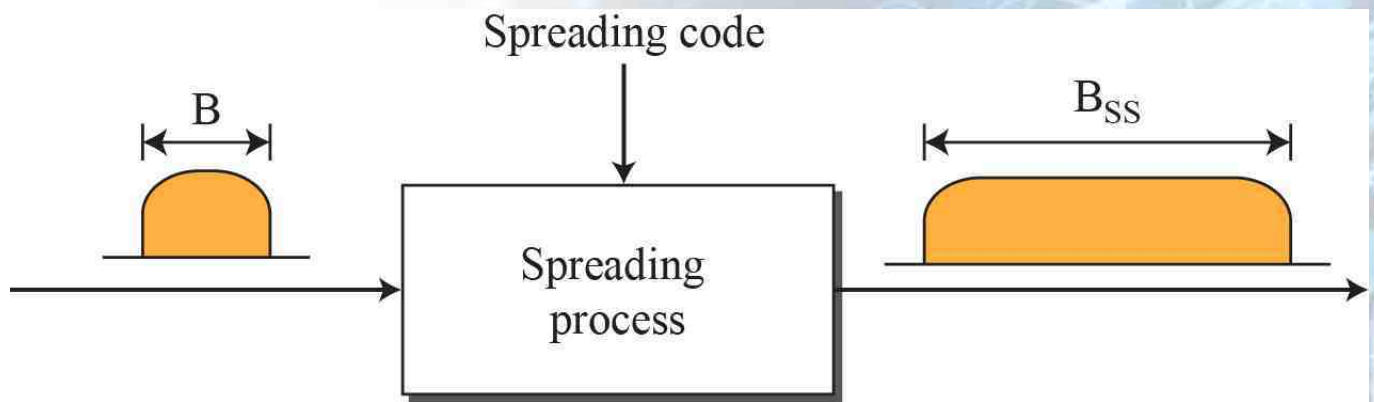
SPREAD SPECTRUM

- In wireless applications, stations must be able to share the medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder
- To achieve these goals, spread spectrum adds redundancy and spread original spectrum needed for each station

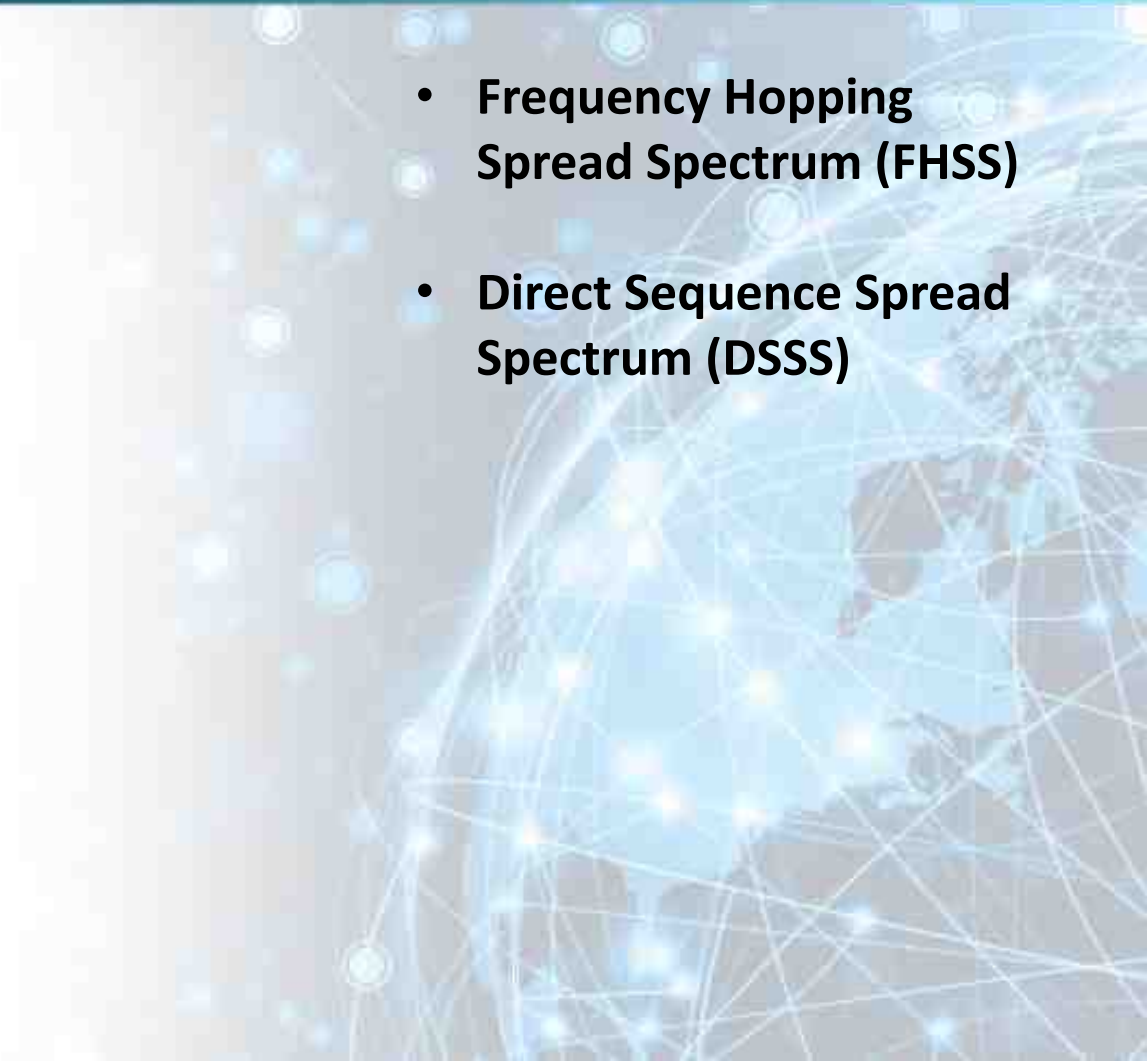
SPREAD SPECTRUM - Principles

- Bandwidth allocated to each station needs to be larger than what is needed to allow Redundancy
- Spreading process should be independent of the original signal

Spread Spectrum



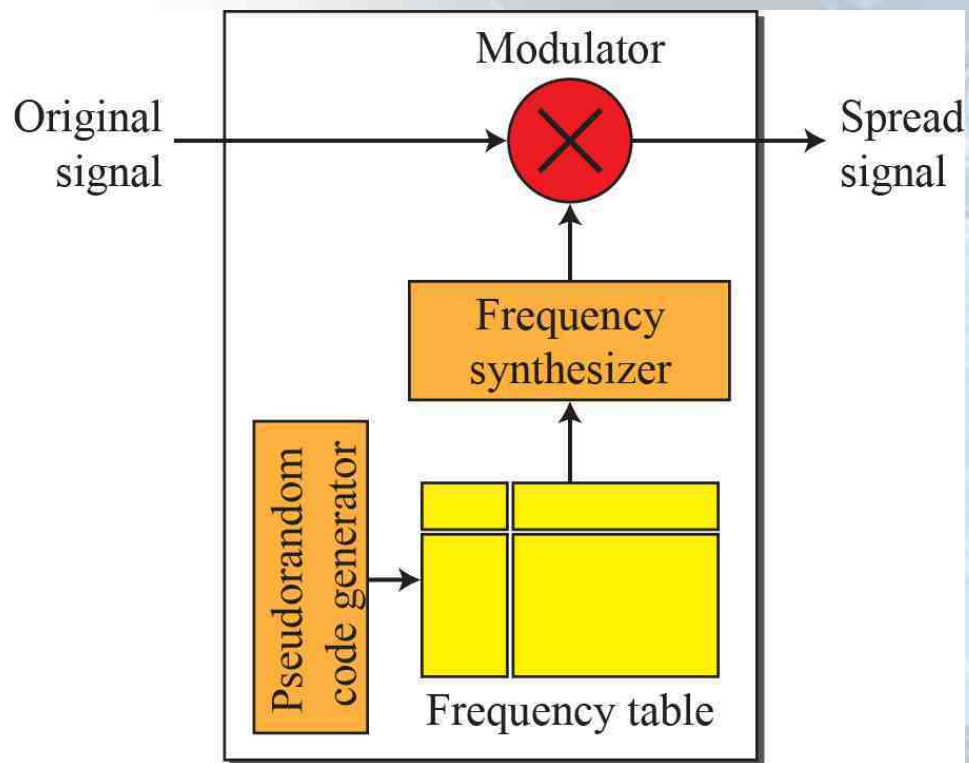
SPREAD SPECTRUM TECHNIQUES

- **Frequency Hopping Spread Spectrum (FHSS)**
 - **Direct Sequence Spread Spectrum (DSSS)**
- 
- A background graphic featuring a stylized globe with a network of glowing blue lines and nodes, suggesting global communication or data flow. The globe is partially obscured by a light blue, semi-transparent grid pattern.

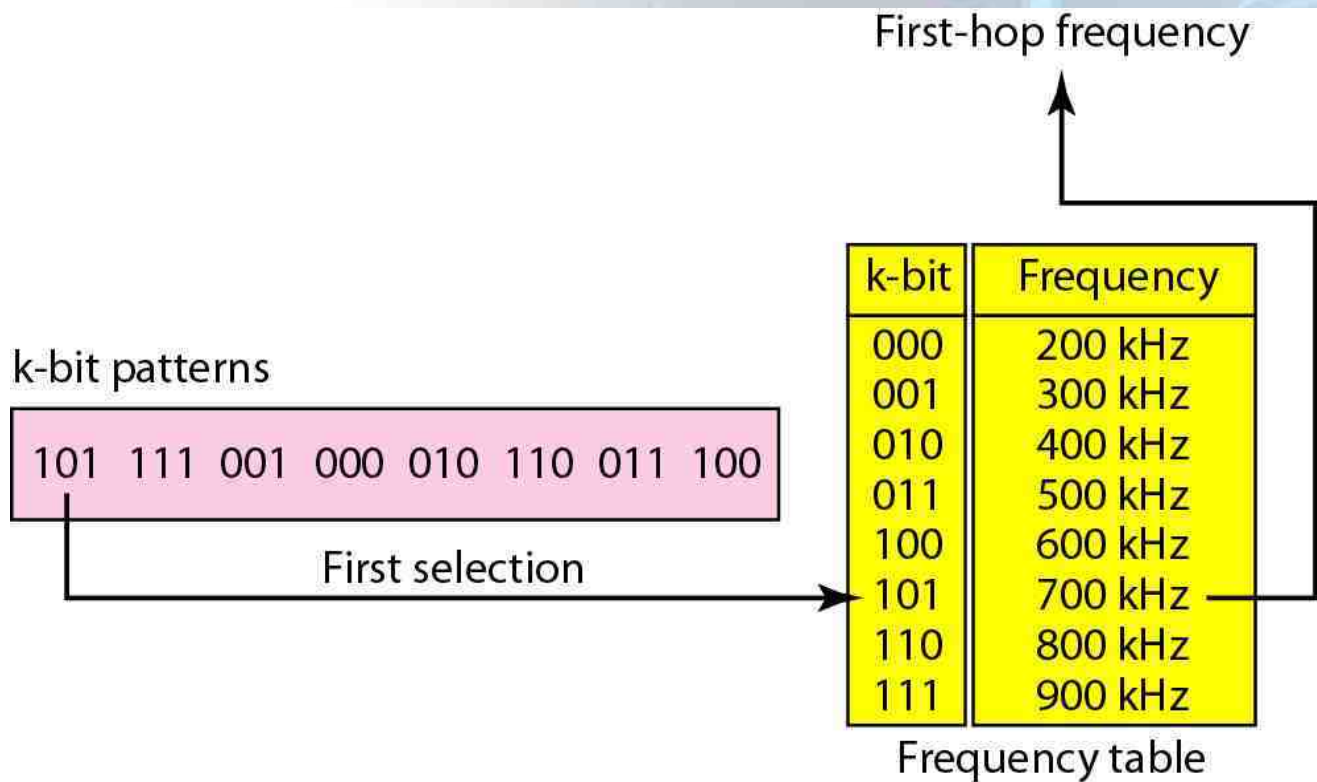
Frequency Hopping Spread Spectrum (FHSS)

- 'M' different carrier frequencies that are modulated by the source signal
- At one moment, signal modulates one carrier frequency and at next moment, it modulates another

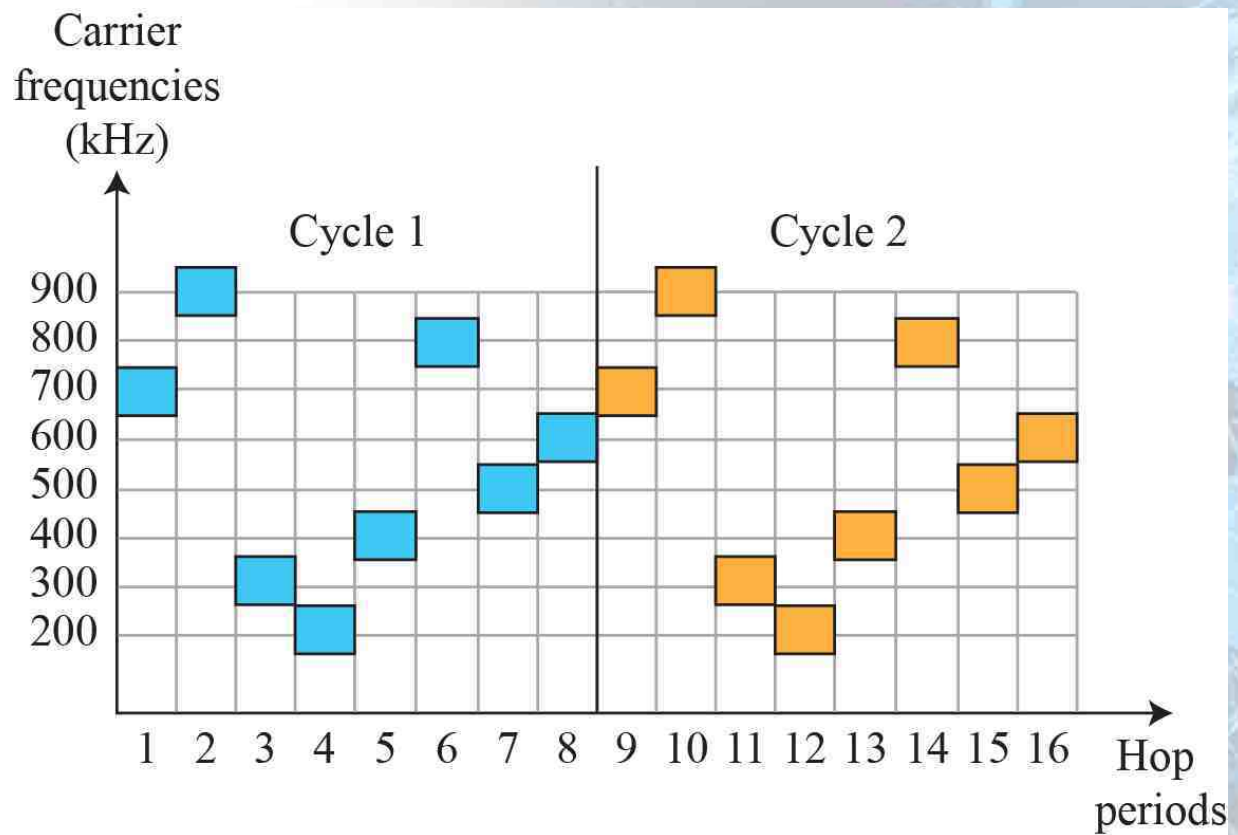
Frequency Hopping Spread Spectrum (FHSS)



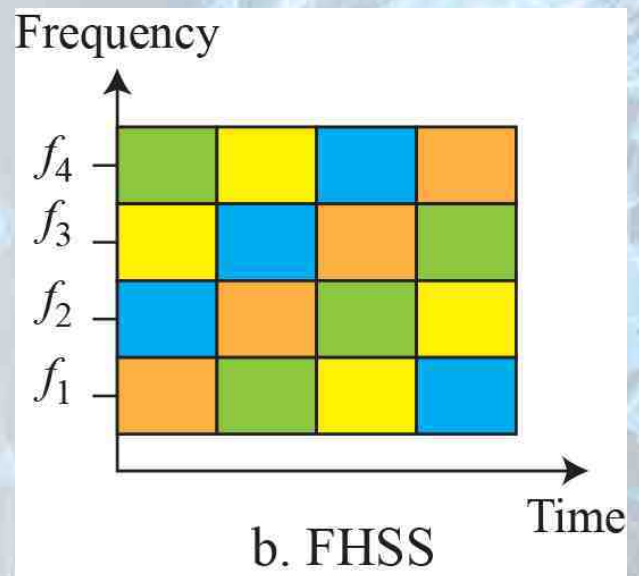
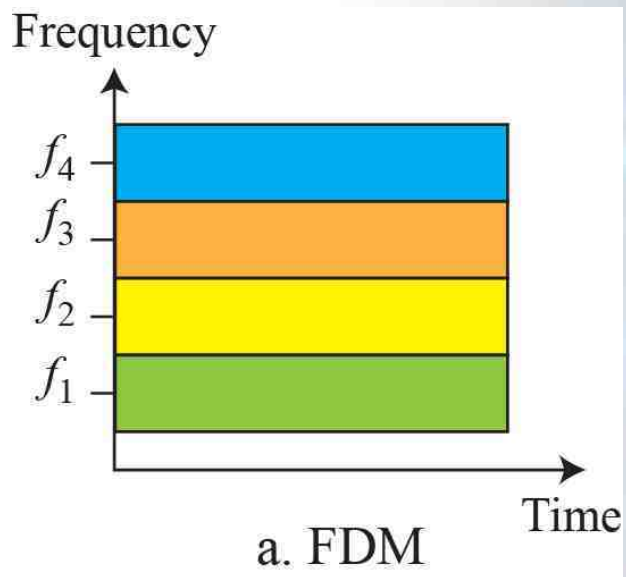
Frequency Selection in FHSS



FHSS Cycles



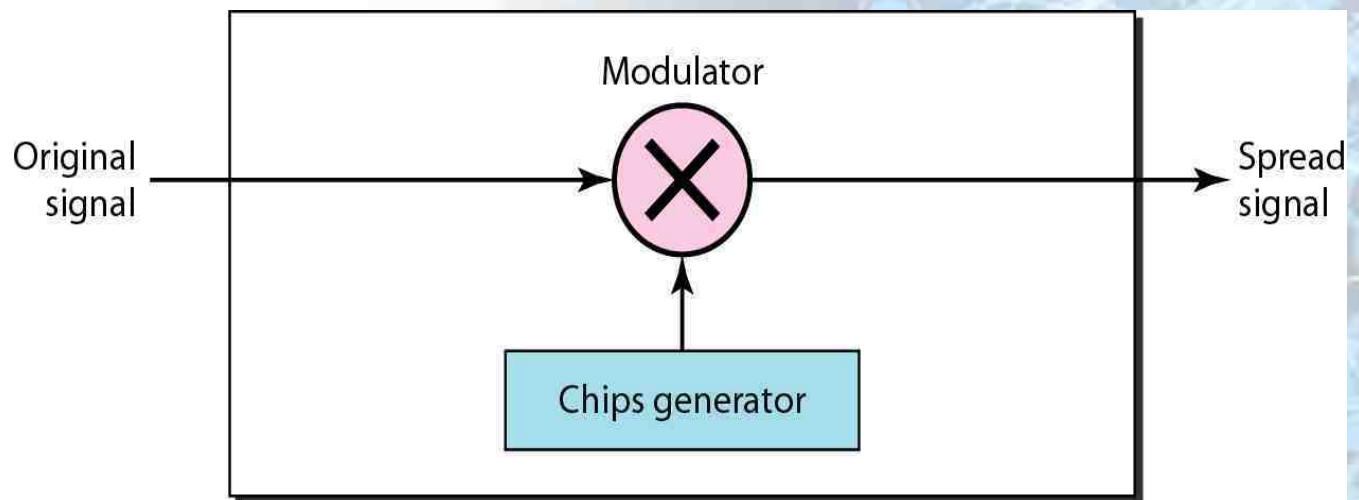
Bandwidth Sharing



DSSS

- DSSS also expands the bandwidth of the original signal, but the process is different
- We replace each data bit with 'n' bits using a spreading code
- Each bit is assigned a code of 'n' bits, called chips, where the chip rate is 'n' times that of the data bit

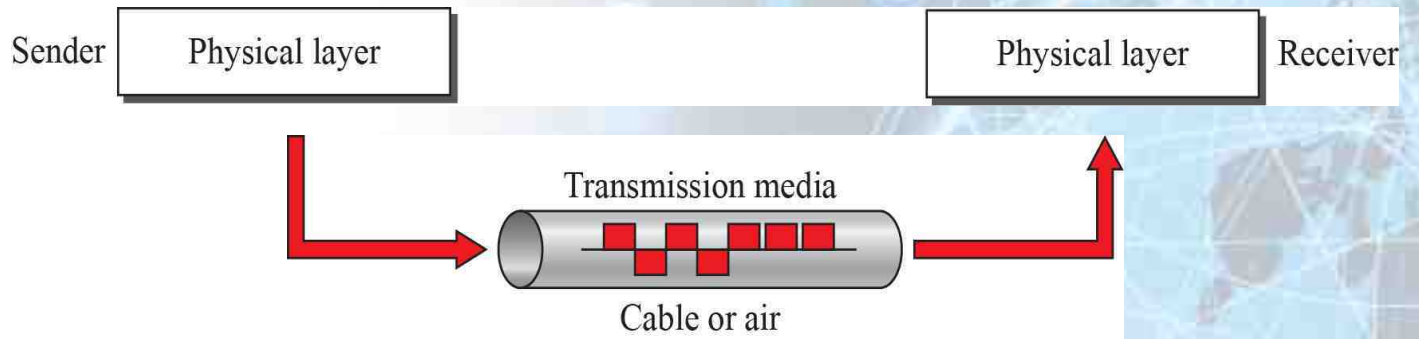
DSSS



Transmission Media

- Located below the physical layer and are directly controlled by the physical layer
- Belong to layer zero
- Metallic Media i.e. Twisted pair and Coaxial Cable
- Optical Fiber Cable
- Free Space i.e. Air, Vacuum

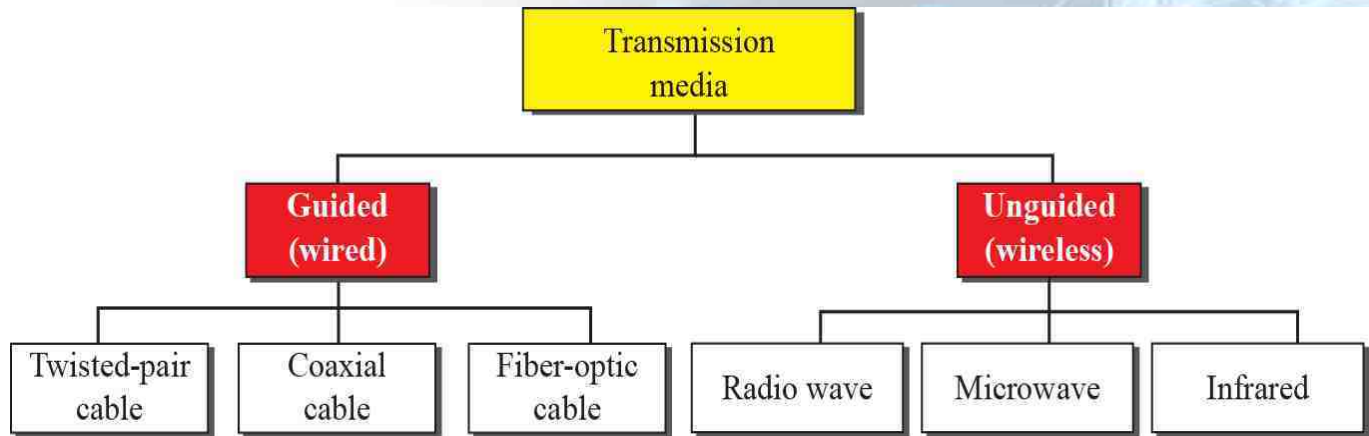
Transmission Media & Physical Payer



Transmission Media

- Located below the physical layer and are directly controlled by the physical layer
- Belong to layer zero
- Metallic Media i.e. Twisted pair and Coaxial Cable
- Optical Fiber Cable
- Free Space i.e. Air, Vacuum

Classes of Transmission Media



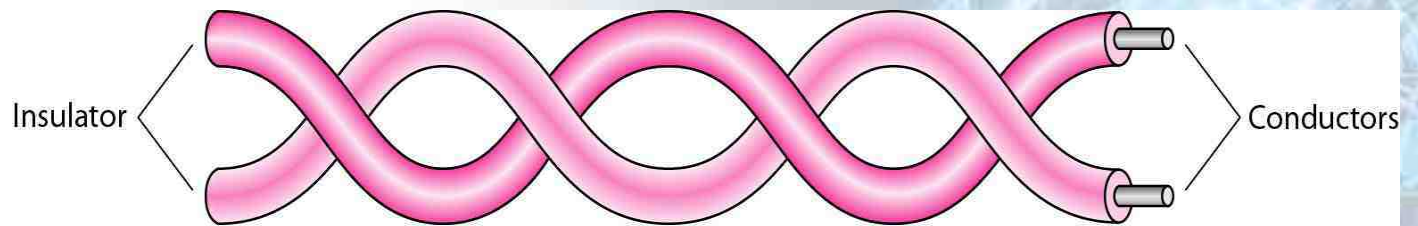
Guided Media

- Media that provides a conduit from one device to another
- Twisted-pair cable, coaxial cable, and fiber-optic cable
- Signal traveling along any of these media is directed and contained by the physical limits of the medium

Twisted-Pair Cable

- Consists of 2 copper conductors, each with its own plastic insulation, twisted together
- One wire carries signals and other is ground reference
- Receiver uses difference between the two
- Interference (Noise) & Crosstalk

Twisted-Pair Cable

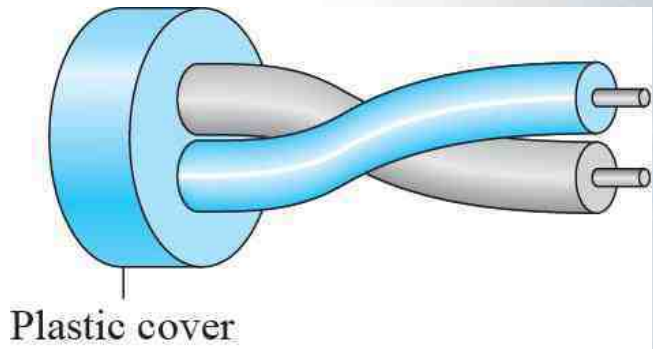


Twisted-Pair Cable

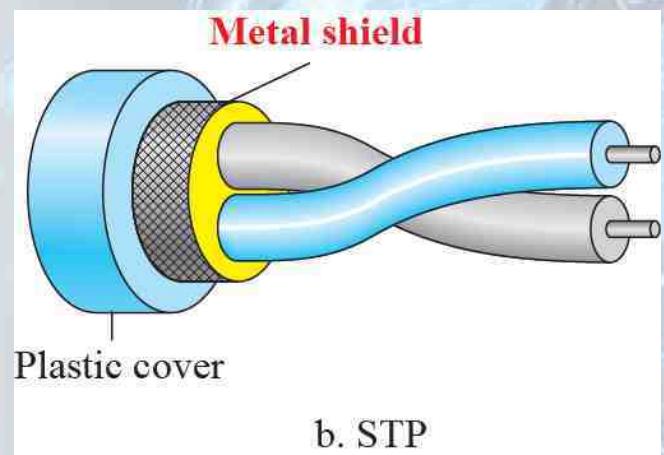
- UTP
- STP



Unshielded vs. Shielded Twisted Pair Cable



a. UTP



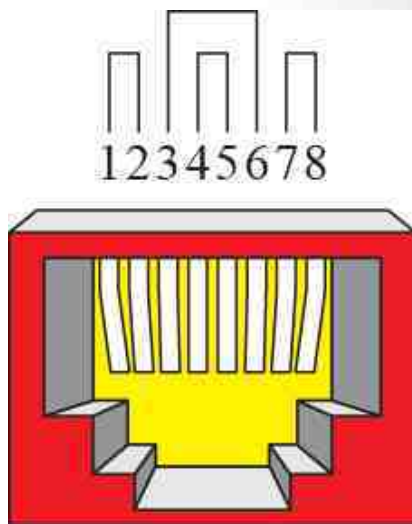
b. STP

Categories of Unshielded Twisted-Pair Cables

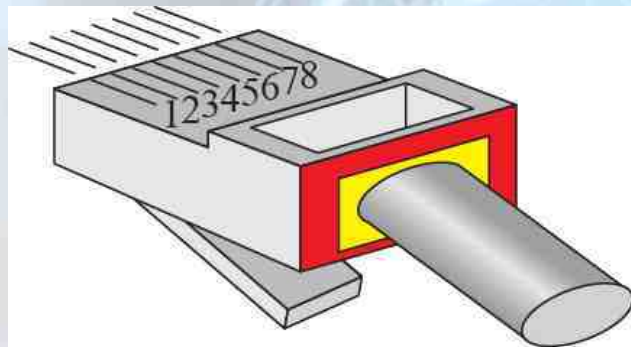
Category	Specification	Data Rate (Mbps)	Use
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs

5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called <i>SSTP (shielded screen twisted-pair)</i> . Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

UTP Connectors

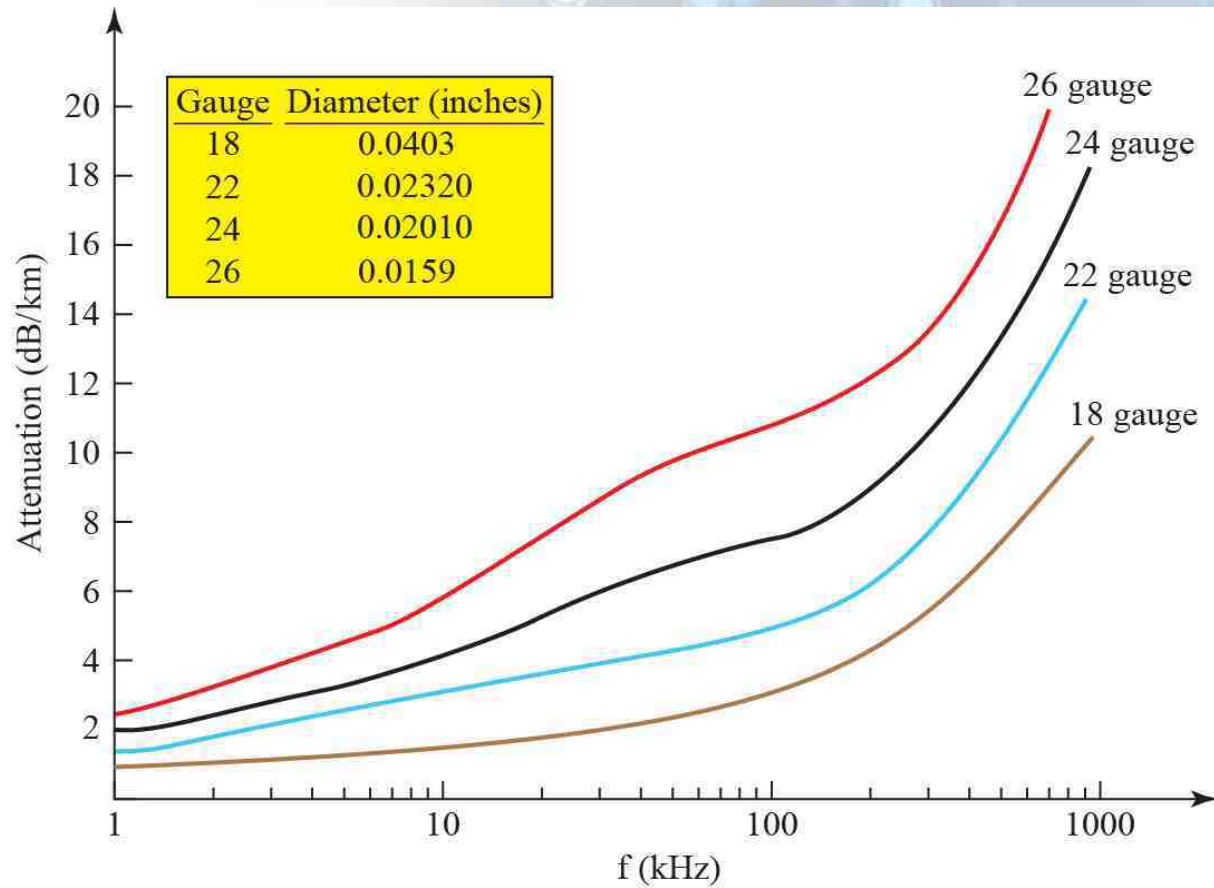


RJ-45 Female



RJ-45 Male

UTP Performance

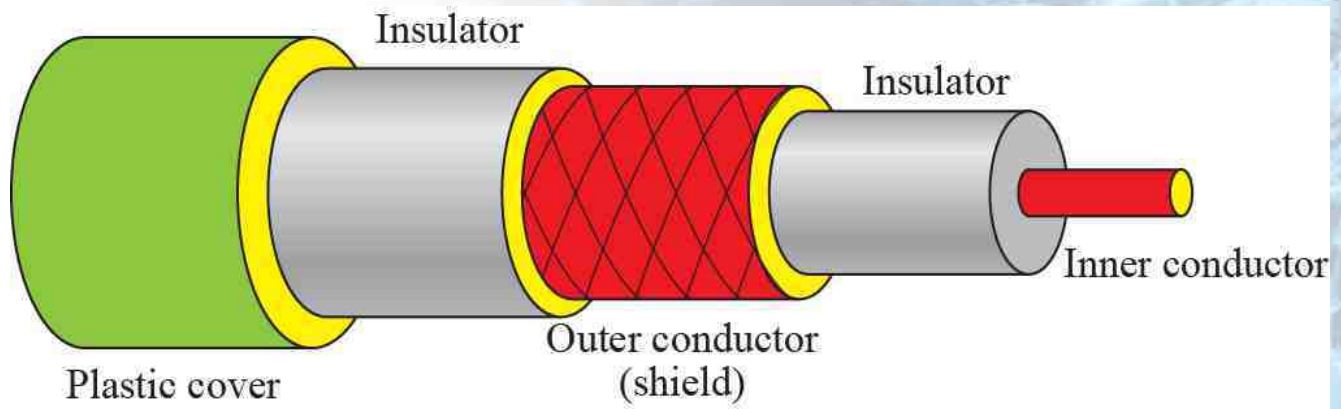


Coaxial Cable

- Carries signals of higher frequency ranges than those in twisted pair cable



Coaxial Cable



Categories of Coaxial Cables

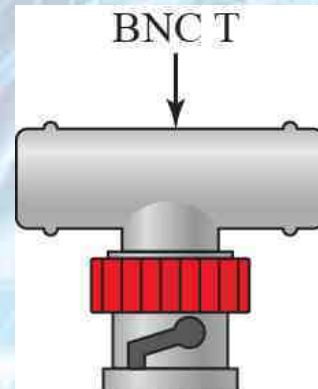
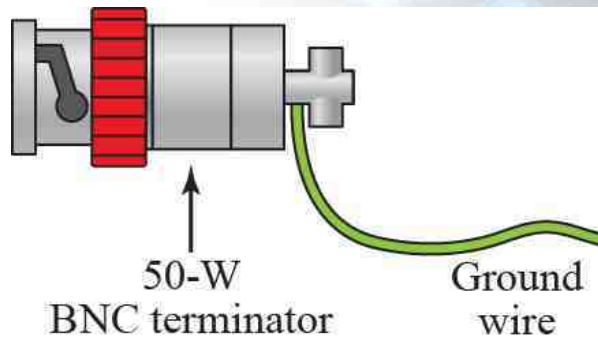
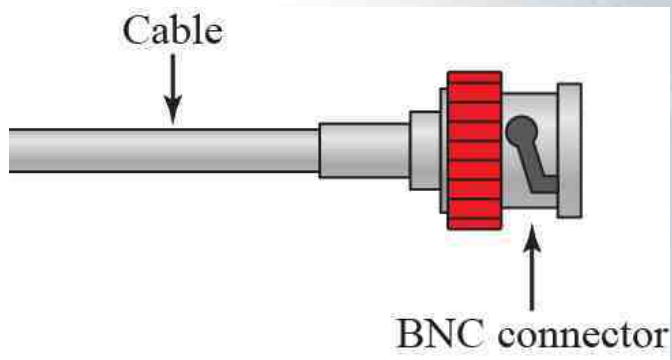
<i>Category</i>	<i>Impedance</i>	<i>Use</i>
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Coaxial Cable

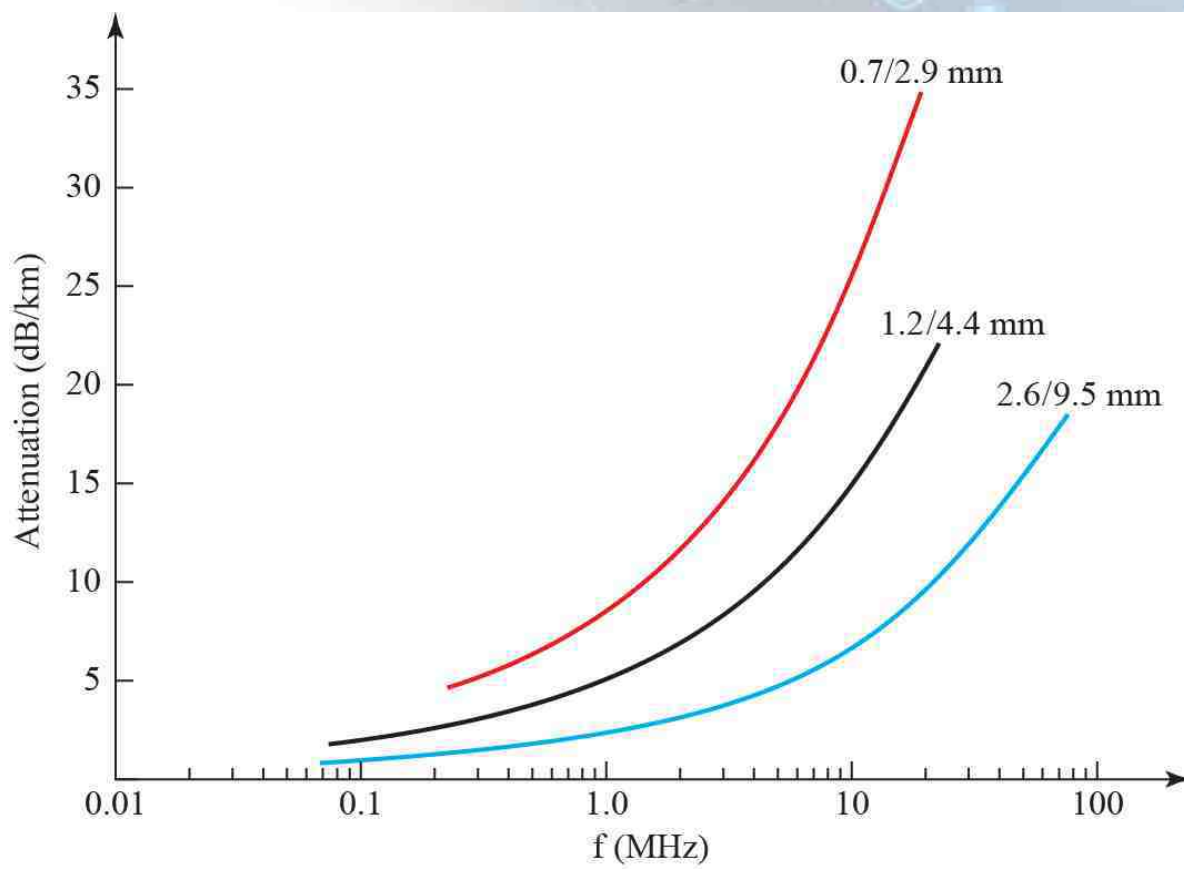
- Carries signals of higher frequency ranges than those in twisted pair cable



BNC Connectors



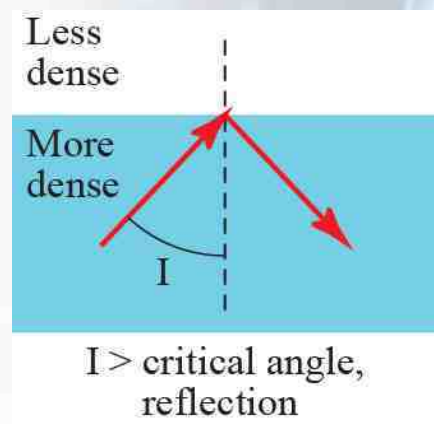
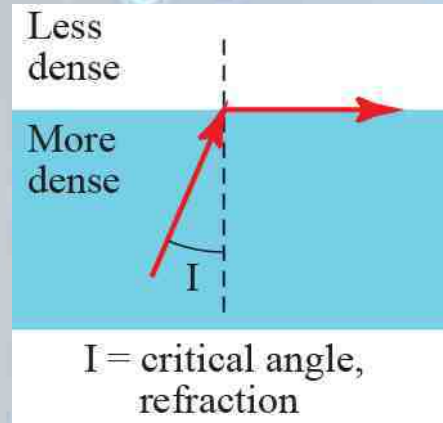
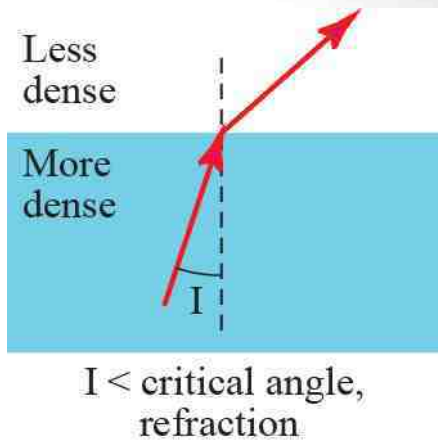
Coaxial Cable Performance



Fiber-Optic Cable

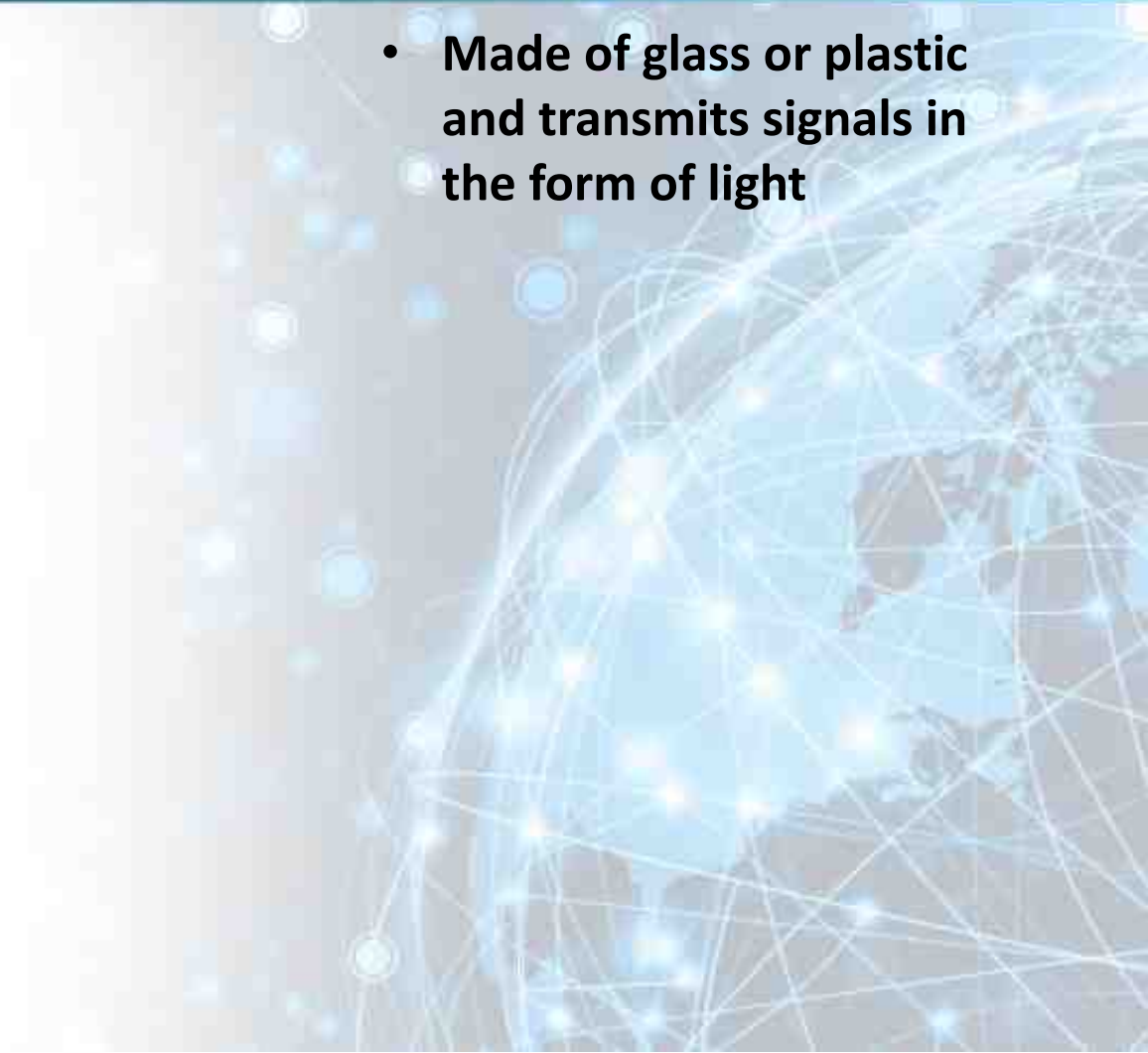
- Made of glass or plastic and transmits signals in the form of light
- Light travels in a straight line as long as it is moving through a single uniform substance
- If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction

Bending of Light Ray

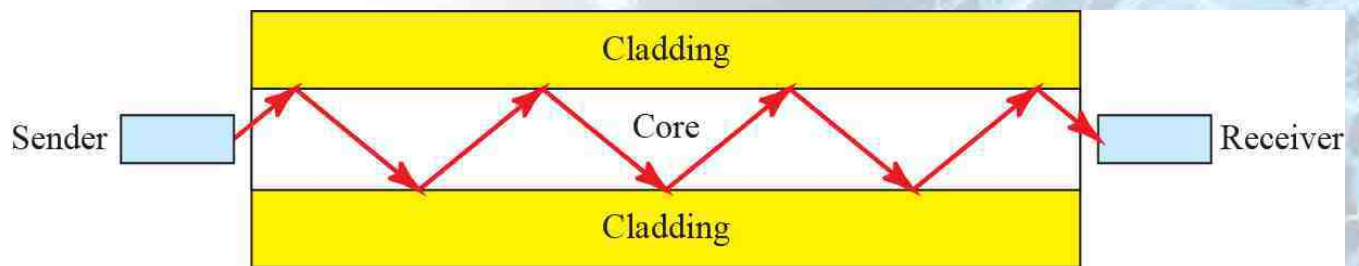


Fiber-Optic Cable

- **Made of glass or plastic and transmits signals in the form of light**

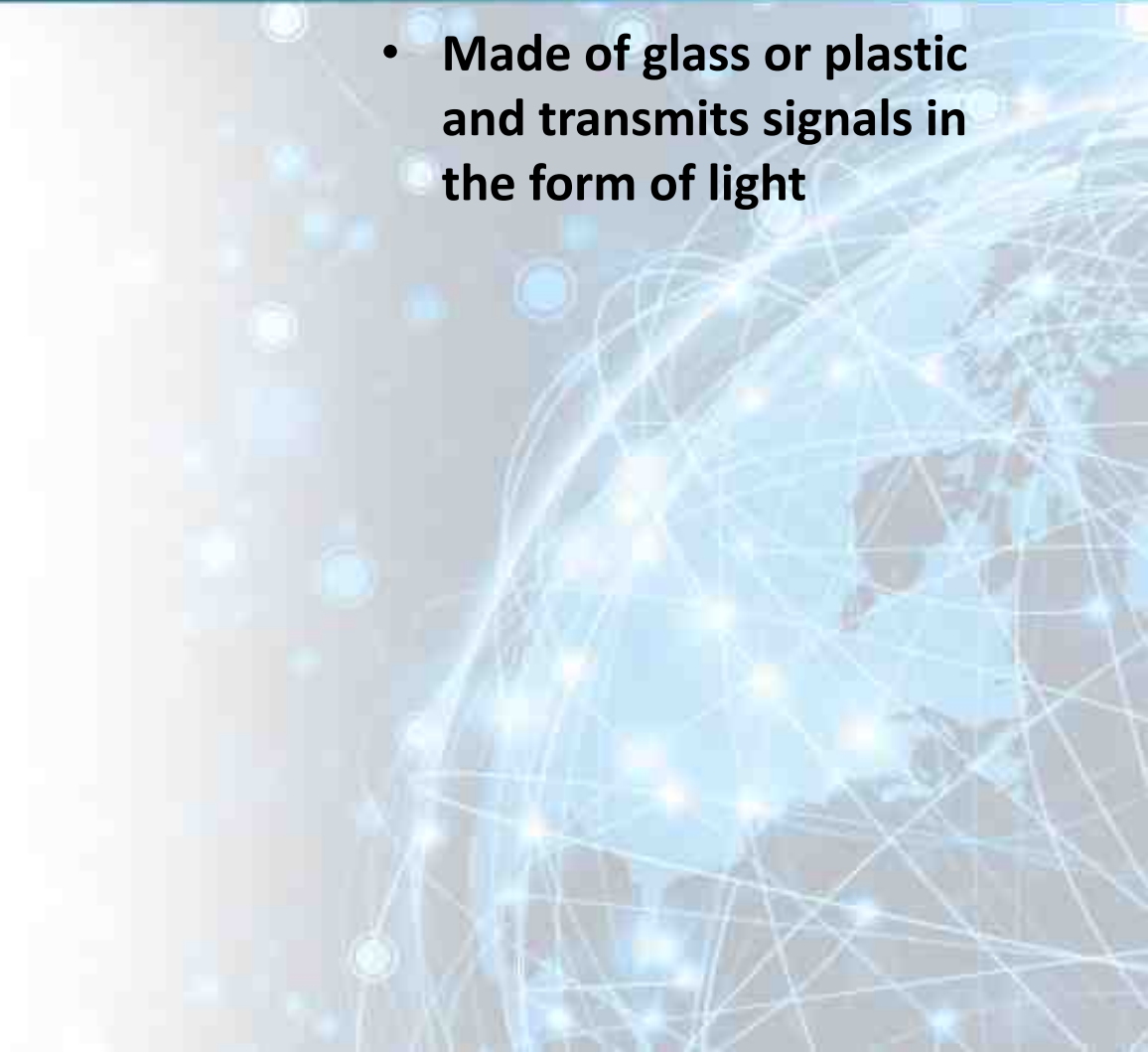


Optical Fiber

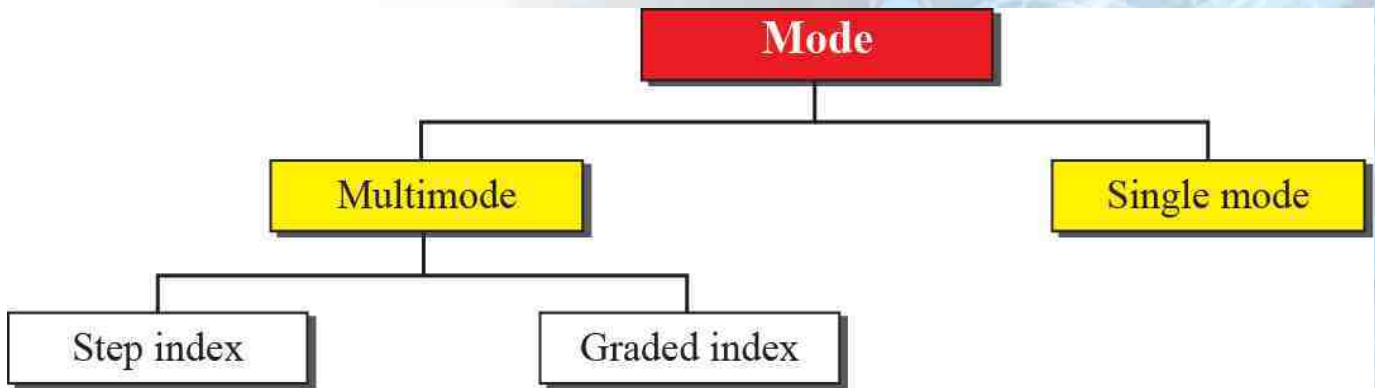


Fiber-Optic Cable

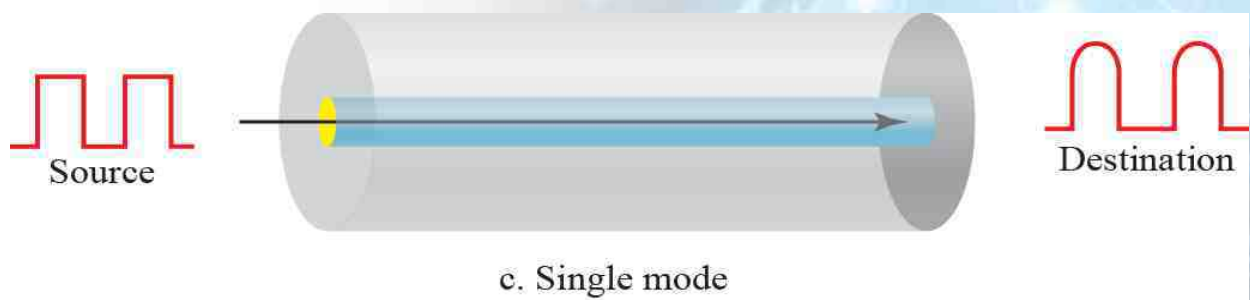
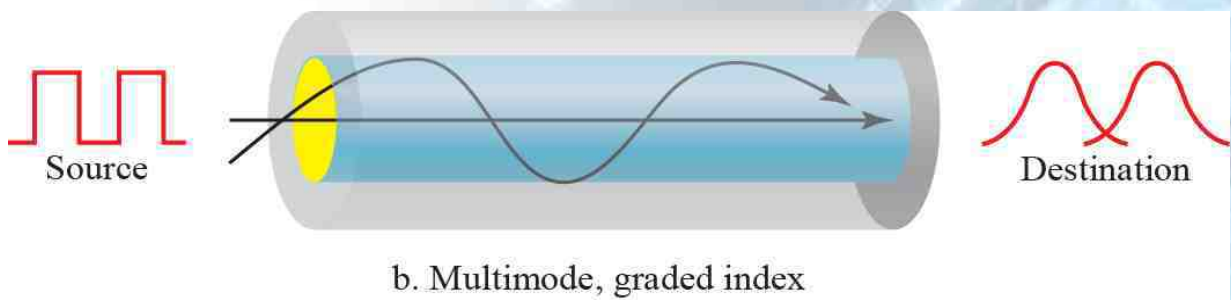
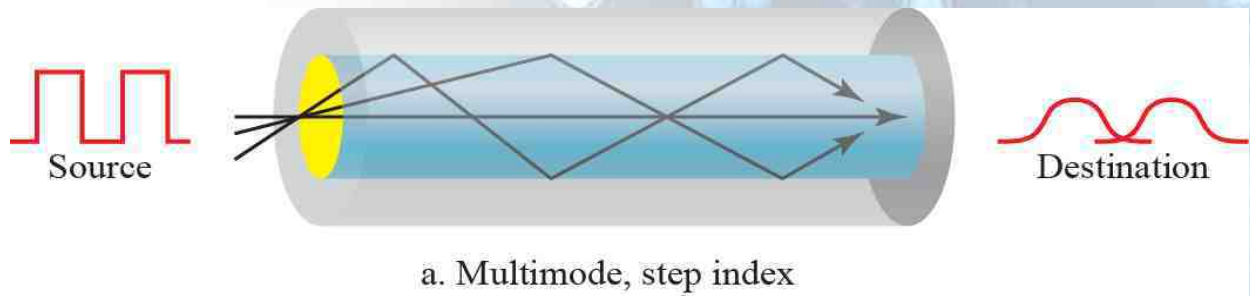
- **Made of glass or plastic and transmits signals in the form of light**



Propagation Modes

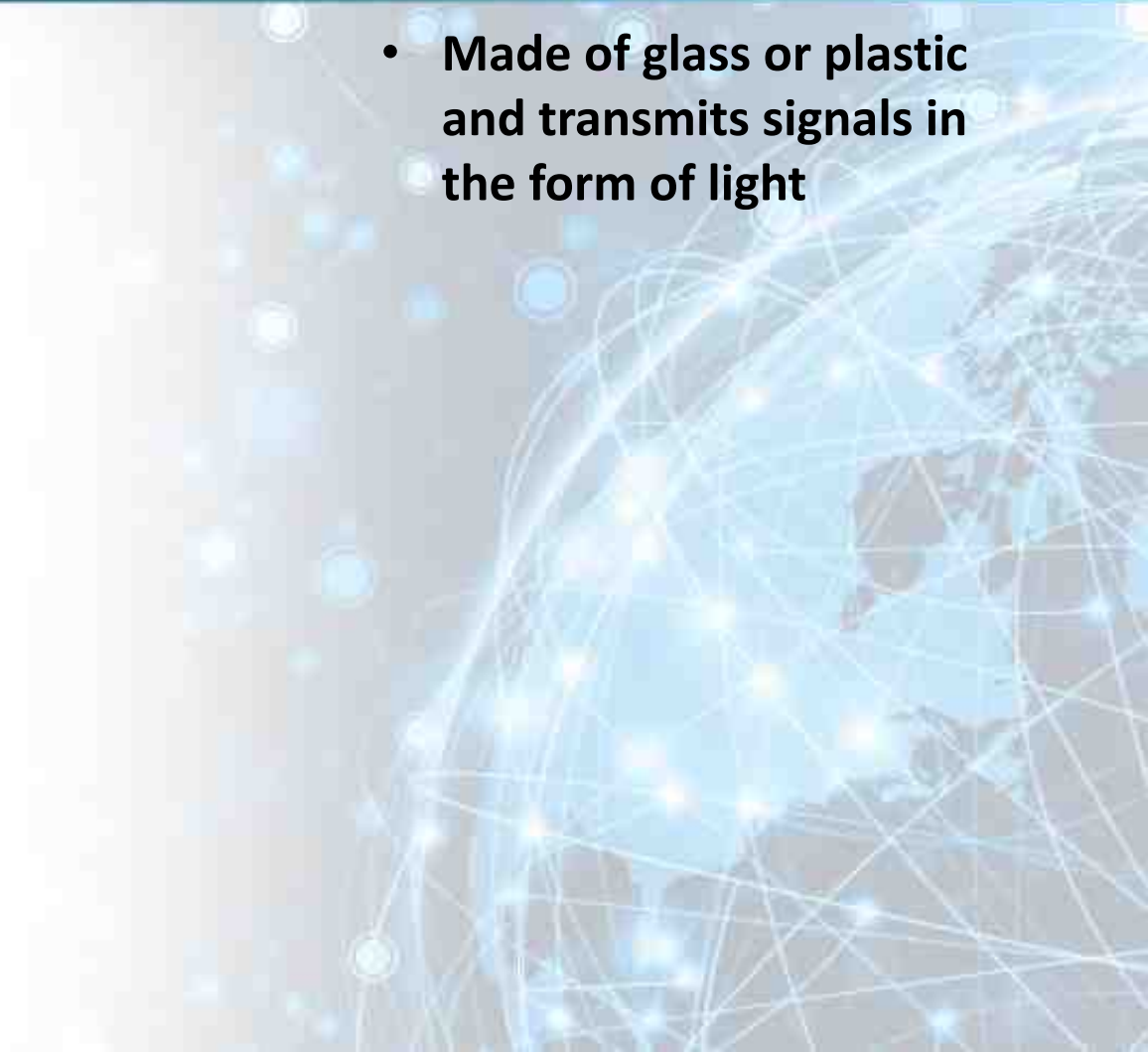


Modes



Fiber-Optic Cable

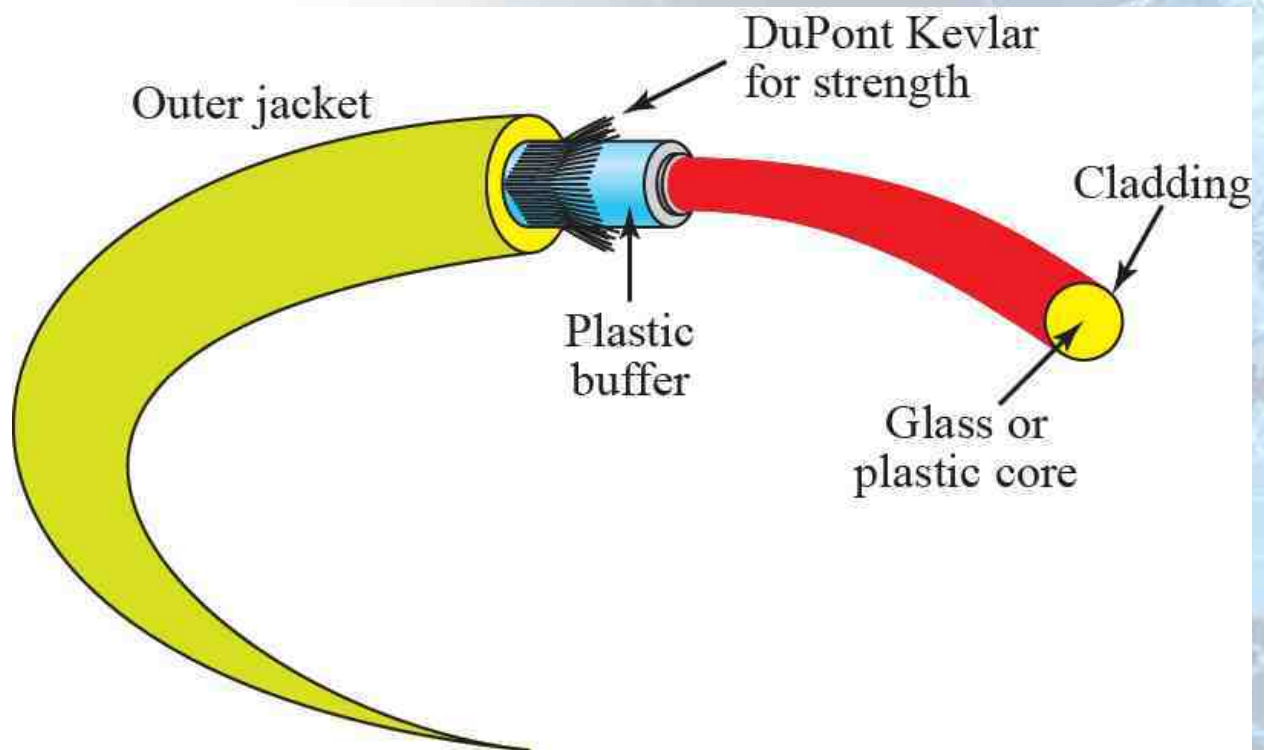
- **Made of glass or plastic and transmits signals in the form of light**



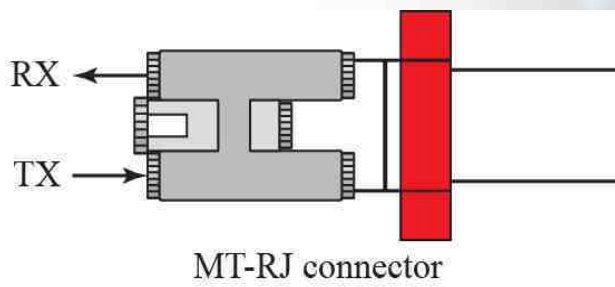
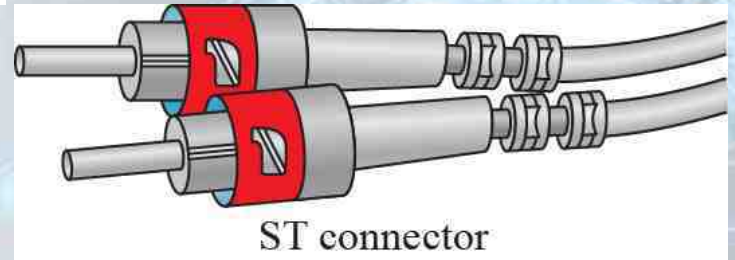
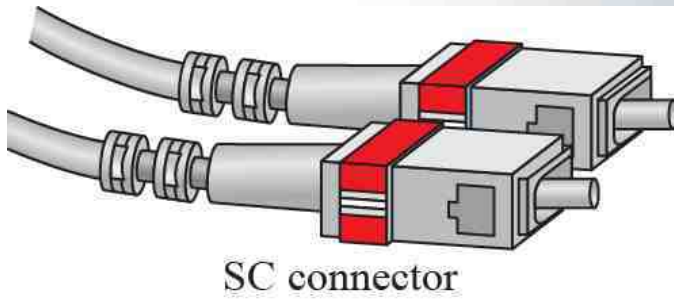
Fiber Types

<i>Type</i>	<i>Core (μm)</i>	<i>Cladding (μm)</i>	<i>Mode</i>
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

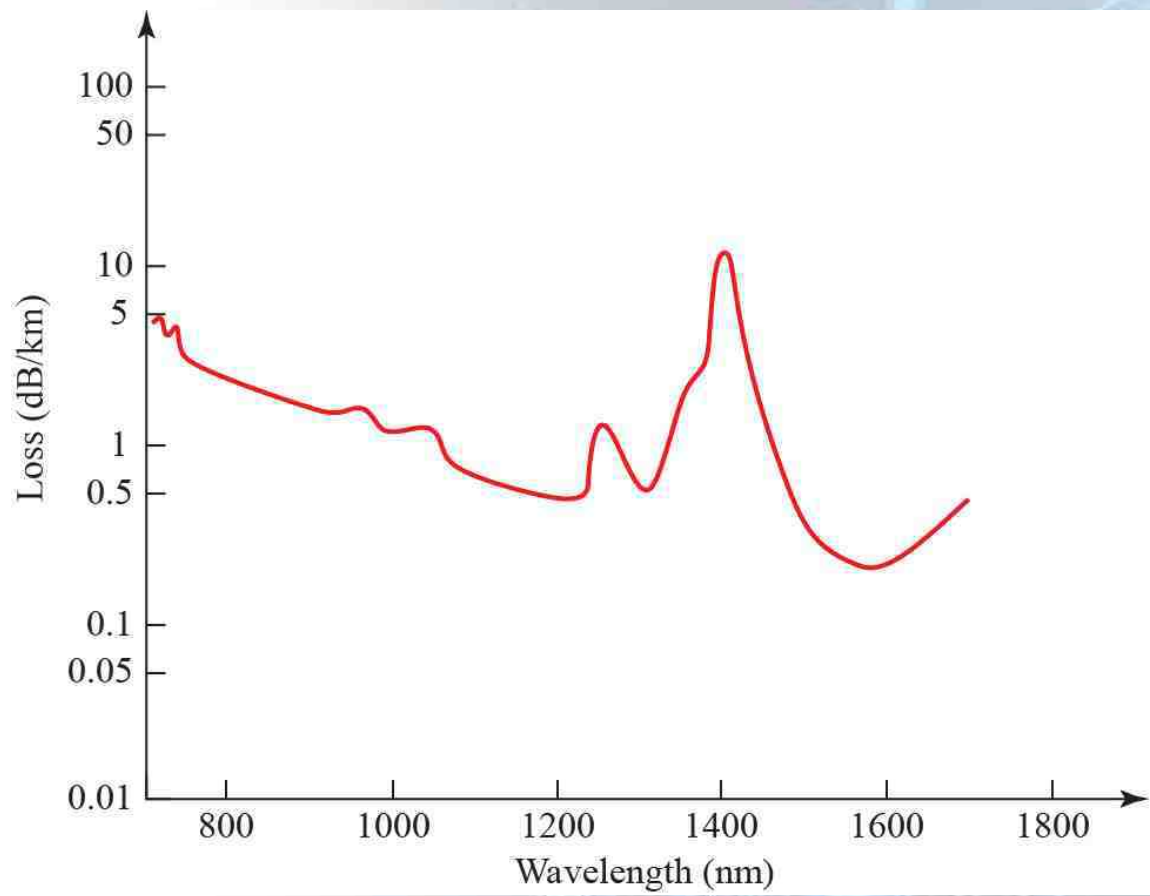
Fiber Composition



Fiber-Optic Cable Connector



Optical Fiber Performance



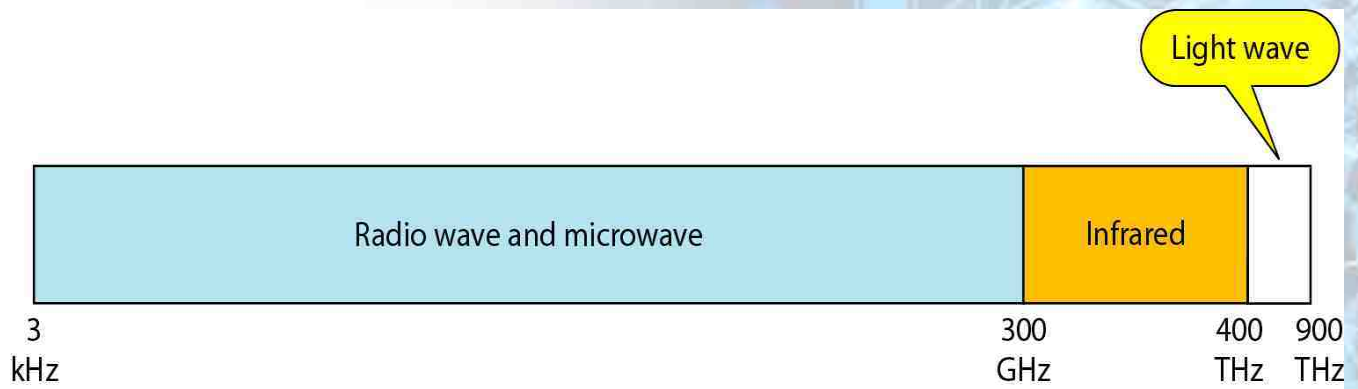
Advantages & Disadvantages

- Higher Bandwidth
- Less Attenuation
- Less EM Interference
- Light Weight
- Less corrosive than copper
- Installation/Maintenance
- Unidirectional
- Cost

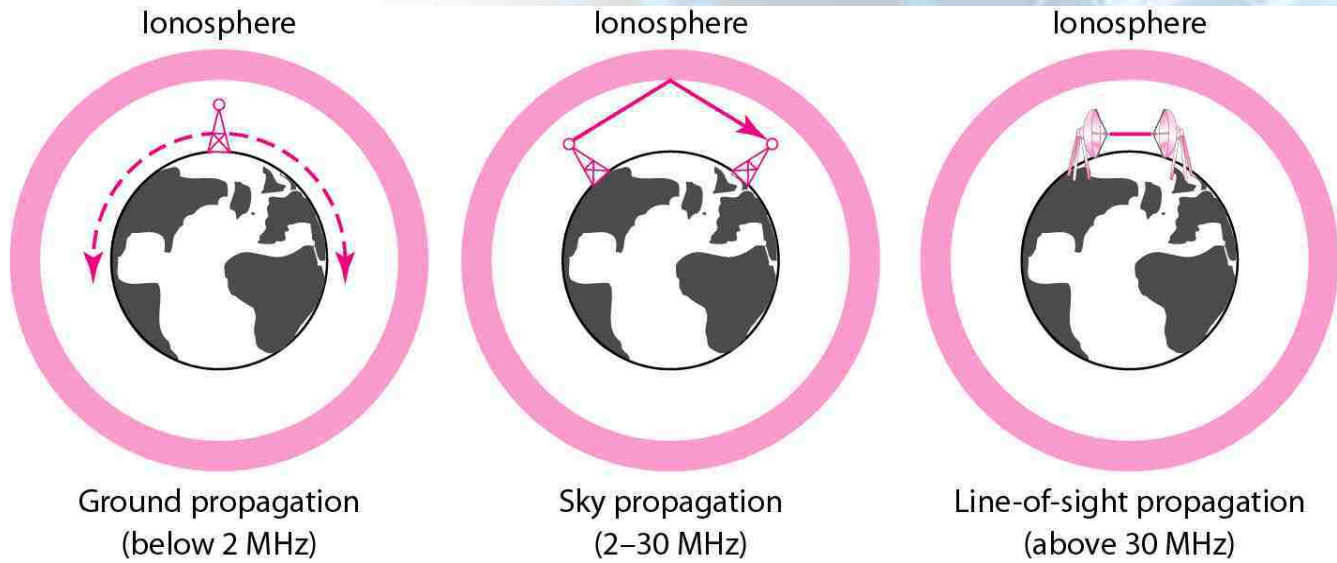
Unguided Media

- Unguided medium transport waves without using a physical conductor
- Often referred to wireless communication
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them

Electromagnetic Spectrum



Propagation Methods



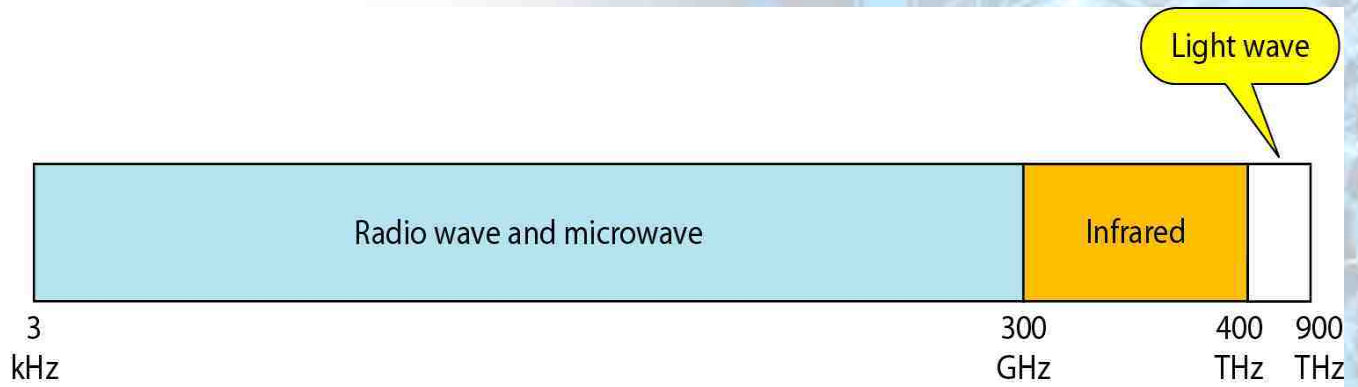
Bands

<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
very low frequency (VLF)	3–30 kHz	Ground	Long-range radio navigation
low frequency (LF)	30–300 kHz	Ground	Radio beacons and navigational locators
middle frequency (MF)	300 kHz–3 MHz	Sky	AM radio
high frequency (HF)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft
very high frequency (VHF)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
ultrahigh frequency (UHF)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
superhigh frequency (SHF)	3–30 GHz	Line-of-sight	Satellite
extremely high frequency (EHF)	30–300 GHz	Line-of-sight	Radar, satellite

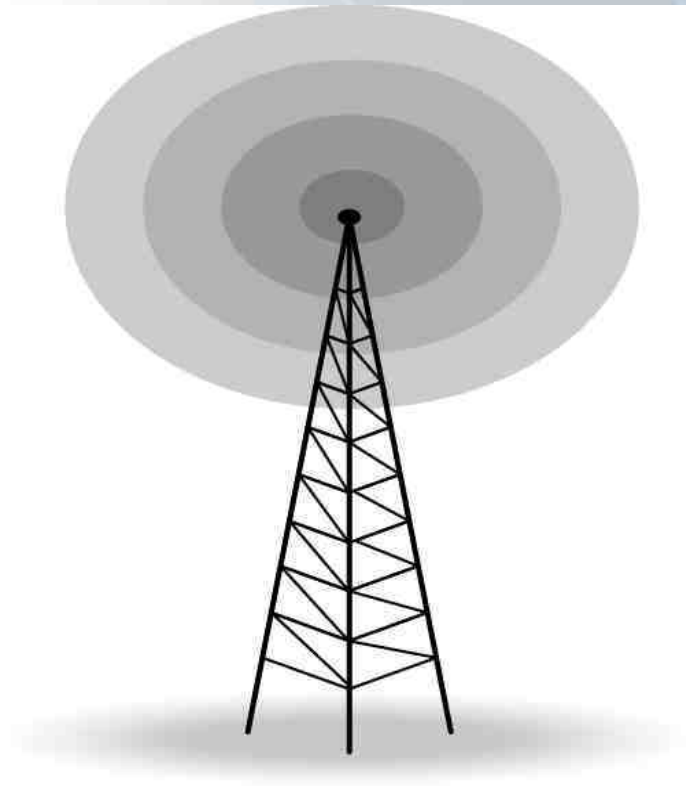
Radio Waves

- **Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves**
- **Electromagnetic waves ranging in frequencies between 1 and 300 GHz are called microwaves**

Electromagnetic Spectrum



Omnidirectional Antenna



Microwaves

- **Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves**
- **Microwaves are unidirectional**
- **When an antenna transmits microwaves, they can be narrowly focused**

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1)The original Ethernet technology with the data rate of 10Mbps is called_____?

a)Standard Ethernet

b)First Ethernet

C)Gigabit Ethernet

d)10 gigabit Ethernet

2)In pure ALOHA, vulnerable time is_____the frame transmission time.

a)same as

b)two times

c)three times

d)four times

3)Collisions in CSMA/CA are avoided through_____.

a)The interframe space

b)The contention window

c)Acknowledgement

d)All of the given

4)Which one of the following is a correct controlled access method?

a)Reservation

b)polling

c) token passing

d) all of the given

5)_____Works with topologies in which one device is designated as a primary station and other devices are secondary stations.

a)polling

b)token

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c) reservation

d) none of the given

6) In _____ Each band is reserved for a specific station, time and it belongs to the station all the time.

a) FDMA

b) TDMA

c) CDMA

d) CSMA

7) In _____ each station is forced to send a frame only at the beginning of time slot.

a) Pure Aloha

b) slotted Aloha

c) Fragmented Aloha

d) none of the given

8) A personal computer or workstation on an Ethernet network must have _____ card

a) TDI

b) NIC

c) PCI

d) None of the given

9) In the context of Ethernet implementations, maximum medium length of 10 Base F was _____ meters.

a) 500

b) 1000

c) 2000

d) 150

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10) What was the main drawback in Aloha Random-Access method?

- a) Redundancy in data
- b) inconsistent data rate
- c) potential collision**
- d) all of the given

11) MAC address is of _____.

- a) 24 bits
- b) 36 bits
- c) 42 bits
- d) 48 bits**

12) Collisions in CSMA/CA are avoided through _____.

- a) The interframe space
- b) Acknowledgements
- c) The contention window
- d) all of the given**

13) Posts stand for

- a) Plain old telephone set
- b) plain old telephone service**
- c) plain old time sharing
- d) Plain old telephone system

14) _____ Random access method was initially designed for a radio wireless LAN but it can be used on any shared medium now.

- a) FDMA
- b) CDMA/CD**

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c)CDMA/CA

d)ALOHA

15)In _____ each station sends a frame whenever it has a frame to send.

a)Pure ALOHA

b)slotted ALOHA

c)Fragmented ALOHA

d)none of the given

16)In CSMA/CA, _____ is the amount of time divided into slots.

a)Contention window

b) interframe window

c) fragmented window

d) collided window

17)Normally the value of K_{max} in pure ALOHA is _____ .

a)10

b)15

c)25

d)30

18)In which controlled-access method all data exchanges must be made through primary devices even when the ultimate destination is a secondary device?

a)Reservation

b)polling

c)Port forwarding

d)token passing

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19) A personal computer or workstation on an Ethernet network must have _____ card

a) TDI

b) NIC

c) PCI

d) none of the given

20) Which of the following is a Random-Access protocol?

a) CDMA/CA

b) ALOHA

c) CDMA/CD

d) TDMA

21) Transmission technology second (based on Ethernet) provides a data rate of 1 billion bits per second.

a) 10Base-T Ethernet

b) standard ethernet

c) Fast Ethernet

d) Gigabit Ethernet

22) In Ethernet the source address field in the MAC frame is the _____ address.

a) Sender's logical

b) Previous station's physical

c) Next destination's physical

d) sender's service port

23) _____ was used as the medium in 10 Base-T.

a) thick coaxial cable

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b) I thin coaxial cable

c) Twisted pair cable

d) two mode fiber cable

24) In telephone network POP stand for _____?

a) Pride Of Place

b) Point Of Presence

c) Post order protocol

d) Proof Of Purchase

25) MAC address is of

a) 24 bits

b) 36 bits

c) 42 bits

d) 48 bits

26) In Ethernet frame, both destination and sender addresses are of length _____.

a) 1 byte

b) 2 byte

c) 4 byte

d) 6 byte

27) When we represent the data in CDMA, if a statement is idle then it sends _____.

a) 0

b) 1

c) -1

d) infinity

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28) _____ use fiber optic cable.

a) 10Base 5

b) 10Base2

c) 10Base-T

d) 10Base-F

29) _____ Uses thick coaxial cable.

a) 10 Base2

b) 10Base5

c) 10Base-T

d) 10 Base-F

39) In _____ each station is allocated a time slot during which it can send data.

a) NDMA

b) FDMA

c) TDMA

d) CDMA

40) In _____ a station monitors the medium after it sends a frame to see if the transmission is successful.

a) CSMA/CD

b) CSMA/CA

c) CSMA

d) None of the given

41) Ethernet address is if _____ bytes.

a) 4

b) 6

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c)8

d)12

42)Fast Ethernet has data rate of _____Mbps.

a)10

b)100

c)1000

d)10000

43)In context of Ethernet address the source address is always a _____ address.

a)Unicast

b)Multicast

c)Broad-cast

d)128 bit address

44)In Hexadecimal notation of Ethernet address, each byte is separated by _____ symbol .

a)Question mark

b)Colon

c)Underscore

d)Hash

45)Normally the value of Kmax in pure ALOHA is _____

a)10

b)15

c)25

d)30

46)Possession of TOKEN gives the station the right to _____ .

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a)Send data

b)receive data

c)delete data

d)modify data

47)In _____ collision are avoided by differing transmission even if the channel is idle.

a)slotted frame

b)The deployment frame

c)The contention window

d)The interframe space

48)DSL stand for _____

a)digital subscribers line

b)digital switched line

c)data subscribers line

d) data switched line

49)_____ is the correct formula to calculate the total number of one system needed to connect in a mesh network.

a) $N(N+1)$

b) $N-1$

c) $N+1$

d) $N \times N$

50)In transmission impairment _____ noise occurs from the random motion of electrons in a wire.

a)Induced noise

b)Thermal noise

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c)Impulse noise

d)Cross talk

51)Identify the correct formula to calculate propagation time.

a)propagation time = distance/propagation speed

b)propagation time = Bandwidth/propagation speed

c)propagation time =delay /propagation speed

d)propagation time = throughput/propagation speed

50)parabolic antenna is used for _____ communication.

a)micro wave

b)light wave

c)infrared wave

d)radio wave

51)At the application layer the object/information is in the form _____.

a)message

b)frame

c)segment

d)packet

52)During transmission _____ is the effect of one wire on the other .

a)Induced noise

b)Thermal noise

c)Crosstalk

d)impulse noise

53)If a digital transmission system is sending five bits in every half a second the bit rate of that system will be _____

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- a)10Hz
- b)0.2bps
- c)5bps
- d)10bps**

54)Two stations are connected via secure link in which messages are encrypted from source to the destination .This encrypted message is called _____.

- a)message
- b)plain text
- c)protocol
- d)cipher text**

55)QAM stand for _____.

- a)Quadrature amplitude modulation**
- b) Quadrature amplitude modifier
- c) Quadrature amplitude modulator
- d) Quadrature amplifier modulation

56)In _____ topology each computer is attached with other through a central device

- a)Star**
- b)mesh
- c)bus
- d)Ring

57)When two peoples are talking with each other in a room then the transmission medium between them is _____.

- a)No medium required
- b)Air**

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c)wireless medium

d)cable

58) In baseband transmission, a digital signal is transmitted as _____

a)digital signal

b)multiplexed signal

c)analog signal

d)modulate signal

59) In optical fiber a glass or a plastic core is surrounded by a _____.

a)insulator

b)outer conductor shield

c)Cladding

d)inner conductor shield

60) _____ signal completes certain pattern in specific amount of time .

a)APeriodic

b)periodic

c)non periodic

d) wavelength

61) The address of transport layer are in the form of _____.

a)Logical address

b)name

c)port numbers

d)link layer address

62) Which digital multiplexing technology is use to allow several connections to share the high bandwidth of a link?

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a)FDM

b)ADM

c)TDM

d)WDM

63)In bidirectional communication each layer perform two _____ tasks in each direction.

a)opposite

b)similar

c)identical

d)complimentary

64)AMI is a popular _____ encoding method.

a)Polar

b)Bi-polar

c)Unipolar

d)substitution

65)In pulse code modulation the term sampling can also be referred as _____

a)Pulse amplitude modulation

b)pulse line modulation

c)pulse frequency modulation

d) pulse sampling modulation

66)Time taken by a periodic signal to complete one cycle is called _____.

a)period

b)routing

c)multiplexing

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d)wavelength

67)_____ does not happen until all the data transferred and all the resources are freed.

a)Tear down

b)acknowledgment

c)switching

d)resource allocation

68)_____ is the example of multipoint topology.

a)Mesh

b)Ring

c)Bus

d)Hub

69)A local telephone network use _____ network .

a)Line switched

b)bit switched

c)circuit switched

d)packet switched

70) _____ is the process of converting binary data to a digital signal.

a)ASK

b)QAM

c)Line coding

d)FSK

71)_____ category of coaxial cable is used for thick Ethernet.

a)RG-11

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b)RG-58

c)RG-47

d)RG-58

72)_____ has less magnetic interface as compared to other guided medium.

a)Unshielded twisted pair

b)Shielded twisted pair

c)Fiber optic

d)Coaxial

73)_____ relatively measure the strength of two signals.

a)decibel

b)signal rate

c)pulse rate

d)bit rate

78)_____ layer is responsible for creating datagrams .

a)physical

b)session

c)network

d)data link

79)Two PCs and one printer are connected in a network within a room, it is an example of _____

a)LAN

b)MAN

c)WAN

d)SONET

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80) In _____ encoding scheme we use three levels positive, zero and negative.

- a) polar
- b) nonpolar
- c) bipolar**
- d) multilevel

81) Which one of the following is not a bipolar encoding level?

- a) Negative
- b) one**
- c) positive
- d) zero

82) According to stats fewer than _____ crossbar point can be used at once.

- a) 14%
- b) 100%
- c) 30%
- d) 25%**

83) TCP/IP protocol suite consists of _____ layers.

- a) 3
- b) 5**
- c) 6
- d) 7

84) In single crossbar switch only _____ row(s) or column(s) is/are active at a time while others are not active.

- a) four
- b) two

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c)one

d)three

85) We can have combination of _____ codes with a 5 bit code.

a)4

b)16

c)32

d)8

86) Bi polar uses _____ voltage level.

a)one

b)two

c)three

d)four

87) In statistical time division multiplexing, number of slots in frame are less than _____

a)output data rate

b)input lines

c)input frequency

d)multiplexed T lines

88) The communication at the Application, Transport and network layer is _____.

a)logical

b)physical

c)real

d)rational

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89) If a digital signal has 'L' number of levels _____ is the number of bits required to represent each level.

a) $\log L$

b) **Log Base 2 L**

c) L^2

d) $\log (2 \cdot L)$

90) Coaxial cables are categorized by their _____ ratings.

a) EIA

b) FM

c) **RG**

d) AM

91) In frequency domain plot, which value is plotted on Y axis.

a) time

b) frequency

c) **amplitude**

d) wavelength

92) A fiber optic cable transmits signal in the form of _____.

a) light

b) sound

c) wave

d) none of the given

93) In frequency modulation the frequency of the oscillator changes according to the _____.

a) output voltage

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b)output frequency

c)input voltage

d)output phase

94)Routing is a function of _____ layer.

a)network

b)physical

c)transport

d)datalink

95)BFSK stand for _____.

a)Baud frequency shift key

b)Binary frequency shift key

c)Barrier frequency shift key

d)Bridge frequency shift key

96)There are _____ basic categories of multiplexing .

a)2

b)3

c)5

d)7

97)Most commonly used connector for twisted pair cable is _____.

a)RJ-11

b)RJ-45

c)RJ-54

d)RJ-44

98)_____ technique does not follow analog to analog conversion .

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a)QAM

b)AM

c)FM

d)PM

99)In TDM based digital hierarchy used by the telephone companies, DS-0 is a single digital channel of _____.

a)1,544Mbps

b)128 kbps

c)64 kbps

d)32 kbps

100)A telephone line analog signal has got the bandwidth of _____.

a)8kHz

b)2kHz

c)16kHz

d)4kHz

101)In _____ TDM slots are dynamically allocated to the connected stations to improve bandwidth efficiency.

a)synchronous

b) statistical

c)Isochronous

d)none of the given

102)_____ Is composed of thousands of interconnected network expanded over large geographical areas.

a)LAN

b)MAN

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c)the internet

d)home network

103) We quantize the sampling out put into certain levels based on range of _____ and required accuracy.

a)frequency

b)amplitude

c)time period

d) none

104) In serial data transmission _____ data transmission mode can be cheap but slower.

a)synchronous

b)Asynchronous

c)Isochronous

d)metaschrounous

105) In pulse code modulation technique technique, which approach will be the most appropriate for sampling _____

a)over sampling

b)under sampling

c)Nyquist sampling

d)Analog sampling

106) _____ signal is represented by the discrete values.

a)analog

b)digital

c)both

d)continuous

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107) Which of the following techniques is used for analog to digital conversion?

a) PCM

b) DM

c) both PCM and DM

d) BDM

108) _____ category of coaxial cable is used for thin Ethernet .

a) RG-58

b) RG-59

c) RG-40

d) RG-1

109) Unipolar encoding scheme uses _____ voltage level.

a) one

b) two

c) three

d) four

110) In _____ scheme all the signal levels are on the side of time axis, either above or below .

a) polar

b) bipolar

c) unipolar

d) multiplexed coding

111) _____ mode of serial transmission guarantees fixed rate data.

a) synchronous

b) Asynchronous

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c)Isochronous

d)Metaschrounous

112)electromagnetic waves ranging from 0z and 300GHz are called _____.

a)microwave

b)light wave

c)Infrared wave

d)radio wave

113)_____ is used where data speed is priority in data transfer.

a)serial data transmission

b)hybrid data transmission

c)parallel data transmission

d)both parallel and serial transmission

114)In transmission impairment _____ occur from a signal with a high energy and with a short period of time.

a)impulse noise

b)thermal noise

c)induced noise

d)cross talk

115)In star based network comprising of four computer and one switch total number of cables needed will be _____.

a)4

b)5

c)6

d)3

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116) The most upper layer of TCP/IP protocol suite is the _____.

a) application

b) transport

c) network

d) data link

117) In analogue hierarchy to carry voice channels, a group can carry _____ voice channels.

a) 60

b) 12

c) 20

d) 10

118) Bandwidth can be measured in _____.

a) Bps

b) phase

c) lambda

d) decibel

119) Unit of phase is _____.

a) bits per second

b) watts

c) baud

d) degree

120) _____ is sometimes called the bit rate.

a) signal rate

b) modulation rate



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c)data rate

d)pulse rate

121) In _____ data moves faster and timing errors are less frequent because the transmitter and receiver's time is synced.

a) Isochronous

b)synchronous

c) Asynchronous

d) Metasynchronous

122) Manchester encoding scheme achieves the same level of synchronization as _____.

a) RZ

b) differential Manchester

c) NRZ-1

d) NRZ

123) _____ is not the advantage of protocol layering.

a) Modularity

b) reduced complexity

c) reduced cost

d) none

124) In TCP/IP model physical layer exchange data in the form of _____.

a) Packet

b) frame

c) bits

d) segments

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125) In amplitude shift keying _____ remains constant .

- a) Phase
- b) frequency and amplitude
- c) frequency and phase
- d) amplitude

126) Multimode technology has _____ main type of light propagating .

- a) two
- b) three
- c) four
- d) five

127) Bandwidth in hertz is the range of _____ contained in a composite signal.

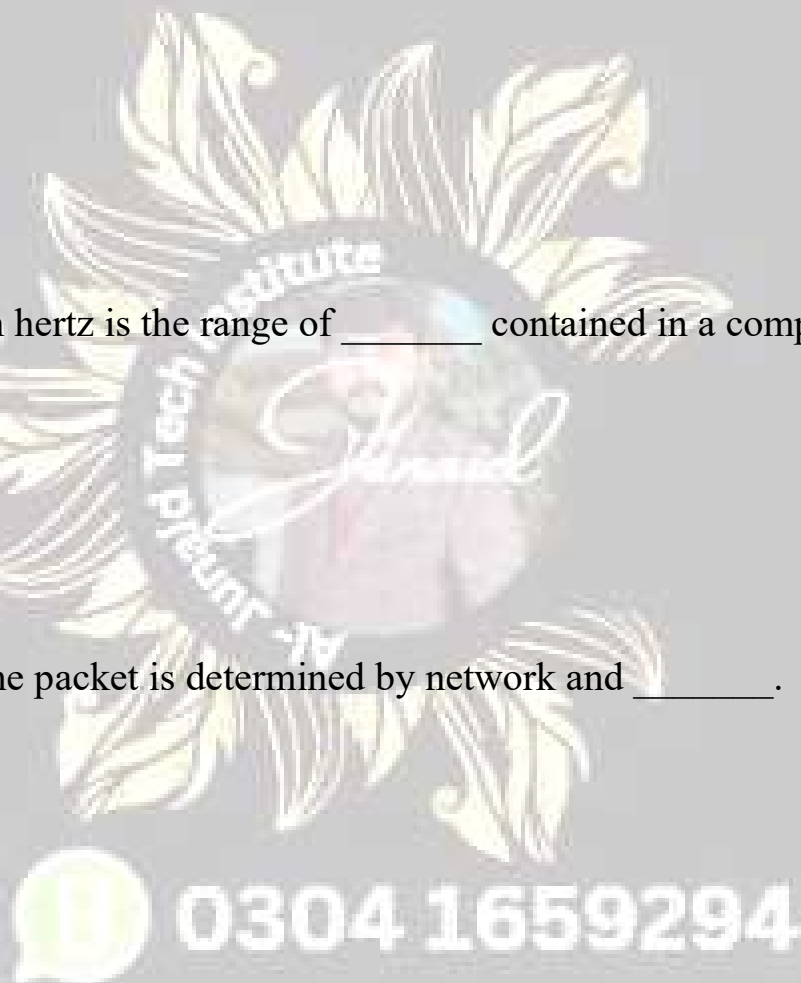
- a) signal
- b) frequencies
- c) amplitude
- d) phase

128) The size of the packet is determined by network and _____.

- a) govern time
- b) instructions
- c) switching time
- d) delay

129) _____ means loss of energy in signals.

- a) attenuation
- b) noise
- c) delay



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d)distortion

130)In circuit switching total delay is combination of connection time, data transfer time and _____.

a)connection tear down time

b)switching time

c)receiving time

d)acknowledgment time

131)In circuit switched network we have low efficiency and minimal _____.

a)speed

b)delay

c)throughout

d)errors

132)There is layer to layer connection between every peer station, this imaginary connection between each layer is termed as _____.

a)Logical connection

b)Physical connection

c)Practical connection

d)None of the given

133)In virtual circuit approach when a frame enters a switch and when it leaves the circuit then its VCI _____.

a)always changes

b)remain same

c)sometimes changes

d)remain the same

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134) When data is received or sent using _____ the data bits are organized in a specific order, since only they can be sent or received one after another .

a) parallel data transmission

b) serial data transmission

c) hybrid data transmission

d) both parallel and serial transmission

135) Cellular telephone uses _____ waves for communication .

a) light wave

b) infrared waves

c) microwaves

d) Radio waves

136) According to the Fourier analysis frequencies obtained after decomposition of digital signals are _____.

a) continuous

b) discrete

c) homogeneous

d) bit length

137) _____ is a multiplexing technique which shift each signal to a different carrier frequency.

a) FDM

b) TDM

c) WDM

d) PDM

138) Packet switching operates on _____ layer.

a) network layer

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b)physical layer

c)data link layer

d)application layer

139)When the bandwidth of a link is greater than the combined bandwidth of a signal we use _____ multiplexing .

a)wavelength division

b)time division

c)frequency division

d)hybrid division

140)If 32 bits are sent in two seconds then the bitrate for that signal is _____.

a)8bps

b)16bps

c)32bps

d)64bps

141)_____ signal can take infinite level of intensity over time .

a)digital

b)discrete

c)analog

d)logical

142)Electromagnetic wave ranging in 3khz and 1GHz are called _____.

a)infrared waves

b)light wave

c) radio wave

d)microwave

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143) Microwaves are _____.

- a) unidirectional
- b) bi directional
- c) Omni directional
- d) all directional

144) A fiber optic cable transmits signals in the form of _____.

- a) light**
- b) sound
- c) wave
- d) none of the given

145) Simultaneous transmission of multiple signals across a single data link is called _____.

- a) Demultiplexing
- b) multiplexing**
- c) modulation
- d) demodulation

146) Multimode technology has _____ main type of light propagating .

- a) two
- b) three**
- c) four
- d) five

147) Set of rules to be followed for effective communication is called _____.

- a) Topology
- b) protocol**

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c)switching

d)Encryption

148) Which multiplexing technique involves signals composed of light beam?

a)FDM

b)TDM

c)WDM

d)PDM

149) Time division multiplexing technique is used in _____ system.

a)analog

b)hybrid

c)digital

d)automated

150) Which of the following technique is used for analog to digital conversion .

a)PCM

b)DM

c)BDM

d)both PC and DM



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HERE IS STARTING GRAND QUIZ FILE

Which one of following is a scrambling coding technique?

- ☒ **B8ZS**
- ☐ B8SZ
- ☐ B08SZ
- ☐ BZ8S

According to the Fourier analysis, frequencies obtained after decomposition of digital signals are_____.

- ☐ Continuous
- ☒ **Discrete**
- ☐ Bit length
- ☐ Homogeneous

Low pass channel has_____bandwidth between two stations.

- ☐ Dedicated
- ☒ **Shared**
- ☐ Multiplexed
- ☐ Infinite

_____is the heart of blocking code.

- ☒ **Coding**
- ☐ Division
- ☐ Substitution
- ☐ Multiplication

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Wireless transmission can be divided into _____ broad groups.

- ☐ Two
- ☒ **Three**
- ☐ Four
- ☐ Five

Wiring used for transmission modes depends upon _____.



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- ☐ **Latency**
- ☐ Data Stream
- ☐ Data sniffing
- ☐ Packet Tracing

If a digital transmission system is sending five bits in every half a second, the bit-rate of the system.

5 bps

- ☐ 10 Hz
- ☐ 0.2 bps
- ☐ 10 bps

If the message going to pass through a packet-switched network, it can be divided into packets of_____.

- ☐ Fixed size
- ☐ Variable size
- ☐ **Both fix and Variable size**
- ☐ None

In an analogue hierarchy to carry voice channels, a group can carry____voice channels.

- ☐ 60
- ☐ **12**
- ☐ 20
- ☐ 10

1. In optical fiber a glass or plastic core is surrounded by a_____.

- ☐ **Cladding**
- ☐ Outer conductor shield
- ☐ Inner Conductor shield
- ☐ Insulator

In transmission impairments_____noise occurs from other sourcelike

- ☐ Impulse
- ☐ **Induced**
- ☐ Cross talk
- ☐ Thermal



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_____ is a digital process that allows connections to share the high bandwidth of a link.

- ☐ FDM
- ☒ **TDM**
- ☐ WDM
- ☐ None of the choices is correct

Bandwidth in hertz in the range of _____ contained in a composite signal.

- ☐ Signals
- ☐ Phases
- ☐ Amplitudes
- ☒ **Frequencies**

LAN stands for _____.

- ☐ Logical area network
- ☐ Long-distance Area Network
- ☒ **Local Area Network**
- ☐ Long Area Network

Demultiplexer is a _____ device.

- ☒ **One to many**
- ☐ Many to many
- ☐ One to one
- ☐ Many to many

Mr. Asif while sitting in Lahore is linking with his friend in Dubai through Skype is a _____.

- ☐ Local Area Network
- ☐ Metropolitan Area Network
- ☒ **Wide Area Network**
- ☐ Home Based Network

In _____ modulation, frequency and amplitude remain constant.

- ☐ Phase
- ☐ Frequency
- ☒ **Amplitude**
- ☐ Quadrature

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In TDM based digital hierarchy used by the Telephone companies, DS-0 is a single digital channel of ____.

- ☐ 1.544 Mbps
- ☐ 128 kbps
- ☒ 64 kbps
- ☐ 32 kbps

The bandwidth-delay product defines the number of bits that can fill the ____.

Link

- ☐ Network
- ☐ Hub
- ☐ Switch

If a digital-signal has four levels, then we need ____ bits to represent each level.

☒ 2

- ☐ 3
- ☐ 4
- ☐ 5

Routing is a function of ____ layer.

Network

- ☐ Physical
- ☐ Transport
- ☐ Datalink

____ relatively measure the strength of two signals.

- ☐ Signal rate
- ☐ Bit rate
- ☒ Decibel
- ☐ Pulse rate

Taking off the header from the message is called ____.

- ☐ Layering

Encryption

- ☐ Decapsulation
- ☐ Stuffing

Which of the following is not a characteristic of a sine wave?

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- o Amplitude

Segmentation

- o Phase
- o Frequency

In frequency domain plot, which value is replaced with frequency?

Wavelength

- o Amplitude
- o Phase
- o Time

_____ is the process of converting binary data to a digital signal.

- o QAM
- o ASK
- o FSK
- o **Line coding**

Low pass channel with _____ bandwidth is not real and is used for theoretical modeling.

- o Multiplex
- o Infinite
- o Ethernet
- o **Narrow**

HDB3 falls under which coding scheme?

Bipolar

- o Bi phase
- o Line
- o Block

Unit of phase is_____.

- o Bits per second
- o Watts
- o Bauds
- o **Degree**

BFSK stands for_____.

- o Baud Frequency Shift Key

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☐ **Binary Frequency Shift Key**

☐ Barrier Frequency Shift Key

☐ Bridge Frequency Shift Key

_____ is NOT an example of connecting device.

☐ Bridge

☐ Router

☐ **TCP**

☐ Switch

Coaxial cables are categorized by their _____ ratings.

☐ Electronic Industries Association (EIA)

☐ Amplitude modulation (AM)

☐ Frequency modulation (FM)

☐ **Radio Government (RG)**

_____ is Shannon capacity formula to determine highest theoretically data rate for a noisy channel.

☐ $\text{Capacity} = \text{Bandwidth} * (1 + \text{SNR})$

☐ $\text{Capacity} = \text{Bandwidth} * \log_2(2 + \text{SNR})$

☐ $\text{Capacity} = \text{Bandwidth} * \log_2(\text{SNR})$

$\text{Capacity} = \text{Bandwidth} * \log_2(1 + \text{SNR})$

Which one of the following is not among the required phases for the actual communication in a circuit switched network?

☐ Setup

☐ Data transfer

Protocol

☐ Connection teardown

A common bipolar encoding scheme is called _____.

AMI

☐ NRZ

☐ RZ

☐ QAM

A telephone line analog signal has got the bandwidth of _____.

☐ 8 kHz

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- ☐ **2 kHz**
- ☐ 16 kHz
- ☐ 4 kHz

In _____ systems, resources are allocated on demand.

- ☐ Frequency switching
- ☐ Line switching
- ☐ **Circuit switching**
- ☐ Packet switching

A fiber-optic cable transmit signals in the form of _____.

- ☐ **Light**
- ☐ Sound
- ☐ Waves
- ☐ None of the given

Composite wave can be used to send _____.

- ☐ Bandwidth
- ☐ Telephone
- ☐ Amplitude
- ☐ **Frequency**

Routing is the function of _____ layer.

- ☐ **Network**
- ☐ Physical
- ☐ Transport
- ☐ Data link

A virtual-circuit network operates on _____ layer.

Datalink

- ☐ Network
- ☐ Physical
- ☐ Application

In TCP/IP model, the Physical Layer exchanges data in the form of _____.

- ☐ Packets
- ☐ Frames
- ☐ **Bits**

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- o Segments

_____ signal completes certain pattern in a specific amount of time.

- o Non-periodic
- o **Periodic**
- o A Periodic
- o Wavelength

Narrow bands of lights in wave division multiplexing are denoted by_____.

- o β
- o $\cap$
- o α
- o **_____**

If a digital signal has four levels, the we need_____ bits to represent.

- o **2**
- o 5
- o 4
- o 5

We can have combination of_____ possible codes with a 5-bit code.

- o 4
- o 8
- o 18
- o **32**

There are_____ categories of multiplexing.

- o 2
- o **3**
- o 7

Top⁵ most organization that provides support for the Internet Standard process is called_____.

- o Internet Society (ISOC)
- o Internet Architecture Board (IAB)

IETF

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- o IRTF

If the message is going to pass through a packet-switched network, it can be divided into packets of_____.

- o Fixed size
- o Variable size
- o **Both-fix and variable size**
- o None

_____data moves faster and timing errors are less frequent because the transmitter and receiver time is synced.

- o Synchronous
- o Asynchronous
- o **Isochronous**
- o Metachronous

In frequency modulation, the frequency of the oscillator changes according to the_____.

- o Output Voltage
- o Output Frequency
- o **Input Voltage**
- o Output Phases

Binary Amplitude Shift Keying also called as:

- o Dual Keying
- o On-In Keying
- o **On-Off Keying**
- o In-Out Keying

_____signals can take infinite levels of intensity over time.

- o Digital
- o Discrete
- o **Analog**
- o Logical

Bi-polar uses_____voltage levels.

- o One
- o **Two**

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- ☐ Three
- ☐ Four

In serial data transmission _____ data transmission mode can be cheap but slower.

- ☐ Synchronous
- ☒ **Asynchronous**
- ☐ Isochronous
- ☐ Metachronous

In star-based network comprising of four computers and one switch, total number of cable needed will be _____.

- ☒ **4**
- ☐ 5
- ☐ 6
- ☐ 3

Parabolic antenna is used for _____ communication.

- ☐ Infrared waves
- ☒ **Microwave**
- ☐ Light wave
- ☐ Radio wave

In circuit switching total Delay is combination of Connection Time and _____.

- ☐ Switching time
- ☐ Acknowledgement time
- ☐ Receiving time
- ☒ **Connection tear down time**

Guard bands are used in _____ multiplexing technique, to avoid overlapping of frequency bands assigned to each user.

- ☐ PDM
- ☐ CSMA
- ☐ TDM
- ☒ **FDM**

Datagram switching is done on _____.

- ☐ Application Layer
- ☐ Datalink Layer

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- o Physical Layer
- o **Network Layer**

Token Ring was devised by_____.

IBM

- o OSI
- o Dell
- o NASA

In TDM based digital hierarchy used by the Telephone companies, DS-0 is a digital channel of_____.

- o **1.544 Mbps**
- o 128 kbps
- o 64 kbps
- o 32 kbps Repeated

_____ can be published using Request for Comments (RFCs).

- o Proposed Standard
- o Draft Standard
- o **Internet Draft**
- o Internet Standard

In transmission impairments _____ noise occurs from the random motion of electrons in a wire.

- o **Thermal noise**
- o Impulse noise
- o Cross talk
- o Induced noise

In ASK correct formula for calculating the bandwidth is as _____-.

- o **$B = (1 + d)S$**
- o $B = (d - 1)S$
- o $B = (d - 5)S$

Which one of the following is among the required phases for the actual communication in a circuit switched network?

- o Setup
- o **Data transfer**

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- o Protocol
- o Connection tear down

According to the Fourier analysis, frequencies obtained after decomposition of non-periodic digital signals are_____.

- o **Bit length**

- o Automatic
- o Heterogenous
- o Continuous

The most common type of connector used by the coaxial cable is_____.

- o **BNC**

- o RJ-45
- o RJ-11
- o RJ-57

When data is sent or received using_____ the data bits are organized in a specific order, since they

- o Parallel data transmission
- o **Serial data transmission**
- o Hybrid data transmission
- o Both parallel and serial data transmission

_____ have established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers.

- o Microwave
- o **Infrared waves**
- o Simple waves
- o Radio waves

In case of Frequency shift key, the difference between two frequency is represented as_____.

- o **2 Δ**

- o 3 Δ
- o 6 Δ
- o 2 $\times$ 5 Δ

_____ category of coaxial cable is used for thin Ethernet.

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☐ **RG-58**

☐ RG-59

☐ RG-1

☐ RG-47

Cellular telephone uses _____ waves for communication.

☐ Light waves

☐ Infrared waves

☐ **Radio waves**

☐ Microwaves

_____ mode of serial transmission guarantees fixed rate data.

☐ Synchronous

☐ Asynchronous

☐ **Isochronous**

☐ Metasochronous

Router is a network device which operates on the _____ layers of the TCP/IP protocolsuit.

☐ Application

☐ Transport

☐ **Network**

☐ Presentation

_____ layer is responsible for the creating of the datagrams.

☐ Physical

☐ Data link

☐ Session

☐ **Network**

Narrow bands of light in waves division multiplexing are denoted by _____.

☐ β

☐ $\cap$

☐ α

☐ λ

Repeated

In virtual circuit approach when a frame enters a switch and when it leaves a switch its VCI always _____.

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- ☐ Always changes
- ☐ Sometimes changes
- ☐ Sometimes Remain Same
- ☒ **Remain the same**

According to stats, fewer than _____ crossbar points can be used at once.

- ☐ 100%
- ☐ 30%

☒ **25%**

If data rate is to be 1 Mbps then what will be the minimum bandwidth required for NRZ-1 coding scheme?

- ☐ 1Khz
- ☐ 200Khz
- ☐ 500Khz
- ☐ 0.5khz

Data format used at Transport layer of TCP/IP protocol suit is called _____.

- ☒ **Packet**
- ☐ Frame
- ☐ Bit
- ☐ Segment

A virtual circuit network operates on _____ layer.

- ☐ Data link
- ☐ Network

☒ **Application** Repeated

If 32 bits are sent in two seconds then the bitrate for that signal is _____.

- ☐ 32 bps
- ☐ 64 bps
- ☐ 8 Hz
- ☒ **16 bps**

Repeated

Cable TV networks use _____ cables.

☒ **Coaxial**

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- o UTP
- o STP
- o Twisted pair Repeated

_____ normally used for long-distance data transfer.

- o Parallel data transmission
- o Serial data transmission
- o Hybrid data transmission
- o **Both parallel and Serial data transmission** Repeated

_____ is a block coding method.

- o NRZ
- o NRZ-1
- o **8B\10B**
- o 8B\12B

TCP/IP was chosen to be the official protocol of internet_____.

- o 1973
- o 1980
- o **1983**
- o 1988

To measure the data rate_____ theoretically formula(s) Were developed.

- o One
- o Two
- o Three
- o **Four** Repeated

In FSK, Baud rate is less than or equal to_____ rate.

- o Signal
- o Decibel
- o Bit
- o **None of these Given** Repeated

_____ signal is represented by the discrete values.

- o Analog
- o **Digital**

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- ☐ Both
- ☐ Continuous

Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are called_____.

- ☐ Infrared waves
- ☒ **Radio waves**
- ☐ PCLight waves
- ☐ Microwaves

Suppose a signal is amplitude then the value od decibel will be_____.

- ☐ 0
 - ☐ 1
 - ☐ Negative
 - ☒ **Positive**
- Repeated

_____technique does not follows analog conversion.

- ☒ **QAM**
- ☐ AM
- ☐ FM
- ☐ PM
- ☐ Repeated

In Pulse Code Modulation (PCM), the sampling is dependent on_____.

- ☐ Time
- ☒ **Amplitude**
- ☐ Frequency
- ☐ Signal Rate

_____is the sub type of Time Division Multiplexing Technique.

- ☐ Amplitude TDM
- ☐ Light TDM
- ☒ **Statistical TDM**
- ☐ Barrier TDM

Modulation of analog signal is needed if the medium is_____in nature.

- ☒ **Bandpass**

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- ☐ Low-pass
- ☐ Bi-pass
- ☐ High-pass

Quadrature Amplitude Modulation (QAM) is the combination of _____ and _____.

- ☐ FSK-PSK
- ☐ PSK-FSK
- ☐ ASK-PSK
- ☒ **ASK-FSK**

There _____ components of data communication system.

- ☐ 4
- ☒ **5**
- ☐ 6
- ☐ 7

QPSK stands for _____.

- ☐ Quality phase shift key
- ☐ Queuing phase shift key
- ☒ **Quadrature phase shift key**
- ☐ Quality physical shift key

To calculate the data rate for noisy channel _____ formula is used.

- ☒ **Shannon**
- ☐ Nyquist
- ☐ Propagation
- ☐ Greedy

Most commonly used connector for twisted pair cable is _____.

- ☐ RJ-11
- ☒ **RJ-45**
- ☐ RJ-44
- ☐ RJ-52

Binary Amplitude Shift Keying is also called as:

- ☐ Dual Keying
- ☐ On-In Keying

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- o **On-Off Keying**

- o In-Out Keying

Some manufacturers provide a special port called the_____.

- o **IrDA**

- o Consol
- o Parallel
- o USB

To calculate the data rate for noiseless channel_____formula is used.

- o Ready
- o Shannon

- o **Nyquist**

- o Propagation

Two PCs and one printer are connected in a network within a room, it is an example of_____.

- o **LAN**

- o WAN
- o MAN
- o SONET

Analog the following cables given,_____cable provides highest bandwidth and less attenuation.

- o UTP
- o Coaxial cable
- o Coaxial cable

- o **Fiber optical**

To calculate the data for noisy channel_____formula is used.

- o **Shannon**

- o Nyquist
- o Propagation
- o Greedy

In circuit switched networks we have low efficiency but minimal_____.

- o **Delay**

- o Speed

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- o Throughput
- o Errors

Signals travel through fiber optic cable are in the form of_____.

- o **Light**
- o Bits
- o Electromagnetic
- o Bytes

Polar encoding scheme uses_____voltage level.

- o 0
- o **1**
- o 2
- o 3

TCP/IP protocol suit consists of_____layers.

- o 3
- o **5**
- o 6
- o 7

Bandwidth in hertz is the range of_____contained in a composite signal.

- o Signals
- o Phases
- o Amplitudes
- o **Frequencies**

_____is an example of a host in a network.

- o Router
- o **Computer**
- o Bridge
- o Hub

Set of rules to be followed for effective communication is called_____.

- o Topology
- o **Protocol**
- o Switching
- o Encryption

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_____ is NOT an example of a connecting device.

- ☐ Bridge
- ☐ Router
- ☒ **TCP**
- ☐ Switch

When an antenna transmits radio waves, they are propagated in_____.

- ☐ All directions
- ☐ Left directions
- ☐ Right directions
- ☒ **Only up and down direction**

Which one of the following is not a bipolar encoding level?

- ☐ Negative
- ☒ **One**
- ☐ Zero
- ☐ Positive

_____ and _____ are the two types of addressing in virtual circuit approach.

- ☒ **Local, Global**
- ☐ Private, Public
- ☐ Variable, Dynamic
- ☐ Dependent, Independent

In transmission implements _____ noise occurs from other sources like motors and appliances etc.

- ☐ Impulse
- ☐ Inductive
- ☒ **Cross talk**
- ☐ Thermal

In computer network the stations are connected to each other in such a way that each station is connected to every other station through dedicated links. This makes_____.

- ☒ **Bus**
- ☐ Ring
- ☐ Star
- ☐ Mesh

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In scrambling coding scheme, the number of pulses replacing the bits will be _____?

- ☒ **Equal to pulses**
- ☐ Double the pulses
- ☐ Half of the pulses
- ☐ Triple the pulses

In the context of analog to analog signal conversation technique. "PM" Stands for _____.

- ☐ Pulse Modulation
- ☒ **Phase modulation**
- ☐ Parse modulation
- ☐ Popular modulation

_____ relatively measurement the strength of two signals.

- ☐ Signals
- ☐ Bit rate
- ☒ **Decibel**
- ☐ Pulse rate

Datagram approach and circuit approach are two popular approaches which lead usto _____.

- ☐ Line switching
- ☐ Circuit switching
- ☐ Dynamic switching
- ☒ **Packet switching**

If data is to be 1 Mbps then what will be the minimum bandwidth required for NRZ-I coding scheme?

- ☐ 1Khz
- ☐ 200Khz
- ☐ 500khz
- ☐ 0.5khz

In virtual circuit approach when frame enters a switch and when it leaves a switch its VCi always _____.

- ☐ Always Changes
- ☐ Sometimes Changes

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- ☐ Sometimes Remain Same
- ☒ **Remain the Same**

A common bipolar coding scheme is called_____.

- ☒ **AMI**
- ☐ NRZ
- ☐ RZ
- ☐ QAM

_____means loss of energy in signal.

- ☐ Noise
- ☐ Delay
- ☒ **Attenuation**
- ☐ Distortion

The conversion of analog sample of the signal into digital form is called _____ process.

- ☐ Quantizing
- ☒ **Sampling**
- ☐ Modulation
- ☐ Quantizing and Sampling

A local telephone network use_____network.

- ☐ Line switched
- ☐ Packet switched
- ☒ **Circuit switched**
- ☐ Bit switched

In Amplitude Shift Keying,_____of the signal is/are changed.

- ☐ Amplitude and Frequency
- ☒ **Amplitude**
- ☐ Frequency and Amplitude
- ☐ Phase

In frequency domain plot, which value is plot on Y-axis?

- ☐ Time
- ☐ Frequency
- ☒ **Amplitude**

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- o Wavelength

In pulse code modulation the term sampling can also be referred as_____.

- o **Pulse Amplitude Modulation**

- o Pulse sample Modulation
- o Pulse line modulation
- o Pulse Frequency Modulation

Switch is a network device which operates on the_____layer of TCP/IP protocolsuit.

- o Application
- o Transport
- o Presentation

- o **Data link**

In analog transmission of digital data, the required bandwidth is always proportional to the signal rate expect in_____a digital to analog conversion technique.

- o **ASK**

- o FSK
- o PSK
- o NSK

WDM stands for_____.

- o Wideband De-Modulation
- o Worst Data Manipulation

- o **Wavelength Division Multiplexing**

- o None of the Choices is correct

_____does not follows analog to analog conversion.

- o **QAM**

- o AM
- o FM
- o PM

On a communication link, different channels are separates by unused strips of bandwidth called as_____.

- o Base band
- o Broad band

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- o Null band
- o **Guard band**
- o

_____ layer converts frames coming from Data Link Layer into bits and sends

Them on the transmission medium.

- o Application
- o **Physical**
- o Network
- o Transport

Wavelength binds the _____ of a simple sine wave to the propagation speed of the medium.

- o Phase
- o **Period of the frequency** book page 61
- o Amplitude
- o Bandwidth

The bandwidth delay product define the number of bits that can fill the _____.

- o Link
- o **Network**
- o Hub
- o Switch

The size of the packet determined by the network and _____.

- o Delays
- o Switching
- o Instructions
- o **Governing protocol**
- o

To improve the performance of line coding _____ coding was.....?

- o **Block coding**
- o MLT-3
- o 2BIQ
- o Double coding

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Which one of the following is not a sampling technique?

- ☐ Flat Top Sampling
- ☒ **Neutral sampling**
- ☐ Natural sampling
- ☐ Ideal sampling

_____ is normally used where speed is priority in data transfer.

- ☐ Serial data transmission
- ☒ **Hybrid transmission**
- ☐ Parallel transmission
- ☐ Both parallel and serial transmission

In a computer network, five stations are connected to each other in such a way that each station is connected to every other station through dedicated links. This makes _____ topology.

- ☐ Bus
- ☐ Ring
- ☐ Star
- ☒ **Mesh**

Some manufacturer provide a special port called the _____ port that allows a wireless keyboard to communicate with a PC.

- ☒ **IrDA**
- ☐ Consol
- ☐ Parallel
- ☐ USB

Current technology supports _____ modes for propagation of light.

- ☒ **Two**
- ☐ Three
- ☐ Four
- ☐ Five

We quantize the sampling output into certain levels based on range of _____ and required accuracy.

- ☐ Frequency
- ☒ **Amplitude**
- ☐ Time period

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- ☐ None

_____ is the Nyquist bit rate formula for noiseless channel.

- ☐ **BitRate=2*Bandwidth*log2 L**

- ☐ BitRate=2*Bandwidth*logL
- ☐ BitRate=2*Bandwidth*log2L
- ☐ BitRate=2*Bandwidth*L

A frequency is called _____, if the rate of change in sine wave Is Instantaneous.

- ☐ **Infinite frequency** book page 59

- ☐ Zero frequency
- ☐ Bandwidth
- ☐ Frequency hertz

Which multiplexing technique involves signals composed of light beams?

- ☐ FDM
- ☐ TDM
- ☐ **WDM**
- ☐ PDM

_____ is a multiplexing technique which shifts each signal to a different carrier frequency.

- ☐ **FDM**
- ☐ TDM
- ☐ WDM
- ☐ PDM

Entire band in United States is regulated by _____ authorities.

- ☐ **FCC**
- ☐ ITU
- ☐ NASA
- ☐ ANSI

Two stations are connected via a secure link in which messages are encrypted from source to the destination message is called-----

- ☐ Plain text
- ☐ Cipher test

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- o Message
- o **Protocol (not confirm)**

Simultaneous transmission of multiple signal across a single data link is called-----

- o Demultiplexing
- o **Multiplexing**
- o Modulation
- o demodulation

in NRZ-1 the signal is inverted if----- is encountered.

- o 0
- o **1**
- o 00
- o 11

----- Protocol suite is being used by the modern internet communication.

- o OSI
- o **TCP/IP**
- o ARPANET
- o Telnet

In the frequency shift keying, ----- remain(s) constant.

- o Frequency
- o **Amplitude**
- o Both amplitude and phase
- o Both phase and frequency

In ----- signal changes its shape or form.

- o Thermal noise
- o **Distortion**
- o Impulse noise
- o Attenuation

----- coding scheme are useful for LAN but not for long ranges.

- o **Biphase** **book page 113**
- o Line
- o Vector
- o Serial

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In ----- TDM, each input connection has an allotment in the output even if is not sending data.

- o Asynchronous
- o Statistical
- o **Synchronous (Book page 164)**
- o Isochronous

----- is a type of serial transmission in which data bits are transmitted as a continues stream in time with a master clock. Start bits, stop bits, and gaps are not used. Transmitter and receiver time is sync.

- o Synchronous
- o Asynchronous
- o **Isochronous**
- o Metasochronous

----- Relative measures the strength of two signals.

- o Signal rate
- o Bit rate
- o **Decibel**
- o Pulse rate

When the bandwidth of a link is greater than the combined bandwidths of the signals, we use -----multiplexing.

- o Wavelength division
- o **Frequency division**
- o Time division
- o Hybrid division

----- is an example of host in a network.

- o Router
- o **Computer**
- o Bridge
- o Hub

On a communication link, different channels are separated by unused strips of bandwidth called as -----.

- o Base band
- o Broad band

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- o Null
- o **Guard band**

The conversion of analog sample of the signal into digital form is called -----process.

- o Quantizing
- o Sampling
- o Modulation
- o **Quantizing and sampling (Book pages 115 to 116)**

In star based network comprising of four computers and on switch, total number of cables needed will be -----

- o **4**
- o 5
- o 6
- o 3

The logical connection between the peer layers is -----connection.

- o Physical
- o **Direct (no strong reference but as per book explanation layer to layer isdirect)**
- o Indirect
- o Tangible

According to stats, lower than ----- crossbar point can be used at once .

- o 14%
- o 100%
- o 30%
- o **25%**

----- Category of coaxial cable is used for cable TV.

- o RG-58



0304 1659294

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- o **RG-59**
- o RG-11
- o RG-47

----- is sometimes called the bit rate.

- o Signal rate
- o Modulation rate
- o **Data rate**
- o Pulse rate

Uni polar, polar and bipolar are the types of -----

- o Line
- o Differential Manchester
- o **NRZ-I**
- o Block

Router is a network device which operates on the -----layer of the TCP/IP protocol.

- o Application
- o Transport
- o **Network**
- o Presentation

According to the Fourier analysis, frequencies obtained after decomposition of no period.

- o Bit length
- o Automatic
- o Heterogeneous
- o **Continuous** (Fourier analysis can be used to decompose a digital signal. If the digital signal is periodic, which is rare in data communications, the decomposed signal has a frequency domain representation with an infinite bandwidth and discrete frequencies. If the digital signal is nonperiodic, the decomposed signal still has an infinite bandwidth, but the frequencies are continuous page 70 Data communication Book 5th generation)

When the bandwidth of a link is greater than combined bandwidths of the

- o Wavelength division
- o **Frequency division**
- o Time division

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- o Hybrid division

The ----- technique expands the bandwidth of a signal by replacing each data bit with n bits using a spreading code.

- o FDM
- o FHSS
- o TDM
- o **DSSS**

----- Cable is used in wavelength division multiplexing.

- o Twisted pair
- o Coaxial
- o **Fiber optic**
- o Ethernet

VCI address is -----address in virtual circuit approach.

- o Private
- o Variable
- o Local
- o **Global**

In baseband transmission, a digital signal is transmitted as -----.

- o Analog signal
- o **Digital signal**
- o Modulation signal
- o Multiplexed signal

telephone network use----- network.

- o Line switched
- o Packet switched
- o **Circuit switched**
- o Bit switched

Entire band in United states is regulated by----- authorities.

- o **FCC**
- o ITU
- o NASA
- o ANSI

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----- are used for multicast communication, such as radio and television.

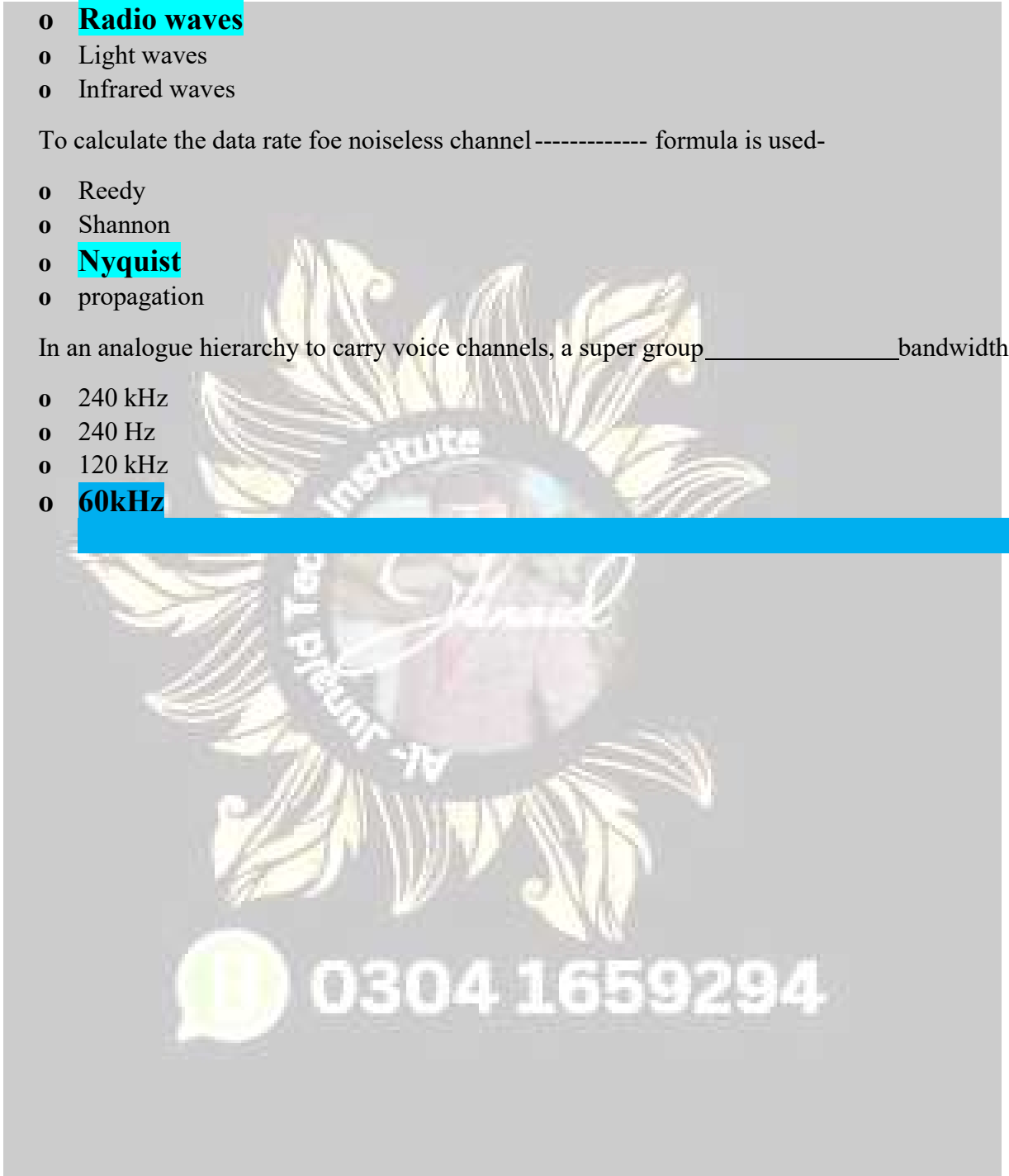
- o Microwaves
- o **Radio waves**
- o Light waves
- o Infrared waves

To calculate the data rate for noiseless channel----- formula is used-

- o Reedy
- o Shannon
- o **Nyquist**
- o propagation

In an analogue hierarchy to carry voice channels, a super group_____bandwidth.

- o 240 kHz
- o 240 Hz
- o 120 kHz
- o **60kHz**





CS-601 Data Communication Update MCQS For Mid Term Solve By Vu Topper RM



80 To 100% Marks



وَتَعَزُّ مِنْ تَشَاءٍ وَتَذِلُّ مِنْ تَشَاءٍ



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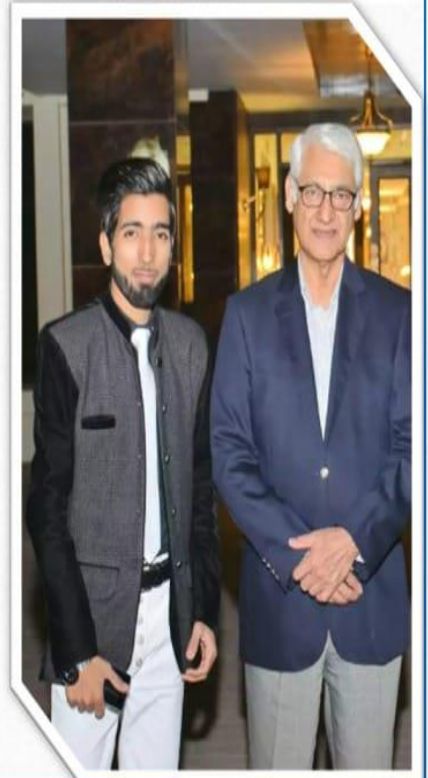
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Question No:1

(Marks:1)

Vu-Topper RM

The most common type of connector used by coaxial cable is _____.

A. BNC

B. RJ-11

C. RJ-57

D. RJ-45

Question No:2

(Marks:1)

Vu-Topper RM

Quadrature Amplitude Modulation is the mechanism of _____ conversion.

A. Digital to analog

B. Analog to Analog

C. Analog to digital

D. Digital to digital

Question No:3

(Marks:1)

Vu-Topper RM

_____ have established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers.

A. Microwaves

B. Radio waves

C. Simple waves

D. Infrared waves

Question No:4

(Marks:1)

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WDM stands for _____.

A. Worst Data Manipulation

B. Wideband De-Modulation

C. Wideband Division Multiplexing

D. Wavelength Division Multiplexing

Question No:5

(Marks:1)

Vu-Topper RM

Voltage controlled oscillator(VCO) generates _____ .

A. Carrier Signals

B. Periodic signals

Google

بري صحبت سے تڙهائي بهتر هے اور تڙهائي سے نيگي صحبت بهتر هے

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- C. Signal elements
- D. Composite Signals

Question No:6

(Marks:1)

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A virtual-circuit network operates on _____ layer.

A. Network **Google**

- B. Physical
- C. Datalink
- D. Application

Question No:7

(Marks:1)

Vu-Topper RM

Top most organization that provides support for the Internet Standard process is called_____.

- A. Internet Society (ISOC)
- B. Internet Architecture Board (IAB)
- C. IETF**
- D. IRTF

Question No:8

(Marks:1)

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In pulse code modulation the term sampling can also be referred as_____.

- A. Pulse Amplitude Modulation**
- B. Pulse sample Modulation
- C. Pulse line modulation
- D. Pulse Frequency Modulation

Question No:9

(Marks:1)

Vu-Topper RM

Unit of phase is _____.

- A. Watts
- B. Bauds
- C. Degree**
- D. Bits per second

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Question No:10

(Marks:1)

Vu-Topper RM

Entire band in United States is regulated by_____ authorities.

- A. ITU
- B. FCC**
- C. ANSI
- D. NASA

Question No:11

(Marks:1)

Vu-Topper RM

In TDM based digital hierarchy used by the Telephone companies, DS-0 is a single digital channel of

- A. 64 kbps**
- B. 32 kbps
- C. 128 kbps
- D. 1.544 Mbps

Question No:12

(Marks:1)

Vu-Topper RM

Radio waves are propagated in_____ whenever they are transmitted by an antenna.

- A. All direction**
- B. Left direction
- C. Right direction
- D. Only up and down direction

Question No:13

(Marks:1)

Vu-Topper RM

A switched network consists of a series of interlinked _____ called switches.

- A. Root
- B. Nodes** **Google**
- C. Points
- D. System

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Question No:14

(Marks:1)

Vu-Topper RM

In circuit switched networks we have low efficiency but minimal _____.

A. Delay

B. Error

C. Speed

D. Throughput

Question No:15

(Marks:1)

Vu-Topper RM

In transmission impairments _____ noise occurs from the random motion of electrons in a wire.

A. Thermal noise

B. Impulse noise

C. Cross talk

D. Induced noise

Question No:16

(Marks:1)

Vu-Topper RM

In Amplitude Shift Keying. _____ of the signal is/are changed.

A. Amplitude and Frequency

B. Amplitude

C. Frequency and Amplitude

D. Phase

Question No:17

(Marks:1)

Vu-Topper RM

A twisted pair cable consists of _____ conductors

A. Two

B. Five

C. Four

D. Three

Question No:18

(Marks:1)

Vu-Topper RM

We quantize the sampling output into certain levels based on range of _____ and required accuracy.

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- A. Frequency
- B. Amplitude**
- C. Time period
- D. None

Question No:19

(Marks:1)

Vu-Topper RM

In star based network comprising of four computers and one switch, total number of cables needed will be_____.

- A. 1
- B. 2
- C. 3
- D. 4**

Question No:20

(Marks:1)

Vu-Topper RM

Which of the following is not a characteristic of a sine wave?

- A. Amplitude
- B. Segmentation**
- C. Phase
- D. Frequency

Question No:21

(Marks:1)

Vu-Topper RM

In a computer network, five stations are connected to each other in such a way that each station is connected to every other station through dedicated links. This makes_____ topology.

- A. Bus
- B. Ring
- C. Star
- D. Mesh**

Question No:22

(Marks:1)

Vu-Topper RM

_____can be published using Request for Comments (RFCs).

- A. Proposed Standard
- B. Draft Standard

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C. Internet Draft

D. Internet Standard

Question No:23

(Marks:1)

Vu-Topper RM

In serial data transmission _____ data transmission mode can be cheap but slower.

A. Synchronous

B. Asynchronous

C. Isochronous

D. Metachronous

Question No:24

(Marks:1)

Vu-Topper RM

QPSK stands for_____.

A. Quality phase shift key

B. Queuing phase shift key

C. Quadrature phase shift key

D. Quality physical shift key

Question No:25

(Marks:1)

Vu-Topper RM

In _____, the instantaneous change in the carrier frequency is proportional to the derivative of the amplitude of the modulating signal.

A. Phase modulation

B. Frequency modulation

C. Amplitude modulation

D. Quadrature modulation

Question No:26

(Marks:1)

Vu-Topper RM

In FSK, Baud rate is less than or equal to _____ rate.

A. Bit

B. Byte

C. Signal

D. Decibel

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Question No:27

(Marks:1)

Vu-Topper RM

Bi-polar uses _____ voltage levels.

A. One

B. Two

C. Three

D. Four

Question No:28

(Marks:1)

Vu-Topper RM

Low pass channel has _____ bandwidth between two stations.

A. Dedicated

B. Shared

C. Multiplexed

D. Infinite

Question No:29

(Marks:1)

Vu-Topper RM

Them on the transmission medium.

A. Application

B. Physical

C. Network

D. Transport

Question No:30

(Marks:1)

Vu-Topper RM

Block coding scheme contains _____ number of steps.

A. 3

B. 4

C. 5

D. 6

Question No:31

(Marks:1)

Vu-Topper RM

BFSK stands for _____.

A. Baud Frequency Shift Key

B. Binary Frequency Shift Key

C. Barrier Frequency Shift Key

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D. Bridge Frequency Shift Key

Question No:32

(Marks:1)

Vu-Topper RM

According to the Fourier analysis, frequencies obtained after decomposition of digital signals are_____.

- A. Continuous
- B. Discrete**
- C. Bit length
- D. Homogeneous

Question No:33

(Marks:1)

Vu-Topper RM

Suppose a signal is amplified then the value of decibel will be_____.

- A. 0
- B. 1
- C. Negative
- D. Positive**

Question No:34

(Marks:1)

Vu-Topper RM

In frequency domain plot, which value is replaced with frequency?

- A. Wavelength**
- B. Amplitude
- C. Phase
- D. Time

Question No:35

(Marks:1)

Vu-Topper RM

We can have _____different groups with a 4 bit block.

- A. 4
- B. 6
- C. 7
- D. 8**

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Question No:36

(Marks:1)

Vu-Topper RM

_____ means loss of energy in signal.

- A. Noise
- B. Delay
- C. Attenuation**
- D. Distortion

Question No:37

(Marks:1)

Vu-Topper RM

Uni polar, polar and bipolar are the types of _____

- A. Line
- B. Differential Manchester
- C. NRZ-I**
- D. Block

Question No:38

(Marks:1)

Vu-Topper RM

If a digital transmission system is sending five bits in every half a second, the bit-rate of the system.

- A. 5 bps**
- B. 10 Hz
- C. 50 bps
- D. 0.2 bps

Question No:39

(Marks:1)

Vu-Topper RM

_____ signal completes certain pattern in a specific amount of time.

- A. Non-periodic
- B. Periodic**
- C. A Periodic
- D. Wavelength

Question No:40

(Marks:1)

Vu-Topper RM

To calculate the data rate for noisy channel _____ formula is used.

- A. Shannon
- B. Nyquist bit rate**

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- C. Propagation
- D. Greedy

Question No:41

(Marks:1)

Vu-Topper RM

_____ mode of serial transmission guarantees fixed rate data.

- A. Synchronous
- B. Asynchronous
- C. Isochronous**
- D. Metasochronous

Question No:42

(Marks:1)

Vu-Topper RM

In TCP\IP model, the Physical Layer exchanges data in the form of ____.

- A. Packets
- B. Frames
- C. Bits**
- D. Segments

Question No:43

(Marks:1)

Vu-Topper RM

_____ signal is represented by the discrete values.

- A. Analog
- B. Digital**
- C. Both
- D. Continuous

Question No:44

(Marks:1)

Vu-Topper RM

A common Bipolar encoding scheme is called _____.

- A. AMI**
- B. NRZ
- C. RZ
- D. QAM

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Question No:45

(Marks:1)

Vu-Topper RM

_____ relatively measures the strength of two signals.

- A. Signal rate
- B. Bit rate
- C. Decibel**
- D. Pulse rate

Question No:46

(Marks:1)

Vu-Topper RM

Two PCs and one printer are connected in a network within a room; it is an example of _____.

- A. LAN**
- B. WAN
- C. MAN
- D. ALN

Question No:47

(Marks:1)

Vu-Topper RM

In analog transmission of digital data, the required bandwidth is always proportional to the signal rate except in _____, a digital to analog conversion technique.

- A. ASK**
- B. NSK
- C. FSK
- D. PSK

Question No:48

(Marks:1)

Vu-Topper RM

A local telephone network use _____ network.

- A. Line switched
- B. Circuit switched**
- C. Bit switched
- D. Packet switched

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Question No:49

(Marks:1)

Vu-Topper RM

In an analogue hierarchy to carry voice channels, a super group _____ bandwidth.

A. 60kHz

B. 240 Hz

C. 240 kHz

D. 120 kHz

Question No:50

(Marks:1)

Vu-Topper RM

Narrow bands of light in wave division multiplexing are denoted by _?

A. Π

B. B

C. A

D. Lambda λ

Question No:51

(Marks:1)

Vu-Topper RM

Which digital multiplexing technology is use to allow several connections to share the high bandwidth of a l

A. TDM

B. FDM

C. ADM

D. WDM

Question No:52

(Marks:1)

Vu-Topper RM

In optical fiber cable, the ratio of the diameter of their core to the diameter of their cladding, both expressed in _____.

A. Meters

B. Millimeters

C. Centimeters

D. Micrometers

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Question No:53

(Marks:1)

Vu-Topper RM

Wireless transmission can be divided into _____ broad groups.

- A. One
- B. Two
- C. Three**
- D. None of these

Question No:54

(Marks:1)

Vu-Topper RM

_____ is the sub type of Time Division Multiplexing Technique.

- A. Light TDM
- B. Barrier TDM
- C. Statistical TDM**
- D. Amplitude TDM

Question No:55

(Marks:1)

Vu-Topper RM

Simultaneous transmission of multiple signals across a single data link is called _____

- A. Multiplexing**
- B. Demultiplexing
- C. Modulation
- D. Demodulation

Question No:56

(Marks:1)

Vu-Topper RM

Cellular telephone uses _____ waves for communication.

- A. Radio waves**
- B. light wave
- C. infrared waves
- D. microwaves

Question No:57

(Marks:1)

Vu-Topper RM

Time division multiplexing is used in _____ systems.

- A. Digital**
- B. Analog

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- C. Hybrid
- D. Automated

Question No:58

(Marks:1)

Vu-Topper RM

A telephone line analog signal has got the bandwidth of _____.

- A. 8kHz
- B. 4kHz**
- C. 16kHz
- D. 12kHz

Question No:59

(Marks:1)

Vu-Topper RM

In _____ modulation process is used by the transmitting devices to shift their data to a specific carrier frequency for multiplexing.

- A. FDM**
- B. ADM
- C. DAM
- D. SDM

Question No:60

(Marks:1)

Vu-Topper RM

A fiber-optic cable transmit signals in the form of _____.

- A. Light**
- B. Waves
- C. Sound
- D. None of the given

Question No:61

(Marks:1)

Vu-Topper RM

If the message is going to pass through a packet-switched network, it can be divided into packets of ____.

- A. Fixed size
- B. Variable size
- C. Both fix and variable size**
- D. None of the these

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Question No:62

(Marks:1)

Vu-Topper RM

Paging system in data communication uses _____ waves

A. Radio waves

B. light wave

C. infrared waves

D. microwaves

Question No:63

(Marks:1)

Vu-Topper RM

According to stats, fewer than _____ crossbar points can be used at once.

A. 14%

B. 25%

C. 30%

D. 100%

Question No:64

(Marks:1)

Vu-Topper RM

Virtual Circuit Identifier is the _____ of packet.

A. Length

B. Header

C. Size

D. Address

Question No:65

(Marks:1)

Vu-Topper RM

In unidirectional antenna, the common point where all the lines intersect each other is _____.

A. Axis

B. Focus

C. Origin

D. Center

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Question No:66

(Marks:1)

Vu-Topper RM

The size of the packet is determined by the network and _____.

A. Governing Protocol

B. Instructions

C. Switching time

D. Delay

Question No:67

(Marks:1)

Vu-Topper RM

In virtual circuit approach when a frame enters a switch and when it leaves the circuit then its VCI ____

A. Remain same

B. Sometimes changes

C. Remain the same

D. Always changes

Question No:68

(Marks:1)

Vu-Topper RM

In _____ TDM slots are dynamically allocated to the connected stations to improve bandwidth efficiency.

A. Statistical

B. Isochronous

C. Synchronous

D. None of the given

Question No:69

(Marks:1)

Vu-Topper RM

In synchronous _____, the data flow of each input connection is divided into units, where each input occupies one input time slot.

A. FDM

B. TDM

C. WDM

D. DWDM

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Question No:70

(Marks:1)

Vu-Topper RM

When the bandwidth of a link is greater than the combined bandwidths of the signals, we use _____ multiplexing.

A. Frequency division

B. Time division

C. Hybrid division

D. Wavelength division

Question No:71

(Marks:1)

Vu-Topper RM

Routing is a function of _____ layer.

A. Network

B. Physical

C. Transport

D. Datalink

Question No:72

(Marks:1)

Vu-Topper RM

_____ is the process of converting binary data to a digital signal.

A. QAM

B. ASK

C. FSK

D. Line coding

Question No:73

(Marks:1)

Vu-Topper RM

_____ signals can take infinite levels of intensity over time.

A. Digital

B. Discrete

C. Analog

D. Logical

Question No:74

(Marks:1)

Vu-Topper RM

During transmission, _____ is the effect of one wire on the other.

A. Crosstalk

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- B. Induced noise
- C. Thermal noise
- D. Thermal noise

Question No:75

(Marks:1)

Vu-Topper RM

Latency is made of-----components.

- A. Two
- B. Five
- C. Four**
- D. Three

Question No:76

(Marks:1)

Vu-Topper RM

If a digital signal has four levels, then we need _____ bits to represent each level.

- A. 2**
- B. 4
- C. 6
- D. 8

Question No:77

(Marks:1)

Vu-Topper RM

_____ is a digital process that allows connections to share the high bandwidth of a link.

- A. TDM**
- B. FDM
- C. ADM
- D. WDM

Question No:78

(Marks:1)

Vu-Topper RM

Transmission medium are of _____ types.

- A. One
- B. Two**
- C. Six
- D. Five

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Question No:79

(Marks:1)

Vu-Topper RM

_____ are used for multicast communications, such as radio and television.

A. Radio waves

B. light wave

C. infrared waves

D. microwaves

Question No:80

(Marks:1)

Vu-Topper RM

We can have combination of _____ possible codes with a 5-bit code.

A. 30

B. 32

C. 34

D. 36

Question No:81

(Marks:1)

Vu-Topper RM

When two people are talking to each other in a room then transmission medium between them is:

A. Air

B. Cable

C. Wireless medium

D. No medium required

Question No:82

(Marks:1)

Vu-Topper RM

Manchester encoding scheme achieves the same level of synchronization as _____.

A. ARZ

B. RZ

C. ARZ1

D. IMP

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Question No:83

(Marks:1)

Vu-Topper RM

Coaxial cables are categorized by their _____ ratings.

A. Radio Government (RG)

B. Frequency modulation (FM)

C. Amplitude modulation (AM)

D. Electronic Industries Association (EIA)

Question No:84

(Marks:1)

Vu-Topper RM

Microwaves are _____.

A. Bi directional

B. Unidirectional

C. Multi directional

D. Omni directional

Question No:85

(Marks:1)

Vu-Topper RM

A virtual-circuit network is a cross between a _____ network and a datagram network.

A. Circuit-switched

B. Packet switched

C. Message switched

D. Both message and packet switched network

Question No:86

(Marks:1)

Vu-Topper RM

In _____ multiplexing technique, guard bands are used to avoid overlapping of frequency bands assigned to each user.

A. FDM **Google**

B. TDM

C. PDM

D. CSMA

Question No:87

(Marks:1)

Vu-Topper RM

In _____ TDM, each input connection has an allotment in the output even if it is not sending data.

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- A. Statistical
- B. Isochronous
- C. Synchronous**
- D. Asynchronous

Google

Question No:88

(Marks:1)

Vu-Topper RM

Cable TV uses the _____ category of coaxial cable.

- A. RG-47
- B. RG-58
- C. RG-59**
- D. RG-11

Question No:89

(Marks:1)

Vu-Topper RM

Connection setup means using _____ channels in circuit switched network.?

- A. Satellite**
- B. Both
- C. Communication
- D. None of the these

Question No:90

(Marks:1)

Vu-Topper RM

Modulation of analog signal is needed if the medium is _____ in nature.

- A. Bi-pass
- B. Low-pass
- C. Bandpass**
- D. High-pass

Question No:91

(Marks:1)

Vu-Topper RM

Quadrature Amplitude Modulation (QAM) is the combination of _____ and _____.

- A. PSK, FSK
- B. FSK, PSK**

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- C. ASK, FSK.
- D. ASK, PSK

Question No:92

(Marks:1)

Vu-Topper RM

_____ is the scope of VCI variable in virtual circuit approach.

- A. Local**
- B. Global
- C. Both
- D. None of these

Question No:93

(Marks:1)

Vu-Topper RM

$N = S \times r$ is the formula for finding _____ .?

- A. Bit rate
- B. Baud rate** Google
- C. Data element
- D. Signal element

Question No:94

(Marks:1)

Vu-Topper RM

In _____ TDM, a round of data units from each input connection is collected into a frame.

- A. Statistical
- B. Non-Periodic
- C. Synchronous**
- D. Asynchronous

Question No:95

(Marks:1)

Vu-Topper RM

VCI address is _____ address in virtual circuit approach.

- A. Local
- B. Global**
- C. Private
- D. Variable

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Question No:96

(Marks:1)

Vu-Topper RM

Virtual-Circuit Networks and datagram networks are the sub categories of _____.

A. Packet-switched networks

B. Circuit Switched Networks

C. Message switched networks

D. Both circuit and message switched networks

Question No:97

(Marks:1)

Vu-Topper RM

Electromagnetic waves ranging in frequencies between 1 and 300 GHz are called _____.

A. Light wave

B. Microwaves

C. Radio waves

D. Infrared waves

Question No:98

(Marks:1)

Vu-Topper RM

_____ has less electromagnetic interference as compared to other guided medium.

A. Coaxial

B. Fiber Optic

C. Shielded Twisted Pair

D. Unshielded Twisted Pair

Question No:99

(Marks:1)

Vu-Topper RM

In an analogue hierarchy to carry voice channels, a group can carry _____ voice channels.

A. 60

B. 12

C. 10

D. 20

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Question No:100

(Marks:1)

Vu-Topper RM

In ASK correct formula for calculating the bandwidth is as _____.

A. $B=(1+d)S$

B. $B=(1*d)S$

C. $B=(d-1)S$

D. $B=(d-5)S$

Question No:101

(Marks:1)

Vu-Topper RM

Switching at the _____ layer in the traditional telephone network uses the circuit-switching approach.

A. Data Layer

B. Session Layer

C. Physical Layer

D. Network Layer

Google

Question No:102

(Marks:1)

Vu-Topper RM

In case of Frequency shift key, the difference between two frequencies is represented as _____.

A. 2Δ

B. 4Δ

C. 6Δ

D. $2X5 \Delta$

Question No:103

(Marks:1)

Vu-Topper RM

Datagram approach and virtual circuit approach are two popular approaches which lead us to _____.

A. Line switching

B. Packet switching

C. Circuit switching

D. Dynamic switching

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Question No:104

(Marks:1)

Vu-Topper RM

The minimum bandwidth required for QAM transmission is the same as that required for _____ and _____ transmission.

A. ASK, PSK

B. FSK, ASK

C. PSK, FSK

D. FSK, QPSK

Question No:105

(Marks:1)

Vu-Topper RM

In circuit switching total Delay is combination of Connection Time, Data transfer time and _____.

A. Switching time

B. Receiving time

C. Acknowledgement time

D. Connection tear down time

Question No:106

(Marks:1)

Vu-Topper RM

RG-11 having impedance of 50 ohm is used in _____ Ethernet

A. Thin

B. Fast

C. Thick

D. None of these

Question No:107

(Marks:1)

Vu-Topper RM

Line coding scheme is roughly divide into _____ broad categories.?

A. Five

B. One

C. Two

D. Zero

بري صحبت سے تنہائی بہتر ہے اور تنہائی سے نيك صحبت بہتر ہے

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Question No:108

(Marks:1)

Vu-Topper RM

Normally the value of K max in pure ALOHA is _____.

A. 15

B. 14

C. 13

D. 12

Question No:109

(Marks:1)

Vu-Topper RM

YMODEM has _____ Byte of data unit.

A. 1024

B. 2410

C. 2210

D. 1042

Question No:110

(Marks:1)

Vu-Topper RM

In Block coding scheme, number of code words is always _____ data words.

Equal to

Question No:111

(Marks:1)

Vu-Topper RM

Which one of the following is not a Channelization Protocol?

CSMA

Question No:112

(Marks:1)

Vu-Topper RM

_____ is an Authentication Protocol, which uses two-step process to authenticate user information.

PAP

Question No:113

(Marks:1)

Vu-Topper RM

The transmission medium that carries the message is referred to as the _____.

Communication channel

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Question No:114

(Marks:1)

Vu-Topper RM

Which error detection method uses one's complement arithmetic

Checksum

Question No:115

(Marks:1)

Vu-Topper RM

Like 10 Base 5, 10 Base 2 is a _____ topology LAN.

Bus

Question No:116

(Marks:1)

Vu-Topper RM

There are _____ types of serial transmission

2

Question No:117

(Marks:1)

Vu-Topper RM

____ and ____ are the two types of addressing in virtual circuit approach.

Local, Global

Question No:118

(Marks:1)

Vu-Topper RM

Bandwidth can be measured in _____.

bits per second bps

Question No:119

(Marks:1)

Vu-Topper RM

A variation of AMI encoding is called as _____.

Pseudoternary

Question No:120

(Marks:1)

Vu-Topper RM

For a noiseless channel, theoretical formula to calculate the data rate was developed by _____.

Nyquist

Question No:121

(Marks:1)

Vu-Topper RM

In ____ scheme the voltages are on both sides of the time axis and voltage level for 0 can be positive and the voltage level for 1

Polar

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Question No:122 (Marks:1) **Vu-Topper RM**
Polar encoding scheme uses _____ voltage level.
Two different

Question No:123 (Marks:1) **Vu-Topper RM**
In a bidirectional communication each layer performs two _____ tasks in each direction.
Opposite

Question No:124 (Marks:1) **Vu-Topper RM**
In _____ scheme, all the signal levels are on one side of the time axis, either above or below.
Unipolar

Question No:125 (Marks:1) **Vu-Topper RM**
Manchester and differential Manchester encoding schemes are subtypes of _____ encoding scheme.
Double

Question No:126 (Marks:1) **Vu-Topper RM**
In _____ transmission medium start and stop bits are used
Asynchronous

Question No:127 (Marks:1) **Vu-Topper RM**
In Ring topology, each node has a point to point dedicated link with.
Exactly 2 nodes

Question No:128 (Marks:1) **Vu-Topper RM**
Simple sine wave can be used to send ----- .
Amplitude

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Question No:129

(Marks:1)

Vu-Topper RM

_____ layer converts frames coming from Data Link Layer
Into bits and sends

Question No:130

(Marks:1)

Vu-Topper RM

The maximum bandwidth required for QAM transmission is the same as that required for _ and _____ transmission.

ASK, PSK

Question No:131

(Marks:1)

Vu-Topper RM

In _____ transmission mode(s), multiple bits are sent simultaneously.

Parallel transmission

Question No:132

(Marks:1)

Vu-Topper RM

Which of following technique(s) is used for Analog-to-Digital Conversion?

Pulse Code modulation (PCM)

Question No:133

(Marks:1)

Vu-Topper RM

_____ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

$N*(N-1)$

Question No:134

(Marks:1)

Vu-Topper RM

In ___ encoding scheme, the voltage level oscillates between a positive and a negative value although it may remain at zero level between the two values.

Polar

Question No:135

(Marks:1)

Vu-Topper RM

_____ is the number of bits sent in 1 second.

Bit rate

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Question No:136

(Marks:1)

Vu-Topper RM

Wavelength binds the ----- of a simple sine wave to the propagation speed of the medium.

Period of the frequency

Page 61

Question No:137

(Marks:1)

Vu-Topper RM

The Full form of AMI encoding scheme is _____.

Alternate mark inversion

Question No:138

(Marks:1)

Vu-Topper RM

AMI is a popular _____ encoding method.

Bi-polar

Question No:139

(Marks:1)

Vu-Topper RM

LF(low frequency) band is used in _____ propagation

Ground

Question No:140

(Marks:1)

Vu-Topper RM

Cable TV networks use _____ cables.

Coaxial cables

Question No:141

(Marks:1)

Vu-Topper RM

In circuit switching, when two end devices “A” and “B” connects with each other they send _____ request.

Acceptance

Question No:142

(Marks:1)

Vu-Topper RM

In virtual Circuit network all packets follow _____ path.

Dedicated

Question No:143

(Marks:1)

Vu-Topper RM

Among the following cables given below _____ cable has the cost as compared to others

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Medium

Question No:144

(Marks:1)

Vu-Topper RM

Among the following cables given below _____ cable has the lowest cost as compared to others.

Coaxial cable

Question No:145

(Marks:1)

Vu-Topper RM

Among the following cables given below _____ cable has the highest cost as compared to others.

Fiber optic cable

Question No:146

(Marks:1)

Vu-Topper RM

Header size _____ while moving down the layers at sending side.

Decreases

Question No:147

(Marks:1)

Vu-Topper RM

Set of rules to be followed for effective network communication is called _____.

Protocol

Question No:148

(Marks:1)

Vu-Topper RM

Datagram (packet) travels across _____ in a network.

Switching

Question No:149

(Marks:1)

Vu-Topper RM

The bandwidth-delay product defines the number of bits that can fill the _____.

Link

Question No:150

(Marks:1)

Vu-Topper RM

In Pulse Code Modulation (PCM), the sampling is dependent on _____.

Amplitude

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Question No:151

(Marks:1)

Vu-Topper RM

Composite wave can be used to send _____.

Frequency

Question No:152

(Marks:1)

Vu-Topper RM

_____ is normally used for long-distance data transfer.

Both parallel and Serial data transmission

Question No:153

(Marks:1)

Vu-Topper RM

The conversion of _____ sample of the signal into _____ form is called quantizing process

Sampling and quantizing

Question No:154

(Marks:1)

Vu-Topper RM

When data is sent or received using _____, the data bits are organized in a specific order, since they can only be sent or received one after another.

Serial data transmission

Question No:155

(Marks:1)

Vu-Topper RM

Multipoint channel can be shared in _____ manner.

Both temporal and spartial.

Question No:156

(Marks:1)

Vu-Topper RM

Communication between two users in a Cellular (Mobile) network is an example of _____ communication.

Full duplex

Question No:157

(Marks:1)

Vu-Topper RM

_____ is normally used where speed is priority in data transfer.

Hybrid transmission

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Question No:158

(Marks:1)

Vu-Topper RM

Wiring used for transmission modes depends upon _____.

Latency

Question No:159

(Marks:1)

Vu-Topper RM

If a digital signal has “L” number of levels, _____ is the number of bits required to represent each level.

Log base 2L

Question No:160

(Marks:1)

Vu-Topper RM

In scrambling coding scheme, the number of pulses replacing the bits will be _____

Equal to pulses

Question No:161

(Marks:1)

Vu-Topper RM

_____ is a type of serial transmission in which data bits are transmitted as a continuous stream in time with a master clock. start bits, stop bits, and gaps are not used. Transmitter and receiver time is sync.

Synchronous

Google

Question No:162

(Marks:1)

Vu-Topper RM

TCP/IP was presented by _____ in a research paper.

Vin Cerf and bob Kahn

Question No:163

(Marks:1)

Vu-Topper RM

HDB3 falls under which coding scheme?

Bipolar

Question No:164

(Marks:1)

Vu-Topper RM

A frequency is called _____, if the rate of change in sine wave Is Instantaneous.

Infinite frequency

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Question No:165

(Marks:1)

Vu-Topper RM

The bipolar encoding scheme was developed as an alternative of _____.

NRZ

Question No:166

(Marks:1)

Vu-Topper RM

In _____ TCP/IP was declared as the official protocol of internet.

1983

Question No:167

(Marks:1)

Vu-Topper RM

If 32 bits are sent in two seconds then bitrate for that signal is _____.

16 bps

Question No:168

(Marks:1)

Vu-Topper RM

In _____ data moves faster and timing errors are less frequent because the transmitter and receiver time is synced.

Synchronous

Question No:169

(Marks:1)

Vu-Topper RM

PSK is _____ susceptible to noise as compared to ASK.

False

Ture

Question No:170

(Marks:1)

Vu-Topper RM

There are _____ components of data communication system.

5

Question No:171

(Marks:1)

Vu-Topper RM

Which one of following is a scrambling coding technique?

B8ZS

بري صحبت سے تنهائي بهتر ہے اور تنهائي سے نيك صحبت بهتر ہے

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Question No:172 (Marks:1) **Vu-Topper RM**
In NRZ-I the signal is inverted if _____ is encountered.

1

Question No:173 (Marks:1) **Vu-Topper RM**
_____ layer is responsible for creating datagrams.

Network

Question No:174 (Marks:1) **Vu-Topper RM**
_____ is the Nyquist bit rate formula for noiseless channel.

BitRate=2*Bandwidth*log2 L

Question No:175 (Marks:1) **Vu-Topper RM**
Signals travel through fiber optic cable are in the form of _____.

Light

Question No:176 (Marks:1) **Vu-Topper RM**
When data is sent using _____ , multiple data bits are transmitted over multiple channels at the same time.

Parallel transmission

Question No:177 (Marks:1) **Vu-Topper RM**
In Binary ASK, the peak amplitude of one signal level is 0 and the other is the same as the _____ of the carrier.

Amplitude

Question No:178 (Marks:1) **Vu-Topper RM**
_____ category of coaxial cable is used for thick Ethernet.

RG-58

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Question No:179

(Marks:1)

Vu-Topper RM

Time taken by a periodic signal to complete one cycle is called ____.

Period

Question No:180

(Marks:1)

Vu-Topper RM

QAM stands for _____.

Quadrature Amplitude Modulation

Question No:181

(Marks:1)

Vu-Topper RM

According to Nyquist theorem, the sampling rate must be at least _____ times the highest frequency contained in the signal.

Twice

Question No:182

(Marks:1)

Vu-Topper RM

_____ is an example of transmitter.

Modem

Question No:183

(Marks:1)

Vu-Topper RM

_____ is composed of thousands of interconnected networks expanded over large geographical area.

Local area network WAN

Question No:184

(Marks:1)

Vu-Topper RM

In frequency domain plot, which value is plot on Y-axis?

Amplitude

Question No:185

(Marks:1)

Vu-Topper RM

_____ signal completes a certain pattern in a specific amount of time.

Periodic

Question No:186

(Marks:1)

Vu-Topper RM

The baud rate is _____ to the bit rate.

Measure

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Question No:187

(Marks:1)

Vu-Topper RM

The material used for conduction in twisted pair cable is _____.

Copper

Question No:188

(Marks:1)

Vu-Topper RM

In _____ topology, each computer/station is attached to other through a central device.

Star topology

Question No:189

(Marks:1)

Vu-Topper RM

In frequency modulation, the frequency of the oscillator changes according to the _____.

Input Voltage

Question No:190

(Marks:1)

Vu-Topper RM

_____ is the heart of block coding.

Substitution

Google

Question No:191

(Marks:1)

Vu-Topper RM

In transmission impairments _____ noise occurs from other source like

Induced

Question No:192

(Marks:1)

Vu-Topper RM

Demultiplexer is a _____ device.

One to many

Question No:193

(Marks:1)

Vu-Topper RM

Low pass channel with _____ bandwidth is not real and is used for theoretical modeling

Narrow

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Question No:194 (Marks:1) **Vu-Topper RM**
_____ is NOT an example of connecting device.

TCP

Question No:195 (Marks:1) **Vu-Topper RM**
Parabolic antenna is used for_____ communication.

Microwave

Question No:196 (Marks:1) **Vu-Topper RM**
Token Ring was devised by_____.

IBM

Question No:197 (Marks:1) **Vu-Topper RM**
Induced noise Cable TV networks use_____ cables.

Coaxial

Question No:198 (Marks:1) **Vu-Topper RM**
Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are called_____.

Radio waves

Question No:199 (Marks:1) **Vu-Topper RM**
Most commonly used connector for twisted pair cable is _____.

RJ-45

Question No:200 (Marks:1) **Vu-Topper RM**
Binary Amplitude Shift Keying is also called as:

On-Off Keying

Question No:201 (Marks:1) **Vu-Topper RM**
_____ is a multiplexing technique which shifts each signal to a different carrier frequency.

FDM

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Question No:202 (Marks:1) **Vu-Topper RM**
in NRZ-1 the signal is inverted if _____ is encountered.

1

Question No:203 (Marks:1) **Vu-Topper RM**
_____ Protocol suite is being used by the modern internet communication.

TCP/IP

Question No:204 (Marks:1) **Vu-Topper RM**
In _____ signal changes its shape or form.

Distortion

Question No:205 (Marks:1) **Vu-Topper RM**
At the application layer, object/information is in the form of _____.

Segment

Question No:206 (Marks:1) **Vu-Topper RM**
_____ is the first step in PCM technique

Sampling

Question No:207 (Marks:1) **Vu-Topper RM**
A frequency is called _____ if sine wave does not change with time

Zero frequency

Question No:208 (Marks:1) **Vu-Topper RM**
Open system interconnection (OSI) has _____ number of layers

7

Question No:209 (Marks:1) **Vu-Topper RM**
 $N = S \times r$ is the formula for finding

Baud rate

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Question No:210

(Marks:1)

Vu-Topper RM

In context of bandwidth-delay product, the cross section of the pipe represents the _____.

Bandwidth

Question No:211

(Marks:1)

Vu-Topper RM

OSI stands for _____.

Open system interconnection

Question No:212

(Marks:1)

Vu-Topper RM

Writing used for transmission modes depends upon _____.

Data stream

Question No:213

(Marks:1)

Vu-Topper RM

Minimum bandwidth required for Manchester line coding scheme is ____.

1Mhz

Question No:214

(Marks:1)

Vu-Topper RM

Packets travel in _____ layer of TCP/IP protocol suit

Network

Question No:215

(Marks:1)

Vu-Topper RM

_____ protocol suit is being used by the modern internet communication

TCP/IP

Question No:216

(Marks:1)

Vu-Topper RM

Frames travel in _____ layer of TCP/IP Protocol suite.

Data link

Question No:217

(Marks:1)

Vu-Topper RM

There are ____ common scrambling technique

Two

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Question No:218

(Marks:1)

Vu-Topper RM

FSK stands for _____.

Frequency shift keying

Question No:219

(Marks:1)

Vu-Topper RM

Switch is a _____ layer devices

Data link

Question No:220

(Marks:1)

Vu-Topper RM

In PCM technique Quantization is an _____ process

Un-balanced

Question No:221

(Marks:1)

Vu-Topper RM

The layer of TCP/IP protocol suite which is on the top is known as _____ layer.

Application

Question No:222

(Marks:1)

Vu-Topper RM

Manchester encoding scheme uses and inversion at the _____ of each bit

Middle

Question No:223

(Marks:1)

Vu-Topper RM

High-Level Data Link Control(HDLC) defines _____ types of frames to deal with the transfer modes.

3

Question No:224

(Marks:1)

Vu-Topper RM

_____ are used to transport user data and control information relating to user data

Information frames

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Question No:225

(Marks:1)

Vu-Topper RM

_____ are used to exchange session management and control information between connected devices.

U-frames

Question No:226

(Marks:1)

Vu-Topper RM

The _____ is the number of signal elements sent per unit time.

Bit rate

Question No:227

(Marks:1)

Vu-Topper RM

Error correction is more _____ than the error detection

Difficult

Question No:228

(Marks:1)

Vu-Topper RM

One of the most common protocols for point-to-point access is _____.

PPP

Question No:229

(Marks:1)

Vu-Topper RM

Traditionally MANs have been implemented using one of the 2 technologies, Circuit Switching and Packet Switching

Ture

False

Question No:230

(Marks:1)

Vu-Topper RM

Which one is not the function of data link layer?

Line discipline

Question No:231

(Marks:1)

Vu-Topper RM

Is not a function of Data Link Control.

Modulation

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Question No:232

(Marks:1)

Vu-Topper RM

The _____ protocol uses both flow and error control.

Stop-and-Wait

Question No:233

(Marks:1)

Vu-Topper RM

The _____ technique expands the bandwidth of a signal by replacing each data bit with n bits using a spreading code.

DSSS

Question No:234

(Marks:1)

Vu-Topper RM

Digital signals are referred to be _____.

Discrete

Question No:235

(Marks:1)

Vu-Topper RM

The original Ethernet technology with the data rate of 10 Mbps is called _____?

Standard Ethernet

Question No:236

(Marks:1)

Vu-Topper RM

The extra bits added with the original data for error detection/correction are called _____.

Redundant bits

Question No:237

(Marks:1)

Vu-Topper RM

In _____, each station sends a frame whenever it has a frame to send.

Slotted ALOHA

Question No:238

(Marks:1)

Vu-Topper RM

In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as _____

Packetizing

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Question No:239

(Marks:1)

Vu-Topper RM

What is the period of a Sine wave having frequency of 5 Hz?

0.2

Question No:240

(Marks:1)

Vu-Topper RM

In Pure ALOHA, the vulnerable time is _____ the frame transmission time.

Two times

Question No:241

(Marks:1)

Vu-Topper RM

Analog refers to something that is continuous in _____.

Time

Question No:242

(Marks:1)

Vu-Topper RM

_____ encoding is almost obsolete today

Unipolar

Question No:243

(Marks:1)

Vu-Topper RM

Which of the following is most affected by noise?

ASK

Question No:244

(Marks:1)

Vu-Topper RM

_____ is sometimes called the bit rate.

Data rate

Question No:245

(Marks:1)

Vu-Topper RM

_____ contains a repeater.

Active hub

Question No:246

(Marks:1)

Vu-Topper RM

Asynchronous transmission is _____.

Slow

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Question No:247

(Marks:1)

Vu-Topper RM

In ASK, both _____ and _____ remain constant.

Frequency, Phase

Question No:248

(Marks:1)

Vu-Topper RM

DLC in Data Link Layer stands for _____.

Data Link Control

Question No:249

(Marks:1)

Vu-Topper RM

In _____, each station is allocated a time slot during which it can send data.

TDMA

Question No:250

(Marks:1)

Vu-Topper RM

Asynchronous TDM is efficient only when the size of the time slot is kept relatively _____.

Large

Question No:251

(Marks:1)

Vu-Topper RM

Human voice is example of _____ signal

Analog

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For More Important Notes
Channel Name = #VuTopperRM**

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	CS601-Data Communication Latest Solved Mcqs from Midterm Papers			May 07,2011
Lectures 1-22	Mc100401285	moaaz.pk@gmail.com	Moaaz Siddiq	Latest Mcqs

MIDTERM EXAMINATION Spring 2010 Question No: 1 (Marks: 1) - Please choose one Effective network mean that the network has fast delivery, timeliness and ➤ high bandwidth ➤ duplex transmission ➤ accurate transmission (P6) ➤ low bandwidth Question No: 2 (Marks: 1) - Please choose A set of devices connected by communication links is called networking ➤ True (P12) ➤ False Question No: 3 (Marks: 1) - Please choose one Which one best describes the given statement: "To allow multiple users to share total capacity of a transmission medium" ➤ congestion control ➤ exchange management ➤ multiplexing (P17) Question No: 4 (Marks: 1) - Please choose one Line configuration refers to the way two or more devices attach to a _____. ➤ Link (P25) ➤ Circuit ➤ Route ➤ Node Question No: 5 (Marks: 1) - Please choose one _____requires the maximum number of I/O ports. ➤ Bus ➤ Star				
---	--	--	--	--

- **Mesh (P29)**
- Ring

Question No: 6 (Marks: 1) - Please choose one

_____ is used to define the direction of the signal flow between the linked devices.

- **transmission mode (P34)**
- transmission method
- transmission impairments

Question No: 7 (Marks: 1) - Please choose one

Unidirectional traffic movement is overcome by dual ring technology.

- **True (P33)**
- False

Question No: 8 (Marks: 1) - Please choose one

Trailer is only added at _____ layer of OSI model.

- **Data link (P41)**
- Physical
- Network
- Application

Question No: 9 (Marks: 1) - Please choose one

Data link layer deals with mechanical and electrical specifications of transmission medium and interface

- **True (P44)**
- False

Question No: 10 (Marks: 1) - Please choose one

A signal that repeats a pattern over a regular interval of time is called_____.

- **periodic signal (P58)**
- analog signal
- composite signal

Question No: 11 (Marks: 1) - Please choose one

If there is _____ in voltage then the frequency is infinite.

- instantaneous change
- **continuous change (P61)**
- no change

Question No: 12 (Marks: 1) - Please choose one

In a time-domain plot, the horizontal axis is measure of _____.

- signal amplitude
- frequency
- phase
- **time (P63)**

Question No: 13 (Marks: 1) - Please choose

DC component is also termed as Direct current component or a component with non-zero frequency.

- **False (P72)**
- True

Question No: 14 (Marks: 1) - Please choose one

Quantization means measuring amplitudes of signal at equal intervals.

- True
- **False (P80)**

Question No: 15 (Marks: 1) - Please choose one

Amplitude in ASK is more resistive to EMI and Noise.

- **True (P87)**
- False

Question No: 16 (Marks: 1) - Please choose one

ASK, PSK, FSK and QAM are examples of _____ modulation.

- digital-to-digital
- **digital-to-analog (P85)**
- analog-to-analog
- analog-to-digital

Question No: 17 (Marks: 1) - Please choose one

The bit rate always equals the baud rate in which type of signal?

- **FSK (P86)**
- QAM
- 4-PSK
- PSK

Question No: 18 (Marks: 1) - Please choose one

Asynchronous transmission is _____

➤ **Slow (P99)**

- costly
- non-effective
- fast

Question No: 19 (Marks: 1) - Please choose one

A _____ converts an analog signal into a digital signal.

➤ **Demodulator (P110)**

- Modulator
- Digital-to-analog converter

Question No: 20 (Marks: 1) - Please choose one

If a station does not receive its signal up to a specified time that system issues an alarm to _____

➤ **another node (P188)**

- network manager
- switch
- network analyzer

MIDTERM EXAMINATION

Spring 2010

Question No: 1 (Marks: 1) - Please choose one

The information to be communicated in a data communications system is the

- ▶ Medium
- ▶ Protocol
- ▶ **Message (P 7)**
- ▶ Transmission

Question No: 2 (Marks: 1) - Please choose one

Which is not an element of protocol

- ▶ semantics
- ▶ timing
- ▶ **communication service module (P19)**

Question No: 3 (Marks: 1) - Please choose one

Line configuration refers to the way two or more devices attach to a _____.

▶ **Link (P 25)**

▶ Circuit

▶ Route

▶ Node

Question No: 4 (Marks: 1) - Please choose one

_____requires the maximum number of I/O ports.

▶ Bus

▶ Star

▶ **Mesh (P29)**

▶ Ring

Question No: 5 (Marks: 1) - Please choose one

Which topology requires a central controller or hub?

▶ Mesh

▶ **Star (P 30)**

▶ Bus

▶ Ring

Question No: 6 (Marks: 1) - Please choose one

Layers 5, 6 and 7 also called as network support layers.

▶ True

▶ **False (P 42)**

Question No: 7 (Marks: 1) - Please choose one

Internet with small “i” specifies the world wide Network the actual internet.

- ▶ True
- ▶ False (P 39)

Question No: 8 (Marks: 1) - Please choose one

Data link layer deals with mechanical and electrical specifications of transmission medium and interface

- ▶ True (P 44)
- ▶ False

Question No: 9 (Marks: 1) - Please choose one

Data synchronization is a function related with _____

- ▶ session layer (P 51)
- ▶ presentation layer
- ▶ transport

Question No: 10 (Marks: 1) - Please choose one

OSI model was developed prior to TCP/IP model.

- ▶ True (P 54)
- ▶ False

Question No: 11 (Marks: 1) - Please choose one

Fourier transform tells us that any digital signal can be decomposed into infinite number of periodic signals

- ▶ True
- ▶ False (P58)

Question No: 12 (Marks: 1) - Please choose one

A signal that repeats a pattern over a regular interval of time is called_____.

▶ **periodic signal (P58)**

▶ analog signal

▶ composite signal

Question No: 13 (Marks: 1) - Please choose one

Frequency band in which 99% of the total power resides is called_____

▶ **power bandwidth (P 67)**

▶ half power bandwidth

▶ 3dB bandwidth

▶ F.M.

Question No: 14 (Marks: 1) - Please choose one

In differential Manchester we require two transition or signal changes to represent binary_____

▶ zero

▶ **one (P76)**

▶ both zero and one

Question No: 15 (Marks: 1) - Please choose one

In Alternate mark inversion the term mark comes from_____

▶ **telegraphy (P77)**

▶ telephony

▶ digital telephony

Question No: 16 (Marks: 1) - Please choose one

A 56k modem can download at a rate of _____ Kbps and upload at a rate of _____ Kbps.

▶ 33.6: 33.6

▶ **33.6: 56.6**

▶ 56.6: 33.6

▶ 56.6: 56.6

Question No: 17 (Marks: 1) - Please choose one

The bit rate always equals the baud rate in which type of signal?

▶ **FSK (P 86)**

▶ QAM

▶ 4-PSK

▶ PSK

Question No: 18 (Marks: 1) - Please choose one

Asynchronous transmission is _____

▶ **slow (P 99)**

▶ costly

▶ non-effective

▶ fast

Question No: 19 (Marks: 1) - Please choose one

At the _____ layer, a DCE takes data generated by a DTE.

- ▶ **physical (P 101)**
- ▶ transport
- ▶ data link
- ▶ application

Question No: 20 (Marks: 1) - Please choose one
_____ requires more bandwidth.

- ▶ FSK
- ▶ **ASK (P 112)**
- ▶ PSK
- ▶ QAM

MIDTERM EXAMINATION Spring 2010

Question No: 1 (Marks: 1) - Please choose one
Which topology requires a central controller or hub?

- ▶ Mesh
- ▶ **Star (P 30)**
- ▶ Bus
- ▶ Ring

Question No: 2 (Marks: 1) - Please choose one
Unidirectional traffic movement is overcome by dual ring technology.

- ▶ **True (P33)**
- ▶ False

Question No: 3 (Marks: 1) - Please choose one

Which one best describes the given statement:

"To allow multiple users to share total capacity of a transmission medium"

- ▶ congestion control
- ▶ exchange management
- ▶ **multiplexing (P17)**

Question No: 4 (Marks: 1) - Please choose one

Which one is not among standard creation committee.

- ▶ **internet society and IETF (P 24)**
- ▶ ITU-T
- ▶ IEEE

Question No: 5 (Marks: 1) - Please choose one

Set of rules that govern communication is called

- ▶ OSI model
- ▶ **Protocol (P 8)**
- ▶ TCP/IP model
- ▶ Datagram

Question No: 6 (Marks: 1) - Please choose one

Frequency of a failure and network recovery time after a failure is measures of the _____ of a network.

- ▶ **Performance (P 14)**
- ▶ Reliability
- ▶ Security
- ▶ Feasibility

Question No: 7 (Marks: 1) - Please choose one

3dB is also called as power bandwidth.

▶ True

▶ **False (P66)**

Question No: 8 (Marks: 1) - Please choose one

ASK, PSK, FSK and QAM are examples of _____ modulation.

▶ digital-to-digital

▶ **digital-to-analog (P85)**

▶ analog-to-analog

▶ analog-to-digital

Question No: 9 (Marks: 1) - Please choose one

Packet switching is a connection oriented service.

▶ True

▶ **False (P38)**

Question No: 10 (Marks: 1) - Please choose one

Data synchronization is a function related with _____

▶ **session layer (P51)**

▶ presentation layer

▶ transport

Question No: 11 (Marks: 1) - Please choose one

Fourier transform tells us that any digital signal can be decomposed into infinite number of periodic signals

▶ True

▶ False (P 58)

Question No: 12 (Marks: 1) - Please choose one

The internet model consists of _____ layers.

▶ three (P239)

▶ two

▶ five

▶ seven

Question No: 13 (Marks: 1) - Please choose one

Analog to digital conversion is also termed as modulating an analog signal.

▶ True

▶ False (P 77)

Question No: 14 (Marks: 1) - Please choose one

PCM is the first process of PAM.

▶ True

▶ False (P80)

Question No: 15 (Marks: 1) - Please choose one

A modulator converts a(n) _____ signal to a(n) _____ signal.

- ▶ FSK: PSK
- ▶ **Digital: analog (P 110)**
- ▶ Analog: digital
- ▶ PSK: FSK

Question No: 16 (Marks: 1) - Please choose one

The _____ layer changes bits into electromagnetic signals.

- ▶ **physical**
- ▶ data link
- ▶ transport
- ▶ none of given

http://www.ee.surrey.ac.uk/Projects/CAL/networks/Physical_Layer.htm

Question No: 17 (Marks: 1) - Please choose one

We need _____ to decompose a composite signal into its components.

- ▶ **fourier transform (P 64)**
- ▶ nyquist theorem
- ▶ shannon capacity

Question No: 18 (Marks: 1) - Please choose one

The bit rate always equals the baud rate in which type of signal?

- ▶ **FSK (P 86)**
- ▶ QAM

► 4-PSK

► PSK

Question No: 19 (Marks: 1) - Please choose one

There are _____ possible variations of 16 QAM.

► five

► two

► **three (P93)**

► four

Question No: 20 (Marks: 1) - Please choose one

Number of signal units per second that are required to represent a bit is called bit rate.

► True

► **False (P 68)**

MIDTERM EXAMINATION Spring 2010

Please choose one (Marks: 1)

Effective network mean that the network has fast delivery, timeliness and

- high bandwidth
- duplex transmission
- **accurate transmission (P6)**
- low bandwidth

Question No: 2 (Marks: 1) - Please choose one

A set of devices connected by communication links is called networking

- **True (P12)**
- False

Question No: 3 (Marks: 1) - Please choose one
_____ is a multipoint topology.

- Ring
- Mesh
- Tree
- **Bus**

Question No: 4 (Marks: 1) - Please choose one
_____ requires the maximum number of I/O ports.

- Bus
- Star
- **Mesh (P29)**
- Ring

Question No: 5 (Marks: 1) - Please choose one
Unidirectional traffic movement is overcome by dual ring technology.

- **True (P33)**
- False

Question No: 6 (Marks: 1) - Please choose one
As the data packet moves from the lower to the upper layers, headers are

- _____.
- added
 - **subtracted**
 - rearranged
 - modified

Question No: 7 (Marks: 1) - Please choose one
Data chunk at data link layer is called _____

- frame
- **packet (P38)**
- datagram

Question No: 8 (Marks: 1) - Please choose one
The _____ layer is the layer closest to the transmission medium.

- **Physical (P45)**
- data link
- network
- transport

Question No: 9 (Marks: 1) - Please choose one
Data synchronization is a function related with _____

- **session layer (P 51)**
- presentation layer
- transport

Question No: 10 (Marks: 1) - Please choose one

Which layer is responsible for source to destination delivery of entire message?

- **transport layer (P59)**
- network layer
- session layer
- application layer

Question No: 11 (Marks: 1) - Please choose one

Encryption and encoding are the same terms.

- True
- **False (P52)**

Question No: 12 (Marks: 1) - Please choose one

TCP/IP model Application layer incorporate OSI Application and presentation

- layers
- **True**
- False

Question No: 13 (Marks: 1) - Please choose one

A sine wave must be an analog signal.

- **True (P58)**
- False

Question No: 14 (Marks: 1) - Please choose one

A periodic signal can always be decomposed into _____

- exactly an odd number of sine waves
- a set of sine waves
- set of sine waves, one of the which must have a phase of 0o
- **none of the given (P58)**

Question No: 15 (Marks: 1) - Please choose one

Manchester is a type of _____ encoding.

- **Biphase (P75)**
- polar
- biphase & polar
- none of the given

Question No: 16 (Marks: 1) - Please choose one

The powerful carrier signal is called as the _____

- **carrier frequency (P86)**
- base signal
- carrier frequency and base signal
- none of given

Question No: 17 (Marks: 1) - Please choose one

Modulation of an analog signal can be accomplished through changing the _____ of the carrier signal.

- amplitude
- frequency
- phase
- **all of the given (P58)**

Question No: 18 (Marks: 1) - Please choose one

In _____ transmission, bits are transmitted over a single wire, one at a time.

- Asynchronous serial
- Synchronous serial
- **Parallel (P99)**
- Asynchronous & Synchronous serial

Question No: 19 (Marks: 1) - Please choose one

In RS 422 Balanced mode two lines carry _____ signals which are not identical to each other.

- **Same (P109)**
- different
- digital
- analog

Question No: 20 (Marks: 1) - Please choose one

A _____ converts an analog signal into a digital signal.

- **Demodulator (P110)**
- Modulator
- Digital-to-analog converter

MIDTERM EXAMINATION

Spring 2009

Question No: 1 (Marks: 1) - Please choose one

In RS 422 Balanced mode two lines carry _____ signals which are not identical to each other.

- ▶ **same (P109)**
- ▶ different
- ▶ digital
- ▶ analog

Question No: 2 (Marks: 1) - Please choose one

The _____ generates the data and passes it along with any control information to a _____.

- ▶ **DTE; DCE (P 101)**
- ▶ DCE; DTE
- ▶ DCE; node
- ▶ DTE; application

Question No: 3 (Marks: 1) - Please choose one

In _____ transmission, a start bit and a stop bit frame a character byte.

- ▶ **Asynchronous serial (P99)**
- ▶ Synchronous serial (P99)
- ▶ Parallel
- ▶ Asynchronous & Synchronous serial

Question No: 4 (Marks: 1) - Please choose one

In _____ transmission, bits are transmitted simultaneously, each across its own wire.

- ▶ Asynchronous serial
- ▶ Synchronous serial
- ▶ Parallel
- ▶ Asynchronous & Synchronous serial

Question No: 5 (Marks: 1) - Please choose one

Asynchronous transmission is _____

- ▶ **slow (P99)**
- ▶ costly
- ▶ non-effective
- ▶ fast

Question No: 6 (Marks: 1) - Please choose one

In _____ transmission the timing of the signal is unimportant.

▶ **Asynchronous (P99)**

- ▶ Synchronous
- ▶ Polar
- ▶ Bi-polar

Question No: 7 (Marks: 1) - Please choose one

In 8QAM each signal shift or one baud represents _____

- ▶ 4 bits
- ▶ 2 bits
- ▶ 5 bits
- ▶ **3 bits (P93)**

Question No: 8 (Marks: 1) - Please choose one

PCM is the first process of PAM.

- ▶ **True**
- ▶ False

Question No: 9 (Marks: 1) - Please choose one

Bi phase encoding is a type of bipolar encoding in which we use two voltage levels.

- ▶ **True (P73)**
- ▶ False

Question No: 10 (Marks: 1) - Please choose one

The amplitude of a digital signal depends upon the_____to represent a bit.

- ▶ phase
- ▶ voltage
- ▶ wavelength

Question No: 11 (Marks: 1) - Please choose one

Time domain plot show changes in signal phase with respect to time.

- ▶ True
- ▶ **False (P63)**

Question No: 12 (Marks: 1) - Please choose one

OSI model was developed prior to TCP/IP model.

- ▶ True
- ▶ **False(P54)**

Question No: 13 (Marks: 1) - Please choose one

_____layer is the layer closest to the transmission medium.

- ▶ **Physical (P45)**
- ▶ Data link
- ▶ Network
- ▶ Transport

Question No: 14 (Marks: 1) - Please choose one

Data chunk at data link layer is called_____

- ▶ frame
- ▶ **packet (P38)**
- ▶ datagram

Question No: 15 (Marks: 1) - Please choose one

The internet model consists of _____layers.

- ▶ **Three (P239)**
- ▶ Two
- ▶ Five
- ▶ Seven

Question No: 16 (Marks: 1) - Please choose one

_____representation of links that connect nodes is called as physical topology.

- ▶ **geometrical (P28)**
- ▶ logical
- ▶ physical

Question No: 17 (Marks: 1) - Please choose one

_____ is a multipoint topology.

- ▶ Ring
- ▶ Mesh
- ▶ Tree
- ▶ **Bus (P31)**

Question No: 18 (Marks: 1) - Please choose one

In mesh topology, if there are five nodes then there will be _____ links.

- ▶ 5
- ▶ **10(P10)**
- ▶ 15
- ▶ 20

Question No: 19 (Marks: 1) - Please choose one

Non proprietary standard is also termed as de facto standard.

- ▶ **True (P23)**
- ▶ False

Question No: 20 (Marks: 1) - Please choose one

Effective network mean that the network has fast delivery, timeliness and

- ▶ high bandwidth
- ▶ duplex transmission
- ▶ **accurate transmission (P6)**
- ▶ low bandwidth

MIDTERM EXAMINATION

Spring 2009

Question No: 1 (Marks: 1) - Please choose one

Frequency of a failure and network recovery time after a failure is measures of the _____ of a network.

- **Performance (P 14)**
- Reliability
- Security
- Feasibility

Question No: 2 (Marks: 1) - Please choose one
Secondary hub in a tree must be a passive hub.

- **True (P31)**
- False

Question No: 3 (Marks: 1) - Please choose one
In mesh topology, if there are five nodes then there will be _____ links.

- 5
- **10 (P29)**
- 15
- 20

Question No: 4 (Marks: 1) - Please choose one
Internet with small "i" specifies the world wide Network the actual internet.

- True
- **False (P39)**

Question No: 5 (Marks: 1) - Please choose one
Data link layer deals with mechanical and electrical specifications of transmission medium and interface

- **True (P 44)**
- False

Question No: 6 (Marks: 1) - Please choose one
When data are transmitted from device A to device B, the header from A's layer 4 is read by B's _____ layer.

- physical
- **transport**
- application
- none of the given

Question No: 7 (Marks: 1) - Please choose one
Time domain plot show changes in signal phase with respect to time.

- True
- **False (63)**

Question No: 8 (Marks: 1) - Please choose one
Frequency band in which 99% of the total power resides is called _____

- **power bandwidth (P67)**
- half power bandwidth
- F.M.

Question No: 9 (Marks: 1) - Please choose one

The inversion of the level at 1 bit is called as _____

- NRZ-L
- **NRZ-I (P73)**
- RZ

Question No: 10 (Marks: 1) - Please choose one

Which of the encoding schemes have bandwidth problems?

- Differential Manchester
- **AMI (P77)**
- RZ

Question No: 11 (Marks: 1) - Please choose one

Bi phase and bipolar are same terms.

- True
- **False (P78)**

Question No: 12 (Marks: 1) - Please choose one

Quantization means measuring amplitudes of signal at equal intervals.

- True
- **False (P80)**

Question No: 13 (Marks: 1) - Please choose one

The last process in PCM is _____ digital data into digital signal.

- **Encoding (P70)**
- decoding
- modulating

Question No: 14 (Marks: 1) - Please choose one

Which modulation technique involves tribits, eight different phase shifts, and one mplitude?

- FSK
- 8-PSK
- ASK
- 4-PSK

Question No: 15 (Marks: 1) - Please choose one

Asynchronous transmission is _____

- **slow (P99)**
- costly
- non-effective
- fast

Question No: 16 (Marks: 1) - Please choose one

Synchronous transmission have _____.

- a start bit
- a stop bit
- gaps between bytes
- **none of the given (P99)**

Question No: 17 (Marks: 1) - Please choose one Asynchronous transmission
The _____ generates the data and passes it along with any control information to a _____.

- **DTE; DCE (P101)**
- DCE; DTE
- DCE; node
- DTE; application

Question No: 18 (Marks: 1) - Please choose one
A human brain is considered as a _____

- driver
- machine
- **DTE (P101)**
- DCE

Question No: 19 (Marks: 1) - Please choose one
The last modified version of EIA-232 standard is _____

- **EIA-232 D (P102)**
- EIA-232 A
- EIA-232 B
- EIA-232 C

Question No: 20 (Marks: 1) - Please choose one
A _____ converts an analog signal into a digital signal.

- **Demodulator (P110)**
- Modulator
- Digital-to-analog converter

Moaaz.pk@gmail.com

Cs601 Mid term

Topic 1 to 113 cover in this quiz

Orange Monkey Team

1. At the application layer, object/information is in the form of ____.

Message...confirm

2. _____ is the first step in PCM technique

Sampling...confirm

3. Communication between two user in a cellular (Mobile) network is an example of __ communication

Full duplex....confirm

4. Transmission medium are of _____ types.

Two....confirm

5. A frequency is called ____ if sine wave changes with time

Infinte frequency...confirm

6. A frequency is called ____ if sine wave does not change with time

Zero frequencyconfirm

7. Open system interconnection (OSI) has __ number of layers.

7...confirm

8. In unidirectional antenna, the common point where all the lines intersect each other is ____.

Focus...confirm

9. To calculate the data rate for noisy channel ____ formula is used.

Shannon....confirm

10. Television broadcast where user can only view the information sent in the form of video is an example of ____ communication.

Simplex

11. $N = S \cdot r$ is the formula for finding

Baud rate.....confirm

12. TCP/IP protocol suit has ____ layers.

5....confirm

13. Unit of phase is ____.

Degree....confirm

14. For a noiseless channel theoretical formula to calculate the data rate was developed by.

Nyquist.....confirm

15. In context of bandwidth-delay product the cross section of the pipe represents the ____.

Bandwidth...confirm

16. Uni polar Polar and Bipolar are types of ____ coding scheme.

Line...confirm

17. In ____ scheme the voltages are on both sides of the time axis and voltage level for 0 can be positive and the voltage level for 1 can be negative.

Polar....confirm

18.In _____ transmission medium start and stop bits are used

Asynchronous....confirm from net

19.In layered architecture of network logical connection between layers means.

They are connected in peers.....confirm

20.OSI stands for _____.

Open system interconnection....confirm

21.Quadrature Amplitude Modulation is the mechanism of _____ conversion.

Digital to analog...confirm

22.Writing used for transmission modes depends upon _____.

Data stream...confirm

23.The maximum bandwidth required for QAM transmission is the same as that required for _ and _____ transmission.

ASK, PSK

24.In optical fiber cable the ratio of the diameter of their core to the diameter of their cladding, both expressed in _____.

Micrometers...confirm

25.Radio waves are propagated in _____ whenever they are transmitted by an antenna.

All direction ...confirm from net

26._____ component of data communication governs the overall communication on the network.

Transmission medium...confirm

27. In binary ASK, the peak amplitude of one signal level is 0 and the other is the same as the _____.

Amplitude

28. In _____ scheme, all the signal levels are on one side of the time axis, either above or below.

Unipolar...confirm from net

29. Simple sine wave can be used to send.

Amplitude

30. Paging system in data communication uses _____ waves.

Radio waves...confirm from net

31. Multipoint channel can be shared in _____ manner.

Both temporal and spartial...confirm

32. _____ is an example of multipoint topology

Bus...confirm

33. Bandwidth can be measured in _____.

Bps...confirm

34. In _____ encoding scheme, we use three levels positive negative and zero

Bipolar...confirm from net

35. Header size _____ while moving down the layers at sending side

Decreases

36. _____ signals have limited number of discrete value

Digital....confirm

37. _____ is a block coding method

NRZ-1....confirm

38. Technique that allows the simultaneous transmission of multiple signals across a single data link, is known as _____.

Multiplexing....confirm

39. Simultaneous transmission of multiple signals across a single data link is called.

Multiplexing...confirm

40. Minimum bandwidth required for Manchester line coding scheme is _____.

1Mhz...confirm

41. AMI is a popular _____ encoding method.

Bi-polar....confirm

42. TCP/IP was presented by _____ in a research paper

Vin Cerf and bob Kahn...confirm

43. Among the following cables given below _____ cable has the cost as compared to others.

Medium....confirm

44. Among the following cables given below _____ cable has the lowest cast as compared to others.

Coaxial cable...confirm

45. Among the following cables given below _____ cable has the highest cast as compared to others.

Fiber optic cable...confirm

46. _____ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

n-1....confirm

47.A twisted pair cable consists of _____ conductors

2confirm

48.Packets travel in _____ layer of TCP/IP protocol suit

Networkconfirm from net

49.HDB3 falls under which coding scheme?

Bipolar...confirm

50.According to the Nyquist theorem, the sampling rate must be at least _____ times the highest frequency contained in the signal.

2.....confirm

51. _____ protocol suit is being used by the modern internet communication

TCP/IP...confirm

52.LF(low frequency) band is used in _____ propagation

Ground....confirm

53. _____ is a type of serial transmission in which data bits are transmitted as a continuous stream in time with a master clock, start bits, stop bits and gaps are not used. Transmitter and receiver time is sync.

Synchronous....confirm

54.Identify the correct formula to calculate Propagation time.

Propagation time= Distance/ propagation speed....confirm

55.The communication of the application, transport and network layer is _____

Physical...confirm

56.Frames travel in _____ layer of TCP/IP Protocol suite.

Data link

57.Manchester encoding scheme achieves the same level of synchronization as _____.

RZ.....confirm

58.In Manchester and differential Manchester encoding schemes, the signal rate is _____ than that of NRZ.

Double...confirm from net

59.In a bidirectional communication each layer performs two _____ tasks in each direction.

Opposite...confirm

60._____ is composed of thousands of interconnected networks expanded over large geographical area.

Local area network

61.Time taken by a periodic signal to complete one cycle is called.

Period...confirm from net

62.There are __ common scrambling technique

Twoconfirm

63.The bipolar encoding scheme was developed as an alternative of _____.

NRZ...confirm from net

64.FSK stands for _____.

Frequency shift keying....confirm

65. In ___ encoding scheme, the voltage level oscillates between a positive and a negative value although it may remain at zero level between the two values.

Polar...confirm from net

66. _____ is an example of transmitter.

Modem

67. The material used for conduction in twisted pair cable is _____.

Copper

68. RG-11 having impedance of 50 ohm is used in _____ Ethernet

Thin....confirm

69. _____ encoding scheme is somewhat a combination of the RZ and NRZ-I scheme

Manchester encoding...confirm from net

70. _____ has less electromagnetic interference as compared to other guided medium.

Unshielded Twisted Pair

71. Electromagnetic waves ranging in frequencies between _____ are called microwaves.

1 to 300 GHz

72. Switch is a _____ layer devices

Data link...confirm

73. _____ is Shannon capacity formula to determine highest theoretically data rate for a noisy channel.

Capacity = Bandwidth * $\log_2(1+SNR)$confirm

74. _____ is the range of frequencies contained in a composite signal.

Bandwidth...confirm

75. In PCM technique Quantization is an _____ process

Un-balanced

76. In Ring topology, each node has a point to point dedicated link with.

Exactly 2 nodes....confirm

77. In _____ modulation process is used by the transmitting devices to shift their data to a specific carrier frequency for multiplexing.

Fdm...confirm

78. Five stations will have _____ full duplexed dedicated links among them to be connected in a Mesh topology.

10....confirm

79. Which of the following technique(s) is used for Analog-to-Digital Conversion.

Both Pulse Code Modulation(PCM) and Delta Modulation(DM)...confirm

80. The human voice normally frequency range from _____.

0 to 3000Hz

81. The layer of TCP/IP protocol suite which is on the top is known as _____ layer.

Application...confirm

82. In _____ type of error, two or more bits in the data unit will be changed.

Burst....confirm

83. _____ is the process of converting digital data to digital signal.

Line coding.....confrim

84. Manchester encoding scheme uses and inversion at the _____ of each bit.

Middle....confirm

85. FF:FF:FF:FF:FF:FF is a _____ link layer address.

Broadcast.....confrim

86. A3 34 45 11 92 F1 is a _____ link layer address.

Data, unicast....confirm

87. High-Level Data Link Control(HDLC) defines _____ types of frames to deal with the transfer modes.

3.....confirm

88. _____ is the number of bits sent in 1 second

Bit rate(bps)...confirm

89. _____ Time required to send 1 single bit.

Bit interval...confirm

90. _____ are used to transport user data and control information relating to user data

Information frams(I-frames).....confirm

91. _____ are used only to transport control information

Supervisory frames(S-frames).....confirm

92. _____ are reserved for system Management

Unnumbered frames(U-frames).....confirm

93. _____ are used to exchange session management and control information between connected devices.

U-frames....confrim

94. In order to find Hamming distance between two pair of words _____ logical operation is used.

XOR....confrim

95. Data Link layer receives services from the _____ layer.
Physical

96. The _____ is the number of signal elements sent per unit time.

Bit rate.....confrim

97. In order to resolve link layer address of the next node, the first generated ARP request is always in _____ mode.

Broadcast

98. In Block coding we divide our message into blocks of k bits called as _____.

Data words.....confirm

99. To control and manage the transfer of data, a protocol called point to point (PPP) is used at the _____ layer

Data Link

100. In character oriented protocol, each character of frame is encoded using _____ coding system.

ASCII

101. Error detection and correction are the services provided by _____ layer.

Data link layerconfirm from net

102. _____ is used in the stop-and-wait protocol to overcome the issue of duplication.

Frame sequencing and Acknowledgment numbering...confirm from net

103. Stop-and-weight protocol is used for _____.

Both flow and error control....confirm

104. Error correction is more _____ than the error detection
Difficult...confirm from net

105. Payload field of the PPP frame is of variable size and it can carry maximum data up to _____.

1500 bytes....confirm from net

106. If duration of noise is decreased during a transmission over noisy channel. Fixed data rate of the channel will cause _____ number of bits being impacted by the noise.

More....confirm from net

107. One of the most common protocols for point-to-point access is _____.

PPP...confirm

108. In linear Block codes _____ of two valid code words creates another valid code word.

XOR...confirm

109. Which layer is concerned with the syntax and semantics of info exchange between two systems?

Presentation

110. Traditionally MANs have been implemented using one of the 2 technologies, Circuit Switching and Packet Switching

false

111. In OSI model headers are added at layer _____.

6,5,4,3,2 only

112. _____ works with topologies in which one device is designated as a primary station and the other devices are secondary stations.

Polling

113. Signals travel through fiber optic cable are in the form of _____.

Light...confirm from net

114. PSK is susceptible to the noise degradation.

False

115. What is the frequency of a sine wave if it completes one cycle in 5 seconds?

0.2

116. Which one is not the function of data link layer?

Line discipline

117. Error detection and correction are the services provided by _____ layer.

Data link layer

118. _____ is not a function of Data Link Control.

Modulation

119. No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between _____.

Two adjacent nodes

120. The inversion of the level represents a 1 bit in _____ encoding

NRZ-I

121. Guard bands are used in _____ multiplexing technique, guard bands to avoid overlapping of frequency bands assigned to each user.

FDM...confirm

122. The _____ protocol uses both flow and error control.

Stop-and-Wait

123. Each computer on the Internet has a unique numeric address called a(n) _____.

IP address

124. WDM stands for _____.

Wave Division Multiplexing

125. A fiber -optic cable transmit signals in the form of _____.

Light

126. CDMA stands for _____.

Code Division Multiple Access

127. Quantization is a method of assigning _____ values in a specific range to sampled instances.

Integral

128. STP is more expensive than UTP and is less susceptible to noise.

True

129. In circuit switched networks we have low efficiency but minimal _____.

Delay....confirm

130. Which level of the TCP/IP reference model routes data/information across a network channel?

Network Layer

131. A portion of the path that carries TX between a given pair of devices is known as _____.

Channel

132. Digital signals are referred to be _____.

Discrete

133. BSC supports _____ transmission using _____ flow control.

Half-duplex, Stop and wait ARQ

134. _____ is the rate of change with respect to time.

Amplitude

135. The original Ethernet technology with the data rate of 10 Mbps is called _____?

Standard Ethernet

136. If we need to correct a single error in an 8-bit data unit, we need to consider _____ possible error locations.

16

137. In Pulse Code Modulation (PCM), the sampling is dependent on _____.

Amplitude...confirm

138. The extra bits added with the original data for error detection/correction are called _____.

Redundant bits

139. Which of the following sublayer resolves the contention for the shared media.

MAC

140. PAP and CHAP are used by _____ as Authentication Protocols

PPP...confirm

141. Media Access Control (MAC) and Data link (DLC) constitute _____ layer

Data Link....confirm

142. In _____, each station sends a frame whenever it has a frame to send.

Slotted ALOHA

143. What is the period of a Sine wave having frequency of 10 Hz?

0.1

144. Is there any difference between Data communication and telecommunication are network?

No

145. In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as _____.

Packetizing

146. _____ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

$N-1$

147. Following characteristics fully describe _____,
Amplitude
Period/Frequency
Phase

Sine Waves

148. What is the period of a Sine wave having frequency of 5 Hz?

0.2

149. The inversion of the level at 1 bit is called as _____.

NRZ-I

150. In _____, both peak amplitude and frequency remains constant as the phase changes.

PSK

151. At the CRC generator, _____ added to the data unit before the divisional process.

0s are

152. In Pure ALOHA, the vulnerable time is _____ the frame transmission time.

Two times

153. In Block coding we divide our message into blocks of k bits called as _____.

Data words

154. Analog refers to something that is continuous in _____.

Time

155. _____ of a signal is the collection of all the component frequencies it contains.

Frequency spectrum

156. _____ encoding is almost obsolete today

Unipolar

157. Which of the following is most affected by noise?

ASK

158. In _____, a signal is directed straight from antenna to antenna.

Line of sight propagation

159. To control and manage the transfer of data, a protocol called Point to Point Protocol (PPP) is used at the _____ layer.

Network

160. The level of signal is always positive in NRZ encoding.

false

161. _____ signal can take infinite levels of intensity over time.

Analog....confirm from net

162. _____ is sometimes called the bit rate.

Data rate...confirm from net

163. _____ contains a repeater.

Active hub

164. _____ is/are un-reliable protocol(s).

UDP

165. Low pass channel has _____ bandwidth between two stations.

Shared...confirm from net

166. The message 1110 sent by a source is received by a destination as 1011. This is _____ type of error.

Burst

167. TROPOSHERE is the layer of atmosphere above the troposphere but below space

False

168. In Pure ALOHA, the vulnerable time is _____ the frame transmission time.

Two times

169. Asynchronous transmission is _____.

Slow

170. Radio is an example of _____ signal conversion.

Analog to analog

171. Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?

Longitudinal Red Check (LRC)

172. Session layer is responsible for _____.

Maintaining the connection

173. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause _____ number of bits being impacted by the noise.

More

174. S-frames in High-Level Data Link Control (HDLC) are only used to transfer _____ information.

Control

175. In ASK, both _____ and _____ remain constant.

Frequency, Phase

176. Encoding digital data into digital signals is called _____.

Digital-to-Digital Conversion

177. GO-BACK-N protocol is one of the _____ layer protocols, which is used to control the flow of data during the transmission.

Network

178. In ASK correct formula for calculating the bandwidth is as _____.

$B=(1+d)S$confirm

179. The type of ALOHA which improves the efficiency of Pure ALOHA is _____.

Slotted ALOHA

180. In which controlled-access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device?

Polling

181. The _____ generates the data and passes it along with any control information to a _____.

DTE, DCE

182. _____ is used in the Stop-and-Wait protocol to overcome the issue of duplication.

Frame sequencing and Acknowledgement numbering

183. Transmission Control Protocol is a(n) _____ layer protocol.

Network

184. The Internet is _____.

a global network of computers networks

185. We can have _____ different groups with a 4 bit block.

8

186. Analog To Analog Conversion Methods include _____.

AM, PM and FM

187. DLC in Data Link Layer stands for _____.

Data Link Control

188. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 200, 400, 600 and 800?

600

189. _____ category of coaxial cable is used for thin Ethernet.

RG-58....confirm

190. _____ category of coaxial cable is used for thin Ethernet.

RG-8, RG-9, RG-11confirm

191. _____ category of coaxial cable is used for thin Ethernet.

RG-8, RG-9, RG-11confirm

192. _____ category of coaxial cable is used for Cable TV

RG59....confirm

193. In _____, each station is allocated a time slot during which it can send data.

TDMA

194. _____ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.

10BASE-T Ethernet

195. Time domain plot show changes in signal phase with respect to time.

false

196. dB is negative if a signal is _____.

Attenuated

197. Which of the following is not a characteristic of a sine wave?

Segmentation...confirm

198. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 300, 600, 900 and 1200?

900

199. Asynchronous TDM is efficient only when the size of the time slot is kept relatively _____.

Large

200. A sine wave is defined by _____ characteristics.

3

Switch is a network device which operates on the _____ layer of the TCP/IP protocol suite.

Data link

201. In even parity check code, the value of syndrome is _____ if the number of 1s is even.

0

202. 4Collisions in CSMA/CA are avoided through _____.

All of the given

203. Human voice is example of _____ signal.

Analog

204. Time domain shows changes in _____.

signal amplitude w.r.t time

205. Normally the value of Kmax in pure ALOHA is _____.

15

206. Central Hub in a Tree is an _____.

Active hub

207. In selective-reject ARQ, only the specific damaged or lost frame is _____.

Retransmitted

208. If users must take turns using the link, then it is called Spatial Sharing.

false

209. At the CRC checker, _____ means that the data unit is damaged.

A nonzero remainder

210. Digitizing an Analog Signal is called _____.

Analog-to-Digital Conversion

211. WDM stands for _____.

Wavelength Division Multiplexing

212. What was the main draw back in ALOHA Random-Access method?

All of the given

213. YMODEM has _____ Byte of data unit.

1024

214. Like Data link layer, _____ is also responsible for Flow control.

a. Transport Layer

215. In Block coding scheme, number of code words is always _____ data words.

Equal to

216. RZ stands for _____.

Return of Zero

217. Line Configuration refers to the way two or more devices attach to a _____.

Link

218. The flag in PPP is a byte that needs to be _____ whenever it appears in the data section of the frame.

Escaped

219. At the Application layer, object/information is in the form of _____.

Packet

220. The term _____ means that only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.

Single Bit Error

221. What is the frequency of a sine wave if it completes one cycle in 2 seconds?

0.5

222. POST stand for _____?

Plain Old Telephone Service

223. To make sure that source does not overwhelm destination by sending data faster than it can be handled and processed is called?

Flow Control

224. If a digital signal has "L" number of levels. _____ is the number of bits required to represent each level.

Log base 2 L....confirm

225. In ASK, Noise usually affects the _____.

Amplitude

226. In order to find the location of two errors in an eight bit stream, we have to see _____ different combinations.

28

227. Which one of the following is not a Channelization Protocol?

CSMA

228. Which one of the following is a correct-access method?

Reservation

229. By using _____ system, change in the file contents during file transmission can be avoided.

Error Detection and Correction

230. Radio wave transmission utilizes _____ different types of propagation.

Five

231. A broadcast signal is received by the orbiting satellites which rebroadcasts _____.

none of the given

232. What is the period of a Sine wave having frequency of 2 Hz?

0.5

233. _____ mode of serial transmission guarantees fixed rate of data.

Isochronous...confirm from net

234. Time division multiplexing is used in _____ systems.

Digital....confirm from net

235. The most common type of connector used by coaxial cable is _____.

BNC

236. Which of the following statement is correct?

Digital signals are less prone to Noise

237. In an analog hierarchy to carry voice channels, a group can carry _____ voice channels.

12(4kHz)....confirm

238. In an analog hierarchy to carry 12 voice channels, 5 group _____

48kHz....confirm

239. In an analog hierarchy to carry 60 voice channels, a 10 super group _____ bandwidth

240kHz....confirm

240. In an analog hierarchy to carry 600 voice channels, a 6 Master group _____

2.52Mhz....confirm

241. In an analog hierarchy to carry 3600 voice channels, a Jumbo group _____

16,984MHzconfirm

242. In line discipline after the data transmission, the sending system finishes with an _____ frame.

EOT

243. Data is transmitted using light through a _____ cable.

fiber-optic

244. Data Link layer provides services to the _____ layer.

Network

245. If the Hamming distance between sent and received code word is _____, then it shows that received data is corrupted.

$\neq 0$

246. _____ is responsible for governing node to node communication.

Data Link layer

247. Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of _____.

Metropolitan Area Network

248. In frequency shift keying, _____ remain(s) constant.

Both Peak amplitude and Phase ...confirm

249. In Amplitude shift keying, _____ remain(s) constant.

Both frequency and phase...confirm

250. In Phase shift keying, _____ remain(s) constant.

Both peak amplitude and frequency...confirm

251. Analog signals are refers to be _____

Continuous

252. Baud rate is greater than or equal to the bit rate.

False

253. In statistical time division multiplexing, number of slots in frame are less than _____.

Input lines.....confirm

254. _____ are used to exchange session management and control information between connected devices.

U-frames

255. Terminal, microprocessor, computer, printer or any other device that generates or consumes digital data is known as _____

DTE

256. There are three types of Hubs that exist. Which of the following options correctly describes these three?

Passive, Active, Intelligent

257. Which of the following consist of just one redundant bit per data unit?

Simple Parity Check

258. In the encoding where positive to negative transition is one and vice versa is zero.

None of given option

259. Error correction is more _____ than the error detection.

Difficult

260. If the window size is 63 in go-back N ARQ then what is the range of sequence number?

0 to 63

261. _____ is normally used where speed is priority in data transfer.

Hybrid transmission

262. In Synchronous TDM, Time slots are not same in number as input devices.

False

263. In analog transmission, the base signal is called the_____.

Carrier Signal

264. In _____ a dedicated communication path is established between two stations through the nodes of the network

Circuit switching

265. Which one of the following is a Random-Access protocol?

CDMA/CD

266. A _____ modem is used to connect two DTEs directly.

NULL

267. The flag in PPP is a byte that needs to be _____ whenever it appears in the data section of the frame.

Escaped

268. _____ is an Authentication Protocol, which is used by the PPP to authenticate passwords.

PAP

269. In CRC there is no error if the remainder at the receiver is _____.

Zero

270. VCI address is _____ address in virtual circuit approach.

Global...confirm from net

271. _____ is an Authentication Protocol, which uses two-step process to authenticate user information.

PAP

272. Possession of TOKEN gives the station the right to _____.

Send the data

273. The transmission medium that carries the message is referred to as the _____.

communication channel

274. Latency is made of _____ components.

Four....confirm

275. Sampling means measuring _____ of signal at equal intervals

Amplitudes

276. Which error detection method uses one's complement arithmetic.

Checksum

277. To calculate the data rate for noisy channel _____ formula is used.

Shannon.....confirm

278. To calculate the data rate for noiseless channel _____ formula is used.

Nyquist....confirm

279. Checksum is an error-detection technique that can be applied to a message of _____ length.

Any

280. In _____ each band is reserved for a specific station, and it belongs to the station all the time.

FDMA

281. Flow control is needed to prevent _____.

Overflow of the receiver buffer

282. Quadrature Amplitude Modulation (QAM) is the combination of _____ and _____.

FSK, PSK

283. Like 10 Base 5, 10 Base 2 is a _____ topology LAN.

Bus

284. In the _____ method, the stations in a network are organized in a logical ring.

Token-passing

285. Two or more computers connected so that they can communicate with each other and share information is called a _____.

network

286. The logical connection between the peer layers is _____ connection.

Direct....confirm from net

287. In CSMA/CA, _____ is the amount of time divided into slots.

Contention Window

288. _____ relative measures the strength of two signals.

Decibalconfirm

289. A sine wave is _____.

periodic and continuous

290. There are _____ types of guided media.

3

291. AM signal requires _____ the BW of original signal

Twice

292. CHAP is an Authentication Protocol, which uses _____ process to authenticate user information.

Three-way

293. There are _____ types of serial transmission.

2

294. There are _____ basic categories of multiplexing.

3

295. Congestion Control is a feature of layer(s).

All mentioned

296. Most commonly used connector for twisted pair cable is _____.

RJ-45....confirm from net

297. _____ does not happen until all the data is transferred and all of the resources are freed.

Switching

298. A local telephone network use _____ network.

Circuit switched...confirm from net

299. Microwaves are _____.

Unidirectionalconfrim

300. _____ have established standards for using these signals for communication between devices such as keyboards, mice, PCs and printers

Infrared waves...confirm from net

301. Datagram approach and virtual circuit approach are two popular approaches which lead us to _____.

Circuit switching....confirm

302. In circuit switching when two end devices 'A' and 'B' connects with each other they send _____ request.

Acceptance

303. In optical fiber a glass or plastic core is surrounded by a _____.

Cladding...confirm

304. _____ are used for multicast communication, such as radio and television.

Radio waves...confirm

305. DLC in data link layer stands for _____

Data Link control....confirm

306. In virtual Circuit network all packets follow _____ path

Dedicated

307. Multi mode technology has _____ main types for light propagating.

Two

308. If the message is going to pass through a packet-switched network. It can be divided into packets of _____.

Both fix and variable size

309. According to state, fewer than _____ crossbar points can be used at once.

25%

310. _____ and _____ are the two types of addressing in virtual circuit approach.

Local, Global

311. Cable TV networks use _____ cables

Coaxial....confirm from net

312. The size of the packet is determined by the network and _____.

Governing Protocol

313. In circuit switching total delay is combination of connection time. Data transfer time and _____

Connection tear down time

314. Which one of the following is not among the required phases for the actual communication in a circuit-switched network?

Protocol

315. Electromagnetic waves ranging in frequencies between 1 and 300 GHz are called _____.

Microwaves

316. If we increase the data rate of a transmission over noisy channel, fixed noise duration will cause _____ number of bits being impacted by the noise.

Less

317. In case of higher data rate, the number of impacted bits will be _____.

Higher

318. In baseband transmission, a digital signal is transmitted as _____.

Analog signal....confirm from net

319. _____ Cable is used in wavelength division multiplexing.

Fiber optic....confirm from net

320. The _____ technique expands the bandwidth of a signal by replacing each data bit with n bits using a spreading code.

DSSS...confirm from net

321. When the bandwidth of a link is greater than combined bandwidths of the

Frequency division....confirm from net

322. According to the Fourier analysis, frequencies obtained after decomposition of no period.

Continuous.....confirm from net

323. Router is a network device which operates on the __ layer of the TCP/IP protocol.

Network...confirm from net

324. Uni polar, polar and bipolar are the types of _____

NRZ-I....confirm from net

325. According to stats, lower than _____ crossbar point can be used at once .

25%....confirm from net

326. In _____ TDM, slots are dynamically allocated to the connected stations to improve bandwidth efficiency.

Statistical...confirm from net

327. In star based network comprising of four computers and on switch, total number of cables needed will be _____

4...confirm from net

328. The conversion of analog sample of the signal into digital form is called _____ process.

Sampling and quantizing...confirm from net

329. On a communication link, different channels are separated by unused strips of bandwidth called as _____.

Guard band

330. _____ is an example of host in a network.

Computer...confirm from net

331. In _____ TDM, each input connection has an allotment in the output even if it is not sending data.

Synchronous ...confirm

332. _____ coding scheme are useful for LAN but not for long ranges

Biphase...confirm

333. In _____ signal changes its shape or form.

Distortion...confirm

334. _____ Protocol suite is being used by the modern internet communication.

TCP/IP...confirm from net

335. in NRZ-1 the signal is inverted if _____ is encountered.

1...confirm from net

336. Simultaneous transmission of multiple signal across a single data link is called _____

Multiplexing..confirm from net

337. Two stations are connected via a secure link in which messages are encrypted from source to the destination message is called _____

Protocol

Which one of following is a scrambling coding technique?

- B8ZS

According to the Fourier analysis, frequencies obtained after decomposition of digital signals are_____.

Discrete

_____ is the heart of blocking code.

- Coding

Wireless transmission can be divided into_____ broad groups.

Three...confirm from net

Wiring used for transmission modes depends upon_____.

- Latency

If a digital transmission system is sending five bits in every half a second, the bit-rate of the system.

- 10 bps...confirm

If the message going to pass through a packet-switched network, it can be divided into packets of_____.

- Both fix and Variable size

In an analogue hierarchy to carry voice channels, a group can carry_____ voice channels.

12

In optical fiber a glass or plastic core is surrounded by a_____.

- Cladding

In transmission impairments _____ noise occurs from other source like

Induced...confirm

_____ is a digital process that allows connections to share the high bandwidth of a link.

- TDM

Bandwidth in hertz in the range of_____ contained in a composite signal.

- Frequencies

LAN stands for_____.

• Local Area Network
Demultiplexer is a _____ device.

- One to many

Mr. Asif while sitting in Lahore is linking with his friend in Dubai through Skype is a _____.

• Wide Area Network
In _____ modulation, frequency and amplitude remain constant.

• Amplitude
In TDM based digital hierarchy used by the Telephone companies, DS-1 is a single digital channel of _____.

- 1.544 Mbps...confirm

In TDM based digital hierarchy used by the Telephone companies, DS-2 is a single digital channel of _____.

6.312 Mbps...confirm

In TDM based digital hierarchy used by the Telephone companies, DS-3 is a single digital channel of _____.

44.376MBps...confirm

In TDM based digital hierarchy used by the Telephone companies, DS-4 is a single digital channel of _____.

274.176 MBps.....confirm

The bandwidth-delay product defines the number of bits that can fill the _____.

- Link

If a digital-signals has four levels, then we need _____ bits to represent each level.

- 2

Routing is a function of _____ layer.

- Network

_____ relatively measures the strength of two signals.

- Decibel

Taking off the header from the message is called _____.

- Encryption

In frequency domain plot, which value is replaced with frequency?

- Wavelength

_____ is the process of converting binary data to a digital signal.

- Line coding

Low pass channel with _____ bandwidth is not real and is used for theoretical modeling.

- Narrow or wide...both are confirm

Unit of phase is _____.

- Degree

BFSK stands for _____.

- Binary Frequency Sift Key

_____ is NOT an example of connecting device.

- TCP

Coaxial cables are categorized by their _____ ratings.

Radio Government (RG)....confirm

Which one of the following is not among the required phases for the actual communication in a circuit switched network

- Protocol

A common bipolar encoding scheme is called _____.

- AMI

A telephone line analog signal has got the bandwidth of _____.

4 kHz...confirm

In _____ systems, resources are allocated on demand.

- Circuit switching

A fiber-optic cable transmit signals in the form of _____.

- Light

Composite wave can be used to send _____.

- Frequency

Routing is the function of _____ layer.

- Network

A virtual-circuit network operates on _____ layer.

- Datalink

In TCP/IP model, the Physical Layer exchanges data in the form of _____.

- Bits...confirm

_____ signal completes certain pattern in a specific amount of time.

- Periodic

If a digital signal has four levels, the we need _____ bits to represent.

- 2

We can have combination of _____ possible codes with a 5-bit code.

- 32

There are _____ categories of multiplexing.

- 3

Top most organization that provides support for the Internet Standard process is called _____.

- IETF

If the message is going to pass through a packet-switched network, it can be divided into packets of _____.

- Both-fix and variable size

_____ data moves faster and timing errors are less frequent because the transmitter and receiver time is synced.

Synchronous...confirm

In frequency modulation, the frequency of the oscillator changes according to the _____.

- Input Voltage

Binary Amplitude Shift Keying also called as:

- On-Off Keying

Bi-polar uses _____ voltage levels.

Two

In serial data transmission _____ data transmission mode can be cheap but slower.

- Asynchronous

In star-based network comprising of four computers and one switch, total number of cable needed will be _____.

- 4

Parabolic antenna is used for _____ communication.

- Microwave...confirm

In circuit switching total Delay is combination of Connection Time and _____.

- Connection tear down time

Datagram switching is done on _____.

- Network Layer

Token Ring was devised by _____.

- IBM...confirm

In TDM based digital hierarchy used by the Telephone companies, DS-0 is a digital channel of _____.

- 64 kbps...confirm

_____ can be published using Request for Comments (RFCs).

- Internet Draft....confirm

In transmission impairments _____ noise occurs from the random motion of electrons in a wire.

- Thermal noise...confirm

Which one of the following is among the required phases for the actual communication in a circuit switched network?

- Data transfer

According to the Fourier analysis, frequencies obtained after decomposition of non-periodic digital signals are _____.

- Bit length

The most common type of connector used by the coaxial cable is _____.

- BNC

When data is sent or received using _____ the data bits are organized in a specific order, since they can only be sent or received one after another.

Serial data transmission

_____ have established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers.

- Infrared waves

In case of Frequency shift key, the difference between two frequency is represented as _____.

- 2 delta....confirm
_____ category of coaxial cable is used for thin Ethernet.

- RG-58

Cellular telephone uses _____ waves for communication.

Radio waves

_____ layer is responsible for the creating of the datagrams.

- Network

In virtual circuit approach when a frame enters a switch and when it leaves a switch its VCI always _____.

- Remain the same

According to stats, fewer than _____ crossbar points can be used at once.

- 25%

Data format used at Transport layer of TCP/IP protocol suit is called _____.

- Packet

A virtual circuit network operates on _____ layer.

- Application

If 32 bits are sent in two seconds then the bitrate for that signal is _____.

- 64 bps
- _____ normally used for long-distance data transfer.
- Both parallel and Serial data transmission

_____ is a block coding method.

- 8B\10B

TCP\IP was chosen to be the official protocol of internet _____.

- 1983

To measure the data rate _____ theoretically formula(s) were developed.

- Four

In FSK, Baud rate is less than or equal to _____ rate.

Bit rate.....confirm

_____ signal is represented by the discrete values.

- Digital

Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are called _____.

- Radio waves

Suppose a signal is amplitude then the value of decibel will be _____.

- Positive

_____ technique does not follow analog conversion.

- QAM

In Pulse Code Modulation (PCM), the sampling is dependent on _____.

- Amplitude

_____ is the sub type of Time Division Multiplexing Technique.

- Statistical TDM

Modulation of analog signal is needed if the medium is _____ in nature.

- Bandpass

Quadrature Amplitude Modulation (QAM) is the combination of _____ and _____.

ASK-FSK

There _____ components of data communication system.

5...confirm

QPSK stands for _____.

- Quadrature phase shift key

To calculate the data rate for noisy channel _____ formula is used.

- Shannon

Binary Amplitude Shift Keying(ASK) is also called as:

- On-Off Keying....confirm

Some manufacturers provide a special port called the _____.

- IrDA

To calculate the data rate for noiseless channel _____ formula is used.

- Nyquist

Two PCs and one printer are connected in a network within a room, it is an example of _____.

- LAN...confirm from net

Analog the following cables given, _____ cable provides highest bandwidth and less attenuation.

- Fiber optical

To calculate the data for noisy channel _____ formula is used.

- Shannon

In circuit switched networks we have low efficiency but minimal _____.

- Delay

Polar encoding scheme uses _____ voltage level.

- 2....confirm

TCP/IP protocol suit consists of _____ layers.

- 5.....confirm

Bandwidth in hertz is the range of _____ contained in a composite signal.

- Frequencies....confirm

_____ is an example of a host in a network.

- Computer

Set of rules to be followed for effective communication is called _____.

- Protocol

_____ is NOT an example of a connecting device.

- TCP

When an antenna transmits radio waves, they are propagated in _____.

- Only up and down direction

Which one of the following is not a bipolar encoding level?

- One

_____ and _____ are the two types of addressing in virtual circuit approach.

- Local, Global

In transmission implements _____ noise occurs from the effect of one wire on the other

- Cross talk...confirm

In computer network the stations are connected to each other in such a way that each station is connected to every other station through dedicated links. This makes _____.

- Bus

In scrambling coding scheme, the number of pulses replacing the bits will be _____?

- Equal to pulses.....confirm

In the context of analog to analog signal conversation technique. "PM" Stands for _____.

- Phase modulation

_____ relatively measurement the strength of two signals.

- Decibel

Datagram approach and circuit approach are two popular approaches which lead us to _____.

- Packet switching

In virtual circuit approach when frame enters a switch and when it leaves a switch its VCI always _____.

- Remain the Same

_____ means loss of energy in signal.

Attenuation

A local telephone network use _____ network.

Circuit switched

In Amplitude Shift Keying. _____ of the signal is/are changed.

- Amplitude

In frequency domain plot, which value is plot on Y-axis?

- Amplitude....confirm

In pulse code modulation the term sampling can also be referred as _____.

- Pulse Amplitude Modulation

In analog transmission of digital data, the required bandwidth is always proportional to the signal rate expect in _____ a digital to analog conversion technique.

- ASK

WDM stands for _____.

- Wavelength Division Multiplexing

_____ does not follows analog to analog conversion.

- QAM

On a communication link, different channels are separates by unused strips of bandwidth called as _____.

Guard band

- _____ layer converts frames coming from Data Link Layer into bits and sends

Them on the transmission medium.

- Physical

Wavelength binds the _____ of a simple sine wave to the propagation speed of the medium.

Period of the frequency

book page 61

The bandwidth delay product define the number of bits that can fill the _____.

Network

The size of the packet determined by the network and _____.

Governing protocol
To improve the performance of line coding _____ coding was.....?

- Block coding...confirm

Which one of the following is not a sampling technique?

- Neutral sampling

_____ is normally used where speed is priority in data transfer.

Hybrid transmission

In a computer network, five stations are connected to each other in such a way that each station is connected to every other station through dedicated links. This makes _____ topology.

Mesh...confirm

Some manufacturer provide a special port called the _____ port that allows a wireless keyboard to communicate with a PC.

- IrDA

Current technology supports _____ modes for propagation of light.

- Two

We quantize the sampling output into certain levels based on range of _____ and required accuracy.

- Amplitude....confirm

_____ is the Nyquist bit rate formula for noiseless channel.

- $\text{BitRate} = 2 * \text{Bandwidth} * \log_2 L$...confirm

A frequency is called _____, if the rate of change in sine wave Is Instantaneous.

- Infinite frequency book page 59

Which multiplexing technique involves signals composed of light beams?

- WDM....confirm

_____ is a multiplexing technique which shifts each signal to a different carrier frequency.

- FDM...confirm

Entire band in United States is regulated by _____ authorities.

- FCC





CS601-Data Communication
Latest Solved subjective from Midterm Papers

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Lectures 1-22

Mc100401285

moaaz.pk@gmail.com

Moaaz Siddiq

Latest subjectives

MIDTERM EXAMINATION
Spring 2010

Question No: 21 (Marks: 2)

How many types of Bipolar encoding scheme exist?

Answer: (PAGE NO.76)

Bipolar uses three voltage levels.

There are three types of bipolar encoding:

1. AMI (Alternate Mark Inversion)
2. B8ZS
3. HDB3

Question No: 22 (Marks: 2)

Give methods of Analog to Analog conversion.

Answer: (PAGE 95)

Analog Conversion:

Representation of analog information by an analog signal is Analog –to- analog conversion.

Question No: 23 (Marks: 3)

What is frequency spectrum?

Answer: (PAGE 66)

Frequency spectrum is the collection of all the components frequencies it contains.

Question No: 24 (Marks: 3)

How bandwidth of a Modem is measured?

Answer: (PAGE 112)

Bandwidth of a modem is not measured in bytes, it is measured in bits. Modem Speed –ASK (full-duplex)

Effective bandwidth of a telephone line is 2400 Hz, the maximum baud rate is also 2400.

Baud rate and bit rate are equal for ASK, so maximum bit rate is also 2400 bps.

Question No: 25 (Marks: 5)

What do you understand by the term DCE? Explain DCE. [5]

Answer: (PAGE 101)

Any device that transmits or receives signal through network is DCE.

It is any functional unit that transmits or receives data in the form of an analog or digital signal through a network.

At the physical layer DCE takes data generated by DTE, converts it to the signal.

Question No: 26 (Marks: 5)

Describe different types of NRZ encoding scheme?

Answer:

In NRZ encoding the level of signal is either positive or negative. NRZ is classified into 2 types:

NRZ-L NRZ-I

NRZ-L:

Level of the signal depends on the type of bit it represents.

Positive voltage usually means the bit is a 1 and a negative voltage usually means the bit is a 0.

NRZ-I:

The inversion of the level represents a 1 bit. A bit 0 is represented by no change.

NRZ-I is superior to NRZ-L.(PAGE 73)

MIDTERM EXAMINATION

Spring 2010

Question No: 21 (Marks: 2)

Define Sampling Rate.

Answer: (page82)

The accuracy of any digital reproduction of an analog signal depends upon the o. of samples taken

Question No: 22 (Marks: 2)

Write down the types of digital to analog modulation.

Answer: (PAGE 85)

Types of digital to analog modulation

ASK

FSK

PSK

QAM

Question No: 23 (Marks: 3)

What is periodic signal?

Answer: (page58)

A signal is called Periodic if it completes a pattern within a measurable time frame called a Period and then repeats that pattern over identical subsequent Periods

Question No: 24 (Marks: 3)

How bandwidth of a Modem is measured?

Answer: (page111)

Bandwidth of a modem is not measured in bytes, it is measured in bits.

Modem Speed –ASK (full-duplex)

Effective bandwidth of a telephone line is 2400 Hz, the maximum buad rate is also 2400.

Baud rate and bit rate are equal for ASK, so maximum bit rate is also 2400 bps.

Question No: 25 (Marks: 5)

Briefly describe Return to Zero (RZ) encoding scheme?

Answer: (page75)

- Any time, data contains long strings of 1's or 0's, Rx can loose its timing
- In unipolar, we have seen a good solution is to send a separate timing signal but this solution is both expensive and full of error
- A better solution is to somehow include synch in encoded signal somewhat similar to what we did in NRZ-I but it should work for both strings of 0 & 1
- One solution is RZ encoding which uses 3 values : Positive, Negative and Zero
- Signal changes not b/w bits but during each bit
- Like NRZ-L , +ve voltage means 1 and a -ve voltage means 0, but unlike NRZ- L, half way through each bit interval, the signal returns to zero
- A 1 bit is represented by positive to zero and a 0 is represented by negative to zero transition
- The only problem with RZ encoding is that it requires two signal changes to encode one bit and therefore occupies more BANDWIDTH
- But of the 3 alternatives we have discussed, it is most effective value.

Question No: 26 (Marks: 5)

Compare Phase Modulation (PM) with Frequency Modulation (FM)?

Answer: (page 96-97)

Phase modulation (PM)

- Simpler hardware requirements
- Phase is modulated with the amplitude
- Amplitude & Frequency of the carrier signal remain constant

Frequency Modulation (FM)

- Frequency of carrier signal is changed according to the amplitude of modulating signal
- Amplitude and Phase of the carrier signal remain constant

FM Bandwidth

Bandwidth of FM signal (modulated signal) = 10 * bandwidth of modulating signal
Significant spectrum of FM audio = 15 KHz

Minimum 150 KHz bandwidth

MIDTERM EXAMINATION Spring 2010

Question No: 21 (Marks: 2)

Write down three characteristics of Sine Wave.

Answer : (page no.59)

- ❖ Amplitude
- ❖ Period/Frequency
- ❖ Phase

Question No: 22 (Marks: 2)

A Sine wave has a frequency of 6 Hz. What is its period?

Answer: (page 61)

$F = 6$, $t = ?$

$T = 1/f = 1/6 = 0.17 \text{ sec}$

Question No: 23 (Marks: 3)

Give similarities between 4 QAM & 8 QAM.

Answer: (page 92)

In both case no. of amplitude shifts is more than the no. of phase shifts

Because amplitude changes are susceptible to Noise , number of phase shifts used by QAM is always larger than the amplitude shifts

Question No: 24 (Marks: 3)

What is digital signal? Give an example

Answer: (page 57)

A digital signal is discrete. It can have only a limited number of defined values, often as simple as 1s and 0s
The transition of a digital signal from value to value is instantaneous like a light being switched ON and OFF.

Question No: 25 (Marks: 5)

Difference between Phase Modulation(PM) with Frequency Modulation(FM)?

Answer: (page 96-97)

Phase modulation (PM)

- Simpler hardware requirements
- Phase is modulated with the amplitude
- Amplitude & Frequency of the carrier signal remain constant

Frequency Modulation (FM)

- Frequency of carrier signal is changed according to the amplitude of modulating signal
- Amplitude and Phase of the carrier signal remain constant

FM Bandwidth

Bandwidth of FM signal (modulated signal) = $10 \times$ bandwidth of modulating signal
Significant spectrum of FM audio = 15 KHz

Minimum 150 KHz bandwidth

Question No: 26 (Marks: 5)

Write down the criteria or technique that we follow for assigning integral values to sampled signals and also its conversion into binary format. Explain it?

**MIDTERM EXAMINATION
Spring 2010**

Question No: 21 (Marks: 2)

How does noise affect FSK?

Answer: (P170)

The duration of the noise is normally longer than the duration of a bit which means that when noise affects data, it affects a set of bits.

Question No: 22 (Marks: 3)

Define Bit interval and Bit rate.

Answer: (P68)

Bit Interval and Bit Rate

Most digital signals are aperiodic and thus Period and Frequency are not the appropriate terms to describe them

Bit Interval (seconds)

Time required to send one single bit

Bit Rate (bps)

Number of bits sent per second

Question No: 23 (Marks: 5)

Write down the names of layers present in TCP/IP Protocol Suite?

Answer: (P54)

Physical (physical standards)

Data Link (N/w Interface)

Network (Interconnectivity)

Transport (Transport Functions)

Application (Session, Pres, app of OSI)

Question No: 24 (Marks: 10)

The data points of a constellation are at (4,0) and (6,0). Draw the constellation. Show the amplitude and phase for each point. Is the modulation ASK, PSK, or QAM? How many bits per baud can one send with this constellation? [10]

Answer:

Hint: 2 amplitudes (4,6) , ASK , bit/baud

**MIDTERM EXAMINATION
Spring 2009**

Question No: 21 (Marks: 2)

What is simplex mode of transmission?

Answer:

Communication is Unidirectional

Only one of the two stations can transmit

Other can only receive (P34)

Question No: 22 (Marks: 3)

What is HDB3.

Answer:

Alteration of AMI adopted in Europe and Japan

Introduces changes into AMI, every time four consecutive zeros are encountered instead of waiting for eight zeros as in the case of B8ZS. (P78)

Question No: 23 (Marks: 5)

Write a note on Full duplex mode of transmission.

Answer:

Both stations can transmit and receive simultaneously

Two way street with traffic flowing in both directions at the same time

Signals traveling in either direction share the capacity of the link

The sharing can take place in two ways:

Either the link must contain two physically separate transmission paths:

- ❖ One for sending and
- ❖ One for receiving

Question No: 24 (Marks: 10)

In your view which one is best either B8ZS or HDB3 and why? Write down salient features of both? What is meant by polarity of 1 since the last substitution?

MIDTERM EXAMINATION Spring 2009

Question No: 21 (Marks: 2)

What is purpose of distributed processing?

Answer:

Multiple computers working on a problem can solve a problem faster than a computer working alone.

(P13)

Question No: 22 (Marks: 3)

What is HDB3?

Answer:

Alteration of AMI adopted in Europe and Japan

Introduces changes into AMI, every time four consecutive zeros are encountered instead of waiting for eight zeros as in the case of B8ZS. **(P78)**

Question No: 23 (Marks: 5)

Differentiate between the following terms. [10 marks]

(a) Syntax and Semantics of protocol.

Answer:

Syntax:

Communication between peer entities (P21)

Semantics:

Client/server (P21)

(b) Network management and exchange management.

Answer:

Exchange Management

Error Detection and Correction (P17)

Network management (P17)

Security

(c) Monolithic and structured.

Answer: (P21)

Monolithic

Communications is a complex task

To complex for single unit

Structured

Structured design breaks down problem into smaller units

Layered structure

Question No: 24 (Marks: 10)

Differentiate between the following terms. **a)**

Frequency spectrum and bandwidth.

Answer: (P66)

Frequency Spectrum / Bandwidth

Frequency Spectrum: of a signal is the collection of all the component frequencies it contains

It is shown using a Frequency domain graph

Bandwidth: of a signal is the width of the frequency spectrum

In other words, Bandwidth refers to the range of the component frequencies and Frequency Spectrum refers to the elements within that range

(b) Bandwidth and throughput.

Answer: (P144)

Measurement of how fast data can pass through a point ○ In other words, if we consider any point in the TX Medium as a wall through which bits pass, then throughput is the number of bits that can pass this wall in second

(c) Bit interval and bit rate.

Answer: (P68)

Bit Interval and Bit Rate

Most digital signals are aperiodic and thus Period and Frequency are not the appropriate terms to describe them

Bit Interval (seconds)

Time required sending one single bit

Bit Rate (bps)

Number of bits sent per second

moaaz.pk@gmail.com